

The vaginal microbiome of pregnant people living with HIV on antiretroviral therapy in the Democratic Republic of Congo: A Pilot Study and Global Meta-Analysis

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Figure scripts for the paper, “The vaginal microbiome of pregnant people living with HIV on antiretroviral therapy in the Democratic Republic of Congo: A Pilot Study and Global Meta-Analysis”	

```
if (!require(pacman)){
  install.packages("pacman")
}
```

```
## Loading required package: pacman
```

```
if (!require("veganEx")) remotes::install_github("vegandevs/veganEx")
```

```
## Loading required package: veganEx
```

```
library(pacman)
pacman::p_load(
  tidyverse,
  phyloseq,
  phyloseqCompanion,
  ComplexHeatmap,
  microViz,
  phylosmith,
  circlize,
  broom,
  microbiome,
  data.table,
  MiscMetabar,
  foreach,
  modeest,
  lmerTest,
  parallel,
  ALDEx2,
  ANCOMBC,
  ComplexUpset,
  rstatix,
  ggpubr,
  dtplyr,
  veganEx, ggh4x,
  install = F
```

```
)

#source your functions
source("functions_drc_cvmb.R")
```

Figure 1

B. Heatmap of RCLR abundances in the CQI-PMTCT cohort

```
ps_congo_filt_spp_rclr <- readRDS("data/ps_congo_filt_spp_rclr.rds")
#change CST column in metadata table
meta_congo <- sample.data.frame(ps_congo_filt_spp_rclr)
meta_congo <- meta_congo %>% mutate(CST = if_else(str_starts(CST, "IV-"), "IV", CST))
sample_data(ps_congo_filt_spp_rclr) <- meta_congo
#calculate the top30 most abundance species
top30 <- tax_top(ps_congo_filt_spp_rclr, n = 30)
#subset phyloseq
ps2 <- subset_taxa(ps_congo_filt_spp_rclr, rownames(tax_table(ps_congo_filt_spp_rclr)) %in% top30)

#set colours
CST_col = c("II" = "#F5530D", "III" = "#16FBCD", "IV" = "navy", "V" = "#8D324D")

Viral_load_enrol_detected_col = c("1" = "#712679", "0" = "#2E7926")
ART_cols = c("TDF_3TC_EFV" = "#E7AC18", "TDF_3TC_TLD" = "#1853E7", "TDF_3TC" = "#D02AD5")

## prepare matrix for heatmap
taxa <- tax_table(ps2) %>% as.data.frame() %>% rownames_to_column("asvid") %>% mutate(Taxon = case_when(
  !is.na(Species) ~ paste(sub("^g__", "", Genus), sub("^s__", "", Species), sep = " "),
  !is.na(Genus) ~ (Genus),
  !is.na(Family) ~ Family,
  !is.na(Order) ~ Order,
  !is.na(Class) ~ Class,
  !is.na(Phylum) ~ Phylum,
  TRUE ~ "Unknown"
)) %>% select(-Kingdom, -Species, -Genus, -Family, -Order, -Class, -Phylum)

abundance_df <- phyloseq::otu_table(ps2) %>% t() %>%
  as.data.frame() %>%
  rownames_to_column("asvid") %>%
  gather(key = "sampleID", value = "Abundance", -asvid) %>%
  full_join(taxa, by = "asvid") %>%
  left_join(meta_congo %>% select(sampleID, CST, Viral_load_enrol_detected, ART_regimen), by = "sampleID")

top_taxa_long <- abundance_df %>%
  group_by(sampleID, Taxon) %>% summarise(Abundance = sum(Abundance)) %>%
  mutate(Taxon = na_if(Taxon, ""))

## 'summarise()' has grouped output by 'sampleID'. You can override using the
## '.groups' argument.
```

```

top_taxa_unique <- abundance_df %>% filter(Taxon %in% top_taxa_long$Taxon) %>% distinct(Taxon)

abundance_plot <- top_taxa_long %>% filter(Taxon %in% top_taxa_unique$Taxon) %>%
  pivot_wider(names_from = sampleID, values_from = Abundance) %>%
  column_to_rownames("Taxon")

#set up the annotations for the heatmap
abund_plot_annotations <- meta_congo %>% select(CST, Viral_load_enrol_detected, ART_regimen)

abund_plot_annotations <- abund_plot_annotations %>% mutate(across(where(is.integer), as.character))

Annotation_cols <- list(
  Viral_load_enrol_detected = c("1" = "#712679", "0" = "#2E7926", "NA" = "white"),
  ART_regimen = c("TDF_3TC_EFV" = "#E7AC18", "TDF_3TC_TLD" = "#1853E7", "TDF_3TC" = "#D02AD5", "NA" = "white"),
  CST = c("II" = "#DE7E5E", "III" = "#16FBCD", "IV" = "navy", "V" = "brown"))

#specify cst order
cst_order <- list("II", "III", "IV", "V")
abund_plot_annotations <- abund_plot_annotations[order(factor(abund_plot_annotations$CST, levels = cst_order)), ]
abundance_plot_cstordered <- abundance_plot[, rownames(abund_plot_annotations)]

#order rows by most abundant
rs <- as.vector(rowSums(abundance_plot_cstordered))
set.seed(24)
abundance_plot_cstordered <- abundance_plot_cstordered[order(rs, decreasing = T), , drop = F]

#create matrix for complex heatmap
abundance_plot_mat <- as.matrix(abundance_plot_cstordered)

#plot the heatmap
Heatmap(abundance_plot_mat,
  cluster_rows = F,
  cluster_columns = F,
  show_column_names = F,
  top_annotation = HeatmapAnnotation(df = abund_plot_annotations,
    col = Annotation_cols,
    na_col = "white"),
  rect_gp = gpar(col = "grey"))

```



Figure 2: Ecological statistics of the CVMB in the CQI-PMTCT cohort.

A.CQI-PMTCT cohort:Shannon diversity by CST

```
#load phyloseq object and extract otu_table
ps_congo_filt_spp <- readRDS("data/ps_congo_filt_spp.rds")
otu <- otu_table(ps_congo_filt_spp) %>% as.data.frame() %>% t()

#calculate asymptotic alpha diversity and prepare table for plotting
congo_alpha_div <- get_asymptotic_alpha(otu)
# write_csv(congo_alpha_div, "tables/congo_alpha_div.csv")
#data wrangle
meta_congo <- sample.data.frame(ps_congo_filt_spp)
meta_congo2 <- meta_congo2 %>% mutate(CST = if_else(str_starts(CST, "IV-"), "IV", CST))
congo_alpha_div <- congo_alpha_div %>% rownames_to_column("sampleID")
congo_alpha_div <- congo_alpha_div %>% left_join(meta_congo2, by = "sampleID")

shannon <- congo_alpha_div %>% select(sampleID, `Shannon Entropy`, CST, Viral_load_enrol_detected, ART_
#stats
kruskal.test(shannon$`Shannon Entropy` ~ shannon$CST) #0.002345917

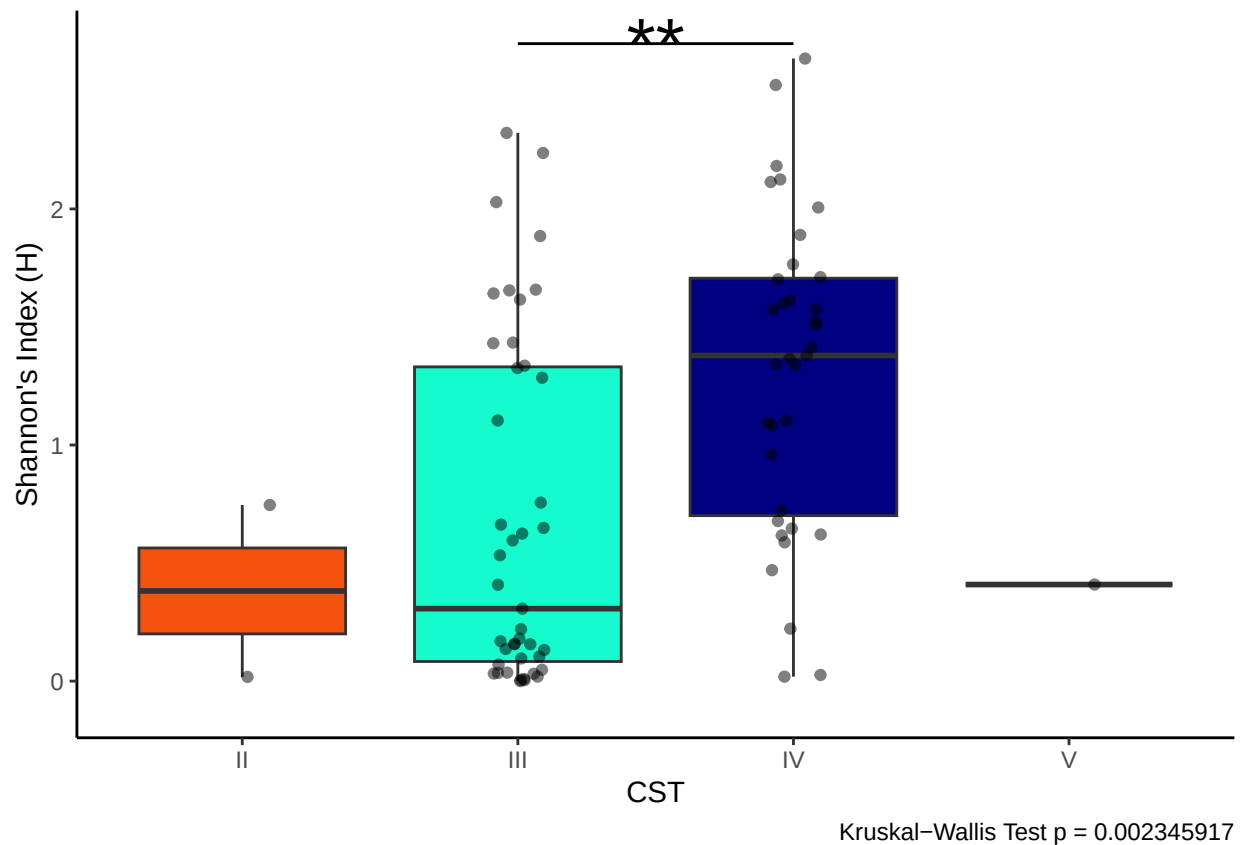
##
## Kruskal-Wallis rank sum test
```

```
##
## data: shannon$`Shannon Entropy` by shannon$CST
## Kruskal-Wallis chi-squared = 14.456, df = 3, p-value = 0.002346
```

```
#posthoc
dunn_test(shannon, `Shannon Entropy` ~ CST, p.adjust.method = "bonferroni")
```

```
## # A tibble: 6 x 9
##   .y.      group1 group2   n1    n2 statistic      p    p.adj p.adj.signif
## * <chr>      <chr> <chr> <int> <int>      <dbl> <dbl> <dbl> <chr>
## 1 Shannon Entr~ II     III      2    43     0.587 5.57e-1 1      ns
## 2 Shannon Entr~ II     IV      2    35     1.71  8.73e-2 0.524  ns
## 3 Shannon Entr~ II     V      2     1     0.174 8.62e-1 1      ns
## 4 Shannon Entr~ III    IV     43    35     3.60  3.24e-4 0.00195 **
## 5 Shannon Entr~ III    V     43     1    -0.210 8.34e-1 1      ns
## 6 Shannon Entr~ IV     V     35     1    -1.02  3.10e-1 1      ns
```

```
shannon <- congo_alpha_div %>% select(sampleID, `Shannon Entropy`, CST, Viral_load_enrol_detected, ART_)
#specify colours
CST_col <- c("II" = "#F5530D", "III" = "#16FBCD", "IV" = "navy", "V" = "#8D324D")
# plot
ggplot(shannon %>% filter(!is.na(CST)), aes(x = CST, y = `Shannon Entropy`, fill = CST)) +
  geom_boxplot() +
  geom_jitter(color = "black", fill = "white", alpha = 0.5, height = 0, width = 0.1) +
  geom_text(aes(x = CST, y = min(shannon$`Shannon Entropy`, na.rm = TRUE) - 0.1, label = ""), vjust = 2) +
  labs(y = "Shannon's Index (H)", x = "CST", caption = paste("Kruskal-Wallis Test p = 0.002345917")) +
  theme(
    legend.position = "none",
    panel.background = element_rect(fill = NA, colour = "white"),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    axis.line = element_line()
  ) +
  scale_fill_manual(values = CST_col) +
  geom_line(data = tibble(x = c(2, 3), y = c(2.7, 2.7)), aes(x = x, y = y), inherit.aes = FALSE) +
  geom_text(data = tibble(x = 2.5, y = 2.7), aes(x = x, y = y, label = "**"), size = 10, inherit.aes = FALSE)
```

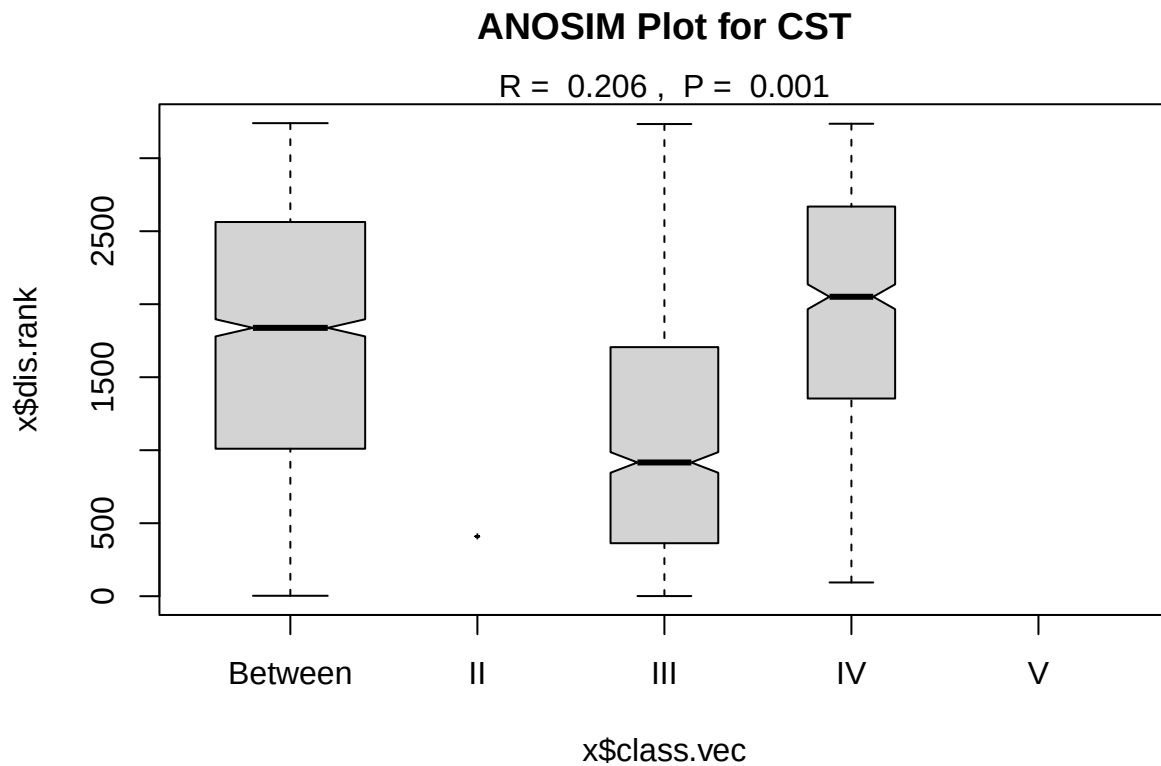


B.CQI-PMTCT cohort:PCA by CST

```
#calculate anosim
perform_anosim(ps_congo_filt_spp_rclr, "CST", "euclidean") #R 0.2058, p=0.001
```

```
## ANOSIM Results:
##
## Call:
## anosim(x = dist_matrix, grouping = grouping_var)
## Dissimilarity: euclidean
##
## ANOSIM statistic R: 0.2058
##      Significance: 0.001
##
## Permutation: free
## Number of permutations: 999
##
## Summary of ANOSIM Results:
##
## Call:
## anosim(x = dist_matrix, grouping = grouping_var)
## Dissimilarity: euclidean
##
```

```
## ANOSIM statistic R: 0.2058
##      Significance: 0.001
##
## Permutation: free
## Number of permutations: 999
##
## Upper quantiles of permutations (null model):
##      90%    95%   97.5%    99%
## 0.0435 0.0571 0.0725 0.0879
##
## Dissimilarity ranks between and within classes:
##      0%  25%  50%  75% 100%   N
## Between 3 1010 1838 2563 3240 1741
## II      410  410  410  410  410    1
## III     1  363  916 1706 3234   903
## IV      94 1354 2051 2669 3236   595
## V       NA   NA   NA   NA   NA    0
```



```
#calculate pairwise_anosim (posthoc)
perform_pairwise_anosim(ps_congo_filt_spp_rclr, "CST", "euclidean", "bonferroni", perm = 999)
```

```
## 'nperm' >= set of all permutations: complete enumeration.
```

```
## Set of permutations < 'minperm'. Generating entire set.
```

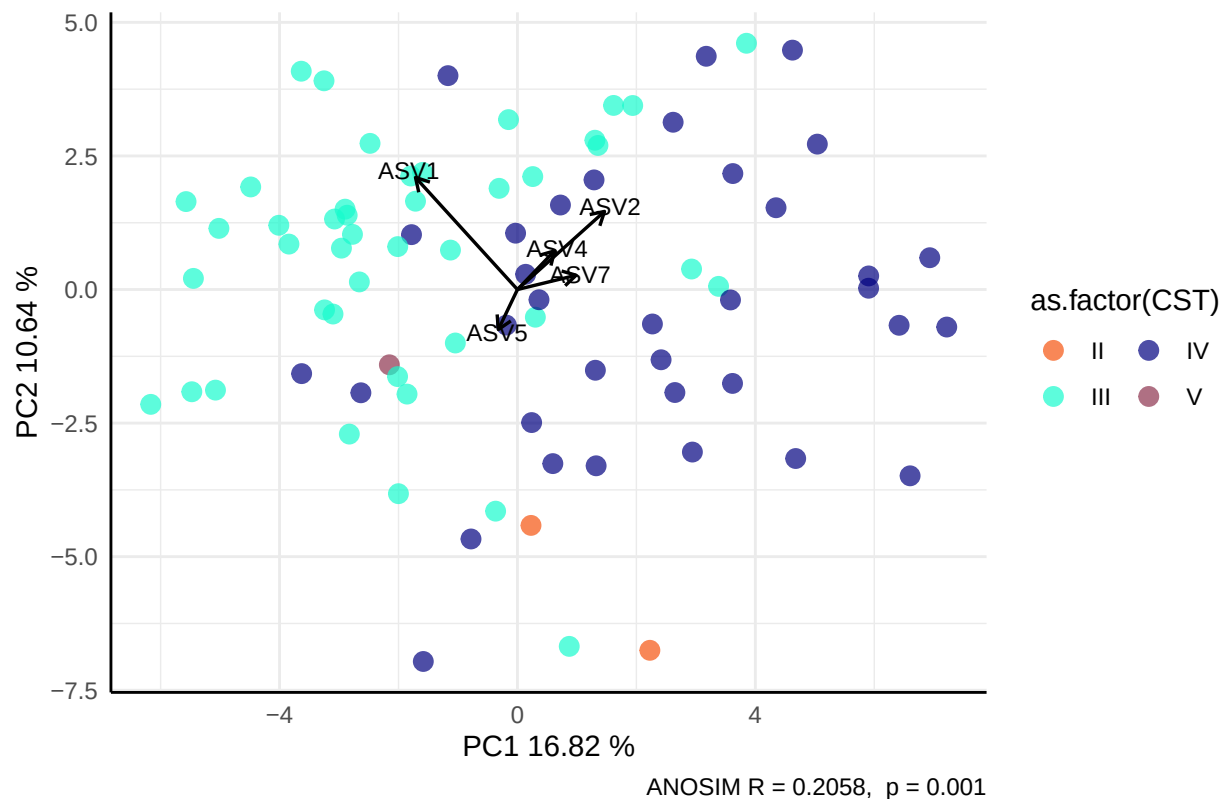


```
## Pairwise Anosim Results:
##      pairs   anosimR   p.value p.adj
## 1 III.vs.IV  0.2188043 0.0010000 0.006
## 2 III.vs.II  0.5497273 0.0240000 0.144
## 3 III.vs.V   -0.4749800 0.9510000 1.000
## 4 IV.vs.II   0.1703739 0.2560000 1.000
## 5 IV.vs.V    -0.5316206 0.9450000 1.000
## 6 II.vs.V    0.0000000 0.6666667 1.000
```

```
#plot
```

```
run_PCA_biplot(ps = ps_congo_filt_spp_rclr,
  batch = "CST",
  scale = FALSE,
  title = "",
  color_palette = CST_col,
  n_loadings = 5) + labs(caption = paste("ANOSIM R = 0.2058, p = 0.001"))
```

```
## Joining with 'by = join_by(sampleID)'
```



```
# ps_congo_filt_spp_rclr %>%
#   microViz::ord_calc(method = "PCA", scale_cc = FALSE) %>%
#   microViz::ord_plot(color = "CST", plot_taxa = 1:5, size = 5) +
#   scale_color_manual(values = CST_col) + labs(caption = paste("ANOSIM R = 0.2058, p = # 0.001"))
```

C.CQI-PMTCT cohort:Shannon diversity by ART regimen

```
#specify colours
ART_cols = c("TDF_3TC_EFV" = "#E7AC18", "TDF_3TC_TLD" = "#1853E7", "TDF_3TC" = "#D02AD5")
```

```
#stats
kruskal.test(shannon$`Shannon Entropy` ~ shannon$ART_regimen) #0.6954
```

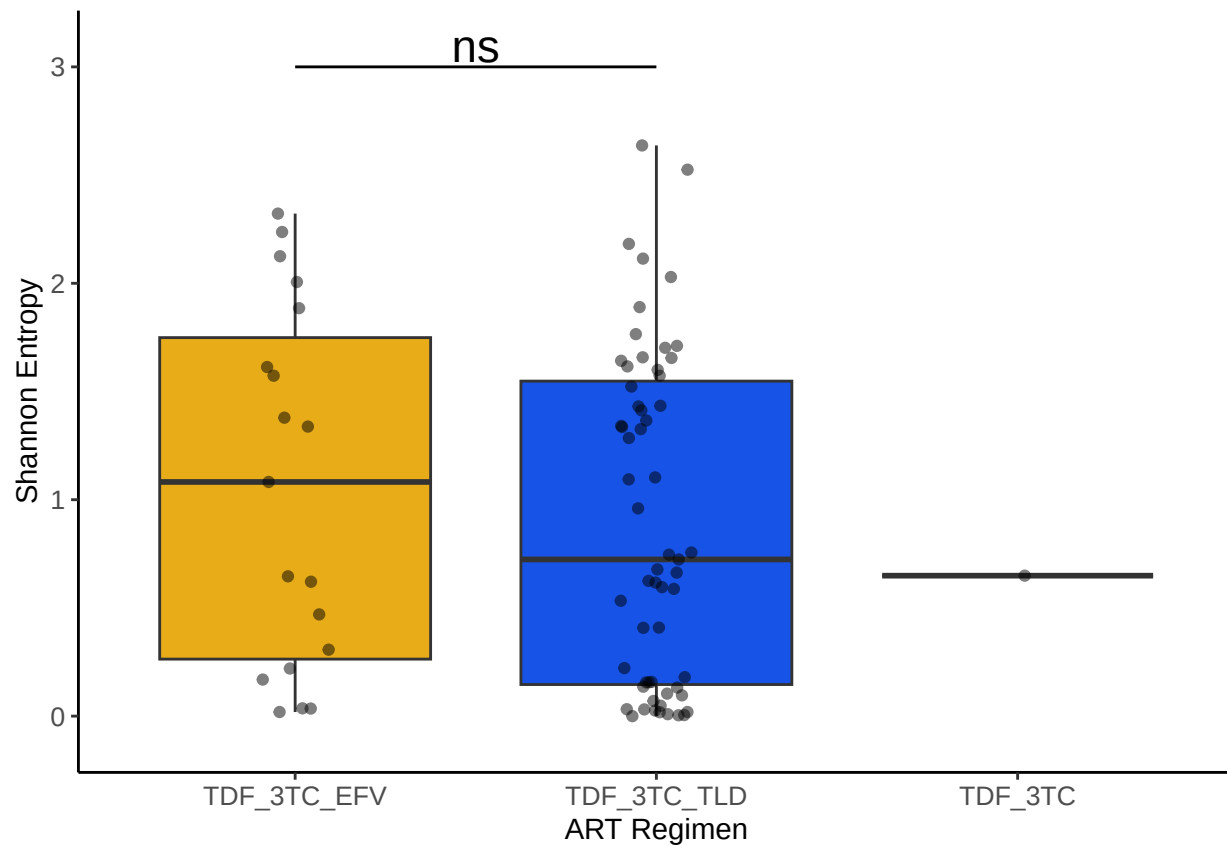
```
##
## Kruskal-Wallis rank sum test
##
## data: shannon$`Shannon Entropy` by shannon$ART_regimen
## Kruskal-Wallis chi-squared = 0.72658, df = 2, p-value = 0.6954
```

```
#posthoc dunns
dunn_test(shannon, `Shannon Entropy` ~ ART_regimen, p.adjust.method = "BH") #0.938
```

```
## # A tibble: 3 x 9
##   .y.      group1  group2    n1    n2 statistic      p p.adj p.adj.signif
## * <chr>      <chr>   <chr>  <int> <int>      <dbl> <dbl> <dbl> <chr>
## 1 Shannon Entropy TDF_3TC TDF_3~      1    19      0.293  0.770 0.938 ns
## 2 Shannon Entropy TDF_3TC TDF_3~      1    59      0.0776 0.938 0.938 ns
## 3 Shannon Entropy TDF_3TC~ TDF_3~     19    59     -0.842  0.400 0.938 ns
```

```
#order
ART_order = c("TDF_3TC_EFV", "TDF_3TC_TLD", "TDF_3TC")
shannon$ART_regimen <- factor(shannon$ART_regimen, levels = ART_order)

ggplot(shannon %>% filter(!is.na(ART_regimen)), aes(ART_regimen, `Shannon Entropy`, fill = ART_regimen))
  geom_boxplot() + geom_jitter(color="black", fill="white", alpha=0.5, height = 0, width = 0.1) +
  geom_text(aes(x = ART_regimen, y = min(shannon$`Shannon Entropy`, na.rm = TRUE) - 0.1, label=""), vjust = -0.5,
    size = 10, color = "black") +
  labs(y = "Shannon Entropy", x = "ART Regimen") +
  theme(legend.position = "none", panel.background = element_rect(fill = NA, colour = "white"), axis.title.x = "",
    axis.title.y = "Shannon Entropy") +
  scale_fill_manual(values = ART_cols) + geom_line(data = tibble(x = c(1,2), y = c(3,3)), aes(x=x, y=y), color = "black",
    size = 2)
```

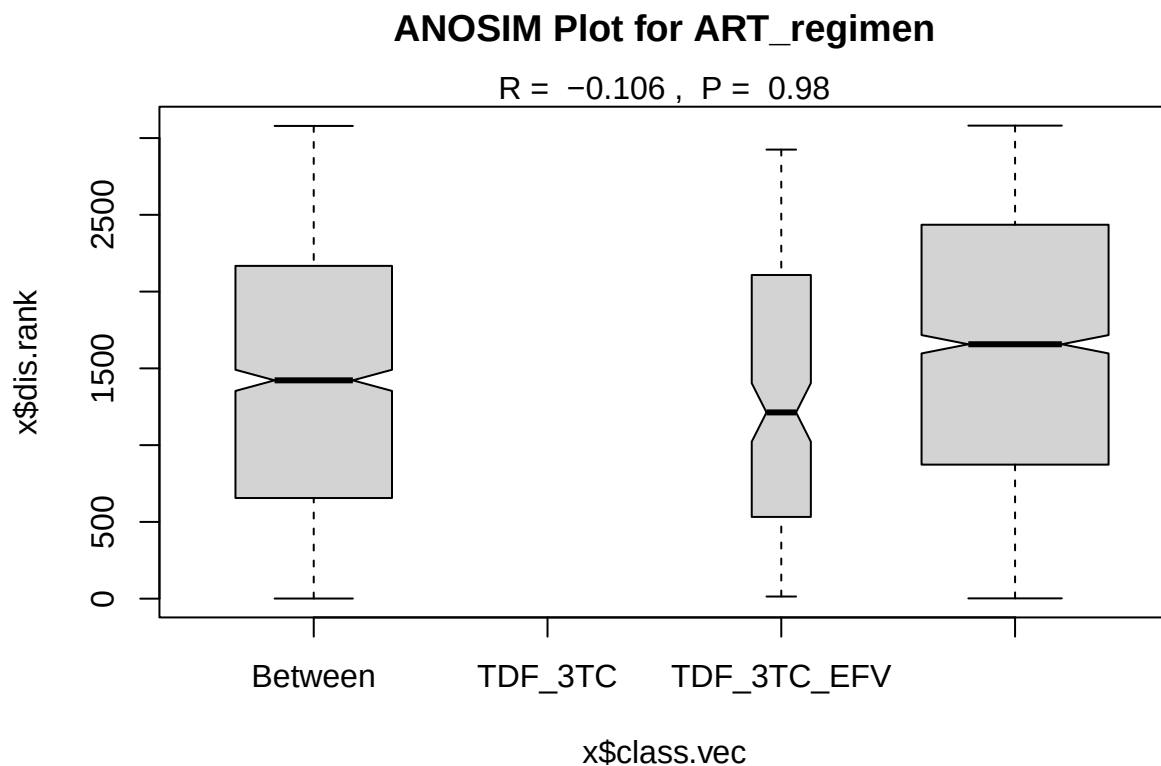


D.CQI-PMTCT cohort: PCA by ART

```
#subset to samples with ART info available
ps_ART_only <- subset_samples(ps_congo_filt_spp, !is.na(ART_regimen))
#transform
ps_ART_only_rclr <- robust_clr_transform2(ps_ART_only)
#stats
perform_anosim(ps_ART_only_rclr, "ART_regimen", "euclidean") # R = -0.07724, p = 0.835
```

```
## ANOSIM Results:
##
## Call:
## anosim(x = dist_matrix, grouping = grouping_var)
## Dissimilarity: euclidean
##
## ANOSIM statistic R: -0.1057
##      Significance: 0.98
##
## Permutation: free
## Number of permutations: 999
##
## Summary of ANOSIM Results:
##
```

```
## Call:
## anosim(x = dist_matrix, grouping = grouping_var)
## Dissimilarity: euclidean
##
## ANOSIM statistic R: -0.1057
##      Significance: 0.98
##
## Permutation: free
## Number of permutations: 999
##
## Upper quantiles of permutations (null model):
##      90%   95%  97.5%   99%
## 0.0698 0.0912 0.1133 0.1379
##
## Dissimilarity ranks between and within classes:
##           0%   25%  50%   75% 100%   N
## Between      1 655.5 1422 2167.5 3079 1199
## TDF_3TC      NA    NA   NA      NA   NA    0
## TDF_3TC_EFV 14 532.5 1213 2108.0 2925  171
## TDF_3TC_TLD  2 873.0 1657 2435.5 3081 1711
```



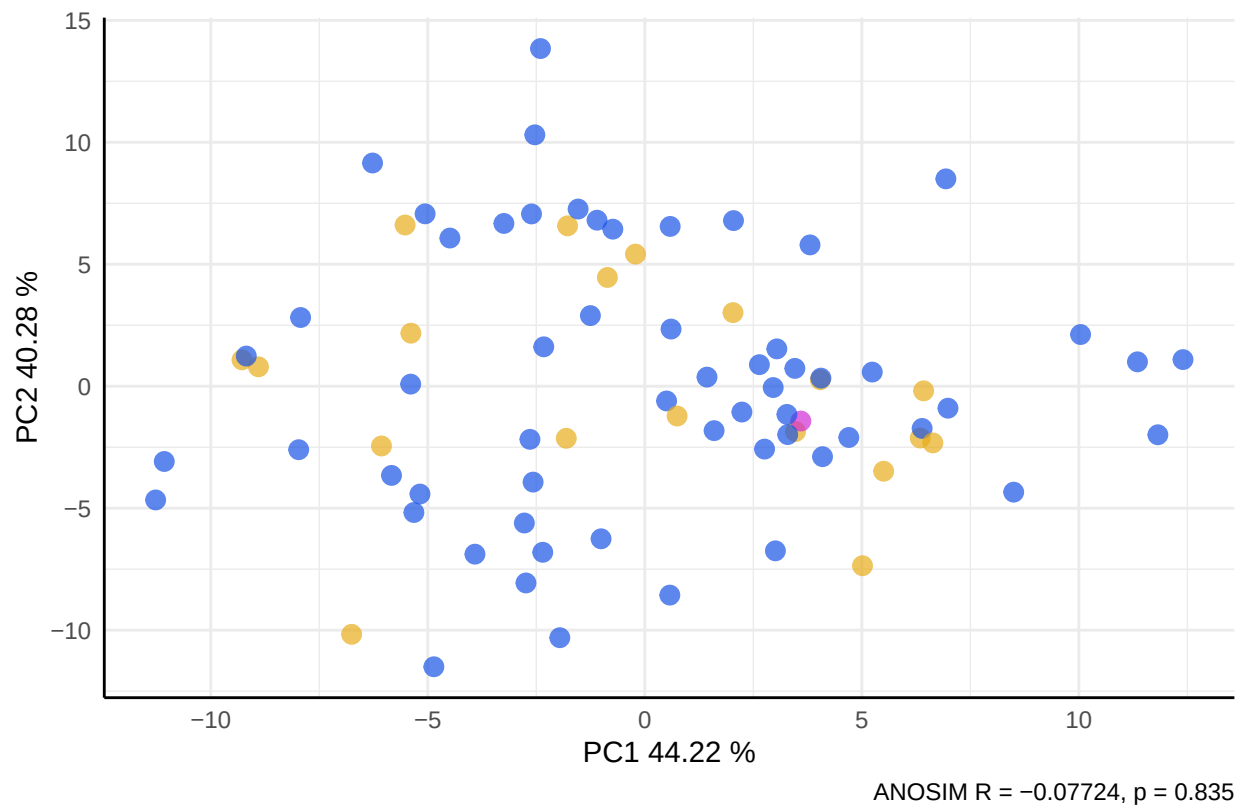
```
#calculate pairwise_anosim (posthoc)
perform_pairwise_anosim(ps_ART_only_rclr, "ART_regimen", "euclidean", "bonferroni", perm = 999)
```

```
## Pairwise Anosim Results:
```

```
##           pairs      anosimR p.value p.adj
## 1 TDF_3TC_EFV.vs.TDF_3TC_TLD -0.0904176  0.960    1
## 2      TDF_3TC_EFV.vs.TDF_3TC -0.2914743  1.000    1
## 3      TDF_3TC_TLD.vs.TDF_3TC -0.3062834  0.877    1
```

```
#plot
run_PCA(ps_ART_only_rclr, "ART_regimen", scale = F, title = "", color_palette = ART_cols) + labs(caption
```

```
## Joining with 'by = join_by(sampleID)'
```

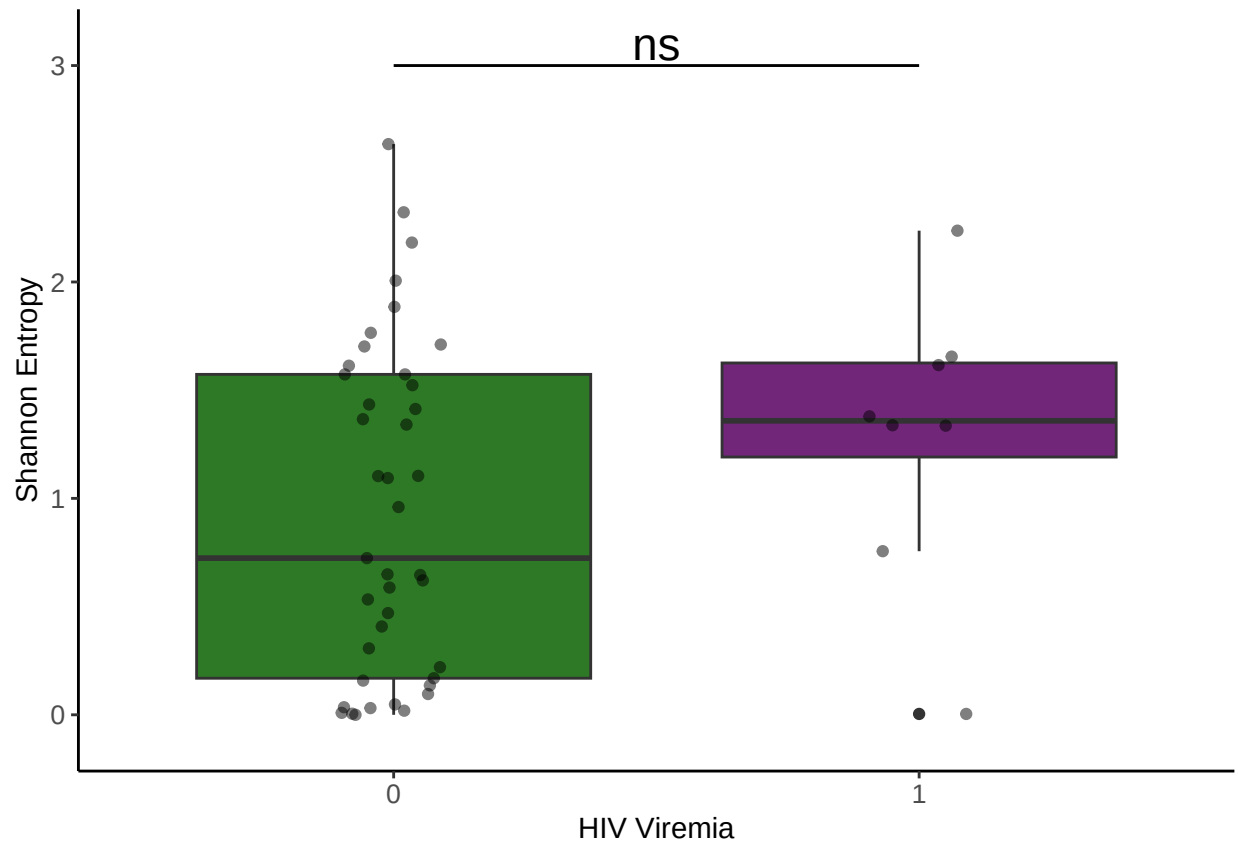


E.CQI-PMTCT cohort: Shannon diversity by viremia

```
#specify colours
VL_col = c("1" = "#712679", "0" = "#2E7926")
#stats
wilcox.test(shannon$`Shannon Entropy` ~ shannon$Viral_load_enrol_detected) #0.2853
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: shannon$`Shannon Entropy` by shannon$Viral_load_enrol_detected
## W = 124, p-value = 0.2853
## alternative hypothesis: true location shift is not equal to 0
```

```
#plot
ggplot(shannon %>% filter(!is.na(Viral_load_enrol_detected)), aes(Viral_load_enrol_detected, `Shannon Entropy`)) +
  geom_boxplot() + geom_jitter(color="black", fill="white", alpha=0.5, height = 0, width = 0.1) +
  geom_text(aes(x = Viral_load_enrol_detected, y = min(shannon$`Shannon Entropy`, na.rm = TRUE) - 0.1, label = "ns"),
  labs(y = "Shannon Entropy", x = "HIV Viremia") +
  theme(legend.position = "none", panel.background = element_rect(fill = NA, colour = "white"), axis.text.x = element_text(angle = 45)) +
  scale_fill_manual(values = VL_col) + geom_line(data = tibble(x = c(1,2), y = c(3,3)), aes(x=x, y=y), linetype="solid", color="black")
```



F.CQI-PMTCT cohort: PCA by viremia

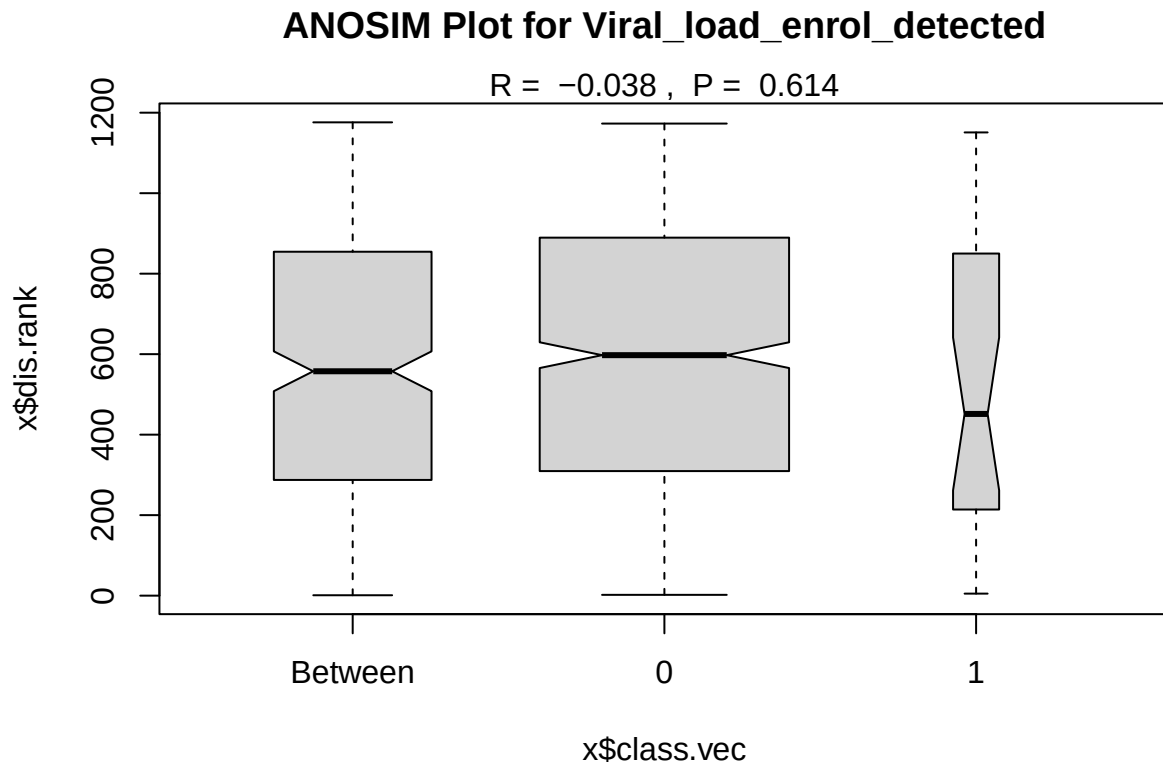
```
VL_col = c("1" = "#712679", "0" = "#2E7926")
#subset to samples with viremia info
ps_VL_only <- subset_samples(ps_congo_filt_spp, !is.na(Viral_load_enrol_detected))
#transform
ps_VL_only_rclr <- robust_clr_transform2(ps_VL_only)
#stats
perform_anosim(ps_VL_only_rclr, "Viral_load_enrol_detected", "euclidean") # R = -0.1062, p = 0.795

## ANOSIM Results:
##
## Call:
## anosim(x = dist_matrix, grouping = grouping_var)
## Dissimilarity: euclidean
```

```

##
## ANOSIM statistic R: -0.03774
##      Significance: 0.614
##
## Permutation: free
## Number of permutations: 999
##
##
## Summary of ANOSIM Results:
##
## Call:
## anosim(x = dist_matrix, grouping = grouping_var)
## Dissimilarity: euclidean
##
## ANOSIM statistic R: -0.03774
##      Significance: 0.614
##
## Permutation: free
## Number of permutations: 999
##
## Upper quantiles of permutations (null model):
##   90%   95% 97.5%   99%
## 0.124 0.160 0.193 0.223
##
## Dissimilarity ranks between and within classes:
##      0%   25%   50%   75% 100%   N
## Between 1 287.75 557.5 852.25 1176 328
## 0       2 309.75 597.5 889.25 1173 820
## 1       5 215.50 451.5 835.50 1151  28

```



```
#calculate pairwise_anosim (posthoc)
perform_pairwise_anosim(ps_VL_only_rclr, "Viral_load_enrol_detected", "euclidean", "bonferroni", perm = 1000)

## Pairwise Anosim Results:
##   pairs      anosimR p.value p.adj
## 1 0.vs.1 -0.03774304  0.636 0.636

#plot
run_PCA(ps_VL_only_rclr, "Viral_load_enrol_detected", scale = F, title = "", color_palette = VL_col) +
  theme_minimal()

## Joining with 'by = join_by(sampleID)'
```

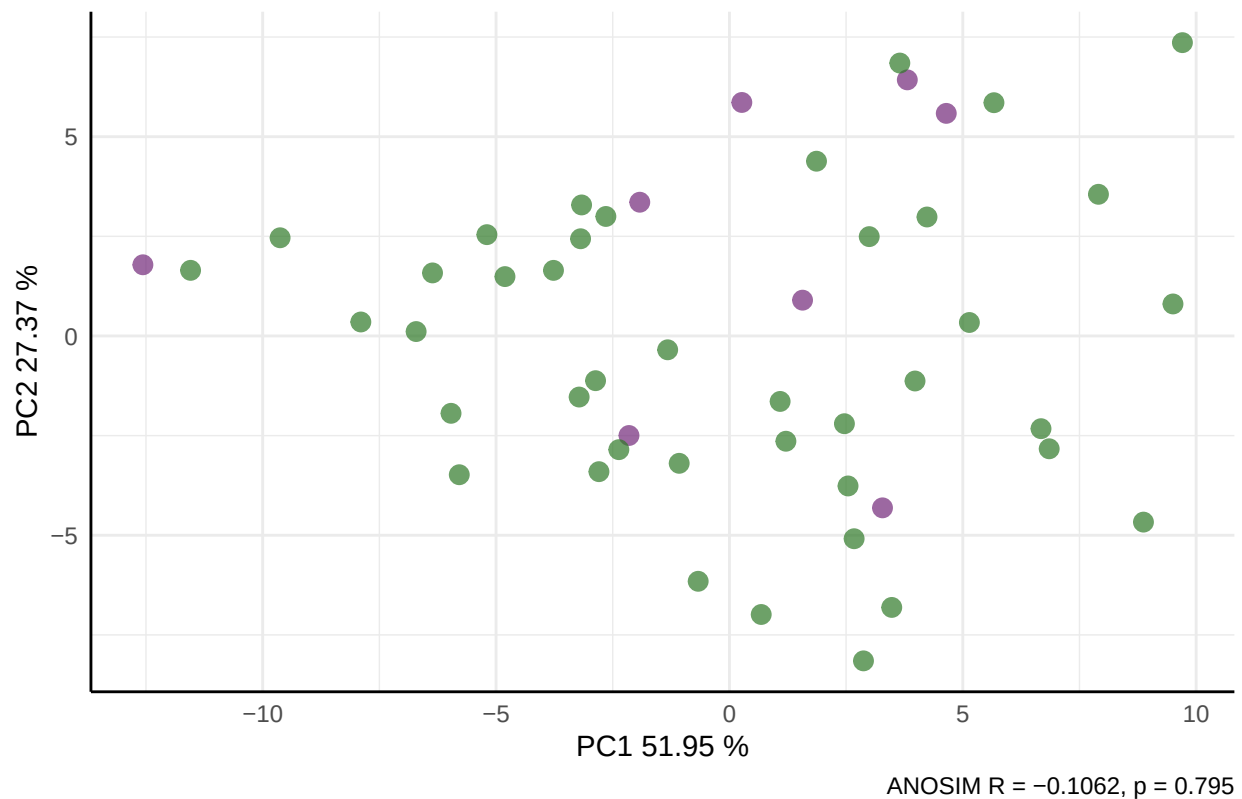



Figure 3: Ecological statistics in the integrated datasets

Calculate alpha diversity

```
# load phyloseq object
load(file= 'data/ps_filt_spp_zero.RData')

# prepare data to calculate alpha diversity
meta_all <- ps_filt_spp_zero %>% sample_tibble(column.id = "sample_id")
otu_df <- ps_filt_spp_zero %>% otu_tibble("id")
ps_melted <- ps_filt_spp_zero %>% psmelt.dplyr() %>% dplyr::rename(id=OTU)

## Joining, by = "Sample"
## Joining, by = "OTU"

CST = read_delim("tables/metadata_final.tsv") %>%
  filter(sampleID %in% meta_all$sample_id) %>%
  dplyr::rename(sample_id = sampleID) %>%
  select(sample_id, CST) %>%
  mutate(CST = factor(CST, levels = c("I", "II", "III", "IV-A", "IV-B", "IV-C", "V"))) %>%
  mutate(CST = fct_recode(CST, "IV" = "IV-A", "IV" = "IV-B", "IV" = "IV-C"))
```

```

## Rows: 1851 Columns: 28
## -- Column specification -----
## Delimiter: "\t"
## chr (18): sampleID, continent, unsdg_region, country, study, condition, sequ...
## dbl (10): age_years, bmikgm2, Birth_outcoome, Asymptomatic, BV, HIV, Antibio...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

meta_all <- meta_all %>% left_join(CST, by = "sample_id")
ps_melted <- ps_melted %>% left_join(CST, by = c("Sample" = "sample_id"))

# using funtion from iNEXT library and a wrapper developed by thomazbastiaan
# Compute alpha diversity using a wrapper around the iNEXT library,
# which implements automatic rarefaction curves.
# ASV by rows and samples by columns
inv_simpson <- ps_filt_spp_zero %>%
  estimate_richness(measures=c("InvSimpson")) %>%
  rownames_to_column("sample_id")

pielou <- otu_df %>%
  column_to_rownames("id") %>%
  evenness(index = "pielou", zeroes = TRUE, detection = 0) %>%
  dplyr::rename(Pielou = pielou) %>%
  mutate(Pielou = if_else(Pielou == "NaN", 0, Pielou)) %>%
  rownames_to_column("sample_id")

ps_nest <- ps_melted %>%
  select(id, Sample, abundance, study) %>%
  group_by(study) %>%
  nest() %>%
  mutate(data = map(data, ~ pivot_wider(.x, names_from = Sample, values_from = abundance) %>%
    column_to_rownames("id")))

alpha_nest <- ps_nest %>%
  mutate(alpha_df = map(data, ~get_asymptotic_alpha(species = .x, verbose = FALSE)))

alpha_df <- alpha_nest$alpha_df %>%
  bind_rows() %>%
  rownames_to_column("sample_id") %>%
  tibble() %>%
  dplyr::left_join(inv_simpson, by = "sample_id") %>%
  dplyr::left_join(pielou, by = "sample_id") %>%
  dplyr::left_join(meta_all %>% dplyr::select(sample_id, study, health_status, CST))

## Joining with 'by = join_by(sample_id)'

# write_csv(alpha_df, file.path(output_folder, "/tables/alpha_diversity_bac.csv"))
# alpha_df <- read_csv(paste0(output_folder, "/tables/alpha_diversity_bac.csv"))

# run stats by health status
kruskal_alpha <- alpha_df %>%

```

```

pivot_longer(-c(sample_id, study, health_status, CST)) %>%
group_by(name) %>%
nest() %>%
mutate(
  model = map(data, ~ .x %>% kruskal_test(value ~ study) %>%
    adjust_pvalue(method = "bonferroni") %>%
    add_significance("p.adj")
  ),
  model = map2(model, name, ~ mutate(.x, variable = paste(.y))),
  pwc = map(data, ~ .x %>% wilcox_test(value ~ study, p.adjust.method = "bonferroni")),
  pwc = map2(pwc, name, ~ mutate(.x, variable = paste(.y)))
)

kruskal_alpha$model %>%
bind_rows() %>%
# mutate(across(where(is.numeric), ~round(., 10))) %>%
# filter(term != "(Intercept)") %>%
filter(p.adj < 0.05)

```

```

## # A tibble: 5 x 9
##   .y.      n statistic    df      p method      p.adj p.adj.signif variable
##   <chr> <int>    <dbl> <int>    <dbl> <chr>    <dbl> <chr>    <chr>
## 1 value   664     275.     5 2.05e-57 Kruskal-W~ 2.05e-57 ****    Chao1
## 2 value   664     218.     5 4.85e-45 Kruskal-W~ 4.85e-45 ****    Simpson~
## 3 value   664     219.     5 2.45e-45 Kruskal-W~ 2.45e-45 ****    Shannon~
## 4 value   664     218.     5 4.22e-45 Kruskal-W~ 4.22e-45 ****    InvSimp~
## 5 value   664     165.     5 7.02e-34 Kruskal-W~ 7.02e-34 ****    Pielou

```

```

stat.test <- kruskal_alpha$pwc %>%
  bind_rows() %>%
  filter(p.adj < 0.05)

#stat.test %>% write_csv("tables/alpha_diversity_bac_stats.csv")
# stat.test <- read_csv(paste0(output_folder, "tables/alpha_diversity_bac_stats.csv"))

```

A.Meta-analysis: Shannon diversity by health status

```

#load phyloseq
ps_filt_spp_clr_zero_adj <- readRDS("data/ps_filt_spp_clr_zero_adj.RDS")
meta_all2 <- sample.data.frame(ps_filt_spp_clr_zero_adj)
#load table
alpha_diversity_bac <- read_csv("tables/alpha_diversity_bac.csv")

## Rows: 664 Columns: 8
## -- Column specification -----
## Delimiter: ","
## chr (3): sample_id, study, health_status
## dbl (5): Chao1, Simpson Index, Shannon Entropy, InvSimpson, Pielou
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

```

```

colnames(alpha_diversity_bac)[1] <- "sampleID"

HS_col <- c("-" = "#005ab5", "+" = "#dc3220")
# HS_col2 <- c("Asymptomatic" = "#005ab5", "HIV" = "#dc3220")
#stats
wilcox.test(`Shannon Entropy`~health_status, data = alpha_diversity_bac) #2.661e-07

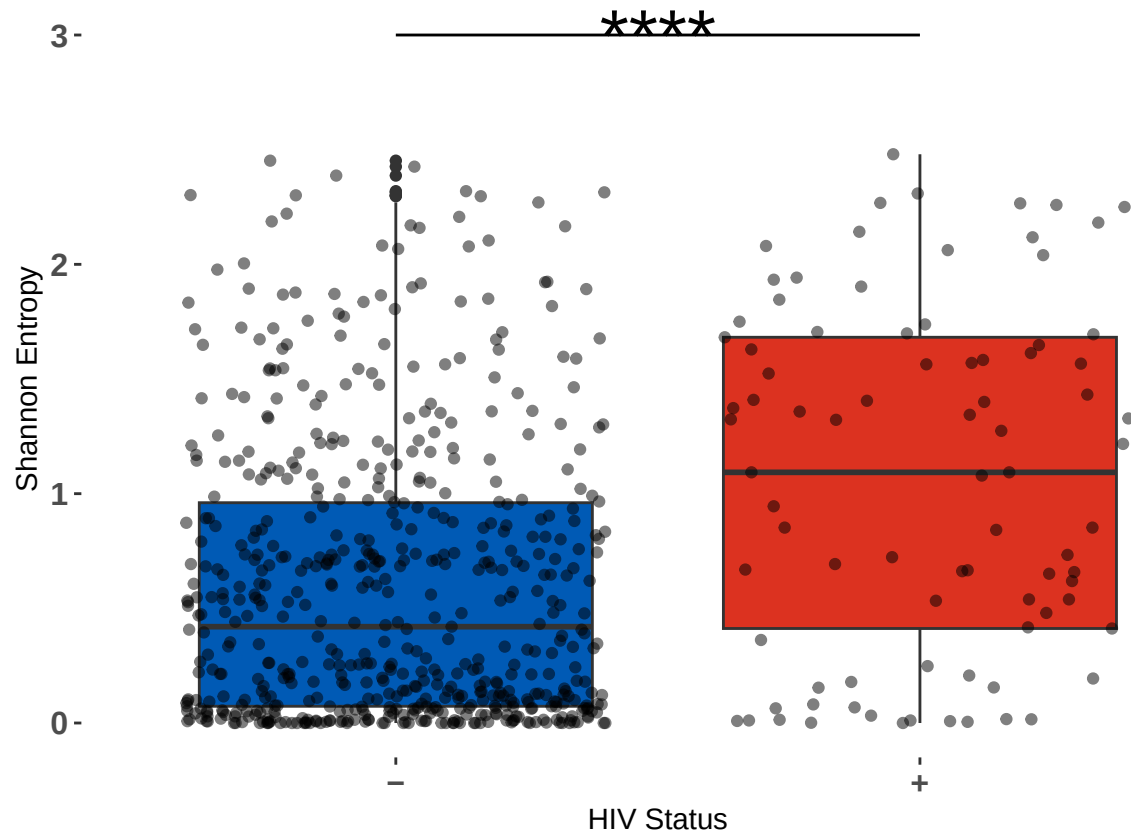
##
## Wilcoxon rank sum test with continuity correction
##
## data: Shannon Entropy by health_status
## W = 16109, p-value = 2.661e-07
## alternative hypothesis: true location shift is not equal to 0

#posthoc
dunn_test(`Shannon Entropy`~health_status, data = alpha_diversity_bac) #2.656451e-07

## # A tibble: 1 x 9
##   .y.      group1 group2    n1    n2 statistic      p    p.adj p.adj.signif
## * <chr>      <chr> <chr> <int> <int>    <dbl>   <dbl>   <dbl> <chr>
## 1 Shannon Entr~ Asymp~ HIV      579    85      5.15 2.66e-7 2.66e-7 ****

alpha_diversity_bac <-alpha_diversity_bac %>% mutate(HS = case_when(health_status == "HIV" ~"+",
                                                                    health_status == "Asymptomatic" ~ "-"))
#plot
ggplot(alpha_diversity_bac %>% filter(!is.na(health_status)), aes(HS, `Shannon Entropy`, fill = HS)) +
  geom_boxplot() + geom_jitter(color="black", fill="white", alpha=0.5, height = 0) +
  labs(y = "Shannon Entropy", x = "HIV Status") + theme(axis.text = element_text(size=12, face = "bold")) +
  theme(legend.position = "none", panel.background = element_rect(fill = NA, colour = "white"))+
  scale_fill_manual(values = HS_col, 0.4) + geom_line(data = tibble(x =c(1,2), y = c(3,3)), aes(x=x, y=

```



B.Meta-analysis: PCA by health status

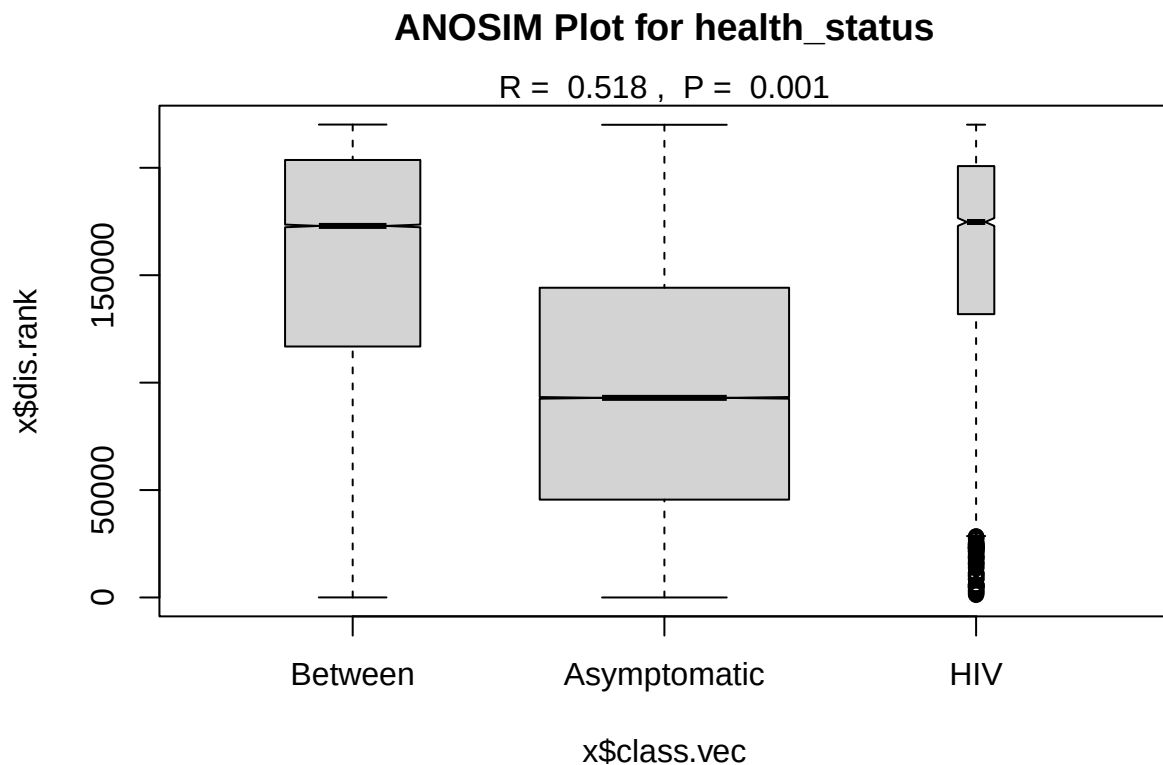
```
#specify shapes
shaps <- c("Kervinen_2022"=21,"Livson_2022"= 22, "Servergnini_2021"=23, "Ho_2021"=24,"PMTCT_CQI"=25,"Mo

cols = c("HIV" = "#DC3220", "Asymptomatic" = "#005AB5")

#calculate permanova
#stats
perform_anosim(ps_filt_spp_clr_zero_adj, "health_status", "euclidean") # R = -0.1062, p = 0.795

## ANOSIM Results:
##
## Call:
## anosim(x = dist_matrix, grouping = grouping_var)
## Dissimilarity: euclidean
##
## ANOSIM statistic R: 0.518
##      Significance: 0.001
##
## Permutation: free
## Number of permutations: 999
##
```

```
##
## Summary of ANOSIM Results:
##
## Call:
## anosim(x = dist_matrix, grouping = grouping_var)
## Dissimilarity: euclidean
##
## ANOSIM statistic R: 0.518
##      Significance: 0.001
##
## Permutation: free
## Number of permutations: 999
##
## Upper quantiles of permutations (null model):
##      90%      95%     97.5%      99%
## 0.0501 0.0646 0.0759 0.0899
##
## Dissimilarity ranks between and within classes:
##           0%      25%      50%      75%     100%      N
## Between      21.5 116817.0 172938 203640.5 220116 49215
## Asymptomatic   4.5  45543.5  92906 144172.5 220025 167331
## HIV           1230.0 131938.5 174769 200778.8 220066  3570
```

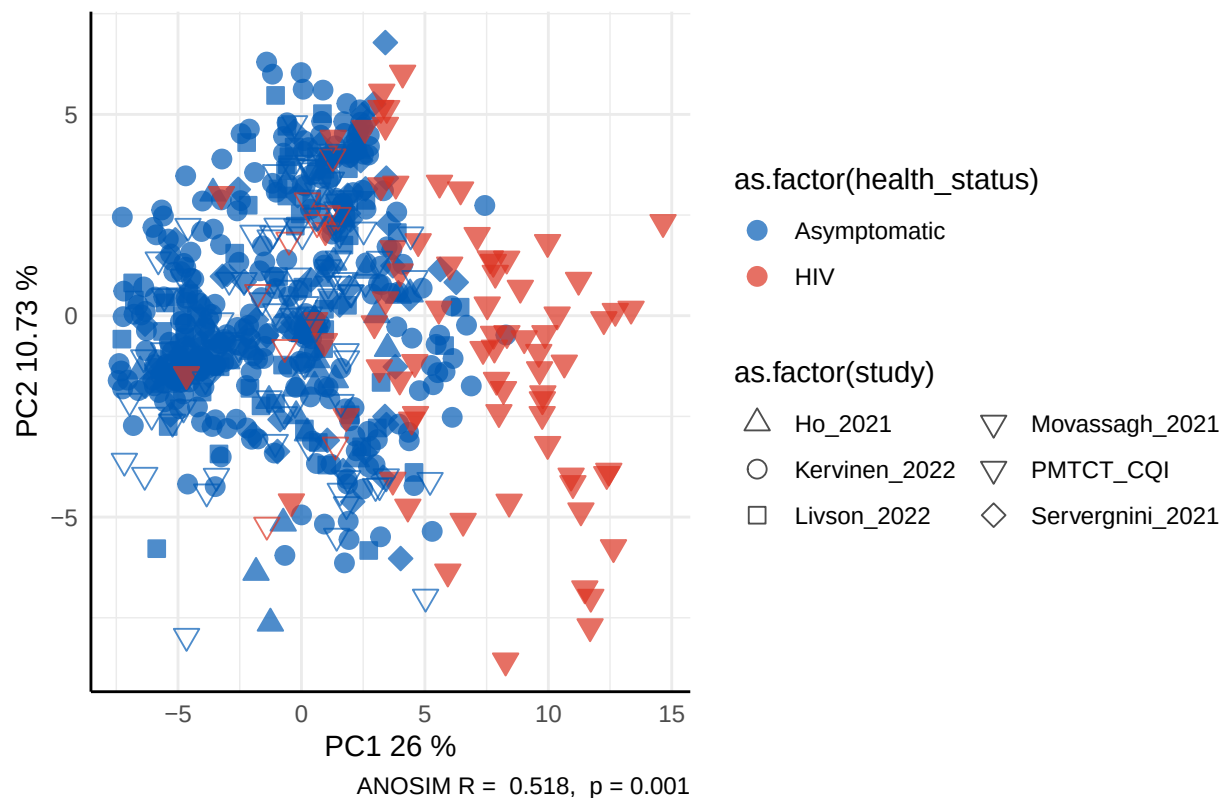


```
#calculate pairwise_anosim (posthoc)
perform_pairwise_anosim(ps_filt_spp_clr_zero_adj, "health_status", "euclidean", "bonferroni", perm = 999)
```

```
## Pairwise Anosim Results:
##           pairs anosimR p.value p.adj
## 1 Asymptomatic.vs.HIV 0.517979 0.001 0.001
```

```
#plot
run_PCA_shape(ps_filt_spp_clr_zero_adj, "health_status", "study", scale = F, title = " ", color_palette
```

```
## Joining with 'by = join_by(sampleID)'
```



C.Metaanaly:Shannon by CST

```
alpha_div_all <- alpha_diversity_bac %>% left_join(meta_all, by = "sampleID")

CST_col= c("I" = "#8035EB", "II" = "#F5530D", "III" = "#16FBCD", "IV" = "navy", "V" = "#8D324D")
#stats
pairwise.wilcox.test(alpha_div_all$`Shannon Entropy`, alpha_div_all$CST, p.adjust.method = "BH")

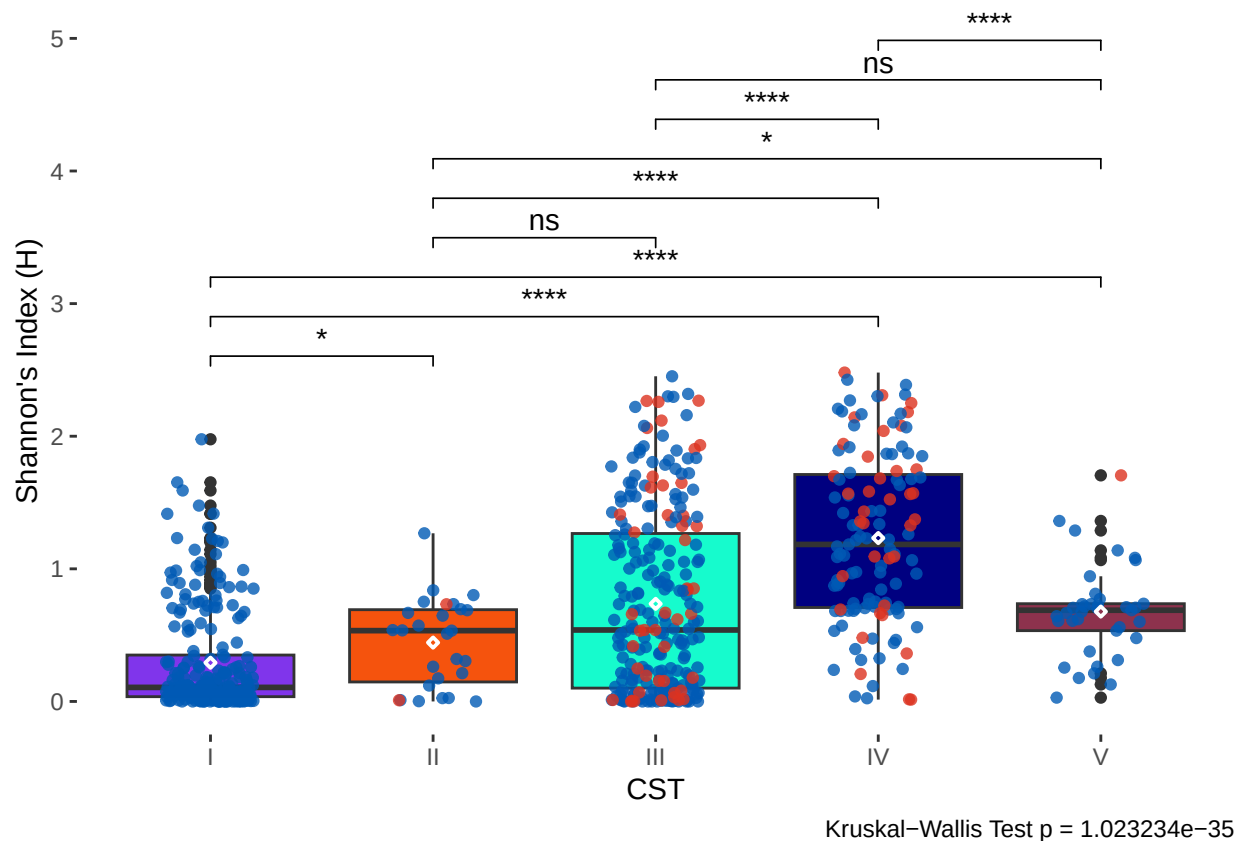
##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data: alpha_div_all$`Shannon Entropy` and alpha_div_all$CST
##
```

```
##      I      II      III      IV
## II  0.026  -      -      -
## III 1.3e-12 0.122  -      -
## IV  < 2e-16 1.5e-08 1.0e-10 -
## V   1.5e-08 0.019  0.448  2.2e-06
##
## P value adjustment method: BH
```

```
my_comparisons <- list(c("I", "II"), c("I", "IV"), c("I", "V"), c("II", "III"), c("II", "IV"), c("II", "V"),
#plot
kruskal.test(`Shannon Entropy` ~ CST, data = alpha_div_all)$p.value #1.023234e-35
```

```
## [1] 1.023234e-35
```

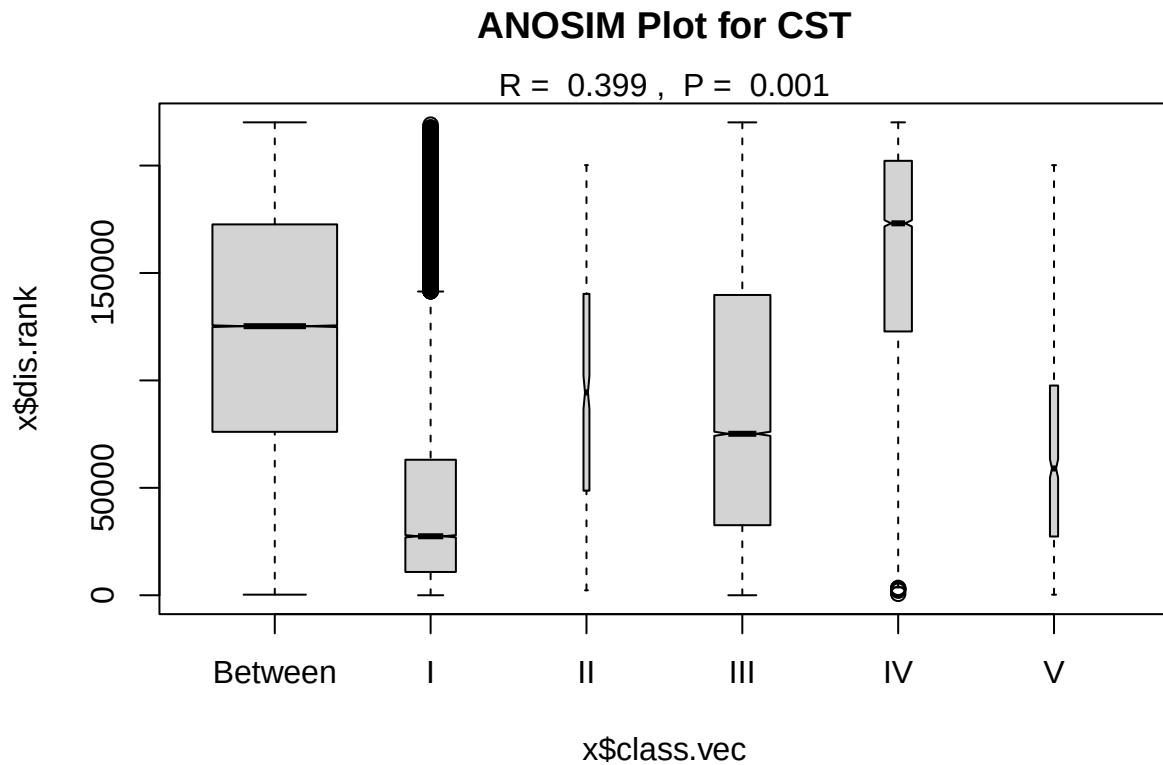
```
ggplot(alpha_div_all %>% filter(!is.na(CST)), aes(CST, `Shannon Entropy`, fill = CST)) +
  geom_boxplot() +
  geom_jitter(aes(color = health_status.x), alpha = 0.8, height = 0, width = 0.2) +
  labs(y = "Shannon's Index (H)", x = "CST", caption = paste("Kruskal-Wallis Test p = 1.023234e-35")) +
  theme(legend.position = "none", panel.background = element_rect(fill = NA, colour = "white")) +
  stat_compare_means(aes(label = ..p.signif..), method = "wilcox", comparisons = my_comparisons) +
  scale_fill_manual(values = CST_col) + scale_color_manual(values = c("HIV" = "#DC3220", "Asymptomatic"
```



D.Meta-analysis: PCA by CST

```
#stats
perform_anosim(ps_filt_spp_clr_zero_adj, "CST", "euclidean") #R 0.3991, p=0.01
```

```
## ANOSIM Results:
##
## Call:
## anosim(x = dist_matrix, grouping = grouping_var)
## Dissimilarity: euclidean
##
## ANOSIM statistic R: 0.3991
##      Significance: 0.001
##
## Permutation: free
## Number of permutations: 999
##
##
## Summary of ANOSIM Results:
##
## Call:
## anosim(x = dist_matrix, grouping = grouping_var)
## Dissimilarity: euclidean
##
## ANOSIM statistic R: 0.3991
##      Significance: 0.001
##
## Permutation: free
## Number of permutations: 999
##
## Upper quantiles of permutations (null model):
##      90%      95%    97.5%      99%
## 0.0179 0.0228 0.0274 0.0335
##
## Dissimilarity ranks between and within classes:
##           0%       25%       50%       75%      100%       N
## Between 277.0 76065.75 125236.5 172619.2 220116 154796
## I         4.5  10842.00  27458.0  63063.0 219071  25425
## II        2296.0 48663.50 94403.0 140282.0 200202   351
## III        4.5  32605.25  75160.0 139758.0 220105 31375
## IV        640.0 122788.50 173113.0 202209.5 220114  7503
## V         269.0 27414.75  59008.5  97577.0 200182   666
```



```
#calculate pairwise_anosim (posthoc)
perform_pairwise_anosim(ps_filt_spp_clr_zero_adj, "CST", "euclidean", "bonferroni", perm = 999)
```

```
## Pairwise Anosim Results:
##      pairs      anosimR p.value p.adj
## 1   I.vs.IV  0.55501274  0.001  0.01
## 2   I.vs.III 0.39309097  0.001  0.01
## 3   I.vs.V   0.64534997  0.001  0.01
## 4   I.vs.II  0.70507056  0.001  0.01
## 5  IV.vs.III 0.34693783  0.001  0.01
## 6  IV.vs.V   -0.05604106  0.839  1.00
## 7  IV.vs.II  -0.15375269  0.988  1.00
## 8  III.vs.V  0.25425460  0.001  0.01
## 9  III.vs.II 0.26104714  0.002  0.02
## 10 V.vs.II   0.47210140  0.001  0.01
```

```
#plot
shaps <- c("Kervinen_2022"=21,"Livson_2022"= 22, "Servergnini_2021"=23, "Ho_2021"=24,"PMTCT_CQI"=25,"Mo
run_PCA_shape(ps_filt_spp_clr_zero_adj, "CST", "study", scale = F, title = " ", color_palette = CST_col
```

```
## Joining with 'by = join_by(sampleID)'
```

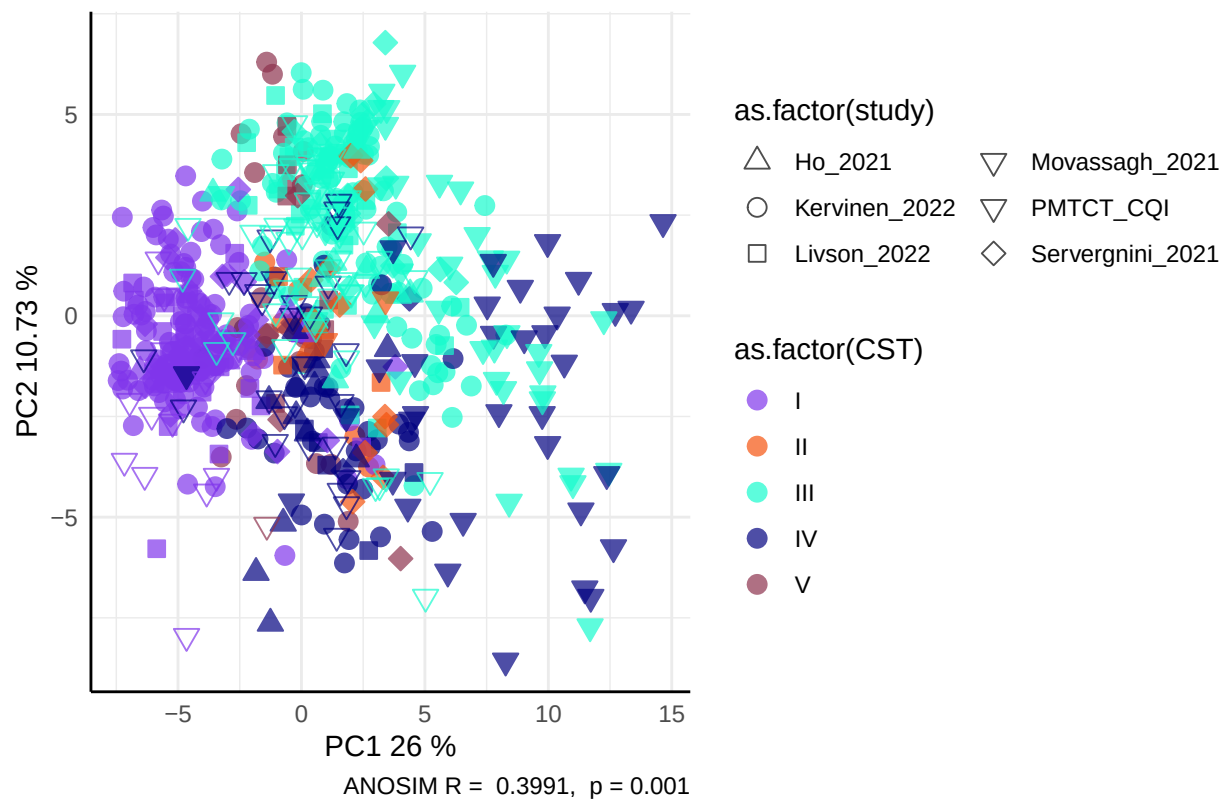


Figure 4: Meta-analysis: Differentially abundant species by health status

perform DAAs

```
#### load data ----
load('data/ps_filt_spp_zero.RData')

pseq.bac <- ps_filt_spp_zero
meta <- pseq.bac %>% sample_tibble(column.id = "sample_id")
tax_df <- pseq.bac %>% tax_tibble("id")
otu_df <- pseq.bac %>% otu_tibble("id")

# cores
n_cores <- detectCores()
cores <- n_cores - 1

####---- LinDA ----####
# INPUTS:
# ASV by row and samples by columns
# meta: samples by rows
linda_input_asv <- pseq.bac %>%
```

```

otu_tibble(column.id = "genus") %>%
column_to_rownames("genus")

linda_meta <- pseq.bac %>%
  sample_tibble(column.id = "sample_id") %>%
  column_to_rownames("sample_id")

# run LinDA
linda.obj <- linda(otu.tab = linda_input_asv,
  meta = linda_meta,
  # phyloseq.obj = NULL,
  formula = "~health_status + study",
  type = 'count',
  adaptive = TRUE,
  imputation = TRUE,
  # pseudo.cnt = 0.65,
  corr.cut = 0.1,
  p.adj.method = 'BH',
  alpha = 0.05,
  prev.cut = 0,
  lib.cut = 1,
  winsor.quan = 0.97,
  n.cores = cores
)

```

Imputation approach is used.

```

# get the results
linda_res <- linda.obj$output %>%
  imap(., ~mutate(., name = .y)) %>%
  map_df(., ~rownames_to_column(., "Species") %>%
    as_tibble %>%
    arrange(padj) %>%
    # reject H0??
    mutate(reject = ifelse(padj < 0.05, "Yes", "No"))) %>%
  mutate(method = "LinDA")

# linda_res %>% filter(padj < 0.05) %>% View()

# save results
# write_csv(linda_res, "tables/DAA/linda_res_bac.csv")

####---- ANCOM-BC ----####
# prepare the phyloseq object
pseq.bac.filt <- pseq.bac
tax_df_filt <- pseq.bac.filt %>%
  tax_tibble("asv") %>%
  mutate(Species = asv) %>%
  column_to_rownames("asv")

tax_table(pseq.bac.filt) <- tax_table(as.matrix(tax_df_filt))

# perform the ANCOM-BC analysis

```

```

ancombc_out <- ancombc2(
  data = pseq.bac.filt,
  assay_name = "counts",
  tax_level = "Species",
  fix_formula = "health_status + study",
  p_adj_method = "fdr",
  #pseudo = 0,
  #pseudo_sens = TRUE,
  prv_cut = 0, # prev filtering has been done above already
  lib_cut = 0,
  group = "health_status",
  struc_zero = TRUE,
  neg_lb = TRUE,
  iter_control = list(tol = 1e-5, max_iter = 20, verbose = FALSE),
  em_control = list(tol = 1e-5, max_iter = 100), # use max_iter >= 100 on real data
  alpha = 0.05,
  global = TRUE, # multi group comparison will be deactivated automatically
  verbose = T,
  n_cl = cores)

```

```
## Checking the input data type ...
```

```
## The input data is of type: phyloseq
```

```
## PASS
```

```
## Checking the sample metadata ...
```

```
## The specified variables in the formula: health_status, study
```

```
## The available variables in the sample metadata: sampleID, continent, unsdg_region, country, study, c
```

```
## PASS
```

```
## Checking other arguments ...
```

```
## The number of groups of interest is: 2
```

```
## The sample size per group is: Asymptomatic = 579, HIV = 85
```

```
## PASS
```

```
## Obtaining initial estimates ...
```

```
## Estimating sample-specific biases ...
```

```
## Conducting sensitivity analysis for pseudo-count addition to 0s ...
```

```
## For taxa that are significant but do not pass the sensitivity analysis,
```

```
## please flag them and proceed with caution, as they are likely false positives.
```

```
## For detailed instructions on performing sensitivity analysis,
```

```
## please refer to the package vignette.
```

```
## ANCOM-BC2 primary results ...
```

```

# store the results in res
ancombc_res <- ancombc_out$res %>%
  as.data.frame() %>%
  pivot_longer(cols = -taxon,
               names_to = "variable",
               values_to = "value") %>%
  # "merge" = preserve any extra characters after the first underscore in the second column
  separate(variable, c("stats", "name"), sep = "_", extra = "merge") %>%
  pivot_wider(id_cols = c(taxon, name),
              names_from = "stats",
              values_from = "value") %>%
  mutate(diff = if_else(diff == 1, TRUE, FALSE),
         method = "ANCOM-BC") %>%
  dplyr::rename(Species = taxon)

```

```

# save results
# write_csv(ancombc_res, file = "tables/DAA/ancombc_res.csv")

```

```

####--- ALDEx2 ----####
df_aldex <- pseq.bac %>%
  otu_tibble("Species") %>%
  column_to_rownames("Species")

```

```

# make model matrix
mm <- model.matrix(~ health_status + study, data = meta)

```

```

# INPUT: ASV by rows and samples by columns
# CLR transformation
x <- aldex.clr(df_aldex, mm, mc.samples=128, denom="all", verbose=F)

```

```
## integer matrix provided
```

```
## using all features for denominator
```

```
## operating in serial mode
```

```
## computing center with all features
```

```

# apply glm
glm.test <- aldex.glm(x, mm)

```

```
## running tests for each MC instance:
```

```
## |------(25%)------(50%)------(75%)-----|
```

```
glm.effect <- aldex.glm.effect(x)
```

```
## operating in serial mode
```

```
## sanity check complete
```

```
## rab.all  complete

## rab.win  complete

## rab of samples complete

## within sample difference calculated

## between group difference calculated

## group summaries calculated

## unpaired effect size calculated

## summarizing output

## operating in serial mode

## sanity check complete

## rab.all  complete

## rab.win  complete

## rab of samples complete

## within sample difference calculated

## between group difference calculated

## group summaries calculated

## unpaired effect size calculated

## summarizing output

## operating in serial mode

## sanity check complete

## rab.all  complete

## rab.win  complete

## rab of samples complete

## within sample difference calculated
```

```
## between group difference calculated

## group summaries calculated

## unpaired effect size calculated

## summarizing output

## operating in serial mode

## sanity check complete

## rab.all  complete

## rab.win  complete

## rab of samples complete

## within sample difference calculated

## between group difference calculated

## group summaries calculated

## unpaired effect size calculated

## summarizing output

## operating in serial mode

## sanity check complete

## rab.all  complete

## rab.win  complete

## rab of samples complete

## within sample difference calculated

## between group difference calculated

## group summaries calculated

## unpaired effect size calculated

## summarizing output
```



```

## operating in serial mode

## sanity check complete

## rab.all complete

## rab.win complete

## rab of samples complete

## within sample difference calculated

## between group difference calculated

## group summaries calculated

## unpaired effect size calculated

## summarizing output

```

```

# make a data frame
aldex2_res <- glm.effect %>%
  imap(., ~mutate(., name = .y)) %>%
  map_df(., ~rownames_to_column(., "Species") %>%
    as_tibble) %>%
  mutate(method = "ALDEx2")

# save results
# write_csv(aldex2_res, file = "tables/DAA/aldex2_res.csv")

genus_df <- pseq.bac %>% otu_tibble("Species")

# put all methods together and count how many taxa were identified
# score = intersection of identified taxa
all_methods <- linda_res %>%
  filter(padj < 0.05) %>%
  dplyr::select(Species, LinDA = reject) %>%
  distinct(Species, .keep_all = TRUE) %>%
  mutate(LinDA = ifelse(LinDA == "No", FALSE, TRUE)) %>%
  full_join(ancombc_res %>%
    filter(q < 0.05) %>%
    dplyr::select(Species, ancombc = diff) %>%
    distinct(Species, .keep_all = TRUE),
    by = "Species",
    relationship = "many-to-many") %>%
  full_join(aldex2_res %>%
    filter(abs(effect) > 0.5) %>%
    dplyr::select(Species, aldex2 = effect) %>%
    distinct(Species, .keep_all = TRUE),
    by = "Species",
    relationship = "many-to-many") %>%

```

```

mutate(aldex2 = if_else(abs(aldex2) >= 0.05, TRUE, FALSE),
      across(-Species, function(x) ifelse(is.na(x), FALSE, x)),
      score = rowSums(across(c(aldex2, ancombc, LinDA))),
)

only_health_status <- linda_res %>%
  filter(padj < 0.05) %>%
  filter(name == "health_statusHIV") %>%
  dplyr::select(Species, LinDA = reject) %>%
  distinct(Species, .keep_all = TRUE) %>%
  mutate(LinDA = ifelse(LinDA == "No", FALSE, TRUE)) %>%
full_join(ancombc_res %>%
  filter(q < 0.05) %>%
  filter(name == "health_statusHIV") %>%
  dplyr::select(Species, ancombc = diff) %>%
  distinct(Species, .keep_all = TRUE),
  by = "Species",
  relationship = "many-to-many") %>%
full_join(aldex2_res %>%
  filter(abs(effect) > 0.5) %>%
  filter(name == "health_statusHIV") %>%
  dplyr::select(Species, aldex2 = effect) %>%
  distinct(Species, .keep_all = TRUE),
  by = "Species",
  relationship = "many-to-many") %>%
mutate(aldex2 = if_else(abs(aldex2) >= 0.05, TRUE, FALSE),
      across(-Species, function(x) ifelse(is.na(x), FALSE, x)),
      score = rowSums(across(c(aldex2, ancombc, LinDA))),
)

# check which variables each genus was significant
variables_all <- linda_res %>%
  filter(padj < 0.05) %>%
  dplyr::select(Species, name) %>%
  mutate(variable = str_extract_all(name, "(health_status|study)")) %>%
  group_by(Species) %>%
  summarise(variable_linda = paste(unique(variable), collapse = " and ")) %>%
  full_join(ancombc_res %>%
    filter(q < 0.05) %>%
    dplyr::select(Species, name) %>%
    filter(name != "(Intercept)") %>%
    mutate(variable = str_extract_all(name, "(health_status|study)")) %>%
    group_by(Species) %>%
    summarise(variable_ancombc = paste(unique(variable), collapse = " and ")),
    by = "Species") %>%
  full_join(aldex2_res %>%
    filter(abs(effect) > 0.5) %>%
    dplyr::select(Species, name) %>%
    mutate(variable = str_extract_all(name, "(health_status|study)")) %>%
    group_by(Species) %>%
    summarise(variable_aldex2 = paste(unique(variable), collapse = " and ")),
    by = "Species")

```

Figure 4

```
# box plot only DAA health Status and detect by at least 2 methods
all_methods <- read_csv("tables/DAA/all_methods.csv")

## Rows: 45 Columns: 8
## -- Column specification -----
## Delimiter: ","
## chr (4): Species, variable_linda, variable_ancombc, variable_aldex2
## dbl (1): score
## lgl (3): LinDA, ancombc, aldex2
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

only_health_status <- all_methods %>%
  filter(if_any(starts_with("variable"), ~str_detect(., "health_status"))) %>%
  filter(score > 1)
daa_spp <- only_health_status %>% distinct(Species) %>% pull(Species)
# load CLR-transformed data
#load('ps_filt_spp_clr_zero.RData') # only CLR
pseq.bac <- ps_filt_spp_clr_zero_adj
meta <- pseq.bac %>% sample_tibble(column.id = "sample_id")
tax_df <- pseq.bac %>% tax_tibble("id")
otu_df <- pseq.bac %>% otu_tibble("id")
ps_melted_clr <- pseq.bac %>% psmelt.dplyr() %>% rename(id = OTU)

## Joining, by = "Sample"
## Joining, by = "OTU"

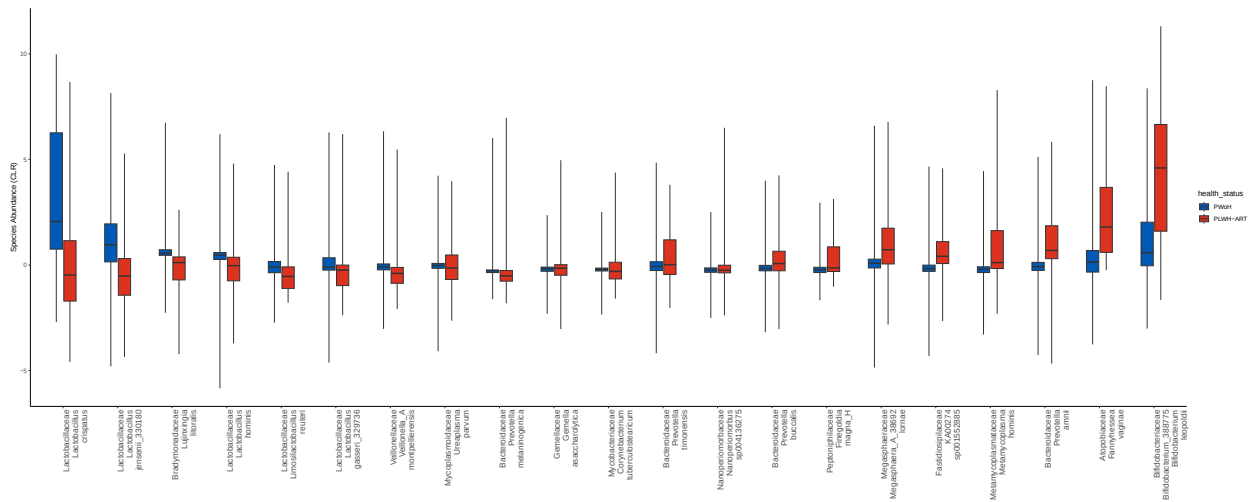
color_study <- c(
  "PMTCT_CQI" = "red",
  "Movassagh_2021" = "green",
  "Kervinen_2022" = "#445002",
  "Livson_2022" = "lavender",
  "Servergnini_2021" = "navy",
  "Ho_2021" = "#AA9F0D"
)

#from reviewer_order by abundnace
species_abundance_diff <- ps_melted_clr %>%
  filter(id %in% daa_spp) %>%
  group_by(id, health_status) %>%
  summarise(mean_abundance = mean(abundance, na.rm = TRUE), .groups = 'drop') %>%
  pivot_wider(names_from = health_status, values_from = mean_abundance) %>%
  # Calculate difference: PWOH abundance - PLWH-ART abundance
  mutate(abundance_diff = Asymptomatic - HIV) %>% # Assuming "Asymptomatic" = PWOH and "HIV" = PLWH-ART
  arrange(desc(abundance_diff)) %>% # Sort by highest PWOH abundance to highest PLWH-ART abundance
  mutate(id_formatted = gsub("f__|g__|s__", " ", id),
         id_formatted = gsub("^ ", "", id_formatted),
         id_formatted = gsub("([ ]+)( )", "\\1\\n", id_formatted))
```

```

ps_melted_clr %>%
  filter(id %in% daa_spp) %>%
  mutate(id_formatted = gsub("f__|g__|s__", " ", id),
         id_formatted = gsub("^ ", "", id_formatted),
         id_formatted = gsub("([ ]+)( )", "\\1\\n", id_formatted)) %>%
  # Convert to factor with levels ordered by abundance difference (PWoH > PLWH-ART)
  mutate(id_formatted = factor(id_formatted,
                              levels = species_abundance_diff$id_formatted)) %>%
  ggplot(aes(x = id_formatted, y = abundance,
            fill = health_status)) +
  geom_boxplot(coef = 100, width = 0.5) +
  scale_fill_manual(values = c("HIV" = "#DC3220", "Asymptomatic" = "#005AB5"),
                  labels = c("HIV" = "PLWH-ART", "Asymptomatic" = "PWoH")) +
  scale_color_manual(values = color_study) +
  theme_classic() +
  labs(y = "Species Abundance (CLR)" +
  guides(colour = guide_legend(override.aes = list(alpha = 1))) +
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
  theme(axis.text.x = element_text(size = 12, angle = 90, hjust = 1),
        strip.background = element_rect(fill = "white", color = "white"),
        strip.text.x = element_text(size = 12),
        axis.title.x = element_blank(),
        plot.title = element_text(size=20))

```



```

#ggsave("box_plot_spp_DAA_option_1.pdf", width = 20, height = 10)

```

Figure 5: CQI-PMTCT: Predicted KEGG pathways differentially abundant by CST

perform DAAs

```

all_congo_df <- vroom::vroom('tables/ko_df_congo_unstrat_with_annotation.tsv')

## Rows: 162 Columns: 85
## -- Column specification -----
## Delimiter: "\t"
## chr (4): KO, pathway_1, pathway_2, pathway_3
## dbl (81): V102, V103, V104, V011, V112, V113, V118, V119, V012, V125, V128, ...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

ko_numeric = all_congo_df %>% dplyr::select(KO, where(is.numeric))
ko_tax = all_congo_df %>% dplyr::select(where(is.character)) %>% mutate(pathway_3 = paste(pathway_1,pathway_2,pathway_3))

melt_congo_ko = all_congo_df %>%
  as.data.table() %>%
  melt(variable.factor = FALSE)
# agglomerate at pathway3 level
path_glom_congo_df = melt_congo_ko %>%
  mutate(pathway_3 = paste(pathway_1,pathway_2,pathway_3, sep = "_")) %>%
  group_by(variable,pathway_3) %>%
  summarise(abundance = sum(value)) %>%
  pivot_wider(id_cols = pathway_3, names_from = "variable", values_from = "abundance")

## 'summarise()' has grouped output by 'variable'. You can override using the
## '.groups' argument.

pathway_tax = path_glom_congo_df %>%
  mutate(KO = paste0("KO_", 1:nrow(.)), .before = 1) %>%
  dplyr::select(KO, pathway_3) %>%
  separate(pathway_3, sep = "_", into = c('pathway_1','pathway_2','pathway_3_clean'), remove = TRUE)

# make a phyloseq object
samples_picrust = colnames(path_glom_congo_df %>% column_to_rownames("pathway_3"))

# remove CST V (only one sample) = sample_id = V057
meta_meta_analysis = sample.data.frame(ps_congo_filt_spp_rclr) %>%
  #left_join(CST, by = "sample_id") %>%
  filter(sampleID != "V057")

# make a phyloseq object
to_otu_table = path_glom_congo_df %>%
  # mutate(across(where(is.numeric), ~abs(.))) %>%
  mutate(KO = paste0("KO_", 1:nrow(.)), .before = 1) %>%
  dplyr::select(-pathway_3) %>%
  column_to_rownames('KO') %>%
  dplyr::select(-V057)
OTU = otu_table(as.matrix(to_otu_table), taxa_are_rows = TRUE)
TAX = tax_table(as.matrix(pathway_tax %>% column_to_rownames('KO')))
META = sample_data(meta_meta_analysis) # %>% column_to_rownames("sample_id")
ps_congo_picrust <- phyloseq(OTU, TAX, META)

```

```

# saveRDS(ps_congo_picrust, file = paste0(data_folder, 'ps_congo_picrust.RDS'))

ps_congo_picrust = readRDS('data/ps_congo_picrust.RDS')
to_otu_table = ps_congo_picrust %>% otu_table() %>% as.data.frame()
meta_meta_analysis = sample.data.frame(ps_congo_picrust)

####---- LinDA ----####
# INPUTS:
# ASV by row and samples by columns
# meta: samples by rows
linda_input_asv <- to_otu_table

linda_meta <- meta_meta_analysis
# run LinDA
linda.obj <- linda(otu.tab = linda_input_asv,
  meta = linda_meta,
  # phyloseq.obj = NULL,
  formula = "~CST",
  type = 'count',
  adaptive = TRUE,
  imputation = TRUE,
  # pseudo.cnt = 0.65,
  corr.cut = 0.1,
  p.adj.method = 'BH',
  alpha = 0.05,
  prev.cut = 0,
  lib.cut = 1,
  winsor.quan = 0.97,
  n.cores = cores
)

```

Pseudo-count approach is used.

```

# get the results
linda_res_congo <- linda.obj$output %>%
  imap(., ~mutate(., name = .y)) %>%
  map_df(., ~rownames_to_column(., "K0") %>%
    as_tibble %>%
    arrange(padj) %>%
    # reject H0??
    mutate(reject = ifelse(padj < 0.05, "Yes", "No"))) %>%
  mutate(method = "LinDA")

# linda_res %>% filter(padj < 0.05) %>% View()
linda_res_congo = linda_res_congo %>%
  left_join(pathway_tax, by = "K0")

# save results
# write_csv(linda_res, "tables/DAA/linda_res_picrust_congo.csv")
linda_res_congo <- read_csv("tables/DAA/linda_res_picrust_congo.csv")

```

Rows: 180 Columns: 15

```
## -- Column specification -----
## Delimiter: ","
## chr (8): KO, reject, name, method, pathway_3, pathway_1, pathway_2, pathway_...
## dbl (7): baseMean, log2FoldChange, lfcSE, stat, pvalue, padj, df
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
####----- ANCOM-BC -----####
```

```
ancombc_out_congo <- ancombc2(
  data = ps_congo_picrust,
  assay_name = "counts",
  # tax_level= "pathway_3",
  fix_formula = "CST",
  p_adj_method = "fdr",
  verbose = T,
  n_cl = cores)
```

```
## Checking the input data type ...
## The input data is of type: phyloseq
## PASS
## Checking the sample metadata ...
## The specified variables in the formula: CST
## The available variables in the sample metadata: sampleID, continent, unsdg_region, country, study, c
## PASS
## Checking other arguments ...
## PASS
## Obtaining initial estimates ...
## Estimating sample-specific biases ...
## Conducting sensitivity analysis for pseudo-count addition to 0s ...
## For taxa that are significant but do not pass the sensitivity analysis,
## please flag them and proceed with caution, as they are likely false positives.
## For detailed instructions on performing sensitivity analysis,
## please refer to the package vignette.
## ANCOM-BC2 primary results ...
```

```
# store the results in res
```

```
ancombc_res_congo <- ancombc_out_congo$res %>%
  as.data.frame() %>%
  pivot_longer(cols = -taxon,
    names_to = "variable",
    values_to = "value") %>%
  # "merge" = preserve any extra characters after the first underscore in the second column
  separate(variable, c("stats", "name"), sep = "_", extra = "merge") %>%
  pivot_wider(id_cols = c(taxon, name),
    names_from = "stats",
    values_from = "value") %>%
  mutate(diff = if_else(diff == 1, TRUE, FALSE),
    method = "ANCOM-BC") %>%
  dplyr::rename(KO = taxon)

ancombc_res_congo = ancombc_res_congo %>%
  left_join(pathway_tax, by = "KO")
```

```

# save results
# write_csv(ancombc_res_congo, "tables/DAA/ancombc_res_picrust_congo.csv")
ancombc_res_congo <- read_csv("tables/DAA/ancombc_res_picrust_congo.csv")

## Rows: 255 Columns: 13
## -- Column specification -----
## Delimiter: ","
## chr (7): KO, name, method, pathway_3, pathway_1, pathway_2, pathway_3_clean
## dbl (5): lfc, se, W, p, q
## lgl (1): diff
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

####--- ALDEx2 ----####
df_aldex_congo <- ps_congo_picrust %>%
  otu_tibble("KO") %>%
  column_to_rownames("KO") %>%
  mutate(across(where(is.numeric), ~round(.)))

# make model matrix
mm <- model.matrix(~ CST, data = meta_meta_analysis)

# INPUT: ASV by rows and samples by columns
# CLR transformation
x <- aldex.clr(df_aldex_congo, mm, mc.samples=128, denom="all", verbose=F)

## integer matrix provided
## using all features for denominator
## operating in serial mode
## computing center with all features

# apply glm
glm.test <- aldex.glm(x, mm)

## running tests for each MC instance:

## |------(25%)------(50%)------(75%)-----|

glm.effect <- aldex.glm.effect(x)

## operating in serial mode
## sanity check complete
## rab.all complete
## rab.win complete
## rab of samples complete
## within sample difference calculated
## between group difference calculated
## group summaries calculated
## unpaired effect size calculated

```



```

## summarizing output
## operating in serial mode
## sanity check complete
## rab.all complete
## rab.win complete
## rab of samples complete
## within sample difference calculated
## between group difference calculated
## group summaries calculated
## unpaired effect size calculated
## summarizing output

# make a data frame
aldex2_res_congo <- glm.effect %>%
  imap(., ~mutate(., name = .y)) %>%
  map_df(., ~rownames_to_column(., "KO") %>%
    as_tibble) %>%
  mutate(method = "ALDEx2")

aldex2_res_congo = aldex2_res_congo %>% left_join(pathway_tax, by = "KO")
# save results
#write_csv(aldex2_res_congo, "tables/DAA/aldex2_res_picrust_congo.csv")
aldex2_res_congo <- read_csv("tables/DAA/aldex2_res_picrust_congo.csv")

## Rows: 178 Columns: 14
## -- Column specification -----
## Delimiter: ","
## chr (7): KO, name, method, pathway_3, pathway_1, pathway_2, pathway_3_clean
## dbl (7): rab.all, rab.win.0, rab.win.1, diff.btw, diff.win, effect, overlap
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

# put all methods together and count how many taxa were identified
# score = intersection of identified taxa
all_methods_congo <- linda_res_congo %>%
  filter(padj < 0.05) %>%
  dplyr::select(KO, LinDA = reject) %>%
  distinct(KO, .keep_all = TRUE) %>%
  mutate(LinDA = ifelse(LinDA == "No", FALSE, TRUE)) %>%
  full_join(ancombc_res_congo %>%
    filter(q < 0.05) %>%
    dplyr::select(KO, ancombc = diff) %>%
    distinct(KO, .keep_all = TRUE),
    by = "KO",
    relationship = "many-to-many") %>%
  full_join(aldex2_res_congo %>%
    filter(abs(effect) > 0.5) %>%
    dplyr::select(KO, aldex2 = effect) %>%
    distinct(KO, .keep_all = TRUE),
    by = "KO",
    relationship = "many-to-many") %>%
  mutate(aldex2 = if_else(abs(aldex2) >= 0.05, TRUE, FALSE),

```

```

        across(-KO, function(x) ifelse(is.na(x), FALSE, x)),
        score = rowSums(across(c(aldex2, ancombc, LinDA))),
    ) %>%
    left_join(pathway_tax, by = "KO")

# put all together for the upset plot
all_methods_congo %>% filter(score >= 2)

## # A tibble: 1 x 9
##   KO      LinDA ancombc aldex2 score pathway_3 pathway_1 pathway_2 pathway_3_clean
##   <chr> <lg1> <lg1>    <lg1> <dbl> <chr>      <chr>      <chr>      <chr>
## 1 KO_3  TRUE  TRUE     FALSE      2 Brite Hi~ Brite Hi~ Signalin~ G Protein-Coup~

#write_csv(all_methods_congo, "tables/DAA/all_methods_picrust_congo.csv")
all_methods_congo <- read_csv("tables/DAA/all_methods_picrust_congo.csv")

## Rows: 25 Columns: 9
## -- Column specification -----
## Delimiter: ","
## chr (5): KO, pathway_3, pathway_1, pathway_2, pathway_3_clean
## dbl (1): score
## lg1 (3): LinDA, ancombc, aldex2
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

```

Figure 5

```

CST_col = c("I" = "#ffd900b9", "II" = "#F5530D", "III" = "#16FBCD", "IV" = "navy" , "V" = "#8D324D")

all_methods_congo <- read_csv("all_methods_picrust_congo.csv")

## Rows: 25 Columns: 9
## -- Column specification -----
## Delimiter: ","
## chr (5): KO, pathway_3, pathway_1, pathway_2, pathway_3_clean
## dbl (1): score
## lg1 (3): LinDA, ancombc, aldex2
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

# get the KO IDs that were found in at least 2 DAA tools
ko_daa_congo = all_methods_congo %>% filter(score >= 2) %>% pull(KO)
ps_congo_picrust_rel = ps_congo_picrust %>% transform("compositional")
path_df_clr_congo = ps_congo_picrust %>%
    otu_tibble("KO") %>%
    column_to_rownames("KO") %>%
    clr_cc()

```

```

## [1] "The DL is equal to 0.176470588235294"
## [1] "The DL is equal to 2"
## [1] "The DL is equal to 0.986"
## [1] "The DL is equal to 0.6875"
## [1] "The DL is equal to 6.86111111111111"
## [1] "The DL is equal to 0.166875"
## [1] "The DL is equal to 0.176470588235294"
## [1] "The DL is equal to 0.208333333333333"
## [1] "The DL is equal to 0.05375"
## [1] "The DL is equal to 1.18181818181818"
## [1] "The DL is equal to 1.208125"
## [1] "The DL is equal to 0.235294117647059"
## [1] "The DL is equal to 0.3125"
## [1] "The DL is equal to 0.125"
## [1] "The DL is equal to 0.277777777777778"
## [1] "The DL is equal to 0.176470588235294"
## [1] "The DL is equal to 0.0526315789473684"
## [1] "The DL is equal to 1.333125"
## [1] "The DL is equal to 0.029"
## [1] "The DL is equal to 0.5"
## [1] "The DL is equal to 49.84"
## [1] "The DL is equal to 0.611111111111111"
## [1] "The DL is equal to 0.25"
## [1] "The DL is equal to 0.043"
## [1] "The DL is equal to 0.029"
## [1] "The DL is equal to 0.363636363636364"
## [1] "The DL is equal to 3.38888888888889"
## [1] "The DL is equal to 0.166666666666667"
## [1] "The DL is equal to 0.036231884057971"
## [1] "The DL is equal to 0.375"
## [1] "The DL is equal to 8.25"
## [1] "The DL is equal to 0.3"
## [1] "The DL is equal to 0.4375"
## [1] "The DL is equal to 4.45833333333333"
## [1] "The DL is equal to 0.222222222222222"
## [1] "The DL is equal to 0.205882352941176"
## [1] "The DL is equal to 0.173478260869565"
## [1] "The DL is equal to 0.029"
## [1] "The DL is equal to 0.117647058823529"
## [1] "The DL is equal to 3.25"
## [1] "The DL is equal to 0.777777777777778"
## [1] "The DL is equal to 2.75"
## [1] "The DL is equal to 10.7628571428571"
## [1] "The DL is equal to 0.117647058823529"
## [1] "The DL is equal to 1.29166666666667"
## [1] "The DL is equal to 0.537222222222222"
## [1] "The DL is equal to 0.192307692307692"
## [1] "The DL is equal to 0.222222222222222"
## [1] "The DL is equal to 0.00886666666666666"
## [1] "The DL is equal to 0.5"
## [1] "The DL is equal to 0.5625"
## [1] "The DL is equal to 0.684782608695652"
## [1] "The DL is equal to 0.104375"
## [1] "The DL is equal to 0.270625"

```

```
## [1] "The DL is equal to 0.375"
## [1] "The DL is equal to 0.03125"
## [1] "The DL is equal to 0.125"
## [1] "The DL is equal to 0.395625"
## [1] "The DL is equal to 0.388888888888889"
## [1] "The DL is equal to 0.125"
## [1] "The DL is equal to 1.6"
## [1] "The DL is equal to 0.1875"
## [1] "The DL is equal to 1.42857142857143"
## [1] "The DL is equal to 0.272727272727273"
## [1] "The DL is equal to 0.428571428571429"
## [1] "The DL is equal to 0.08"
## [1] "The DL is equal to 0.875"
## [1] "The DL is equal to 2.61111111111111"
## [1] "The DL is equal to 20.8888888888889"
## [1] "The DL is equal to 0.8"
## [1] "The DL is equal to 1.375"
## [1] "The DL is equal to 0.5"
## [1] "The DL is equal to 0.611111111111111"
## [1] "The DL is equal to 6"
## [1] "The DL is equal to 0.00971014492753623"
## [1] "The DL is equal to 0.291875"
## [1] "The DL is equal to 0.0825"
## [1] "The DL is equal to 0.067"
## [1] "The DL is equal to 2.375"
## [1] "The DL is equal to 0.0625"
```

```
meta_congo = ps_congo_picrust %>%
  sample_tibble("sample_id") %>%
  select(sample_id, CST)
path_tax_congo = ps_congo_picrust %>%
  tax_tibble("KO")

mean_long_df <- ps_congo_picrust_rel %>%
  psmelt.dplyr() %>%
  group_by(OTU, CST) %>%
  mutate(mean = mean(abundance),
         sd = sd((abundance))) %>%
  dplyr::select(OTU, CST, mean, sd) %>%
  distinct(mean, sd, .keep_all = TRUE) %>%
  pivot_longer(-c(OTU, CST), names_to = "stats", values_to = "mean_value")
  ungroup()
```

```
## Joining, by = "Sample"
## Joining, by = "OTU"
```

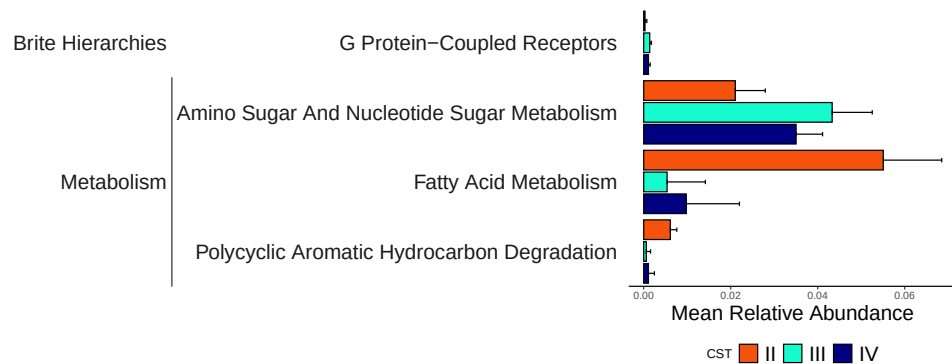
```
mean_df <- mean_long_df %>% filter(!grepl("sd", stats))
mean_df$sd <- mean_long_df %>% filter(grepl("sd", stats)) %>% pull(mean_values)

mean_df %>%
  rename(KO = OTU) %>%
  filter(KO %in% ko_daa_congo) %>%
  left_join(path_tax_congo, by = "KO") %>%
```

```

ggplot(aes(x = desc(CST), y = mean_values, fill = CST)) +
  geom_bar(stat="identity", color="black",
           position=position_dodge()) +
  geom_errorbar(aes(ymin=mean_values, ymax=mean_values+sd), width=.2,
               position=position_dodge(.9)) +
  scale_fill_manual(values = CST_col) +
  coord_flip() +
  ggh4x::facet_nested(pathway_1 + pathway_3_clean~.,
                      scales = "free", switch = "y",
                      nest_line = element_line(colour = "black")) +
  labs(y = "Mean Relative Abundance") +
  theme_classic() +
  theme(strip.text.y.left = element_text(angle = 0, size = 18, hjust = 1),
        # panel.border = element_rect(color = "grey", fill = NA),
        strip.background = element_blank(),
        axis.title.y = element_blank(),
        axis.text.y = element_blank(),
        axis.ticks.y = element_blank(),
        axis.line.y = element_blank(),
        legend.text = element_text(size = 18),
        strip.placement = "outside",
        axis.title.x = element_text(size = 18),
        legend.position = "bottom",
        # aspect.ratio = 0.4,
        axis.line.x = element_line(),
        # plot.margin=grid::unit(c(0,0,0,0), "mm"),
        panel.spacing=unit(0,'lines'))

```



```

#ggsave(paste0(output_folder, "figures/bar_plot_DAA_picrust_congo.pdf"), width = 20)
#ggsave(paste0(output_folder, "figures/bar_plot_DAA_picrust_congo.pdf"), width = 20)

```

Figure 6: Meta-analysis: Predicted KEGG pathways differentially abundant by health status

perform DAAs

```
all_meta_analysis_df <- vroom::vroom(file = 'tables/ko_df_meta_analysis_unstrat_with_annotation.tsv')
```

```
## Rows: 130 Columns: 668
## -- Column specification -----
## Delimiter: "\t"
## chr (4): KO, pathway_1, pathway_2, pathway_3
## dbl (664): EU_PRJEB47492_ERR6799751, EU_PRJEB47492_ERR6799752, EU_PRJEB47492...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
melt_meta_df = all_meta_analysis_df %>%
  as.data.table() %>%
  melt(variable.factor = FALSE)

# agglomerate at pathway3 level
path_glom_meta_df = melt_meta_df %>%
  mutate(pathway_3 = paste(pathway_1, pathway_2, pathway_3, sep = "_")) %>%
  group_by(variable, pathway_3) %>%
  summarise(abundance = sum(value)) %>%
  pivot_wider(id_cols = pathway_3, names_from = "variable", values_from =
```

```
## 'summarise()' has grouped output by 'variable'. You can override using the
## '.groups' argument.
```

```
pathway_tax_meta = path_glom_meta_df %>%
  mutate(KO = paste0("KO_", 1:nrow(.)), .before = 1) %>%
  dplyr::select(KO, pathway_3) %>%
  separate(pathway_3, sep = "_", into = c('pathway_1', 'pathway_2', 'pathway_3_clean'), rem
```

```
# make a phyloseq object
samples_picrust = colnames(path_glom_meta_df %>% column_to_rownames("pathway_3"))
load('data/ps_raw_all.RData')
meta_meta_analysis = ps_raw %>%
  sample_tibble("sample_id") %>%
  # select(-Other_vaginal_infection) %>%
  mutate(Asymptomatic = replace(Asymptomatic, is.na(Asymptomatic), 0)) %>%
  mutate(BV = replace(BV, is.na(BV), 0)) %>%
  mutate(HIV = replace(HIV, is.na(HIV), 0)) %>%
  mutate(HPV = replace(HPV, is.na(HPV), 0)) %>%
  mutate(health_status = case_when(Asymptomatic == 1 ~ "Asymptomatic",
                                   HIV == 1 ~ "HIV",
                                   BV == 1 ~ "BV",
                                   HPV == 1 ~ "HPV",
                                   TRUE ~ "Asymptomatic")) %>%
  filter(sample_id %in% samples_picrust)
```

```
CST = read_delim("tables/metadata_final.tsv") %>%
  filter(sampleID %in% samples_picrust) %>%
  dplyr::rename(sample_id = sampleID) %>%
  dplyr::select(sample_id, CST) %>%
```

```

mutate(CST = case_when(str_detect(CST, "IV-A|IV-B|IV-C") ~ "IV",
                        TRUE ~ as.character(CST))) %>%
mutate(CST = factor(CST, levels = c("I", "II", "III", "IV", "V")))

## Rows: 1851 Columns: 28
## -- Column specification -----
## Delimiter: "\t"
## chr (18): sampleID, continent, unsdg_region, country, study, condition, sequ...
## dbl (10): age_years, bmikgm2, Birth_outcoome, Asymptomatic, BV, HIV, Antibio...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

meta_meta_analysis = meta_meta_analysis %>%
  left_join(CST, by = "sample_id")

# make a phyloseq object
to_otu_table = path_gloom_meta_df %>%
  mutate(across(where(is.numeric), ~abs(.))) %>%
  mutate(KO = paste0("KO_", 1:nrow(.)), .before = 1) %>%
  dplyr::select(-pathway_3) %>%
  column_to_rownames('KO')
OTU_m = otu_table(as.matrix(to_otu_table), taxa_are_rows = TRUE)
TAX_m = tax_table(as.matrix(pathway_tax_meta %>% column_to_rownames('KO')))
META_m = sample_data(meta_meta_analysis %>% column_to_rownames("sample_id"))
ps_meta_picrust <- phyloseq(OTU_m, TAX_m, META_m)

# saveRDS(ps_meta_picrust, 'ps_meta_picrust.RDS')

ps_meta_picrust = readRDS('data/ps_meta_picrust.RDS')
meta_meta_analysis = ps_meta_picrust %>% sample_tibble("sample_id")
####---- LinDA ----####
# Health Status + Study
# INPUTS:
# ASV by row and samples by columns
# meta: samples by rows
linda_input_asv <- ps_meta_picrust %>%
  otu_tibble("KO") %>%
  mutate(across(where(is.numeric), ~round(.))) %>%
  column_to_rownames("KO")

linda_meta <- meta_meta_analysis %>%
  column_to_rownames("sample_id")

# run LinDA
linda.obj_meta <- linda(otu.tab = linda_input_asv,
  meta = linda_meta,
  # phyloseq.obj = NULL,
  formula = "~health_status + study",
  type = 'count',
  adaptive = TRUE,
  imputation = TRUE,
  # pseudo.cnt = 0.65,

```

```

corr.cut = 0.1,
p.adj.method = 'BH',
alpha = 0.05,
prev.cut = 0,
lib.cut = 1,
winsor.quan = 0.97,
n.cores = cores
)

```

Imputation approach is used.

```

# get the results
linda_res_meta <- linda.obj_meta$output %>%
  imap(., ~mutate(.x, name = .y)) %>%
  map_df(., ~rownames_to_column(., "KO") %>%
    as_tibble %>%
    arrange(padj) %>%
    # reject H0??
    mutate(reject = ifelse(padj < 0.05, "Yes", "No"))) %>%
  mutate(method = "LinDA")

# linda_res %>% filter(padj < 0.05) %>% View()

# save results
#write_csv(linda_res_meta, "tables/DAA/linda_res_picrust_meta.csv")
linda_res_meta <- read_csv( "tables/DAA/linda_res_picrust_meta.csv")

```

```

## Rows: 480 Columns: 11
## -- Column specification -----
## Delimiter: ","
## chr (4): KO, reject, name, method
## dbl (7): baseMean, log2FoldChange, lfcSE, stat, pvalue, padj, df
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

```

```

####---- ANCOM-BC ----####
# prepare the phyloseq object
# make a phyloseq object first
# perform the ANCOM-BC analysis
ancombc_out_meta <- ancombc2(
  data = ps_meta_picrust,
  assay_name = "counts",
  # tax_level= "pathway_2",
  fix_formula = "health_status + study",
  p_adj_method = "fdr",
  pseudo = 0,
  pseudo_sens = TRUE,
  prv_cut = 0, # prev filtering has been done above already
  lib_cut = 0,
  group = "health_status",
  struc_zero = TRUE,

```



```

neg_lb = TRUE,
iter_control = list(tol = 1e-5, max_iter = 20, verbose = FALSE),
em_control = list(tol = 1e-5, max_iter = 100), # use max_iter >= 100 on real data
alpha = 0.05,
global = TRUE, # multi group comparison will be deactivated automatically
verbose = T,
n_cl = cores)

```

```

## Checking the input data type ...
## The input data is of type: phyloseq
## PASS
## Checking the sample metadata ...
## The specified variables in the formula: health_status, study
## The available variables in the sample metadata: sampleID, continent, unsdg_region, country, study, c
## PASS
## Checking other arguments ...
## The number of groups of interest is: 2
## The sample size per group is: Asymptomatic = 579, HIV = 85
## PASS
## Obtaining initial estimates ...
## Estimating sample-specific biases ...
## Conducting sensitivity analysis for pseudo-count addition to 0s ...
## For taxa that are significant but do not pass the sensitivity analysis,
## please flag them and proceed with caution, as they are likely false positives.
## For detailed instructions on performing sensitivity analysis,
## please refer to the package vignette.
## ANCOM-BC2 primary results ...

```

```

# store the results in res
ancombc_res_meta <- ancombc_out_meta$res %>%
  as.data.frame() %>%
  pivot_longer(cols = -taxon,
               names_to = "variable",
               values_to = "value") %>%
  # "merge" = preserve any extra characters after the first underscore in the second column
  separate(variable, c("stats", "name"), sep = "_", extra = "merge") %>%
  pivot_wider(id_cols = c(taxon, name),
              names_from = "stats",
              values_from = "value") %>%
  mutate(diff = if_else(diff == 1, TRUE, FALSE),
         method = "ANCOM-BC") %>%
  dplyr::rename(KO = taxon)

# save results
#write_csv(ancombc_res_meta, "tables/DAA/ancombc_res_picrust_meta.csv")
ancombc_res_meta <- read_csv("tables/DAA/ancombc_res_picrust_meta.csv")

```

```

## Rows: 525 Columns: 9
## -- Column specification -----
## Delimiter: ","
## chr (3): KO, name, method
## dbl (5): lfc, se, W, p, q
## lgl (1): diff

```

```
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
####--- ALDEx2 ----####
df_aldex_meta <- ps_meta_picrust %>%
  otu_tibble("K0") %>%
  column_to_rownames("K0") %>%
  mutate(across(where(is.numeric), ~round(.)))

# make model matrix
mmm <- model.matrix(~ health_status + study, data = meta_meta_analysis)

# INPUT: ASV by rows and samples by columns
# CLR transformation
xm <- aldex.clr(df_aldex_meta, mmm, mc.samples=128, denom="all", verbose=F)
```

```
## integer matrix provided
## using all features for denominator
## operating in serial mode
## computing center with all features
```

```
# apply glm
glm.test_meta <- aldex.glm(xm, mmm)
```

```
## running tests for each MC instance:
```

```
## |------(25%)------(50%)------(75%)-----|
```

```
glm.effect_meta <- aldex.glm.effect(xm)
```

```
## operating in serial mode
## sanity check complete
## rab.all complete
## rab.win complete
## rab of samples complete
## within sample difference calculated
## between group difference calculated
## group summaries calculated
## unpaired effect size calculated
## summarizing output
## operating in serial mode
## sanity check complete
## rab.all complete
## rab.win complete
## rab of samples complete
## within sample difference calculated
## between group difference calculated
## group summaries calculated
## unpaired effect size calculated
## summarizing output
```

```

## operating in serial mode
## sanity check complete
## rab.all complete
## rab.win complete
## rab of samples complete
## within sample difference calculated
## between group difference calculated
## group summaries calculated
## unpaired effect size calculated
## summarizing output
## operating in serial mode
## sanity check complete
## rab.all complete
## rab.win complete
## rab of samples complete
## within sample difference calculated
## between group difference calculated
## group summaries calculated
## unpaired effect size calculated
## summarizing output
## operating in serial mode
## sanity check complete
## rab.all complete
## rab.win complete
## rab of samples complete
## within sample difference calculated
## between group difference calculated
## group summaries calculated
## unpaired effect size calculated
## summarizing output
## operating in serial mode
## sanity check complete
## rab.all complete
## rab.win complete
## rab of samples complete
## within sample difference calculated
## between group difference calculated
## group summaries calculated
## unpaired effect size calculated
## summarizing output

```

```

# make a data frame
aldex2_res_meta <- glm.effect_meta %>%
  imap(., ~mutate(., name = .y)) %>%
  map_df(., ~rownames_to_column(., "KO") %>%
    as_tibble) %>%
  mutate(method = "ALDEx2")

# save results
#write_csv(aldex2_res_meta, "tables/DAA/aldex2_res_picrust_meta.csv")
aldex2_res_meta <- read_csv("tables/DAA/aldex2_res_picrust_meta.csv")

```

```

## Rows: 480 Columns: 10
## -- Column specification -----

```

```
## Delimiter: ","
## chr (3): KO, name, method
## dbl (7): rab.all, rab.win.0, rab.win.1, diff.btw, diff.win, effect, overlap
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
# put all methods together and count how many taxa were identified
# score = intersection of identified taxa
```

```
all_methods_meta <- linda_res_meta %>%
  filter(padj < 0.05) %>%
  dplyr::select(KO, LinDA = reject) %>%
  distinct(KO, .keep_all = TRUE) %>%
  mutate(LinDA = ifelse(LinDA == "No", FALSE, TRUE)) %>%
full_join(ancombc_res_meta %>%
  filter(q < 0.05) %>%
  dplyr::select(KO, ancombc = diff) %>%
  distinct(KO, .keep_all = TRUE),
  by = "KO",
  relationship = "many-to-many") %>%
full_join(aldex2_res_meta %>%
  filter(abs(effect) > 0.5) %>%
  dplyr::select(KO, aldex2 = effect) %>%
  distinct(KO, .keep_all = TRUE),
  by = "KO",
  relationship = "many-to-many") %>%
mutate(aldex2 = if_else(abs(aldex2) >= 0.05, TRUE, FALSE),
  across(-KO, function(x) ifelse(is.na(x), FALSE, x)),
  score = rowSums(across(c(aldex2, ancombc, LinDA))),
)
```

```
only_health_status <- linda_res_meta %>%
  filter(padj < 0.05) %>%
  filter(name == "health_statusHIV") %>%
  dplyr::select(KO, LinDA = reject) %>%
  distinct(KO, .keep_all = TRUE) %>%
  mutate(LinDA = ifelse(LinDA == "No", FALSE, TRUE)) %>%
full_join(ancombc_res_meta %>%
  filter(q < 0.05) %>%
  filter(name == "health_statusHIV") %>%
  dplyr::select(KO, ancombc = diff) %>%
  distinct(KO, .keep_all = TRUE),
  by = "KO",
  relationship = "many-to-many") %>%
full_join(aldex2_res_meta %>%
  filter(abs(effect) > 0.5) %>%
  filter(name == "health_statusHIV") %>%
  dplyr::select(KO, aldex2 = effect) %>%
  distinct(KO, .keep_all = TRUE),
  by = "KO",
  relationship = "many-to-many") %>%
mutate(aldex2 = if_else(abs(aldex2) >= 0.05, TRUE, FALSE),
  across(-KO, function(x) ifelse(is.na(x), FALSE, x)),
  score = rowSums(across(c(aldex2, ancombc, LinDA))),
```

```

)

# check which variables each genus was significant
variables_all <- linda_res_meta %>%
  filter(padj < 0.05) %>%
  dplyr::select(KO, name) %>%
  mutate(variable = str_extract_all(name, "(health_status|study)")) %>%
  group_by(KO) %>%
  summarise(variable_linda = paste(unique(variable), collapse = " and ")) %>%
  full_join(ancombc_res_meta %>%
    filter(q < 0.05) %>%
    dplyr::select(KO, name) %>%
    filter(name != "(Intercept)") %>%
    mutate(variable = str_extract_all(name, "(health_status|study)")) %>%
    group_by(KO) %>%
    summarise(variable_ancombc = paste(unique(variable), collapse = " and ")),
    by = "KO") %>%
  full_join(aldex2_res_meta %>%
    filter(abs(effect) > 0.5) %>%
    dplyr::select(KO, name) %>%
    mutate(variable = str_extract_all(name, "(health_status|study)")) %>%
    group_by(KO) %>%
    summarise(variable_aldex2 = paste(unique(variable), collapse = " and ")),
    by = "KO") %>%
  left_join(pathway_tax_meta, by = "KO")

# put all together for the upset plot
all_methods_meta <- all_methods_meta %>% left_join(variables_all)

```

```
## Joining with 'by = join_by(KO)'
```

```
all_methods_meta %>% filter(score > 2)
```

```
## # A tibble: 19 x 12
##   KO      LinDA ancombc aldex2 score variable_linda variable_ancombc
##   <chr> <lg1> <lg1> <lg1> <dbl> <chr>          <chr>
## 1 KO_43 TRUE  TRUE  TRUE     3 study          study
## 2 KO_49 TRUE  TRUE  TRUE     3 study          study
## 3 KO_24 TRUE  TRUE  TRUE     3 study          study
## 4 KO_45 TRUE  TRUE  TRUE     3 study          study
## 5 KO_51 TRUE  TRUE  TRUE     3 study          study
## 6 KO_56 TRUE  TRUE  TRUE     3 study          study
## 7 KO_68 TRUE  TRUE  TRUE     3 study          study
## 8 KO_9  TRUE  TRUE  TRUE     3 study          study
## 9 KO_2  TRUE  TRUE  TRUE     3 study          study
## 10 KO_33 TRUE  TRUE  TRUE     3 study          study
## 11 KO_72 TRUE  TRUE  TRUE     3 study          study
## 12 KO_14 TRUE  TRUE  TRUE     3 study          study
## 13 KO_48 TRUE  TRUE  TRUE     3 study          study
## 14 KO_71 TRUE  TRUE  TRUE     3 study          study
## 15 KO_28 TRUE  TRUE  TRUE     3 study          study

```

```
## 16 KO_22 TRUE TRUE TRUE 3 study study
## 17 KO_59 TRUE TRUE TRUE 3 study study
## 18 KO_15 TRUE TRUE TRUE 3 study study
## 19 KO_74 TRUE TRUE TRUE 3 study study
## # i 5 more variables: variable_al dex2 <chr>, pathway_3 <chr>, pathway_1 <chr>,
## # pathway_2 <chr>, pathway_3_clean <chr>

all_methods_meta = all_methods_meta %>% left_join(pathway_tax_meta, by = "KO")

#write_csv(all_methods_meta, "tables/DAA/all_methods_picrust_meta.csv")
all_methods_meta <- read_csv("tables/DAA/all_methods_picrust_meta.csv")

## Rows: 78 Columns: 12
## -- Column specification -----
## Delimiter: ","
## chr (8): KO, variable_linda, variable_ancombc, variable_al dex2, pathway_3, p...
## dbl (1): score
## lgl (3): LinDA, ancombc, al dex2
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

Figure 6

```
# load files
ps_meta_picrust = readRDS("data/ps_meta_picrust.RDS")
# health Status
all_methods_meta <- read_csv("tables/DAA/all_methods_picrust_meta.csv")

## Rows: 78 Columns: 12
## -- Column specification -----
## Delimiter: ","
## chr (8): KO, variable_linda, variable_ancombc, variable_al dex2, pathway_3, p...
## dbl (1): score
## lgl (3): LinDA, ancombc, al dex2
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

# get the KO IDs that were found in at least 2 DAA tools
ko_daa_meta = all_methods_meta %>%
  filter(score >= 2) %>%
  filter(if_any(starts_with("variable"), ~!str_detect(., "health_status"))) %>%
  pull(KO)

ps_meta_picrust_rel = ps_meta_picrust %>% transform("compositional")
path_tax_meta = ps_meta_picrust %>%
  tax_tibble("KO")
```

```

mean_long_df <- ps_meta_picrust_rel %>%
  psmelt.dplyr() %>%
  group_by(OTU, health_status) %>%
  mutate(mean = mean(abundance),
         sd = sd((abundance))) %>%
  select(OTU, health_status, mean, sd) %>%
  distinct(mean, sd, .keep_all = TRUE) %>%
  pivot_longer(-c(OTU, health_status), names_to = "stats", values_to =

```

```

## Joining, by = "Sample"
## Joining, by = "OTU"

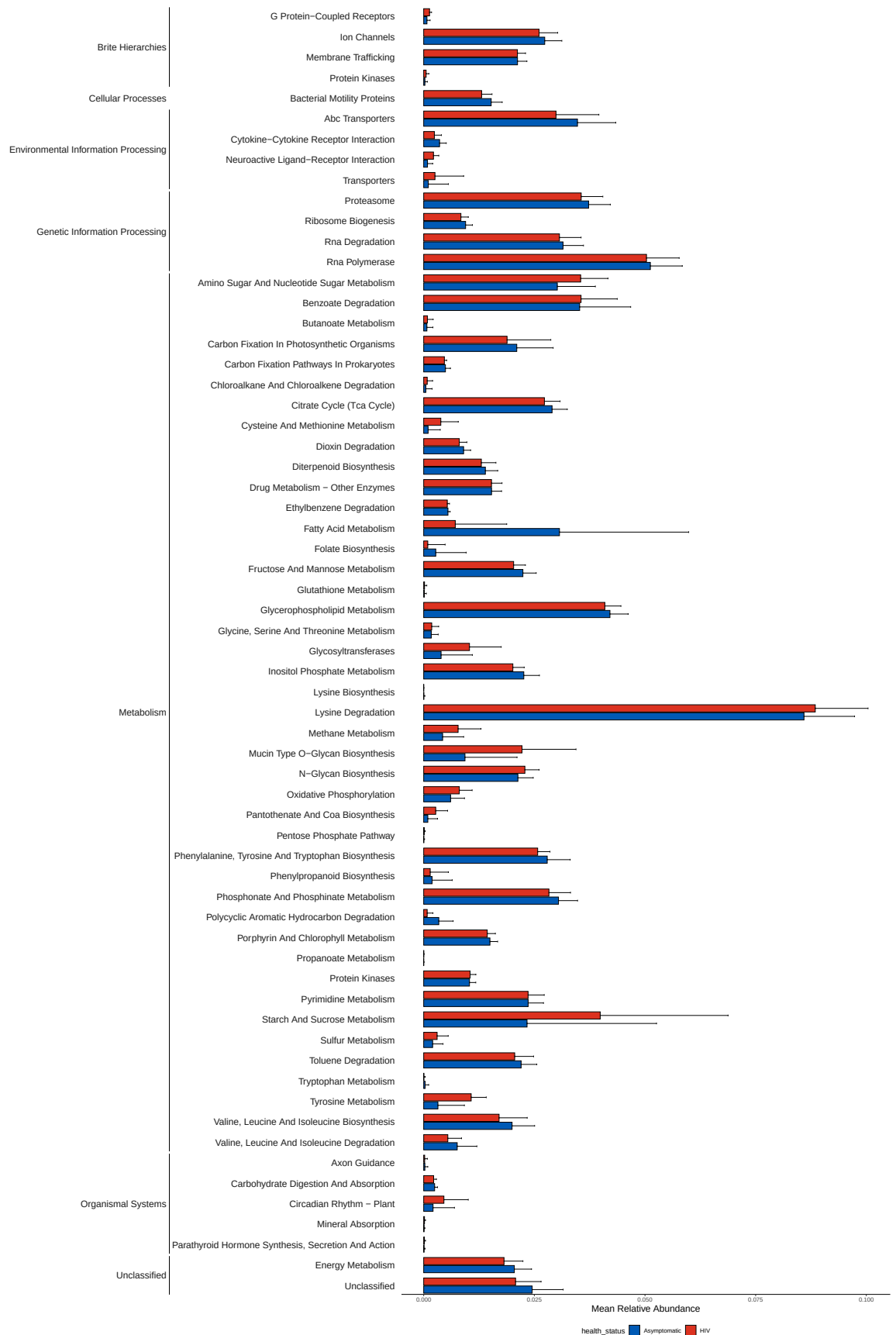
```

```

mean_df <- mean_long_df %>% filter(!grepl("sd", stats))
mean_df$sd <- mean_long_df %>% filter(grepl("sd", stats)) %>% pull(mean_values)

mean_df %>%
  rename(KO = OTU) %>%
  filter(KO %in% ko_daa_meta) %>%
  left_join(path_tax_meta, by = "KO") %>%
  ggplot(aes(x = health_status, y = mean_values, fill = health_status)) +
    geom_bar(stat="identity", color="black",
            position=position_dodge()) +
    geom_errorbar(aes(ymin=mean_values, ymax=mean_values+sd), width=.2,
                  position=position_dodge(.9)) +
    scale_fill_manual(values = c("HIV" = "#DC3220", "Asymptomatic" = "#005AB5")) +
    coord_flip() +
    facet_nested(pathway_1 + pathway_3_clean~,
                  scales = "free", switch = "y",
                  nest_line = element_line(colour = "black")) +
    labs(y = "Mean Relative Abundance") +
    theme_classic() +
    theme(strip.text.y.left = element_text(angle = 0, size = 14, hjust = 1),
          # panel.border = element_rect(color = "grey", fill = NA),
          strip.background = element_blank(),
          axis.title.y = element_blank(),
          axis.text.y = element_blank(),
          axis.ticks.y = element_blank(),
          axis.line.y = element_blank(),
          strip.placement = "outside",
          axis.title.x = element_text(size = 14),
          legend.position = "bottom",
          # aspect.ratio = 0.3,
          axis.line.x = element_line(),
          panel.spacing=unit(0.3,'lines'))

```




```
# ggsave(paste0(output_folder, "figures/bar_plot_DAA_picrust_meta.pdf"), height = 28, width = 20)
```

Figure 7: CQI-PMTCT: Correlation between CVMB and immune mediators

```
#specify the columns that have immune mediator information
enrol_columns <-
  grep("^enrol", colnames(sample_data(ps_congo_filt_spp_rclr)), value = TRUE)
#subset phyloseq
ps_all_cyto <-
  subset_samples(ps_congo_filt_spp_rclr, rowSums(is.na(sample_data(
    ps_congo_filt_spp_rclr
  ))[, enrol_columns])) == 0) #53 samples

otu_cyt <- as.data.frame(otu_table(ps_all_cyto))

meta_cyt_log <- sample.data.frame(ps_all_cyto) %>% select(all_of(enrol_columns)) %>% mutate(across(wher

#meta_cong <- sample.data.frame(ps_congo_filt_spp_rclr)

results_list <- list()
# Loop through each cytokine and OTU pair
for (cytokine_name in colnames(meta_cyt_log)) {
  for (otu_name in colnames(otu_cyt)) {
    # Extract the data for the current pair
    x <- otu_cyt[[otu_name]]
    y <- meta_cyt_log[[cytokine_name]]

    # Perform linear regression
    model <- lm(y ~ x)

    # Extract the Spearman correlation coefficient (rho)
    rho <- cor(x, y, method = "spearman")

    # Extract p-value from the model summary
    p_value <- summary(model)$coefficients[2, 4]

    # Store the results in a list
    results_list[[paste(otu_name, cytokine_name, sep = "_")] <- list(
      OTU = otu_name,
      Cytokine = cytokine_name,
      Rho = rho,
      PValue = p_value
    )
  }
}

# Convert the results list to a data frame
results_df <- do.call(rbind, lapply(results_list, function(res) {
  data.frame(
```

```

    OTU = res$OTU,
    Cytokine = res$Cytokine,
    Rho = res$Rho,
    PValue = res$PValue
  )
}))

# Add taxonomic information to the results
taxa <- tax_table(ps_all_cyto) %>%
  as.data.frame() %>%
  rownames_to_column("OTU") %>%
  mutate(
    Taxon = case_when(
      !is.na(Species) ~ paste(sub("^g__", "", Genus), sub("^s__", "", Species), sep = "_"),
      !is.na(Genus) ~ Genus,
      !is.na(Family) ~ Family,
      !is.na(Order) ~ Order,
      !is.na(Class) ~ Class,
      !is.na(Phylum) ~ Phylum,
      TRUE ~ "Unknown"
    )
  ) %>%
  select(-Kingdom, -Species, -Genus, -Family, -Order, -Class, -Phylum)

# Join taxonomic information with the results
corr_results_df <- results_df %>% left_join(taxa, by = "OTU")

# Adjust p-values using the Benjamini-Hochberg method
corr_results_df$qvalue <- p.adjust(corr_results_df$PValue, method = "BH")

# Save the full results to a CSV file
# write.csv(results_df, "stats/congo/linear_regression_results_log.csv", row.names = FALSE)

###REVIEWER SUGGESTION- add two sided pvalues
results_table_asv <- data.frame(
  Cytokine = character(),
  OTU = character(),
  Taxon = character(),
  Rho = numeric(),
  PValue = numeric(),
  TwoSided_PValue = numeric(),
  Adjusted_PValue = numeric(),
  stringsAsFactors = FALSE
)

# Fill the table with results
for (i in 1:nrow(corr_results_df)) {
  # Extract the cytokine name, rho, and p-value
  cytokine_name <- corr_results_df$Cytokine[i]
  taxon_name <- corr_results_df$Taxon[i]
  otu_name <- corr_results_df$OTU[i]
  rho <- corr_results_df$Rho[i]

```

```

p_value <- corr_results_df$PValue[i]

# Calculate two-sided p-value from correlation test
x <- otu_cyt[[otu_name]]
y <- meta_cyt_log[[cytokine_name]]
cor_test <- cor.test(x, y, method = "spearman") # or "pearson" depending on your method
two_sided_p <- cor_test$p.value

# Add to results table
results_table_asv <- rbind(results_table_asv,
  data.frame(Cytokine = cytokine_name,
    Rho = rho,
    OTU = otu_name,
    Taxon = taxon_name,
    PValue = p_value,
    TwoSided_PValue = two_sided_p,
    Adjusted_PValue = NA, # Will fill this after the loop
    stringsAsFactors = FALSE))
}

# Apply Benjamini-Hochberg correction to the two-sided p-values
results_table_asv$Adjusted_PValue <- p.adjust(results_table_asv$PValue, method = "BH")
# write.csv(results_table_asv, "correlation_results_taxa_cyt.csv", row.names = FALSE)

# Define the top 3 OTUs
top3 <- c("ASV1", "ASV2", "ASV4")

# Filter results for the top 10 OTUs and significant p-values (PValue <= 0.05)
corr_results_df_top3 <- results_table_asv %>%
  filter(OTU %in% top3) %>%
  filter(TwoSided_PValue <= 0.05)

##Shannon

# Load alpha diversity data
alpha_div <- read_csv("tables/congo_alpha_div.csv")

## New names:
## Rows: 81 Columns: 69
## -- Column specification
## ----- Delimiter: "," chr
## (11): sampleID, Age_Bin, Trimester, ART_regimen, subCST, CST, Preg_outco... dbl
## (58): ...1, Chao1, Simpson Index, Shannon Entropy, age_years, Gestationa...
## i Use 'spec()' to retrieve the full column specification for this data. i
## Specify the column types or set 'show_col_types = FALSE' to quiet this message.
## * ' -> '...1'

# Filter and select relevant columns
alpha_div <- alpha_div %>%
  select(sampleID, `Shannon Entropy`, CST) %>%
  filter(sampleID %in% rownames(meta_cyt_log))

```

```

# Initialize an empty list to store results
results_list_alpha <- list()

# Loop through each cytokine
for (cytokine_name in colnames(meta_cyt_log)) {
  # Extract the data for the current pair
  x <- alpha_div$`Shannon Entropy`
  y <- meta_cyt_log[[cytokine_name]]

  # Perform linear regression
  model <- lm(y ~ x)

  # Extract the Spearman correlation coefficient (rho)
  rho <- cor(x, y, method = "spearman")

  # Extract p-value from the model summary
  p_value <- summary(model)$coefficients[2, 4]

  # Store the results in a list
  results_list_alpha[[cytokine_name]] <- list(
    Cytokine = cytokine_name,
    Rho = rho,
    PValue = p_value,
    Model = model
  )
}

# Convert the results list to a data frame
corr_results_alpha <- do.call(rbind, lapply(results_list_alpha, function(res) {
  data.frame(
    Cytokine = res$Cytokine,
    Rho = res$Rho,
    PValue = res$PValue
  )
}))

# Save the results to a CSV file
# write.csv(corr_results_alpha, "stats/congo/shannon_cytokine_correlation_results.csv", row.names = FALSE)

#add two sided pvalues
results_table <- data.frame(
  Cytokine = character(),
  Rho = numeric(),
  PValue = numeric(),
  TwoSided_PValue = numeric(),
  Adjusted_PValue = numeric(),
  stringsAsFactors = FALSE
)

# Fill the table with results
for (i in 1:nrow(corr_results_alpha)) {
  # Extract the cytokine name, rho, and p-value

```

```

cytokine_name <- corr_results_alpha$Cytokine[i]
rho <- corr_results_alpha$Rho[i]
p_value <- corr_results_alpha$PValue[i]

# Calculate two-sided p-value from correlation test
x <- alpha_div$`Shannon Entropy`
y <- meta_cyt_log[[cytokine_name]]
cor_test <- cor.test(x, y, method = "spearman") # or "pearson" depending on your method
two_sided_p <- cor_test$p.value

# Add to results table
results_table <- rbind(results_table,
                        data.frame(Cytokine = cytokine_name,
                                   Rho = rho,
                                   PValue = p_value,
                                   TwoSided_PValue = two_sided_p,
                                   Adjusted_PValue = NA, # Will fill this after the loop
                                   stringsAsFactors = FALSE))
}

# Apply Benjamini-Hochberg correction to the two-sided p-values
results_table$Adjusted_PValue <- p.adjust(results_table$PValue, method = "BH")

# Print the table
#print(results_table)

# Optional: Write to CSV file
#write.csv(results_table, "correlation_results_shannon_cyt.csv", row.names = FALSE)

```

A-B. Correlation of immune factors to shannon diversity

```

# Loop through each row in the results_df_alpha data frame
for (i in 1:nrow(results_table)) {
  # Extract the cytokine name, rho, and p-value
  cytokine_name <- results_table$Cytokine[i]
  rho <- results_table$Rho[i]
  p_value <- results_table$TwoSided_PValue[i]

  # Check if the p-value is significant (p <= 0.06)
  if (p_value <= 0.05) {
    # Extract the data for the current cytokine
    x <- alpha_div$`Shannon Entropy`
    y <- meta_cyt_log[[cytokine_name]]

    # Generate the scatter plot
    p <- ggplot(data = data.frame(x = x, y = y), aes(x = x, y = y)) +
      geom_point() +
      geom_smooth(method = "lm", color = "blue") +
      labs(
        title = paste("Scatter Plot: Shannon Entropy vs", cytokine_name),
        subtitle = paste("r =", round(rho, 3), "| p =", round(p_value, 3)),

```

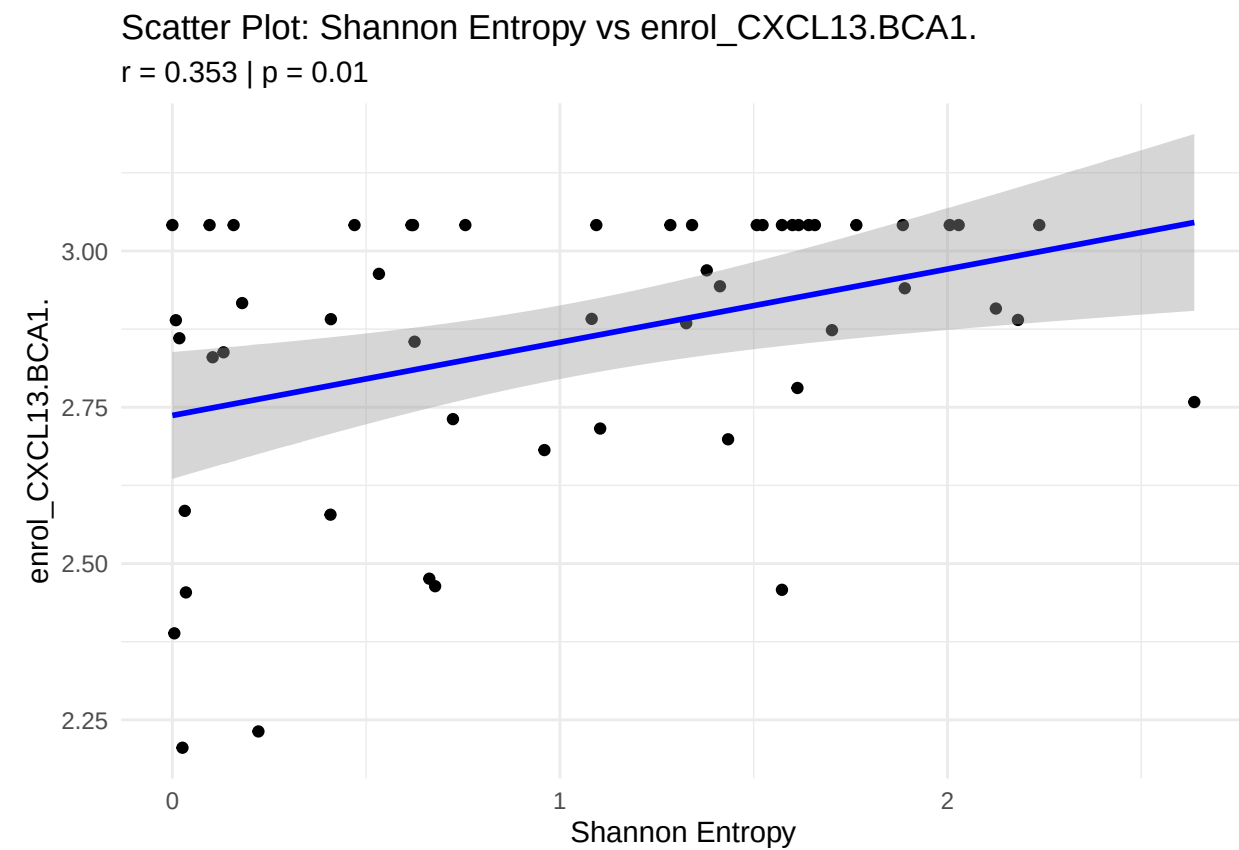
```

    x = "Shannon Entropy",
    y = cytokine_name
  ) +
  theme_minimal()

  print(p)
}
}

```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



C:Correlation of L.iners with sCD163

```

# Loop through each row in the corr_results_df_top3 data frame
for (i in 1:nrow(corr_results_df_top3)) {
  # Extract the ASV name, cytokine name, rho, and p-value
  otu_name <- corr_results_df_top3$OTU[i]
  cytokine_name <- corr_results_df_top3$Cytokine[i]
  rho <- corr_results_df_top3$Rho[i]
  p_value <- corr_results_df_top3$TwoSided_PValue[i]

  # Check if the p-value is significant (p <= 0.09)
  if (p_value <= 0.05) {

```

```

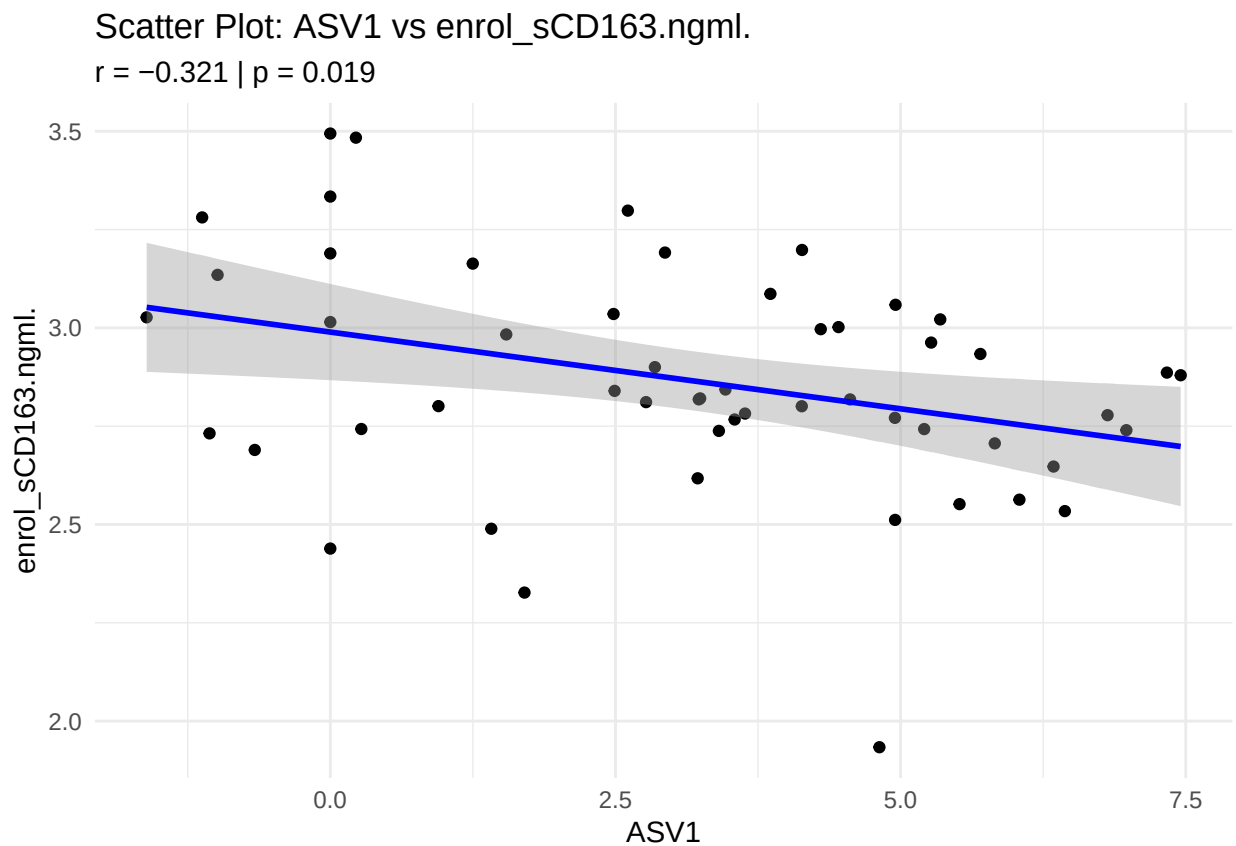
# Extract the data for the current ASV and cytokine
x <- otu_cyt[[otu_name]] # ASV abundance
y <- meta_cyt_log[[cytokine_name]] # Cytokine level

# Generate the scatter plot
p <- ggplot(data = data.frame(x = x, y = y), aes(x = x, y = y)) +
  geom_point() +
  geom_smooth(method = "lm", color = "blue") +
  labs(
    title = paste("Scatter Plot:", otu_name, "vs", cytokine_name),
    subtitle = paste("r =", round(rho, 3), "| p =", round(p_value, 3)),
    x = otu_name,
    y = cytokine_name
  ) +
  theme_minimal()

print(p)
}
}

```

'geom_smooth()' using formula = 'y ~ x'



D:Levels of sTNFRSF1A between CST III and IV

```
meta_cyt_log <- meta_cyt_log %>%
  rownames_to_column(var = "sampleID")
CST_congo <- meta_congo %>% dplyr::select(CST, sampleID=sample_id)
meta_cyt_log2 <- left_join(meta_cyt_log, CST_congo, by = "sampleID")

meta_cyt_log2 <- meta_cyt_log2 %>% filter(CST != "II", CST != "V")
meta_cyt_log2 <- meta_cyt_log2 %>% select(CST, everything())
meta_cyt_log2 <- meta_cyt_log2 %>%
  mutate(across(starts_with("enrol"), as.numeric))
CST_cols = c("III" = "#16FBCD", "IV" = "navy")

corr_results_cst <- list()

# Loop through each cytokine column
for (col in names(meta_cyt_log2)[3:46]) {
  # Perform Wilcoxon test
  wilcox_test <- wilcox.test(meta_cyt_log2[[col]] ~ meta_cyt_log2$CST)
  p_value <- wilcox_test$p.value

  # Store the results in the list
  corr_results_cst[[col]] <- list(
    Cytokine = col,
    PValue = p_value
  )
}

# Convert the results list to a data frame
corr_results_df_cst <- do.call(rbind, lapply(corr_results_cst, function(res) {
  data.frame(
    Cytokine = res$Cytokine,
    PValue = res$PValue
  )
}))

# Save the results to a CSV file
# write.csv(corr_results_df_cst, file = "stats/congo/cytokine_cst_correlations2.csv", row.names = FALSE)

# Filter for significant results (p <= 0.05)
significant_results <- corr_results_df_cst %>% filter(PValue <= 0.05)

# Loop through significant cytokines and generate boxplots
for (col in significant_results$Cytokine) {
  meta_cyt_log2[[col]] <- remove_outliers(meta_cyt_log2[[col]])
  p <- ggplot(meta_cyt_log2 %>% filter(!is.na(CST)),
    aes(x = CST, y = !!sym(col), fill = CST)) + # Corrected aesthetics
    geom_boxplot(outlier.shape = NA) +
    geom_jitter(color = "black", fill = "white", alpha = 0.5, height = 0) +
    labs(
      y = paste(sub("^enrol_", "", col)),

```

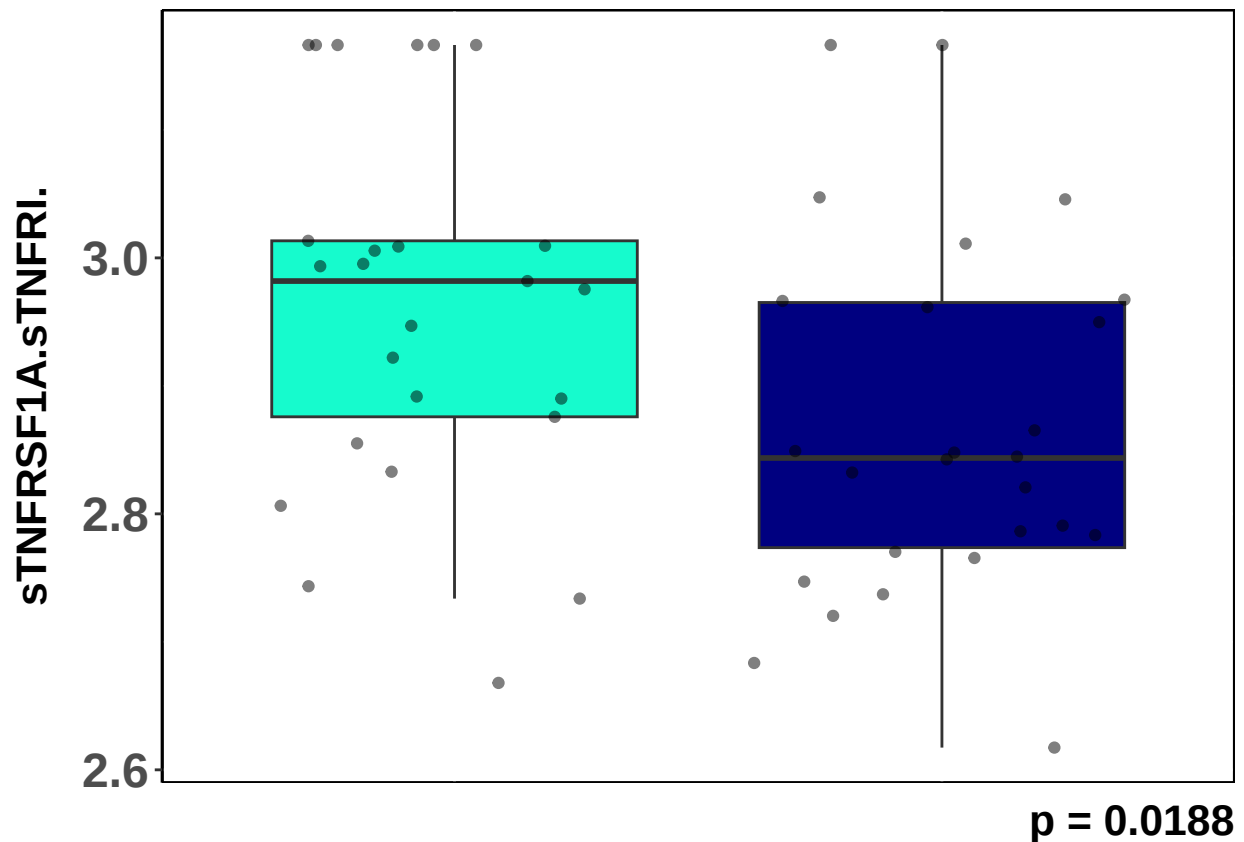


```

x = " ",
caption = paste("p =", round(wilcox.test(meta_cyt_log2[[col]] ~ meta_cyt_log2$CST)$p.value, 4))
) +
theme(
  legend.position = "none",
  panel.background = element_rect(fill = NA, colour = "black"),
  panel.border = element_rect(color = "black", fill = NA),
  axis.line = element_line(color = "black"),
  axis.line.x = element_blank(),
  axis.ticks.x = element_blank(),
  axis.text.x = element_blank(),
  axis.title.x = element_blank(),
  axis.title.y = element_text(margin = margin(t = 0, r = 10, b = 0, l = 0), size = 16, face = "bold", family = "sans"),
  axis.text.y = element_text(size = 18, face = "bold", family = "sans"),
  plot.caption = element_text(size = 16, face = "bold", family = "sans")
) +
scale_fill_manual(values = CST_cols)

print(p)
}

```



Supplementary Figures

Fig S3: Heatmap of correlations between all taxa and immune mediators in the CQI-PMTCT cohort

```
#corr_df <- read.csv("tables/linear_regression_results_log.csv", header = T, sep = ";")
corr_df <- results_table_asv
#count numbe of samples in each taxo
taxa <-
  tax_table(ps_congo_filt_spp_rclr) %>% as.data.frame() %>% rownames_to_column("asvid") %>% mutate(
    Taxon = case_when(
      !is.na(Species) ~ paste(sub("^g_", "", Genus), sub("^s_", "", Species), sep = "_"), !is.na(Genus)
      TRUE ~ "Unknown"
    )
  ) %>% select(-Kingdom, -Species, -Genus, -Family, -Order, -Class, -Phylum)

top10 <- c("Lactobacillus_iners", "Bifidobacterium_388775_Bifidobacterium leopoldii",
  "Bifidobacterium_388775_Bifidobacterium vaginale", "Lactobacillus_crispatus",
  "Fannyhessea_vaginae", "UBA629_sp005465875", "Lactobacillus_gasseri_329736",
  "Megasphaera_A_38692_lornae", "Sneathia_vaginalis", "Prevotella_amnii")
top10<- rev(top10)
# Get the rest of the taxa ordered according to taxa_hclust

taxa_hclust <- corr_df %>%
  dplyr::select(Cytokine, Taxon, Rho) %>%
  tidyr::pivot_wider(
    id_cols = Taxon, names_from = Cytokine, values_from = Rho
  ) %>% mutate_at(vars(-Taxon), ~replace_na(., 0)) %>%
  tibble::column_to_rownames("Taxon") %>%
  as.matrix() %>%
  stats::dist(method = "euclidean") %>%
  hclust(method = "ward.D2")

#taxa_order <- taxa_hclust$labels[taxa_hclust$order]

rest_taxa <- taxa_hclust$labels[taxa_hclust$order]
rest_taxa <- rest_taxa[!rest_taxa %in% top10]

# Combine the top 10 and the rest of the taxa
new_taxa_order <- c(rest_taxa, top10)

# Step 2: Plot the heatmap with the new taxa order
figs2<- corr_df %>%
  ggplot(aes(x = Cytokine, y = Taxon)) +
  geom_tile(aes(fill = Rho), color = "grey40") +
  geom_point(
    data = function(x) filter(x, TwoSided_PValue < 0.05),
    shape = "asterisk", size = 2
  ) +
  geom_point(
    data = function(x) filter(x, Adjusted_PValue < 0.05),
    shape = "circle", size = 3
  )
```

```

) +
scale_y_discrete(limits = new_taxa_order) + # Use the new taxa order here
scale_fill_distiller(palette = "PiYG", limits = c(-0.55, 0.55)) +
theme_minimal() +
theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
labs(
  x = "Cytokine", y = "Species", fill = "Spearman's\nRank\nCorrelation",
  caption = paste(
    "Asterisk indicates p < 0.05, not FDR adjusted",
    "Filled circle indicates FDR corrected p < 0.05", sep = "\n"
  )
)
)
ggsave("figS2_revision1.pdf", figs2, width = 10, height = 10)

```

Fig S2: Covariates explaining microbial variation in the CVMB of the integrated datasets

```

#load phyloseq
ps_filt_spp_clr_zero_adj <- readRDS("data/ps_filt_spp_clr_zero_adj.RDS")
meta_all <- sample.data.frame(ps_filt_spp_clr_zero_adj)
meta_all$racioethnicity[is.na(meta_all$racioethnicity)] <- "unknown"
meta_all2 <- meta_all[, colSums(is.na(meta_all)) < 166]

otu_all <- as.data.frame(otu_table(ps_filt_spp_clr_zero_adj)) %>% t()
# Columns with any NAs

colnames(meta_all2)[colSums(is.na(meta_all2)) > 0]

```

```
## [1] "primer_pair" "Trimester" "PCR_cycles"
```

```
colSums(is.na(meta_all2)) / nrow(meta_all2) * 100
```

```
##          sampleID          continent          unsdg_region
##          0.000000          0.000000          0.000000
##          country          study          condition
##          0.000000          0.000000          0.000000
##          sequencer          sample_loc          sample_loc_simple
##          0.000000          0.000000          0.000000
##          racioethnicity          library_type          primer_pair
##          0.000000          0.000000          10.090361
##          BMI_Bin          Pregnant          Trimester
##          0.000000          0.000000          5.873494
##          Asymptomatic          BV          HIV
##          0.000000          0.000000          0.000000
##          HPV          PCR_cycles          Racioethnic_category
##          0.000000          21.385542          0.000000
##          Age_Bin          health_status          CST
##          0.000000          0.000000          0.000000
```

```
supply(lapply(meta_all2, unique), length)
```

```
##          sampleID          continent          unsdg_region
##          664              3              2
##          country          study          condition
##          5              6              3
##          sequencer          sample_loc          sample_loc_simple
##          2              4              2
##          racioethnicity          library_type          primer_pair
##          8              1              5
##          BMI_Bin          Pregnant          Trimester
##          1              1              5
##          Asymptomatic          BV              HIV
##          2              1              2
##          HPV          PCR_cycles          Racioethnic_category
##          1              4              6
##          Age_Bin          health_status          CST
##          5              2              5
```

```
#remove columns with NAs or 1 level
```

```
meta_all2 <- meta_all2 %>% select(-primer_pair, -PCR_cycles, -condition, -HIV, -Asymptomatic, -library_
```

```
all_dbrda <- c()
```

```
set.seed(2024)
```

```
for (i in 1:ncol(meta_all2)) {
```

```
  cap <- capscale(otu_all ~ meta_all2[,i], distance="euclidean", na.action=na.omit)
```

```
  av <- cap %>%
```

```
    anova.cca %>%
```

```
    as.data.frame
```

```
  pval <- av$`Pr(>F)`[1]
```

```
  Fstat <- av$`F`[1]
```

```
  r2 <- RsquareAdj(cap)[[1]]
```

```
  adjr2 <- RsquareAdj(cap)[[2]]
```

```
  all_dbrda <- rbind(all_dbrda, cbind(Fstat,r2,adjr2,pval))
```

```
  print(all_dbrda)
```

```
}
```

```
##
```

```
## The model is overfitted with no unconstrained (residual) component
```

```
##          Fstat r2 adjr2 pval
```

```
## [1,]      NA  1      NA      NA
```

```
##          Fstat          r2          adjr2          pval
```

```
## [1,]          NA 1.00000000          NA          NA
```

```
## [2,] 16.31136 0.04703237 0.04414895 0.001
```

```
##          Fstat          r2          adjr2          pval
```

```
## [1,]          NA 1.00000000          NA          NA
```

```
## [2,] 16.31136 0.04703237 0.04414895 0.001
```

```
## [3,] 31.13118 0.04491383 0.04347111 0.001
```

```
##          Fstat          r2          adjr2          pval
```

```
## [1,]          NA 1.00000000          NA          NA
```

```
## [2,] 16.31136 0.04703237 0.04414895 0.001
```

```

## [3,] 31.13118 0.04491383 0.04347111 0.001
## [4,] 20.05330 0.10851160 0.10310044 0.001
##      Fstat      r2      adjr2  pval
## [1,]      NA 1.00000000      NA    NA
## [2,] 16.31136 0.04703237 0.04414895 0.001
## [3,] 31.13118 0.04491383 0.04347111 0.001
## [4,] 20.05330 0.10851160 0.10310044 0.001
## [5,] 16.20750 0.10965274 0.10288718 0.001
##      Fstat      r2      adjr2  pval
## [1,]      NA 1.00000000      NA    NA
## [2,] 16.31136 0.04703237 0.04414895 0.001
## [3,] 31.13118 0.04491383 0.04347111 0.001
## [4,] 20.05330 0.10851160 0.10310044 0.001
## [5,] 16.20750 0.10965274 0.10288718 0.001
## [6,] 75.48909 0.10235961 0.10100365 0.001
##      Fstat      r2      adjr2  pval
## [1,]      NA 1.00000000      NA    NA
## [2,] 16.31136 0.04703237 0.04414895 0.001
## [3,] 31.13118 0.04491383 0.04347111 0.001
## [4,] 20.05330 0.10851160 0.10310044 0.001
## [5,] 16.20750 0.10965274 0.10288718 0.001
## [6,] 75.48909 0.10235961 0.10100365 0.001
## [7,] 11.45409 0.04948751 0.04516700 0.001
##      Fstat      r2      adjr2  pval
## [1,]      NA 1.00000000      NA    NA
## [2,] 16.311359 0.047032365 0.044148953 0.001
## [3,] 31.131178 0.044913833 0.043471105 0.001
## [4,] 20.053301 0.108511595 0.103100437 0.001
## [5,] 16.207497 0.109652739 0.102887182 0.001
## [6,] 75.489094 0.102359607 0.101003655 0.001
## [7,] 11.454086 0.049487507 0.045166995 0.001
## [8,]  2.618947 0.003940525 0.002435903 0.009
##      Fstat      r2      adjr2  pval
## [1,]      NA 1.000000000      NA    NA
## [2,] 16.311359 0.047032365 0.044148953 0.001
## [3,] 31.131178 0.044913833 0.043471105 0.001
## [4,] 20.053301 0.108511595 0.103100437 0.001
## [5,] 16.207497 0.109652739 0.102887182 0.001
## [6,] 75.489094 0.102359607 0.101003655 0.001
## [7,] 11.454086 0.049487507 0.045166995 0.001
## [8,]  2.618947 0.003940525 0.002435903 0.009
## [9,]  8.540680 0.083523380 0.073743904 0.001
##      Fstat      r2      adjr2  pval
## [1,]      NA 1.000000000      NA    NA
## [2,] 16.311359 0.047032365 0.044148953 0.001
## [3,] 31.131178 0.044913833 0.043471105 0.001
## [4,] 20.053301 0.108511595 0.103100437 0.001
## [5,] 16.207497 0.109652739 0.102887182 0.001
## [6,] 75.489094 0.102359607 0.101003655 0.001
## [7,] 11.454086 0.049487507 0.045166995 0.001
## [8,]  2.618947 0.003940525 0.002435903 0.009
## [9,]  8.540680 0.083523380 0.073743904 0.001
## [10,] 15.792706 0.070885202 0.066396724 0.001
##      Fstat      r2      adjr2  pval

```

```

## [1,]      NA 1.000000000      NA      NA
## [2,] 16.311359 0.047032365 0.044148953 0.001
## [3,] 31.131178 0.044913833 0.043471105 0.001
## [4,] 20.053301 0.108511595 0.103100437 0.001
## [5,] 16.207497 0.109652739 0.102887182 0.001
## [6,] 75.489094 0.102359607 0.101003655 0.001
## [7,] 11.454086 0.049487507 0.045166995 0.001
## [8,]  2.618947 0.003940525 0.002435903 0.009
## [9,]  8.540680 0.083523380 0.073743904 0.001
## [10,] 15.792706 0.070885202 0.066396724 0.001
## [11,]  5.877101 0.042749669 0.035475731 0.001
##      Fstat      r2      adjr2  pval
## [1,]      NA 1.000000000      NA      NA
## [2,] 16.311359 0.047032365 0.044148953 0.001
## [3,] 31.131178 0.044913833 0.043471105 0.001
## [4,] 20.053301 0.108511595 0.103100437 0.001
## [5,] 16.207497 0.109652739 0.102887182 0.001
## [6,] 75.489094 0.102359607 0.101003655 0.001
## [7,] 11.454086 0.049487507 0.045166995 0.001
## [8,]  2.618947 0.003940525 0.002435903 0.009
## [9,]  8.540680 0.083523380 0.073743904 0.001
## [10,] 15.792706 0.070885202 0.066396724 0.001
## [11,]  5.877101 0.042749669 0.035475731 0.001
## [12,]  6.757066 0.039398181 0.033567517 0.001
##      Fstat      r2      adjr2  pval
## [1,]      NA 1.000000000      NA      NA
## [2,] 16.311359 0.047032365 0.044148953 0.001
## [3,] 31.131178 0.044913833 0.043471105 0.001
## [4,] 20.053301 0.108511595 0.103100437 0.001
## [5,] 16.207497 0.109652739 0.102887182 0.001
## [6,] 75.489094 0.102359607 0.101003655 0.001
## [7,] 11.454086 0.049487507 0.045166995 0.001
## [8,]  2.618947 0.003940525 0.002435903 0.009
## [9,]  8.540680 0.083523380 0.073743904 0.001
## [10,] 15.792706 0.070885202 0.066396724 0.001
## [11,]  5.877101 0.042749669 0.035475731 0.001
## [12,]  6.757066 0.039398181 0.033567517 0.001
## [13,] 66.488563 0.091269193 0.089896488 0.001
##      Fstat      r2      adjr2  pval
## [1,]      NA 1.000000000      NA      NA
## [2,] 16.311359 0.047032365 0.044148953 0.001
## [3,] 31.131178 0.044913833 0.043471105 0.001
## [4,] 20.053301 0.108511595 0.103100437 0.001
## [5,] 16.207497 0.109652739 0.102887182 0.001
## [6,] 75.489094 0.102359607 0.101003655 0.001
## [7,] 11.454086 0.049487507 0.045166995 0.001
## [8,]  2.618947 0.003940525 0.002435903 0.009
## [9,]  8.540680 0.083523380 0.073743904 0.001
## [10,] 15.792706 0.070885202 0.066396724 0.001
## [11,]  5.877101 0.042749669 0.035475731 0.001
## [12,]  6.757066 0.039398181 0.033567517 0.001
## [13,] 66.488563 0.091269193 0.089896488 0.001
## [14,] 44.060036 0.211005355 0.206216313 0.001

```

```

rownames(all_dbrda) <- colnames(meta_all2)
all_dbrda <- as.data.frame(cbind(all_dbrda, padj=p.adjust(all_dbrda[, "pval"], method="BH")), stringsAsFactors=FALSE)
significant <- all_dbrda[all_dbrda[, "padj"] < 0.05, ]
significant <- na.omit(significant[order(significant$adjr2, decreasing = T), ])
print(paste0(nrow(significant), " variables can individually explain part of the (active) microbiome variation in these samples"))

```

```
## [1] "13 variables can individually explain part of the (active) microbiome variation in these samples"
```

```

# Stepwise multivariate RDA, with only significant variables
metadata_curated_filtered <- meta_all2[, rownames(significant), drop=F] # select only significant variables
metadata_curated_filtered <- na.omit(metadata_curated_filtered) # samples with NAs need to be removed from the CLR object

# remove samples with NAs also from the CLR object
reduced_spp_filt_n0_clr <- otu_all[rownames(metadata_curated_filtered), ]

metadata_curated_filtered <- metadata_curated_filtered %>% mutate(across(where(is.numeric), as.factor))

# set limits for the stepwise analysis, from null model (fit0) to a full model with the 20 selected variables
fit0 <- capscale(reduced_spp_filt_n0_clr ~ 1, data=metadata_curated_filtered, distance="euclidean")
fit1 <- capscale(reduced_spp_filt_n0_clr ~ ., data=metadata_curated_filtered, distance="euclidean")

```

```

##
## Some constraints or conditions were aliased because they were redundant. This
## can happen if terms are constant or linearly dependent (collinear):
## 'studyLivson_2022', 'studyMovassagh_2021', 'studyPMTCT_CQI',
## 'studyServergnini_2021', 'sequencernextseq', 'racioethnicityUgandan',
## 'Trimesterthird', 'TrimesterThird', 'sample_locmidvaginal',
## 'sample_locposterior_fornix', 'sample_locvaginal_swab', 'continentEurope',
## 'continentNorthAm', 'unsdg_regionSubSaharanAf',
## 'Racioethnic_categoryBlack/African American',
## 'Racioethnic_categoryHispanic/Latino', 'Racioethnic_categoryOther',
## 'Racioethnic_categoryUnknown', 'Racioethnic_categoryWhite',
## 'sample_loc_simplevaginal_swab'

```

```

# run stepwise analysis between the two limits established
fin_dbrda <- ordiR2step(fit0, fit1, data=metadata_curated_filtered, Pin = 0.05, R2scope = TRUE, pstep = 0.05)

# Print call and anova
fin_dbrda$call

```

```

## capscale(formula = reduced_spp_filt_n0_clr ~ CST + study + Age_Bin,
##           data = metadata_curated_filtered, distance = "euclidean")

```

```
fin_dbrda$anova
```

```

##           R2.adj Df      AIC      F Pr(>F)
## + CST          0.21093  4 2434.9 42.7019 0.002 **
## + study         0.25837  5 2401.1  8.9315 0.002 **
## + Age_Bin       0.26127  4 2402.6  1.6042 0.016 *
## <All variables> 0.26239
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

significant <- as.tibble(cbind(significant, covariates=rownames(significant)))
anova_table <- as.tibble(cbind(fin_dbrda$anova, covariates=gsub(rownames(fin_dbrda$anova), pattern="\\"

significant <- significant %>%
  full_join(anova_table, by="covariates")

significant[is.na(significant$R2.adj), "R2.adj"] <- max(significant$R2.adj, na.rm=T)
significant <- significant %>%
  filter(covariates!="<All variables>")

# gather, define sorting order, and plot
significantL <- gather(significant, key = "Class", value="EffectSize", adjr2, R2.adj)

significantL[significantL$Class=="adjr2", "Class"] <- "Individual"
significantL[significantL$Class=="R2.adj", "Class"] <- "Cumulative, non-redundant"

# Combine the two tables into a list
tables <- list(table1 = significant, table2 = significantL)

# Apply the mutate operation to both tables
tables <- lapply(tables, function(df) {
  df %>%
    mutate(covariates = case_when(
      covariates == "Age_Bin" ~ "Age Bin",
      covariates == "study" ~ "Study",
      covariates == "country" ~ "Country",
      covariates == "sequencer" ~ "Sequencing Platform",
      covariates == "racioethnicity" ~ "Self-reported Racioethnicity",
      covariates == "sample_loc" ~ "Sampling Location",
      covariates == "continent" ~ "Continent",
      covariates == "unsdg_region" ~ "UNSDG Region",
      covariates == "Racioethnic_category" ~ "Racioethnic Bin",
      covariates == "sample_loc_simple" ~ "Sampling Location Bin",
      covariates == "CST" ~ "CST",
      covariates == "Trimester" ~ "Trimester",
      covariates == "health_status" ~ "Health Status (PLWH-ART/ PwOH)"
    ))
})

#extract tables
significant <- tables$table1
significantL <- tables$table2

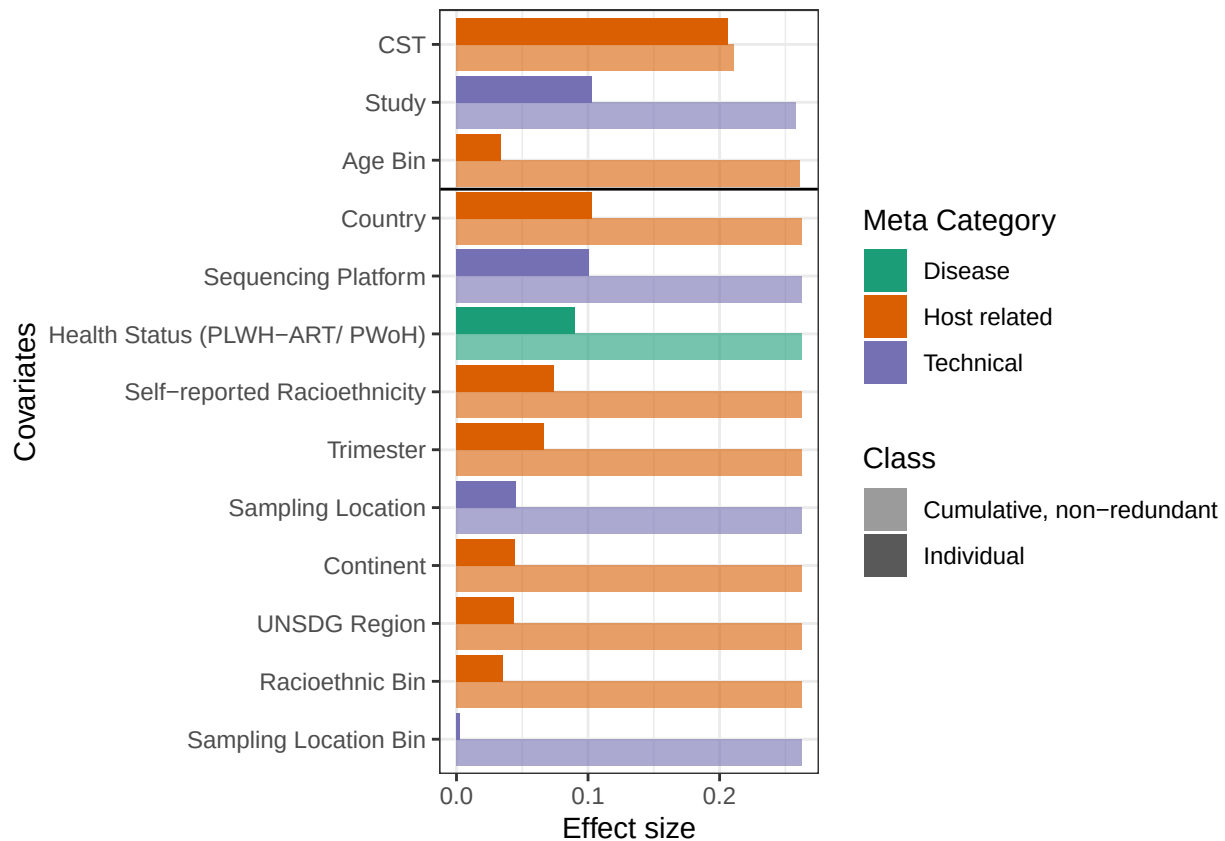
significantL <- significantL %>% #CHANGE FOR EACH TABLE
  mutate("Meta Category" = case_when(
    covariates %in% c("Study", "Sequencing Platform", "Sampling Location Bin", "Sampling Location") ~ "
    covariates %in% c("Continent", "UNSDG Region", "Country", "Self-reported Racioethnicity", "Racioethn
    covariates %in% c("Health Status (PLWH-ART/ PwOH)") ~ "Disease",
    TRUE ~ "Other" # Default category for unmatched metadata
  ))

#Specify order
positions <- rev.default(significant$covariates[order(significant$R2.adj, significant$AIC, -significant$

```



```
#plot
ggplot(significantL, aes(x = covariates, y = EffectSize)) +
  geom_bar(aes(color = NULL, fill = `Meta Category`, alpha=Class),
    stat = "identity", position = position_dodge()) +
  ggpubr::rotate() + theme_bw() +
  scale_x_discrete(limits = positions) +
  scale_alpha_manual(values=c(0.6,1)) +
  geom_vline(xintercept = nrow(significant)-(nrow(anova_table)-1.5))+
  labs(x="Covariates", y="Effect size") + scale_fill_brewer(palette = "Dark2")
```



##Suggested by reviewer: All Cytokine and CST plots

```
library(patchwork)
```

```
##
## Attaching package: 'patchwork'

## The following object is masked from 'package:MASS':
##
## area
```

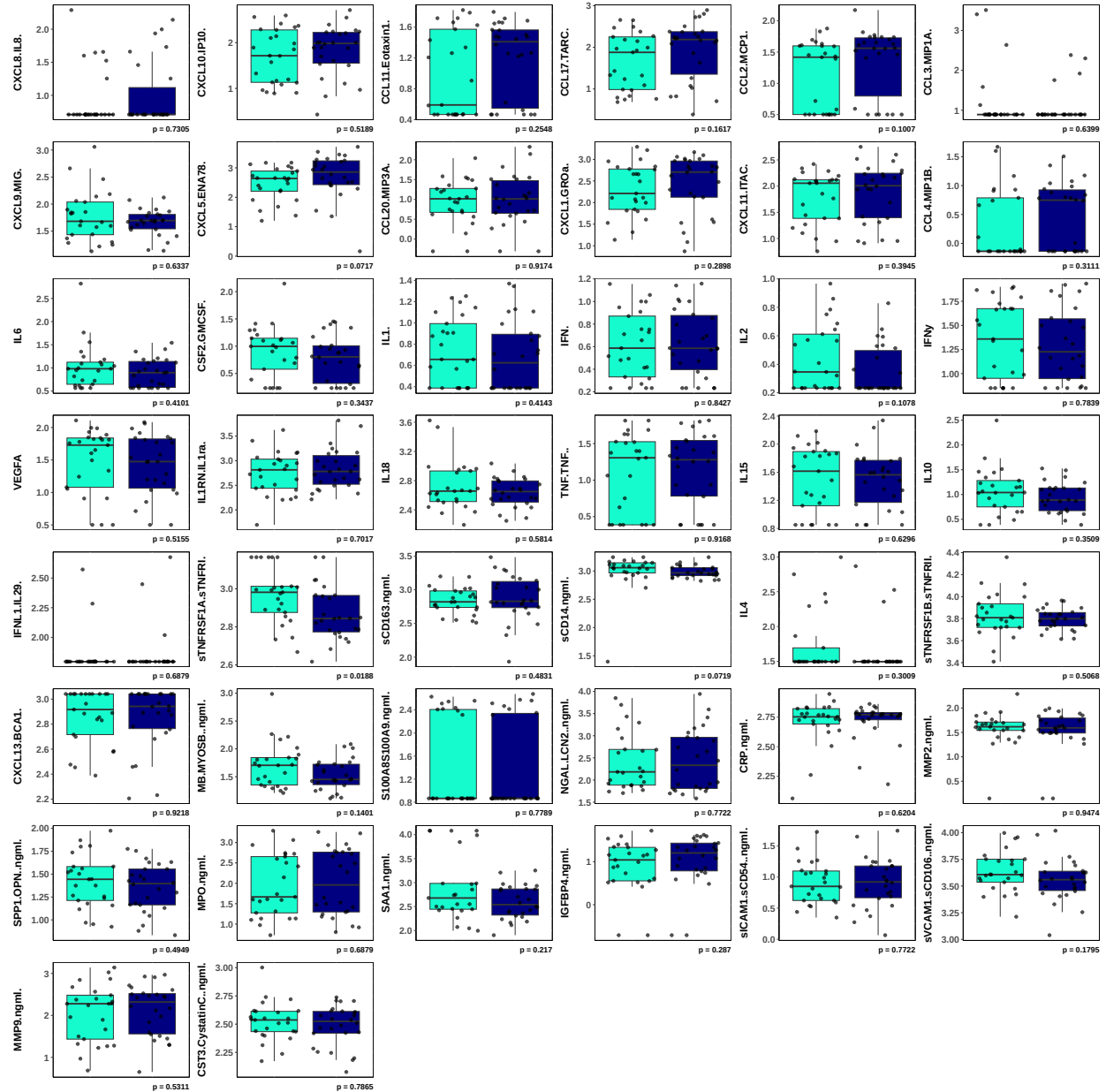
```
plots <- lapply(names(meta_cyt_log2)[3:46], function(col) {
  p <- ggplot(meta_cyt_log2 %>% filter(!is.na(CST)), aes_string(x = "CST", y = col, fill = "CST")) +
    geom_boxplot(outlier.shape = NA) +
```

```

geom_jitter(color = "black", fill = "white", alpha = 0.7, height = 0) +
labs(y = paste(sub("^enrol_", "", col)), x = " ",
      caption = paste("p =", round(wilcox.test(meta_cyt_log2[[col]] ~ meta_cyt_log2$CST)$p.value, 4)),
theme(legend.position = "none",
      panel.background = element_rect(fill = NA, colour = "black"),
      panel.border = element_rect(color = "black", fill = NA),
      axis.line = element_line(color = "black"),
      axis.line.x = element_blank(),
      axis.ticks.x = element_blank(),
      axis.text.x = element_blank(),
      axis.title.x = element_blank(),
      axis.title.y = element_text(margin = margin(t = 0, r = 10, b = 0, l = 0), size = 14, face = "bold"),
      axis.text.y = element_text(size = 12, face = "bold"),
      plot.caption = element_text(size = 10, face = "bold")) +
scale_fill_manual(values = CST_cols)
  p
})

# Combine all plots into one figure
wrap_plots(plots, ncol = 6)

```



Suggested by reviewer: All shannon and cytokine linear plots

```
n_plots <- nrow(corr_results_alpha)
n_cols <- ceiling(sqrt(n_plots))
n_rows <- ceiling(n_plots / n_cols)

# Set up the plotting area
par(mfrow = c(n_rows, n_cols))

for (i in 1:n_plots) {
  # Extract the cytokine name, rho, and p-value
  cytokine_name <- corr_results_alpha$Cytokine[i]
```

```

rho <- corr_results_alpha$Rho[i]
p_value <- corr_results_alpha$PValue[i]

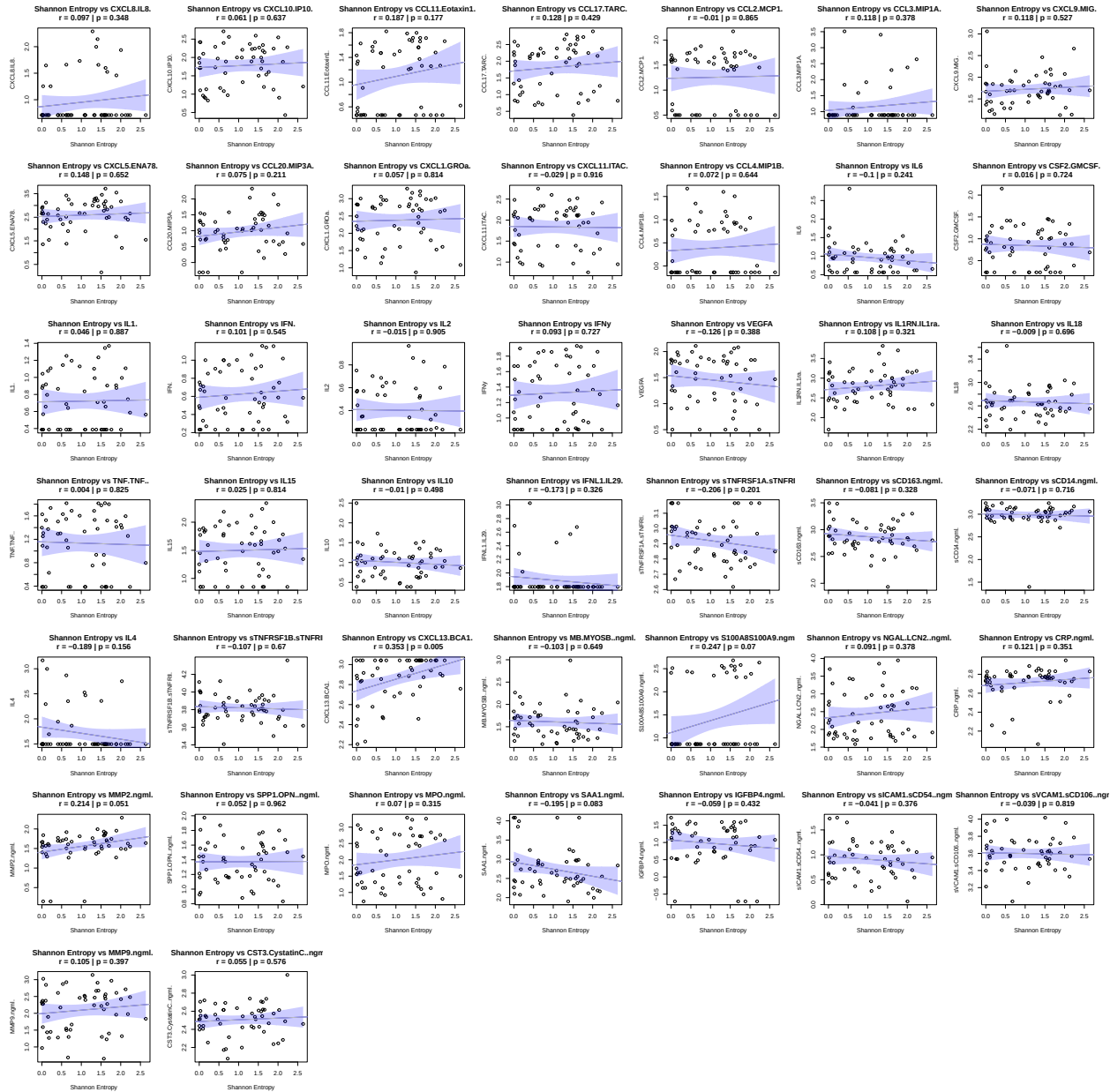
# Remove "enrol_" prefix from cytokine name
clean_cytokine_name <- gsub("^enrol_", "", cytokine_name)

# Extract the data for the current cytokine
x <- alpha_div$`Shannon Entropy`
y <- meta_cyt_log[[cytokine_name]]

# Generate the scatter plot
plot(x, y,
      main = paste("Shannon Entropy vs", clean_cytokine_name,
                   "\nr =", round(rho, 3), "| p =", round(p_value, 3)),
      xlab = "Shannon Entropy",
      ylab = clean_cytokine_name)
abline(lm(y ~ x), col = "blue")

# Add 95% CI
model <- lm(y ~ x)
newx <- seq(min(x), max(x), length.out = 100)
pred <- predict(model, newdata = data.frame(x = newx), interval = "confidence")
polygon(c(newx, rev(newx)), c(pred[, "lwr"], rev(pred[, "upr"])),
        col = rgb(0, 0, 1, 0.2), border = NA)
abline(model, col = "grey")
}

```



Suggested by reviewer: linear plots of all taxon and cytokines

```
n_plots <- nrow(results_table_asv)
n_cols <- ceiling(sqrt(n_plots))
n_rows <- ceiling(n_plots / n_cols)

# Set up the plotting area
par(mfrow = c(n_rows, n_cols))

# Loop through each row in the results_df_alpha data frame
for (i in 1:n_plots) {
  # Extract the cytokine name, rho, and p-value
```

```

cytokine_name <- results_table_asv$Cytokine[i]
otu_name <- results_table_asv$OTU[i]
rho <- results_table_asv$Rho[i]
p_value <- results_table_asv$TwoSided_PValue[i]

# Check if the p-value is significant (p <= 0.06)
if (p_value <= 0.05) {
  # Extract the data for the current cytokine
  x <- otu_cyt[[otu_name]]
  y <- meta_cyt_log[[cytokine_name]]
  clean_cytokine_name <- gsub("^enrol_", "", cytokine_name)
  # Generate the scatter plot

#   plot(x, y,
#         main = paste(otu_name, "vs", clean_cytokine_name,
#                       "\nr =", round(rho, 3), "| p =", round(p_value, 3)),
#         xlab = otu_name,
#         ylab = clean_cytokine_name)
#abline(lm(y ~ x), col = "blue")
#
## Add 95% CI
#model <- lm(y ~ x)
#newx <- seq(min(x), max(x), length.out = 100)
#pred <- predict(model, newdata = data.frame(x = newx), interval = "confidence")
#polygon(c(newx, rev(newx)), c(pred[, "lwr"], rev(pred[, "upr"])),
#        col = rgb(0, 0, 1, 0.2), border = NA)
#abline(model, col = "grey")

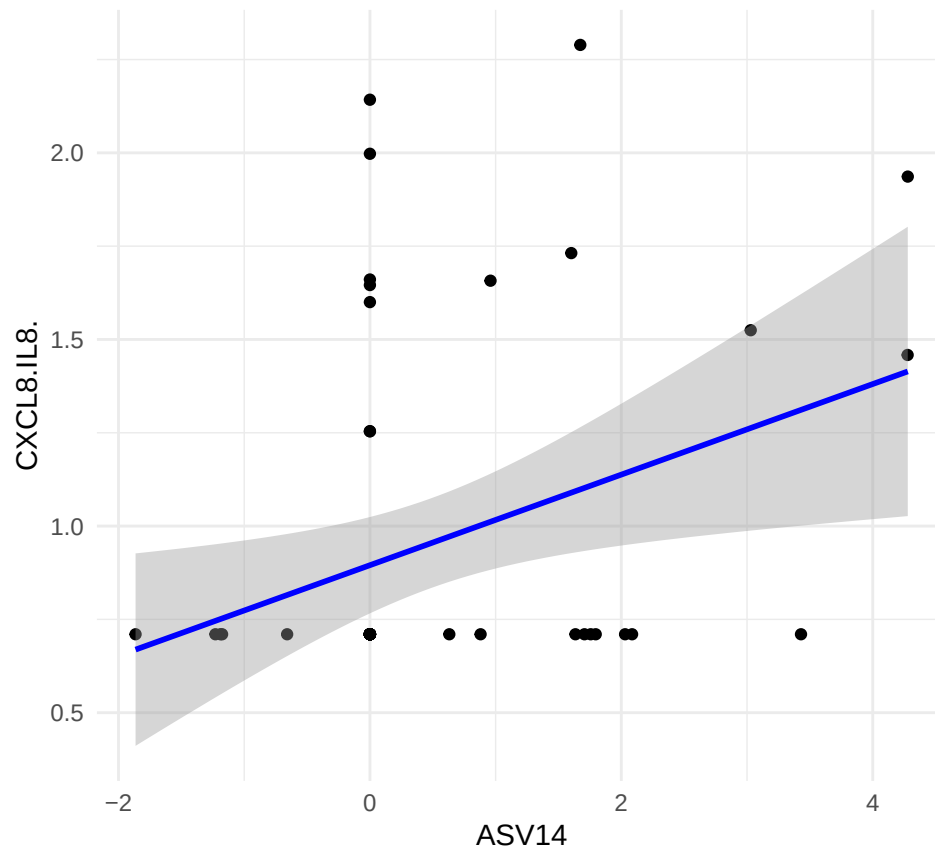
plot(ggplot(data = data.frame(x = x, y = y), aes(x = x, y = y)) +
     geom_point() +
     geom_smooth(method = "lm", color = "blue") +
     labs(
       title = paste(otu_name, "vs", clean_cytokine_name),
       subtitle = paste("r =", round(rho, 3), "| p =", round(p_value, 3)),
       x = otu_name,
       y = clean_cytokine_name
     ) +
     theme_minimal())
}
}

## 'geom_smooth()' using formula = 'y ~ x'

```

ASV14 vs CXCL8.IL8.

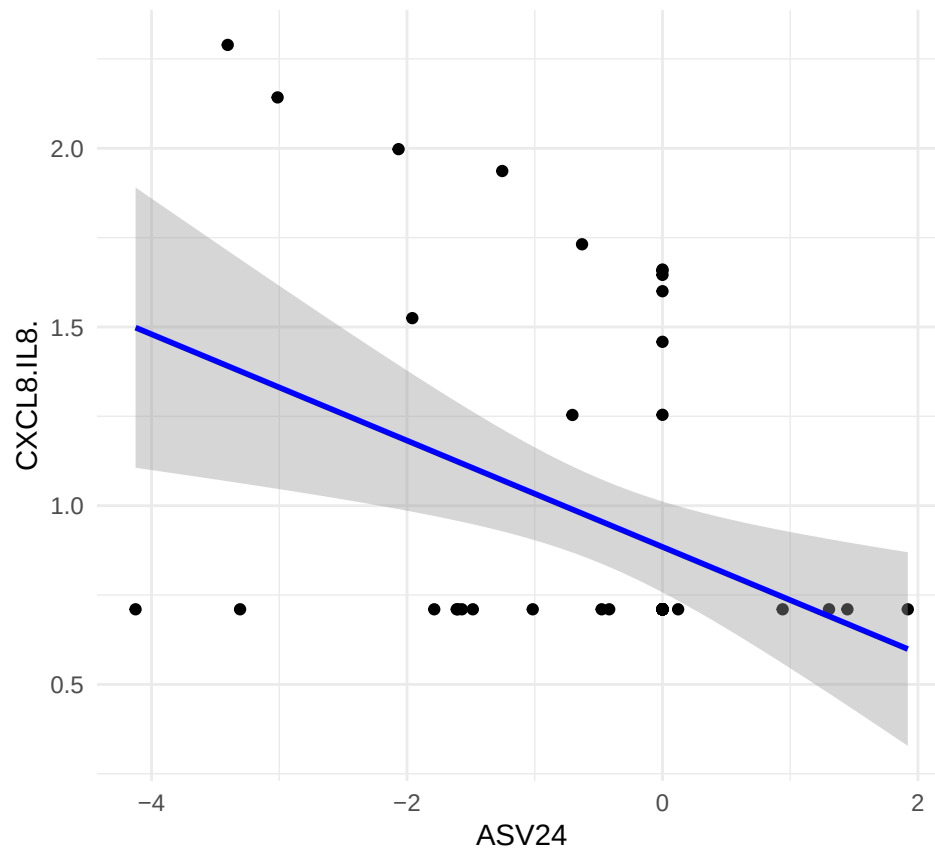
$r = 0.284$ | $p = 0.04$



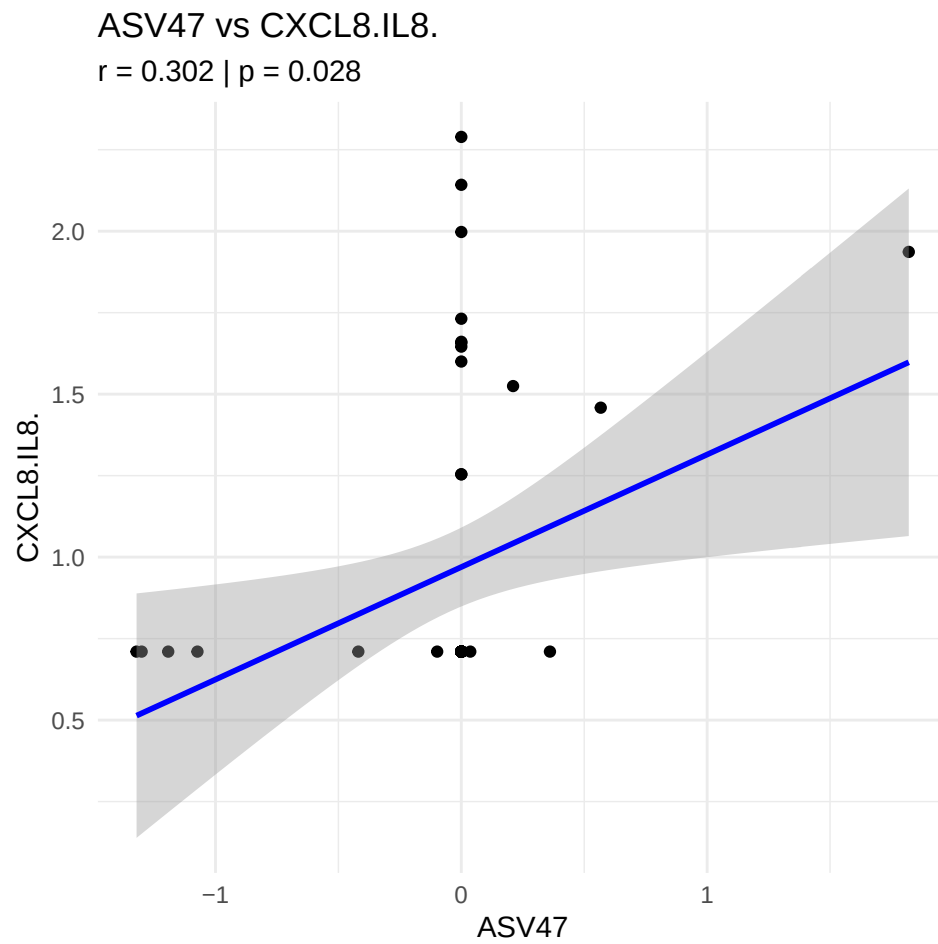
```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV24 vs CXCL8.IL8.

$r = -0.342$ | $p = 0.012$



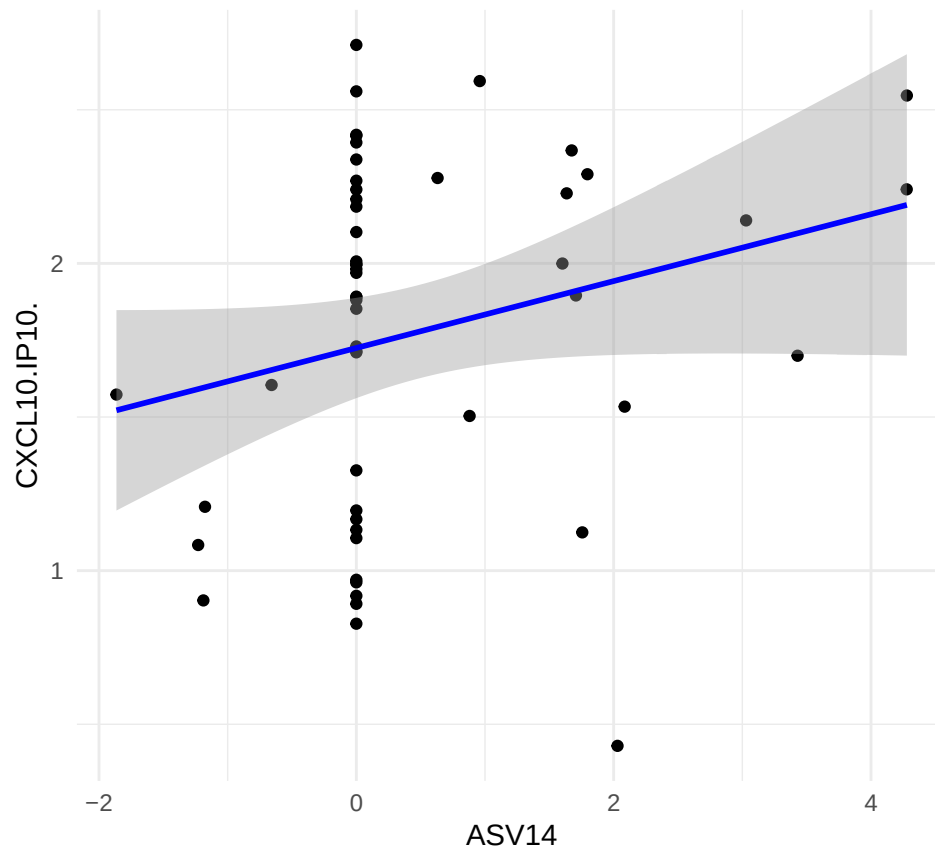
```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV14 vs CXCL10.IP10.

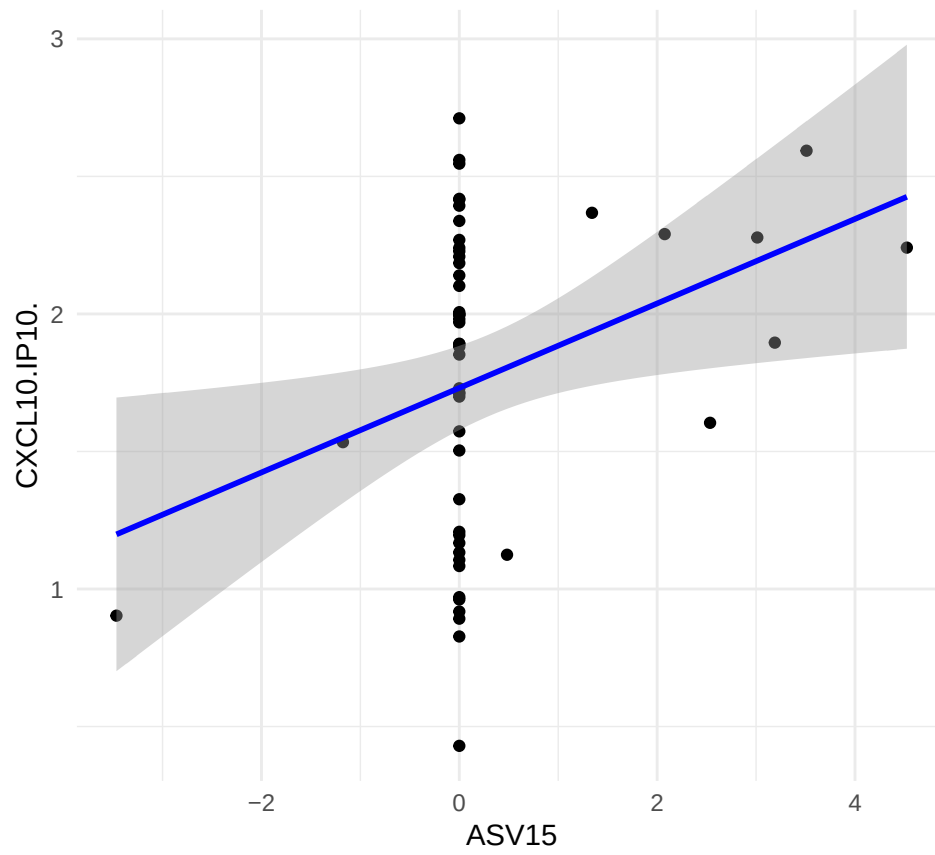
$r = 0.277$ | $p = 0.045$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV15 vs CXCL10.IP10.

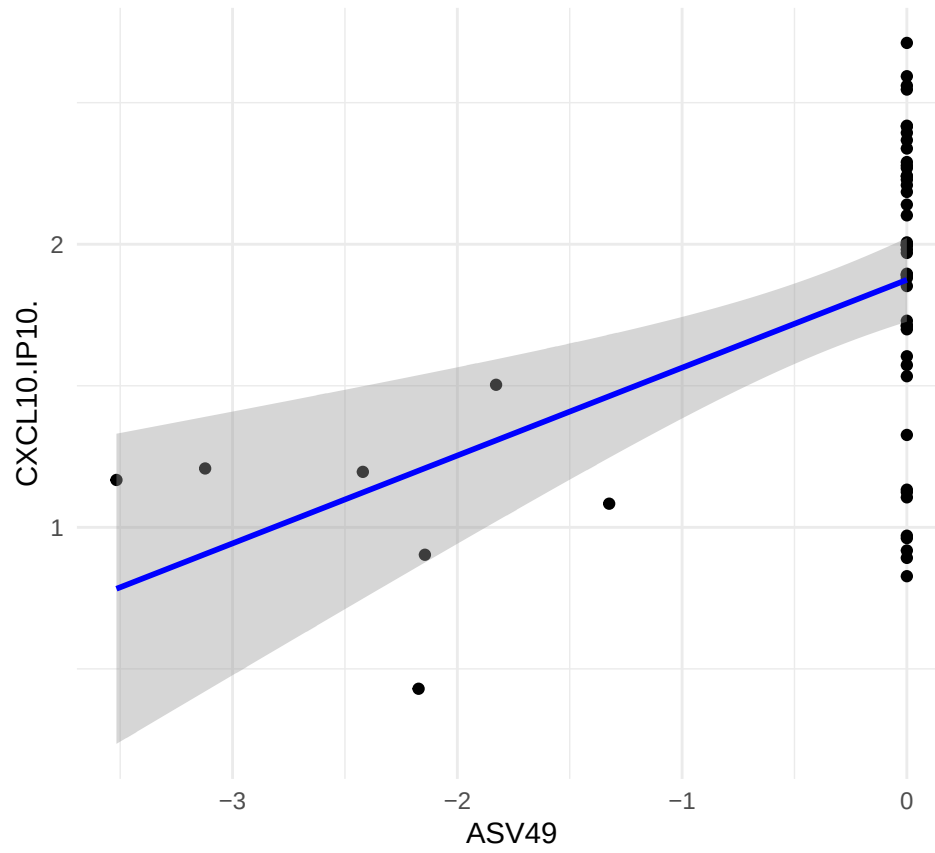
$r = 0.303$ | $p = 0.027$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV49 vs CXCL10.IP10.

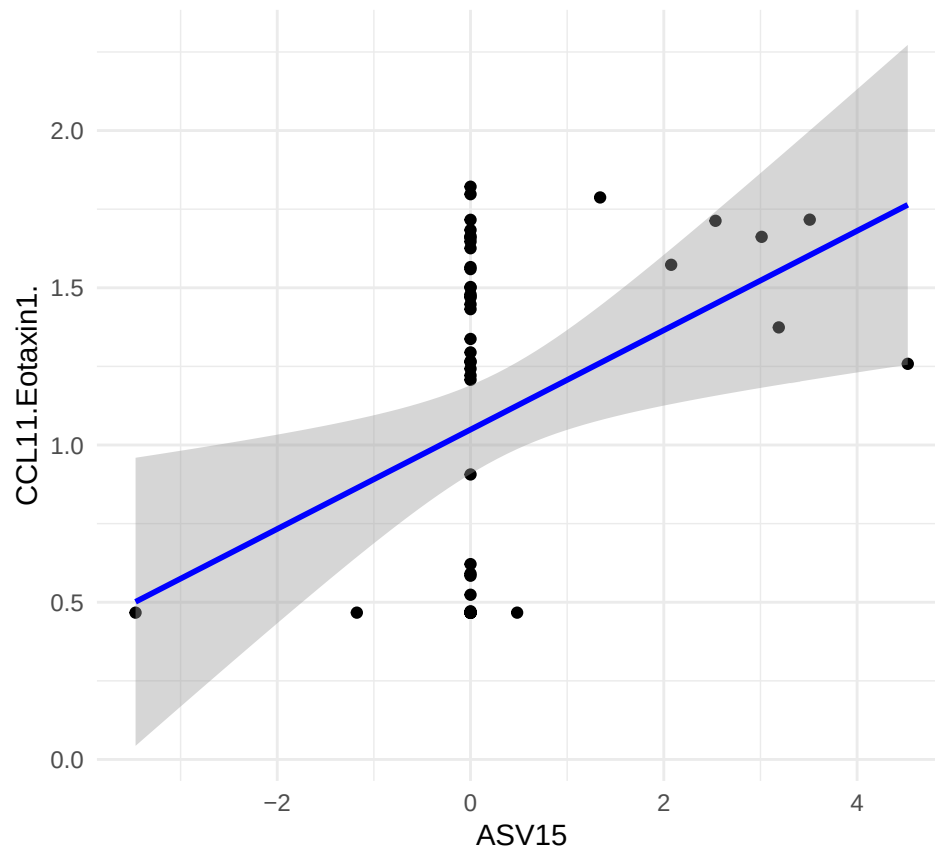
$r = 0.437$ | $p = 0.001$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV15 vs CCL11.Eotaxin1.

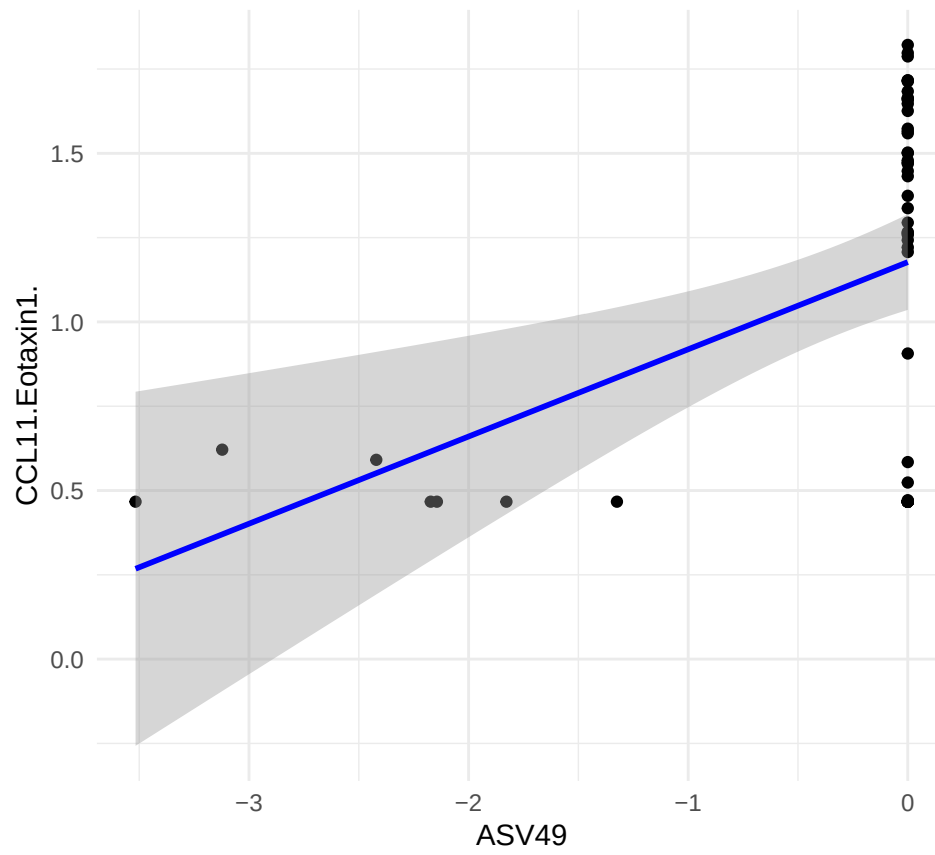
$r = 0.363$ | $p = 0.008$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV49 vs CCL11.Eotaxin1.

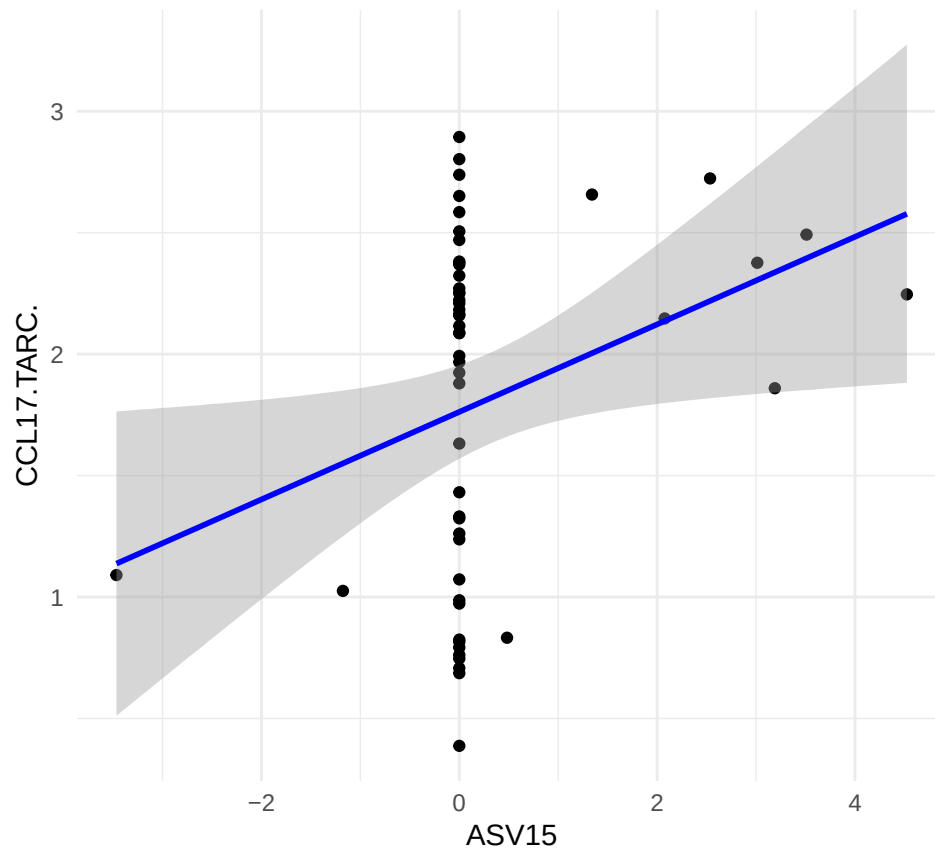
$r = 0.383$ | $p = 0.005$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV15 vs CCL17.TARC.

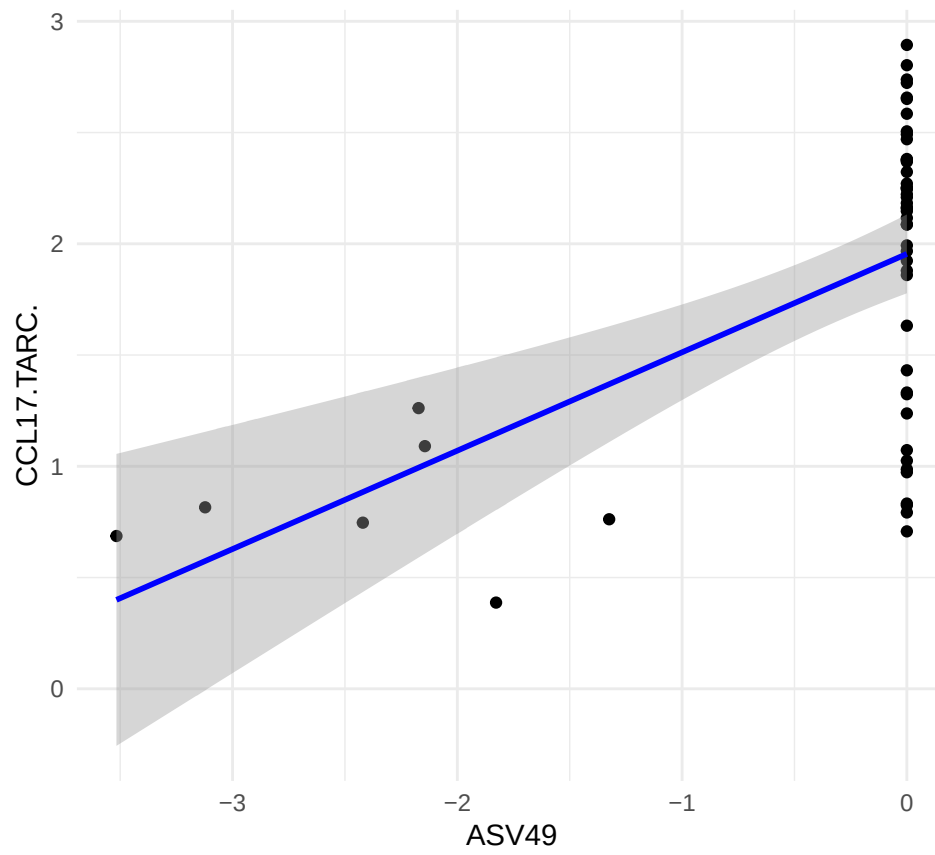
$r = 0.272$ | $p = 0.049$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV49 vs CCL17.TARC.

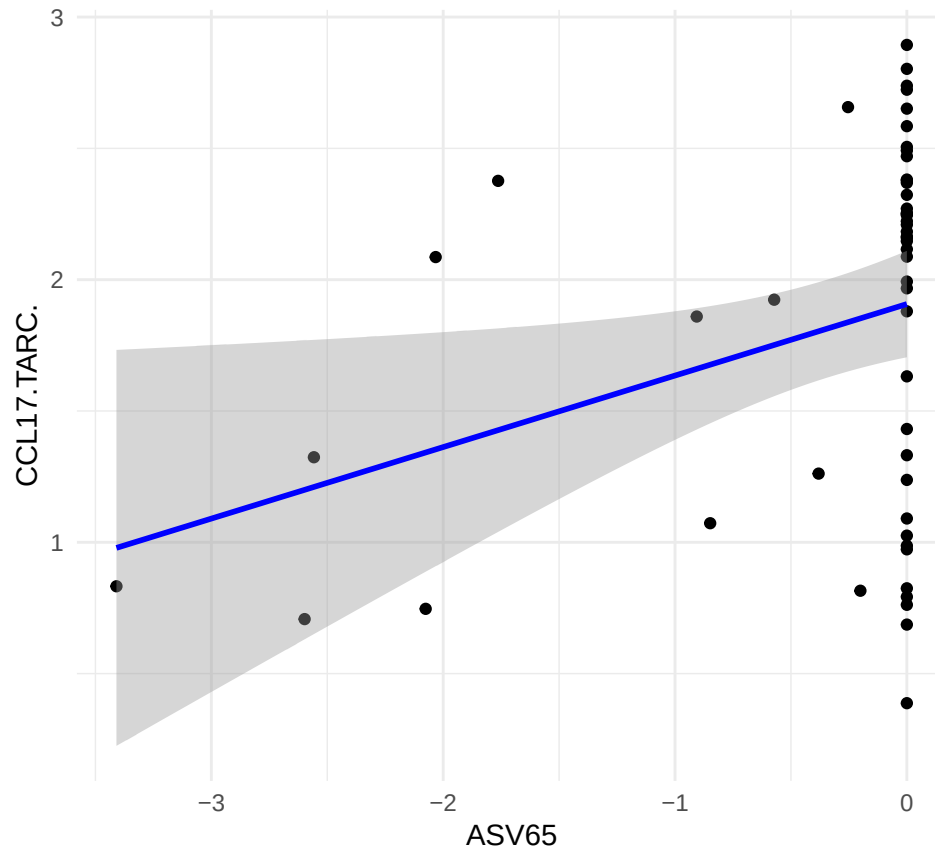
$r = 0.509$ | $p = 0$



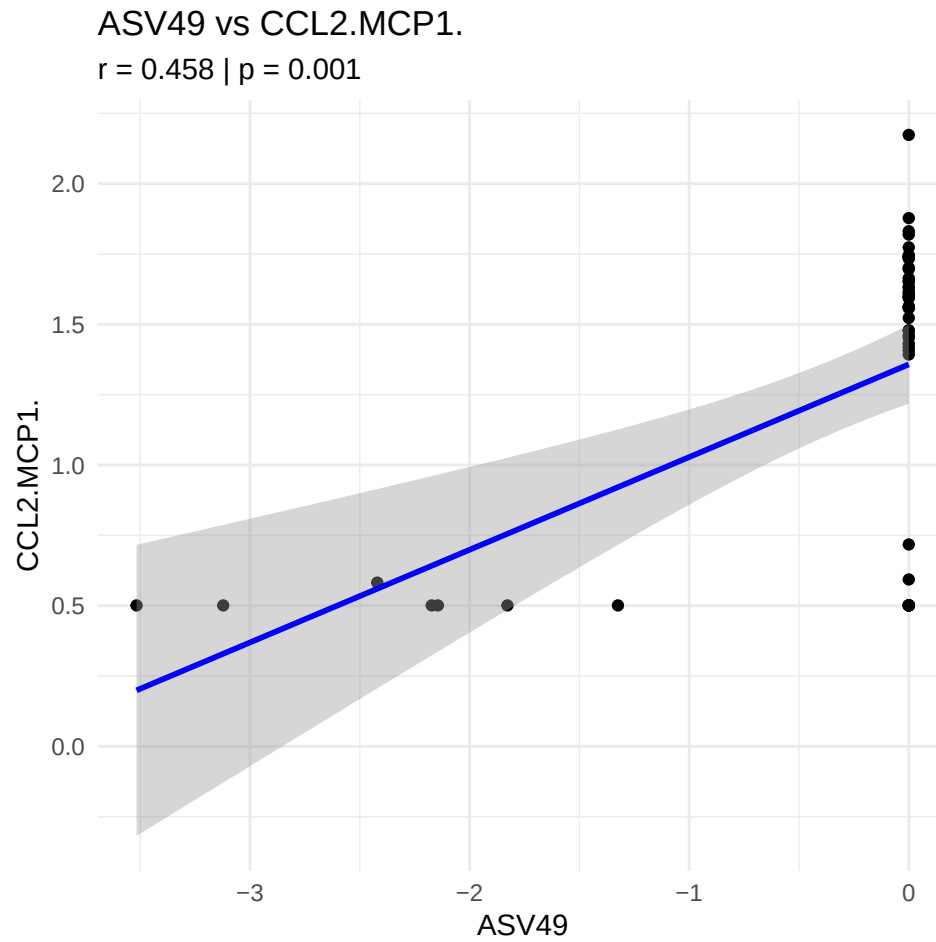
```
## 'geom_smooth()' using formula = 'y ~ x'
```


ASV65 vs CCL17.TARC.

$r = 0.3 \mid p = 0.029$



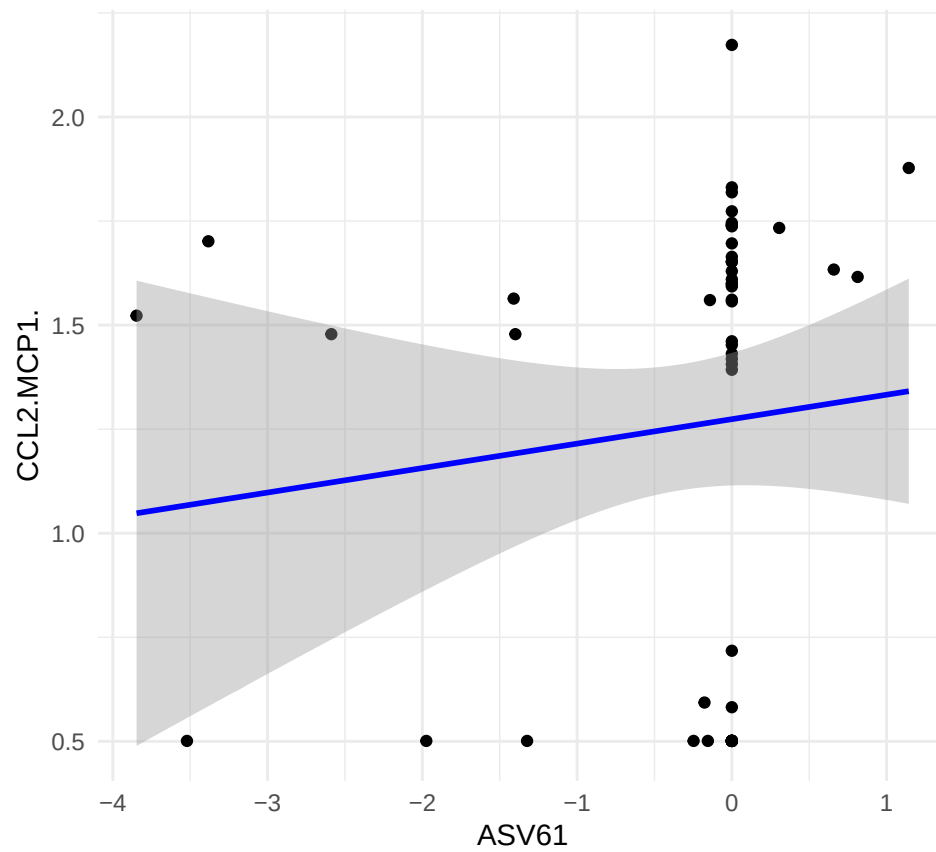
```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV61 vs CCL2.MCP1.

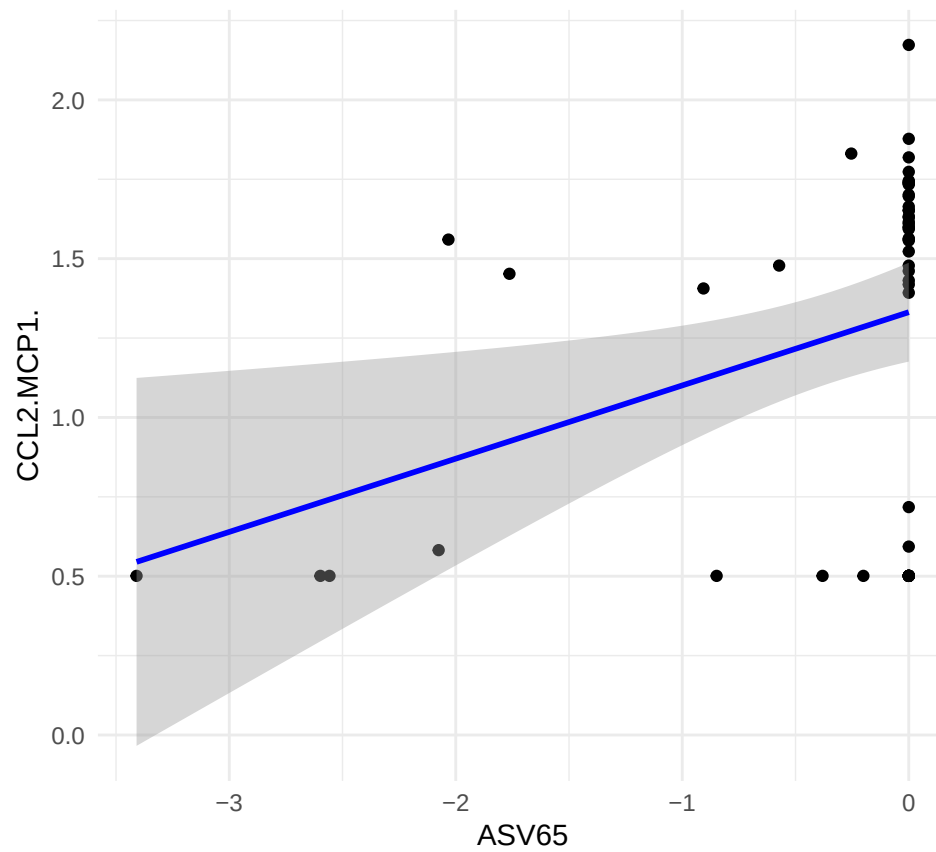
$r = 0.328$ | $p = 0.017$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV65 vs CCL2.MCP1.

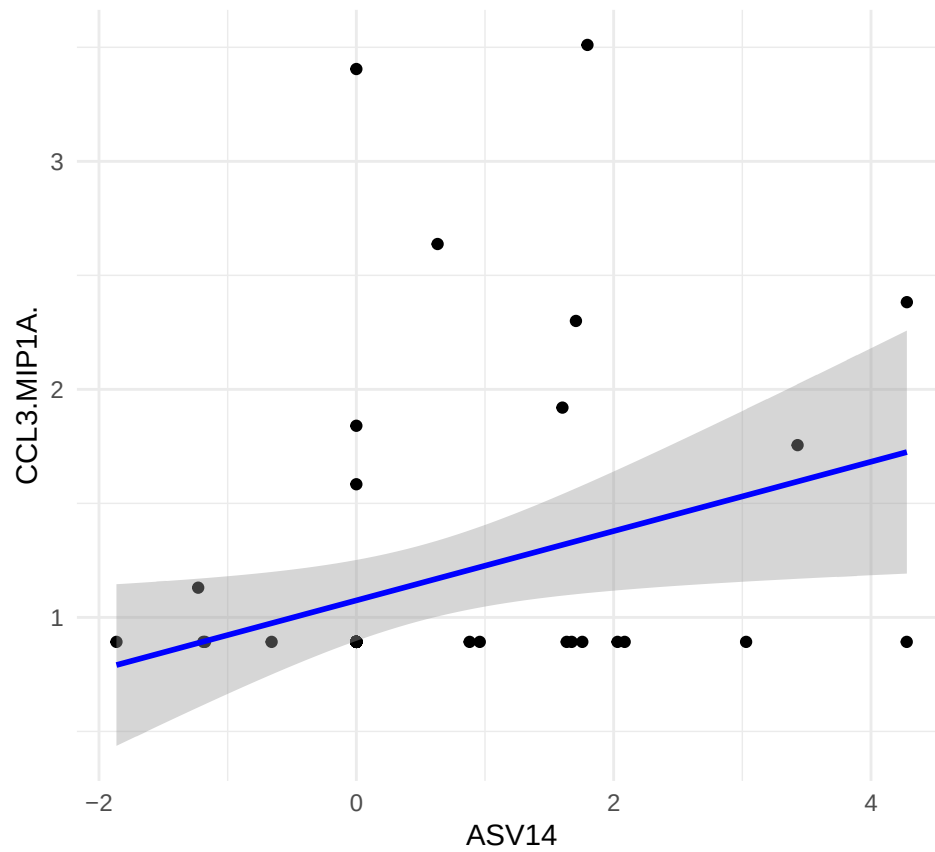
$r = 0.349$ | $p = 0.01$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV14 vs CCL3.MIP1A.

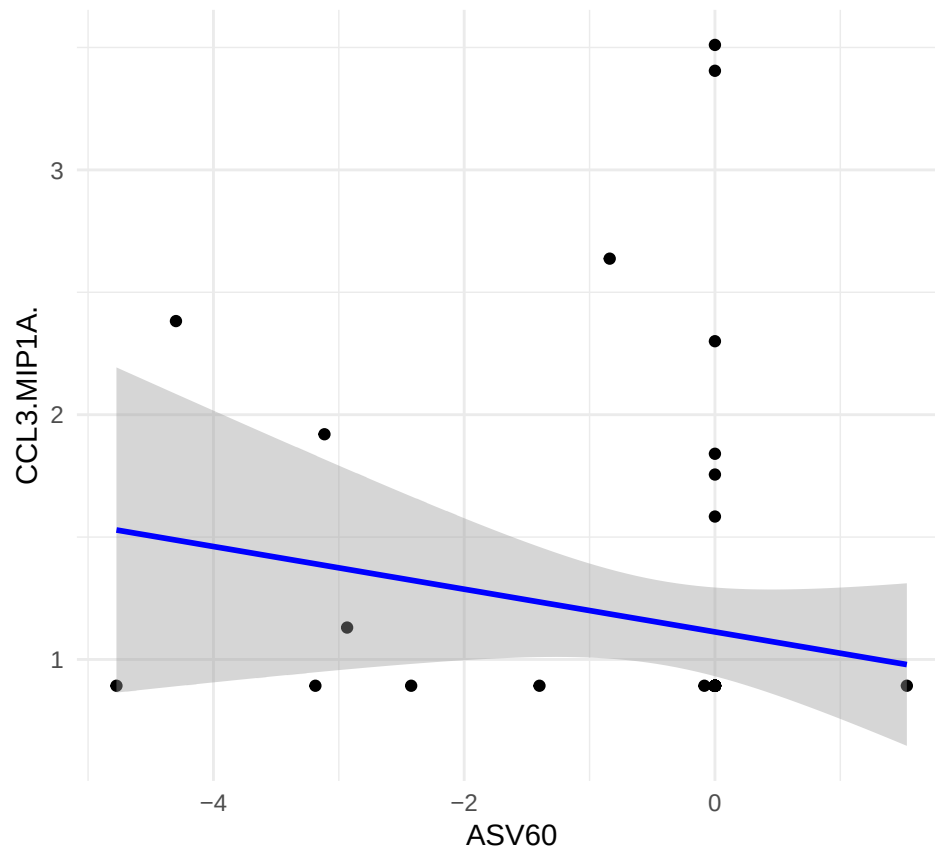
$r = 0.299$ | $p = 0.029$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV60 vs CCL3.MIP1A.

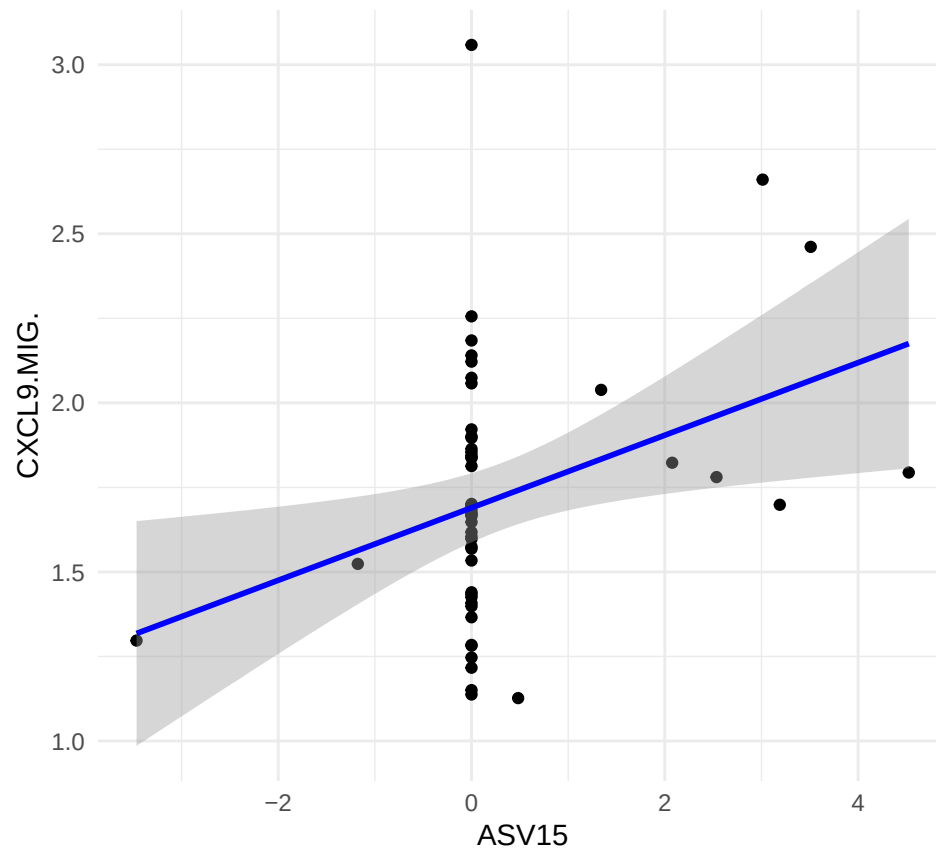
$r = -0.293$ | $p = 0.033$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV15 vs CXCL9.MIG.

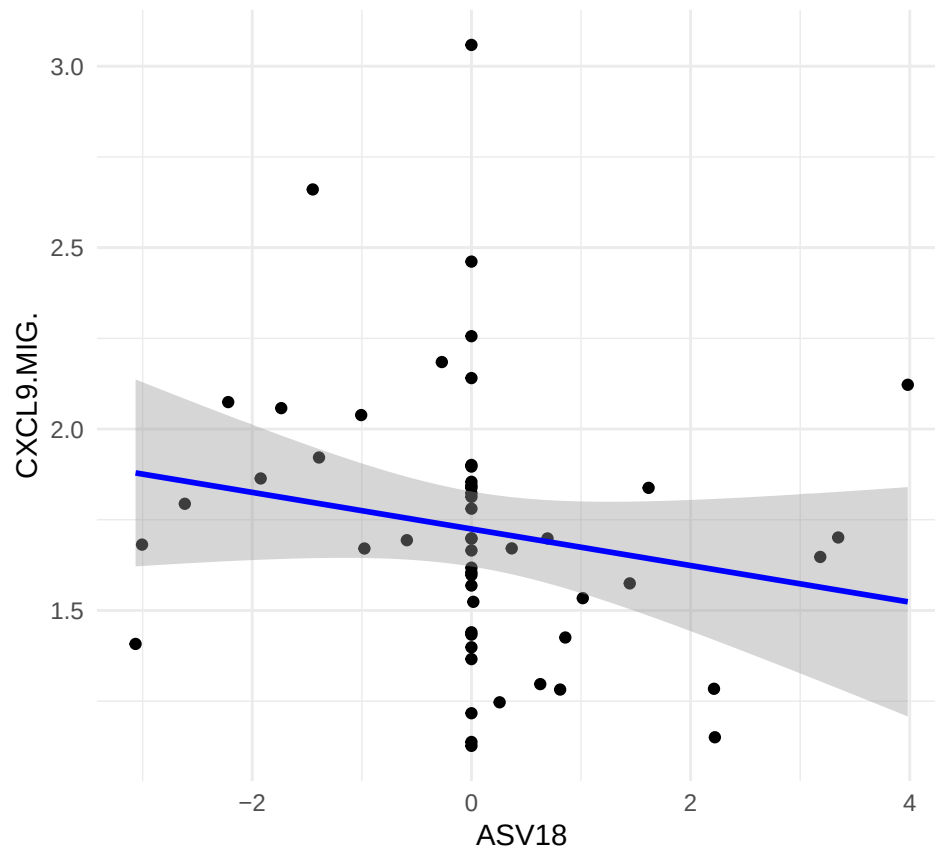
$r = 0.285$ | $p = 0.039$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV18 vs CXCL9.MIG.

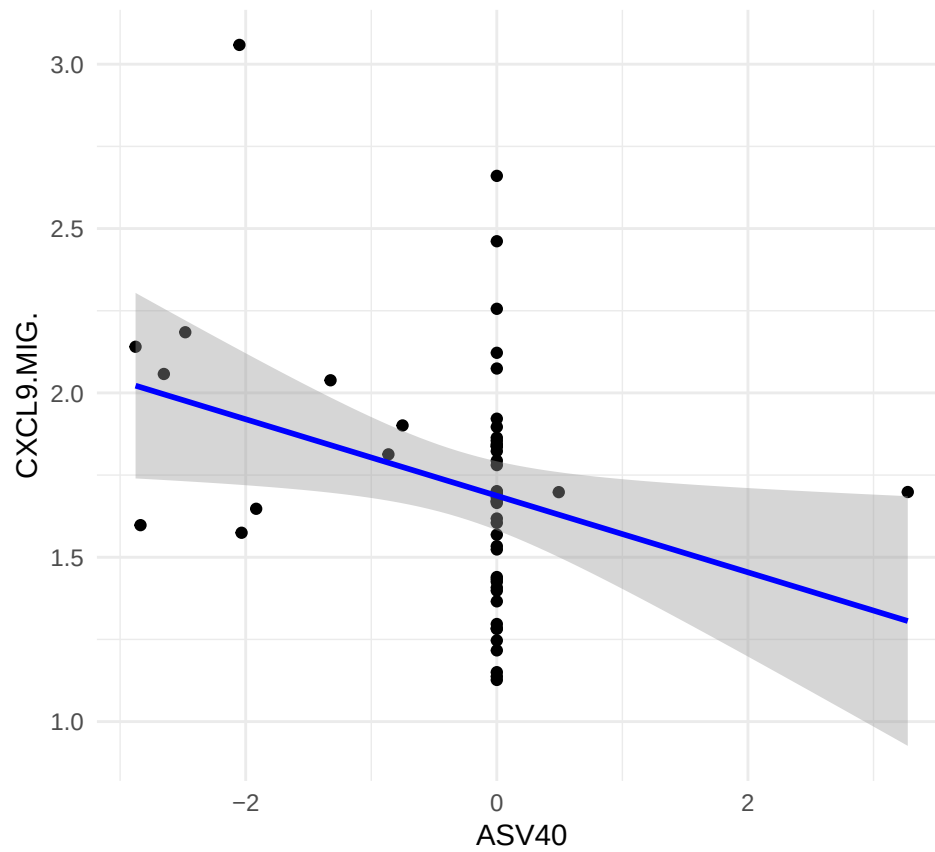
$r = -0.362$ | $p = 0.008$



```
## 'geom_smooth()' using formula = 'y ~ x'
```


ASV40 vs CXCL9.MIG.

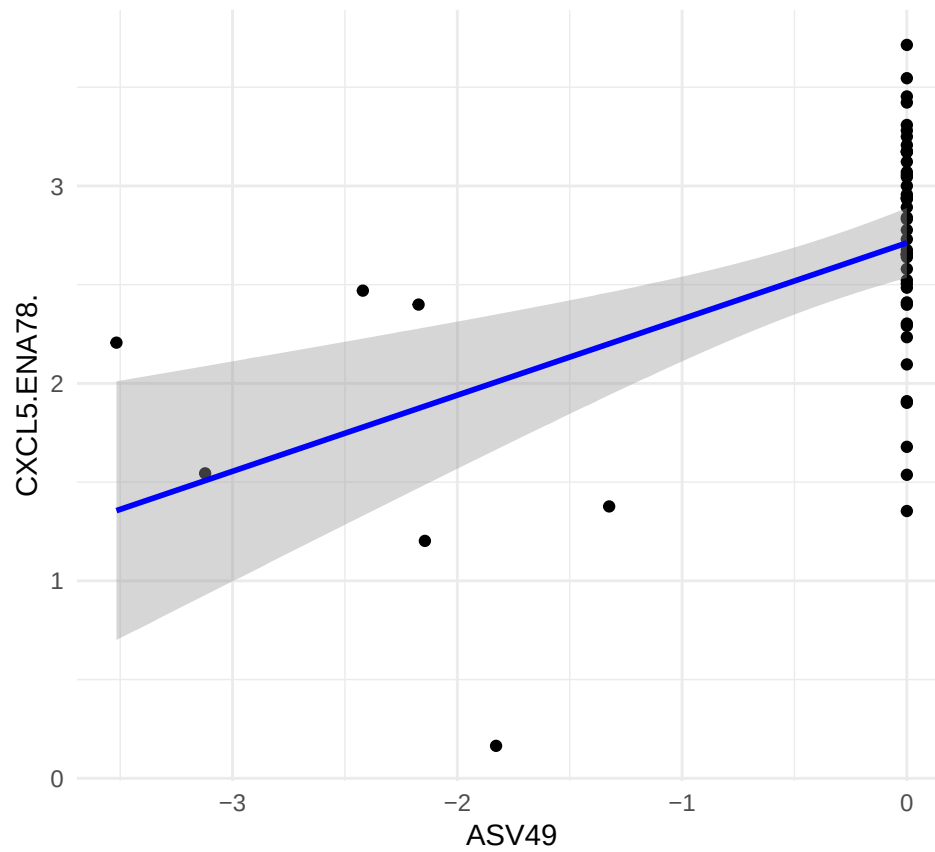
$r = -0.288$ | $p = 0.036$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV49 vs CXCL5.ENA78.

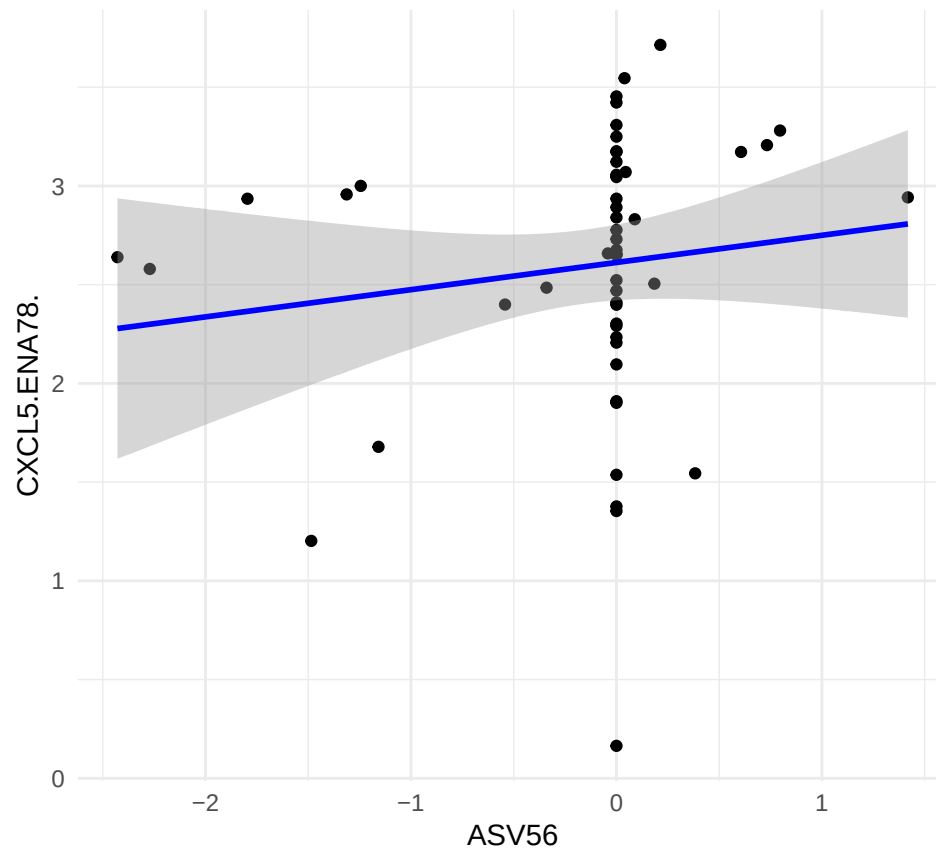
$r = 0.469$ | $p = 0$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV56 vs CXCL5.ENA78.

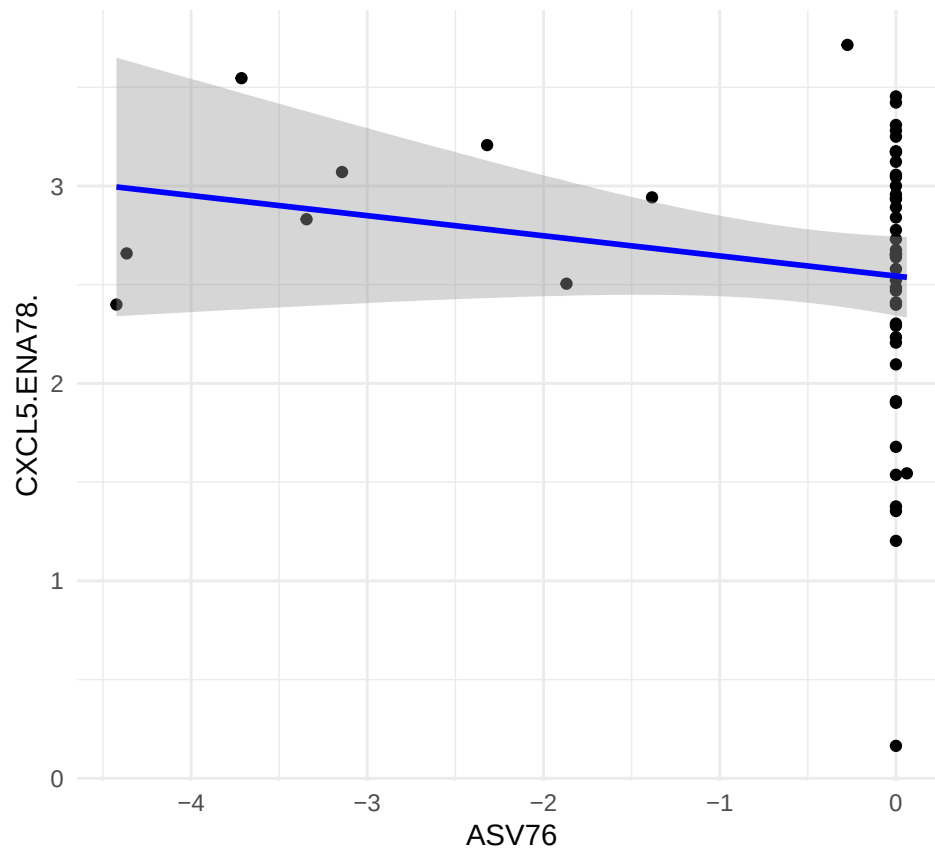
$r = 0.298$ | $p = 0.03$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV76 vs CXCL5.ENA78.

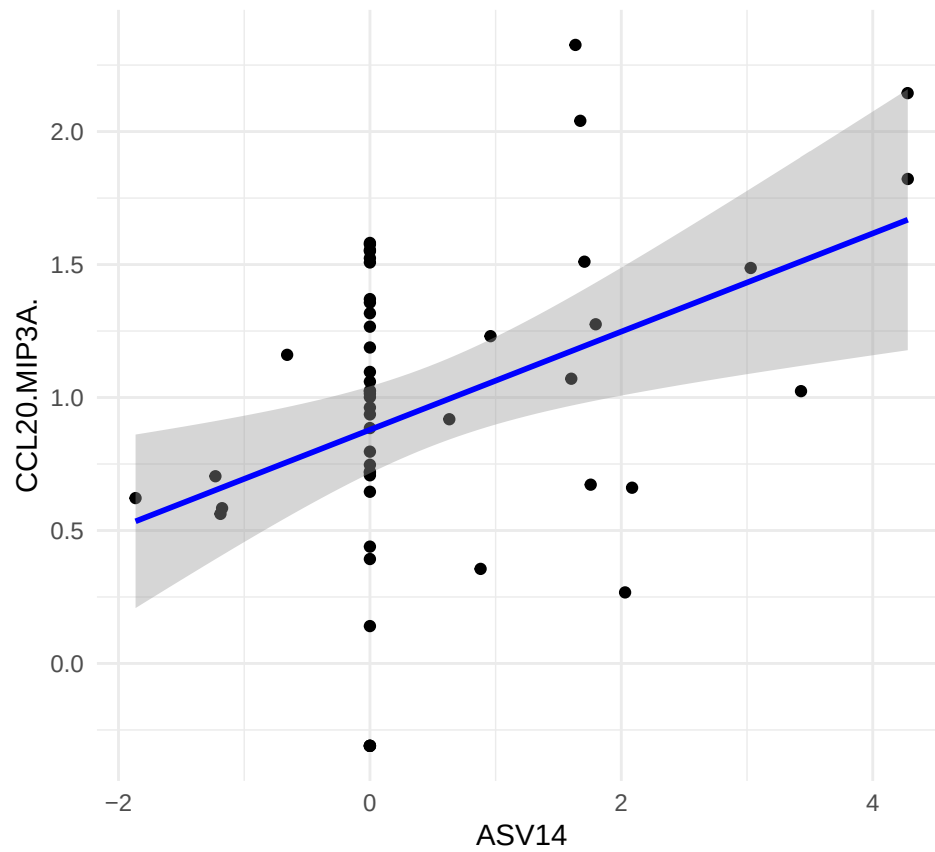
$r = -0.282$ | $p = 0.041$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV14 vs CCL20.MIP3A.

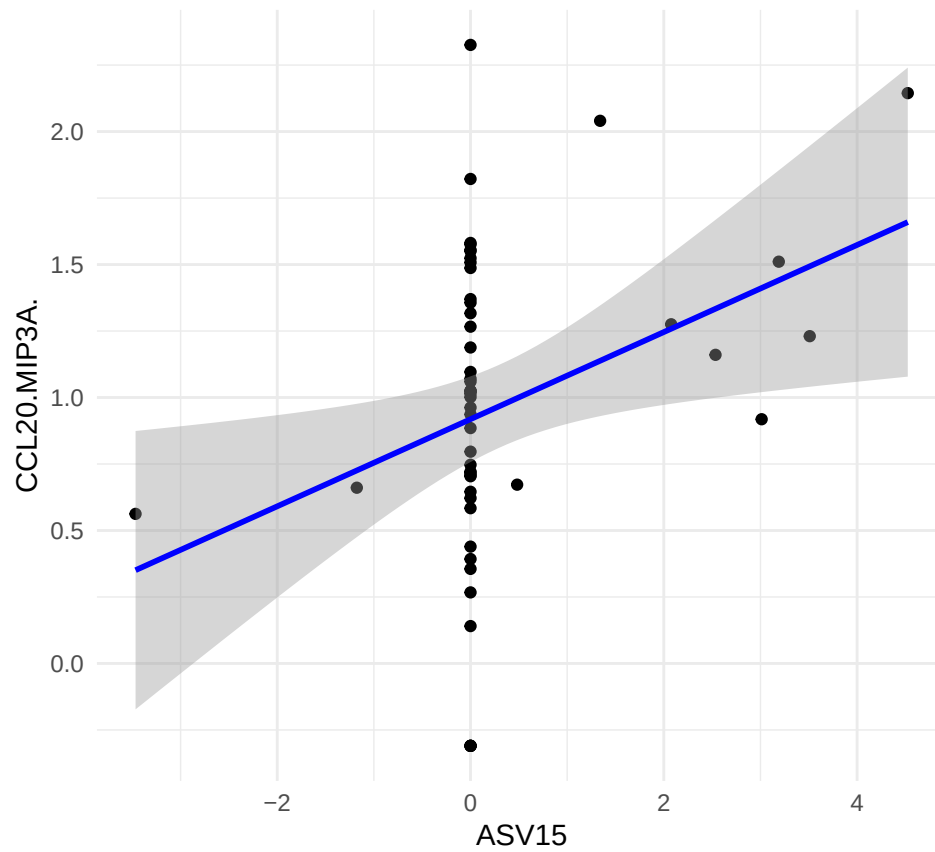
$r = 0.309$ | $p = 0.024$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV15 vs CCL20.MIP3A.

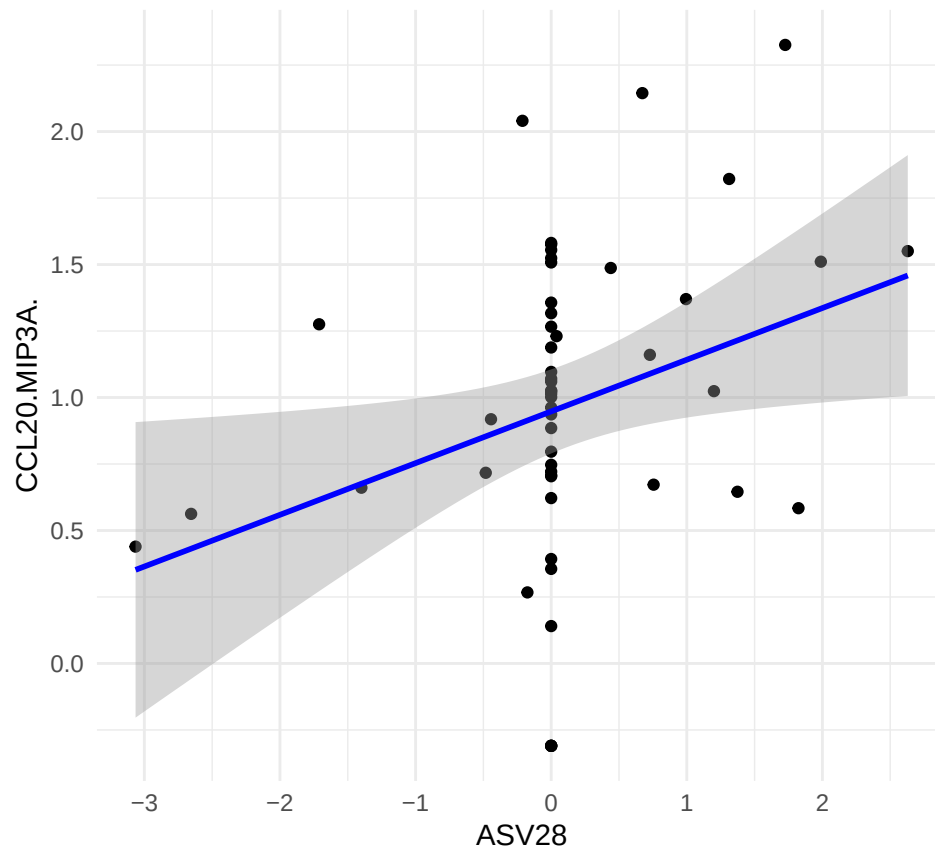
$r = 0.323$ | $p = 0.018$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV28 vs CCL20.MIP3A.

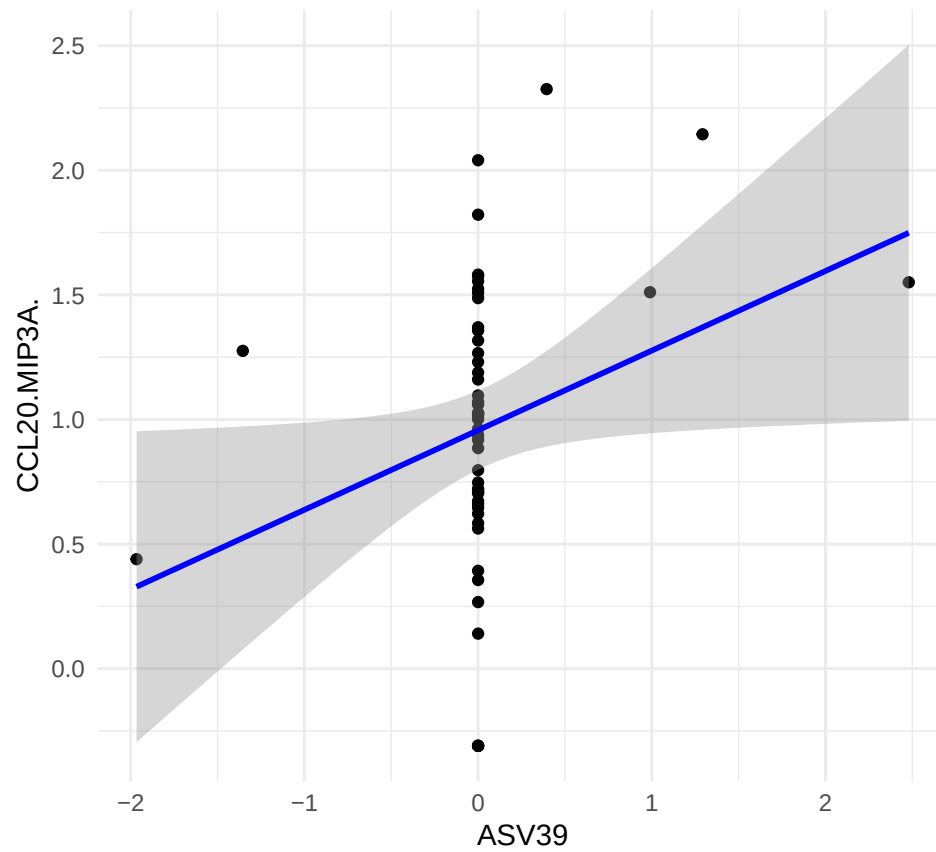
$r = 0.324$ | $p = 0.018$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV39 vs CCL20.MIP3A.

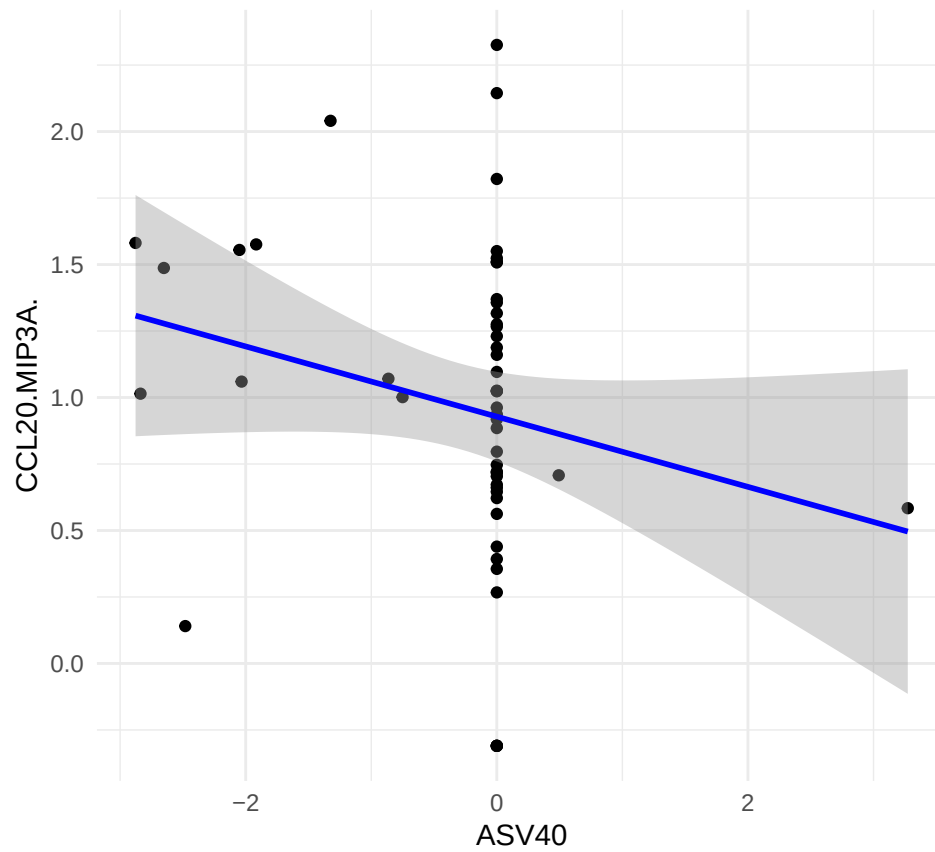
$r = 0.352$ | $p = 0.01$



```
## 'geom_smooth()' using formula = 'y ~ x'
```


ASV40 vs CCL20.MIP3A.

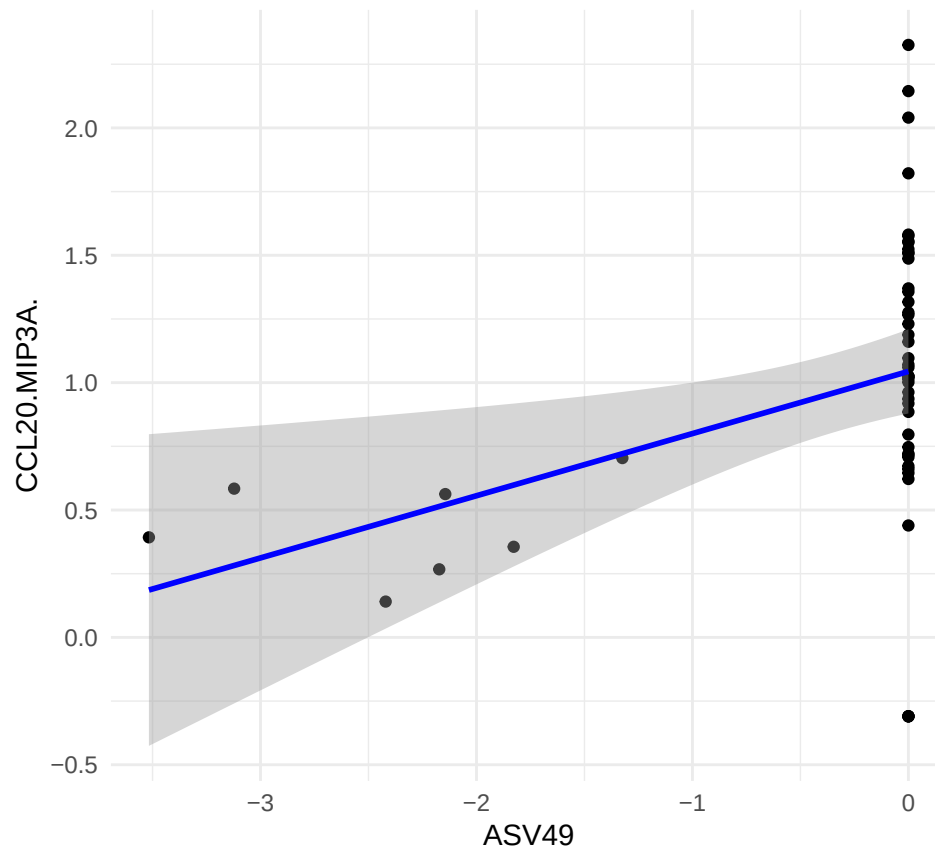
$r = -0.31$ | $p = 0.024$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV49 vs CCL20.MIP3A.

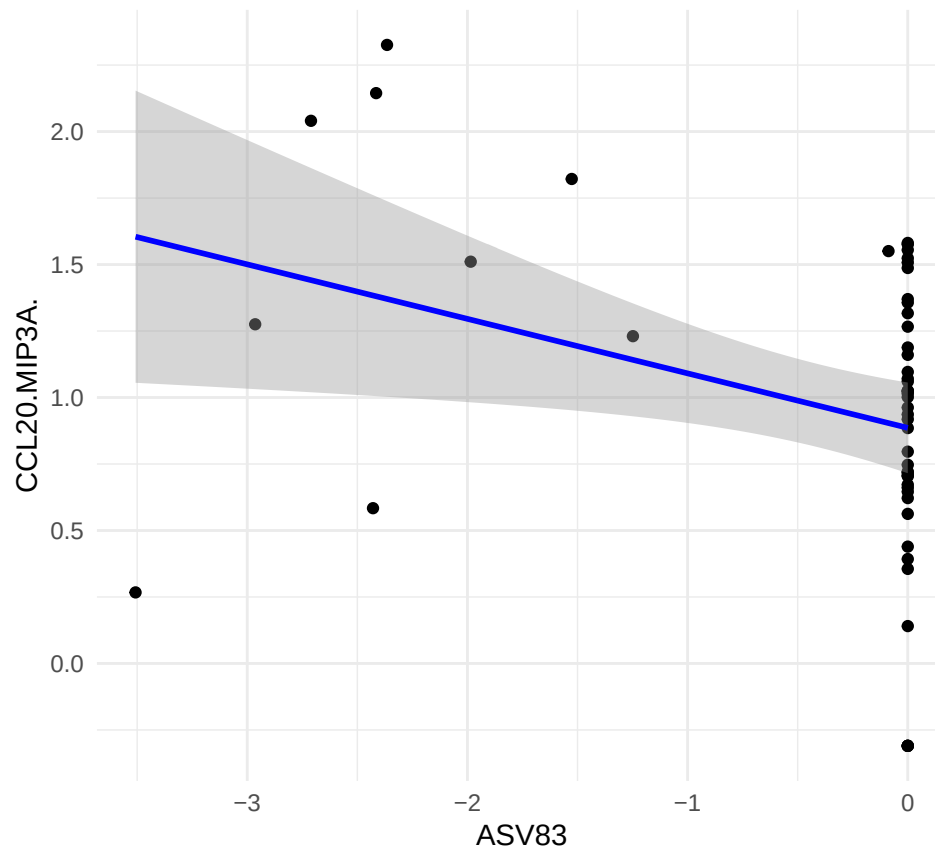
$r = 0.46$ | $p = 0.001$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV83 vs CCL20.MIP3A.

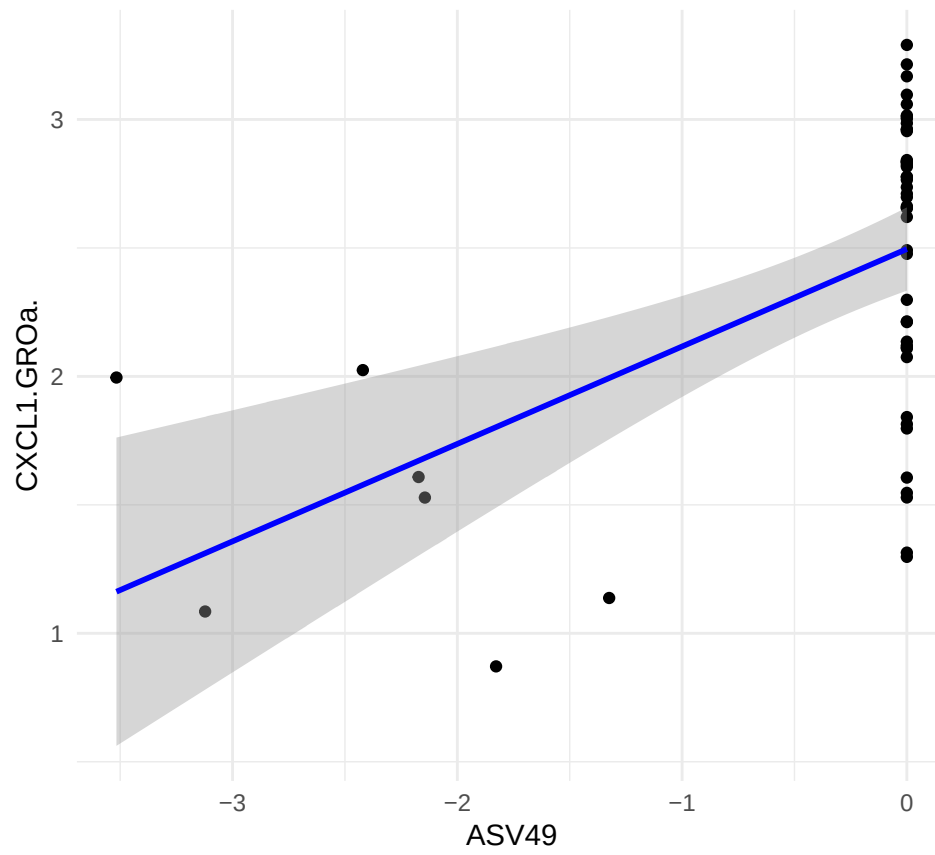
$r = -0.34$ | $p = 0.013$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV49 vs CXCL1.GROa.

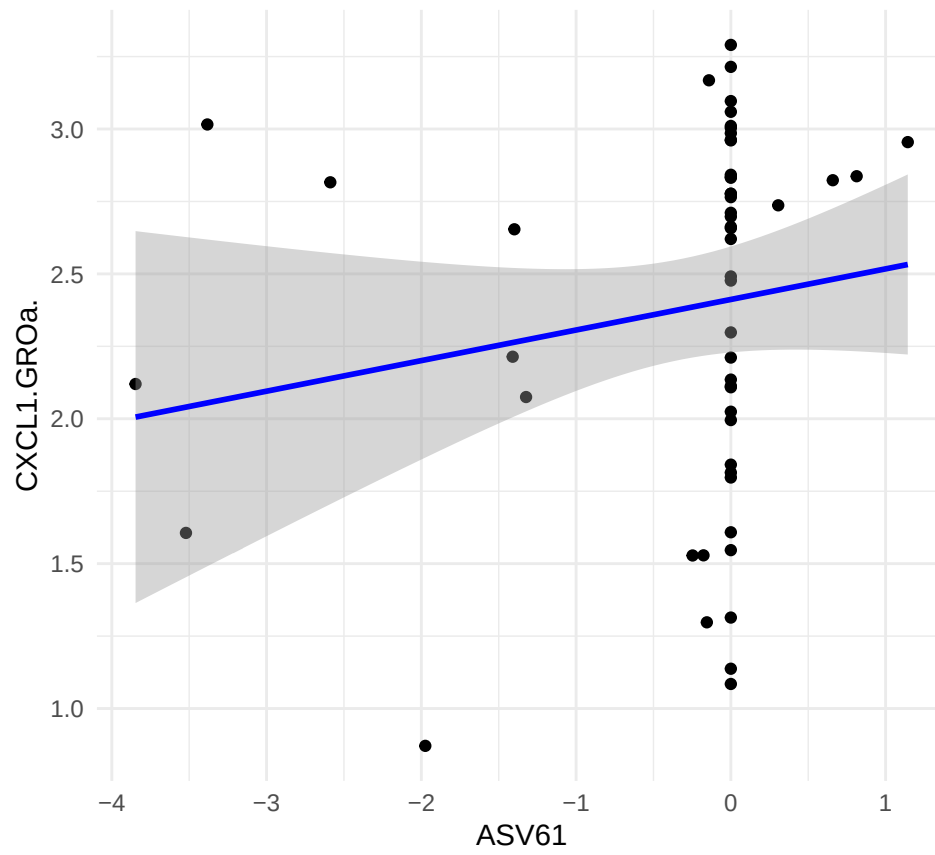
$r = 0.495$ | $p = 0$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV61 vs CXCL1.GROa.

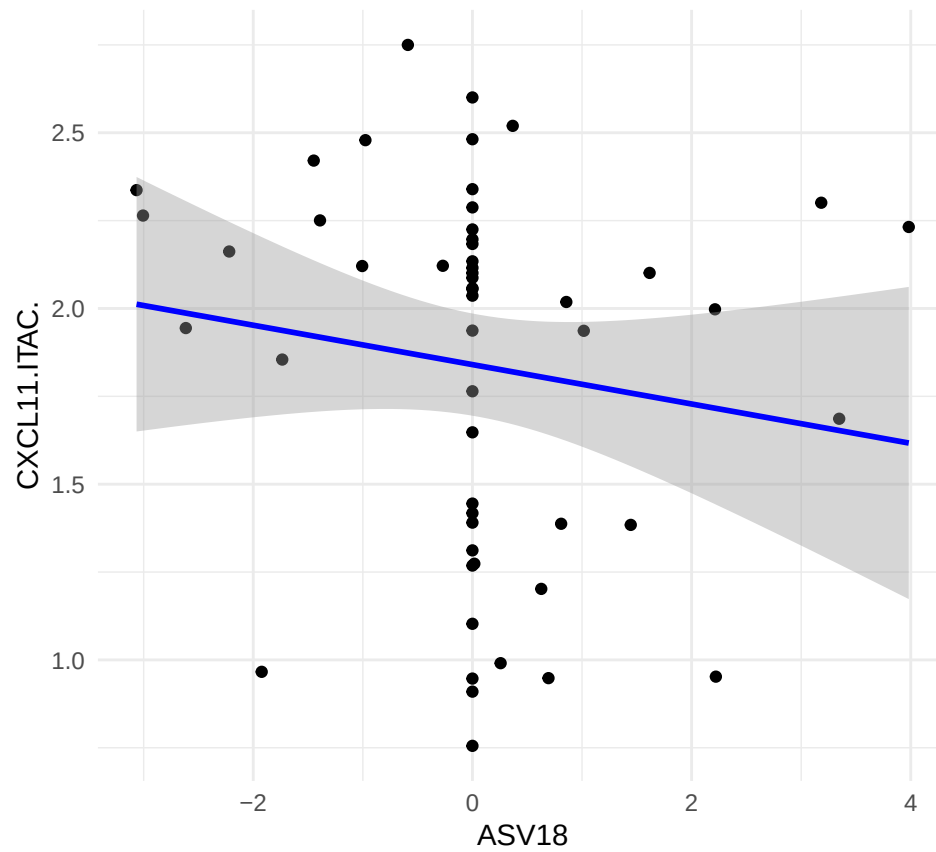
$r = 0.275$ | $p = 0.046$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV18 vs CXCL11.ITAC.

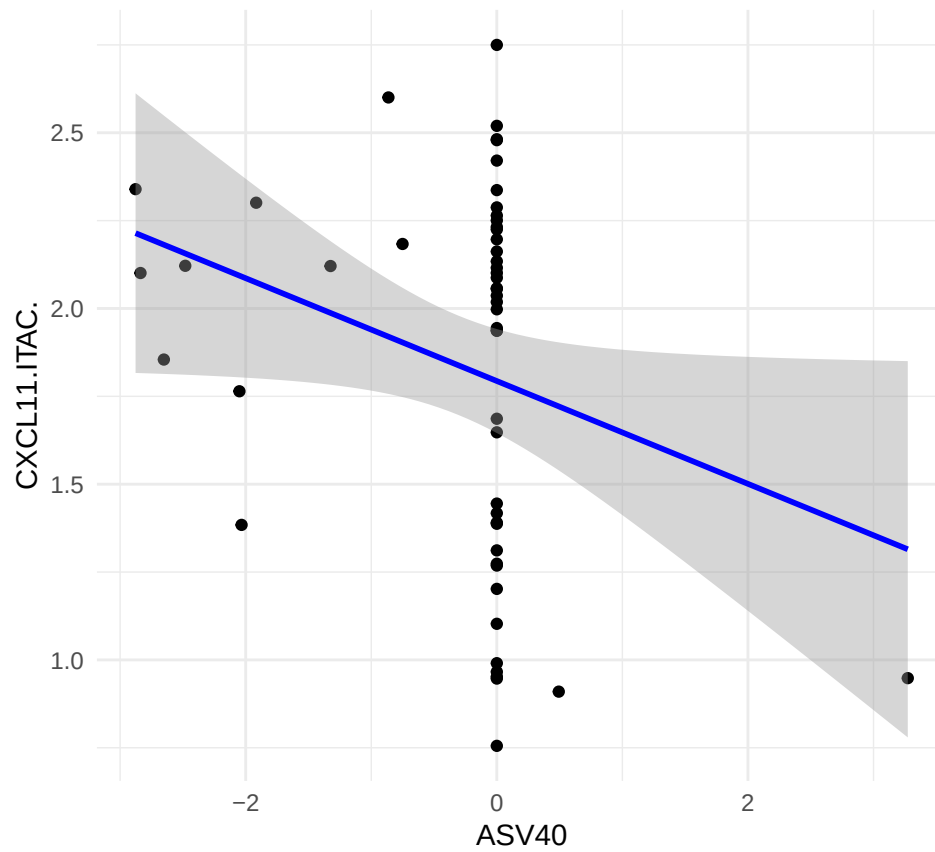
$r = -0.29 \mid p = 0.035$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV40 vs CXCL11.ITAC.

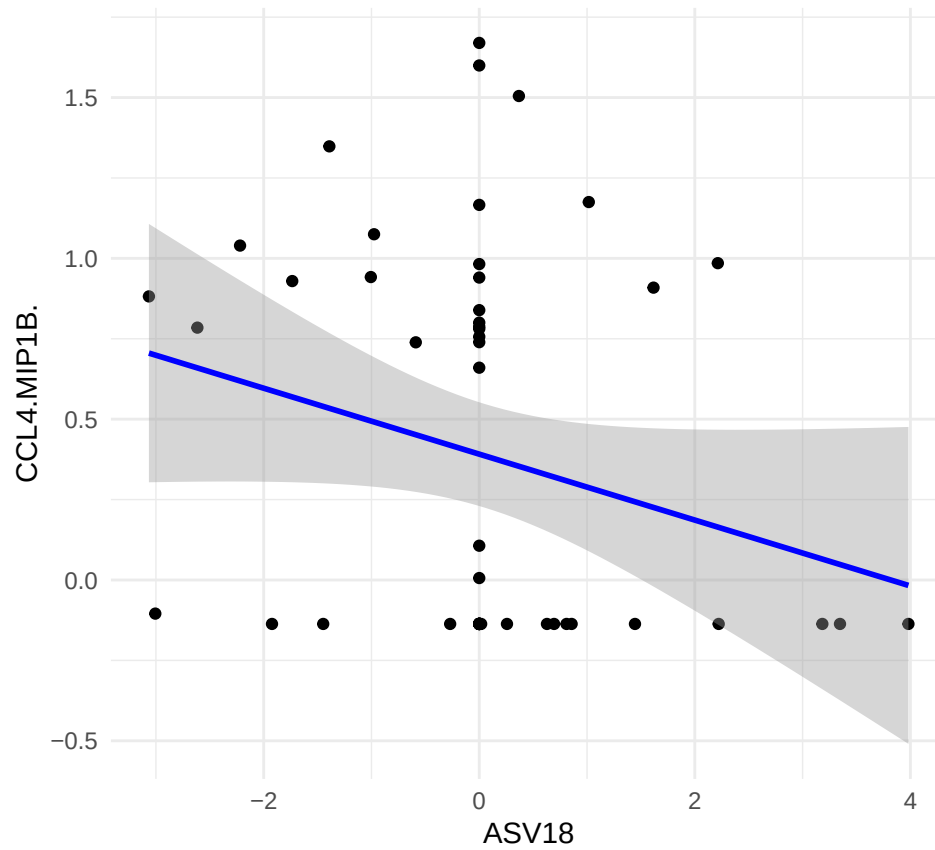
$r = -0.29$ | $p = 0.035$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV18 vs CCL4.MIP1B.

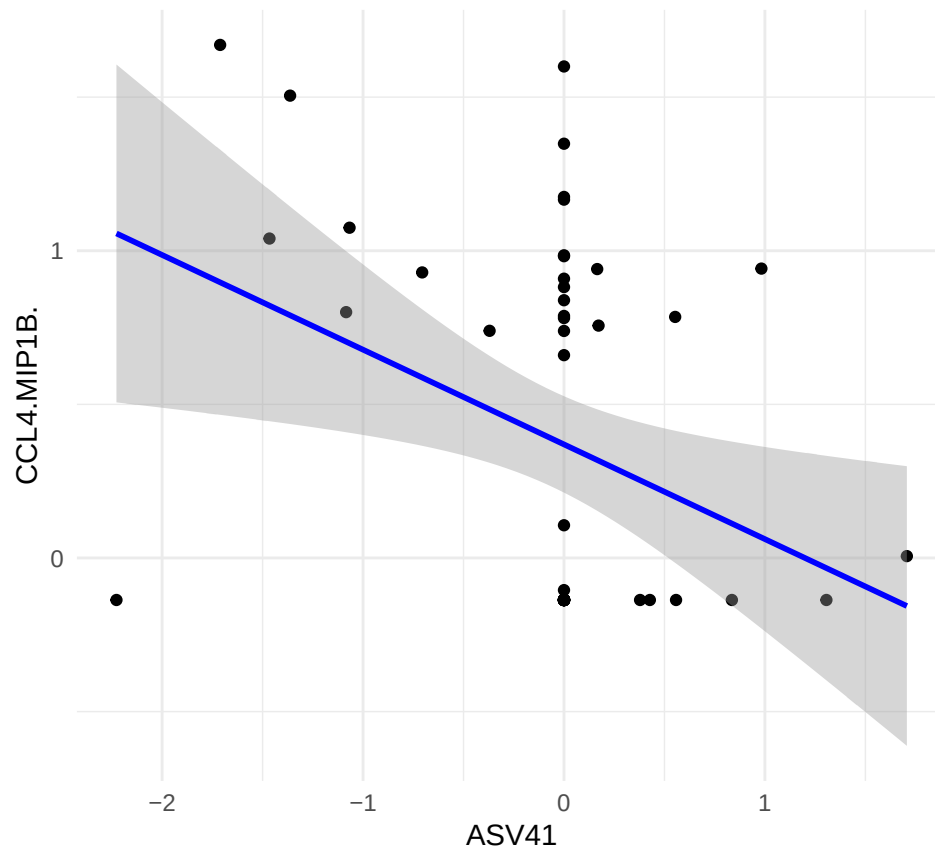
$r = -0.273$ | $p = 0.048$



```
## 'geom_smooth()' using formula = 'y ~ x'
```


ASV41 vs CCL4.MIP1B.

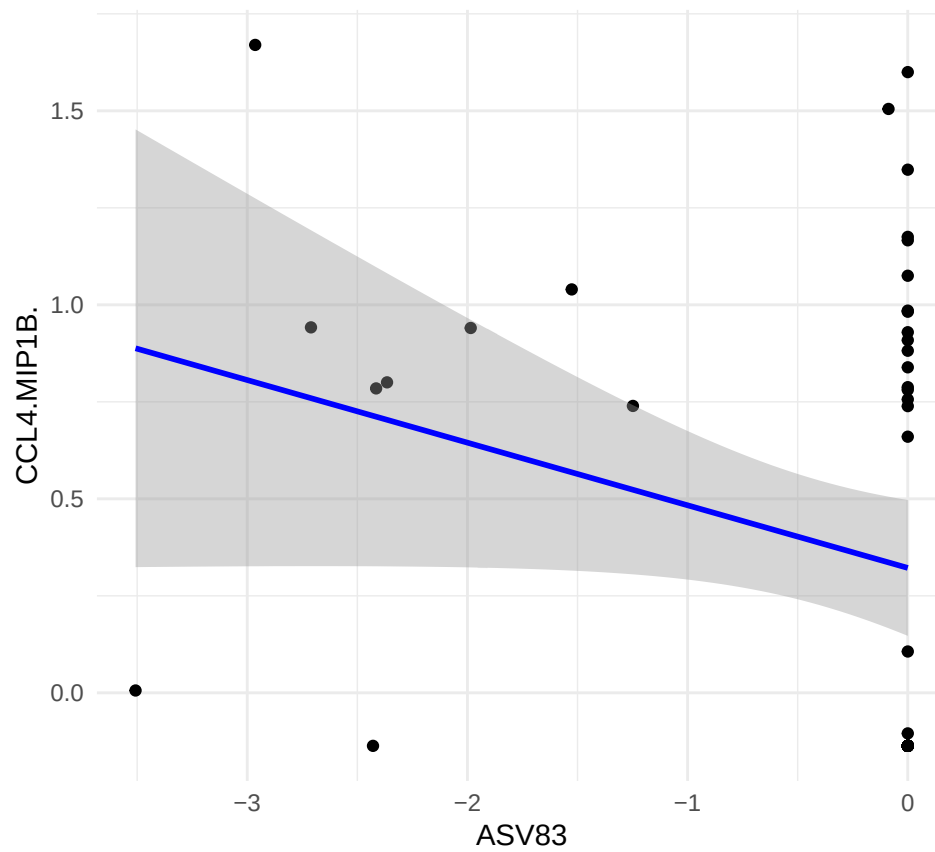
$r = -0.282$ | $p = 0.041$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV83 vs CCL4.MIP1B.

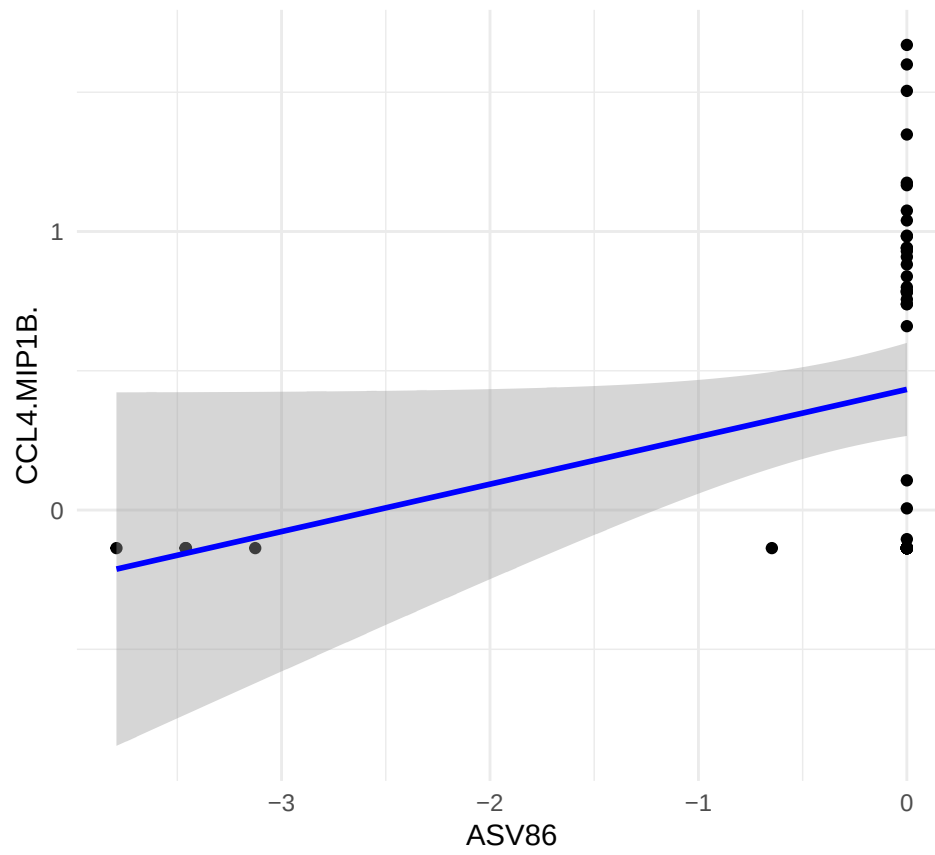
$r = -0.358$ | $p = 0.008$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

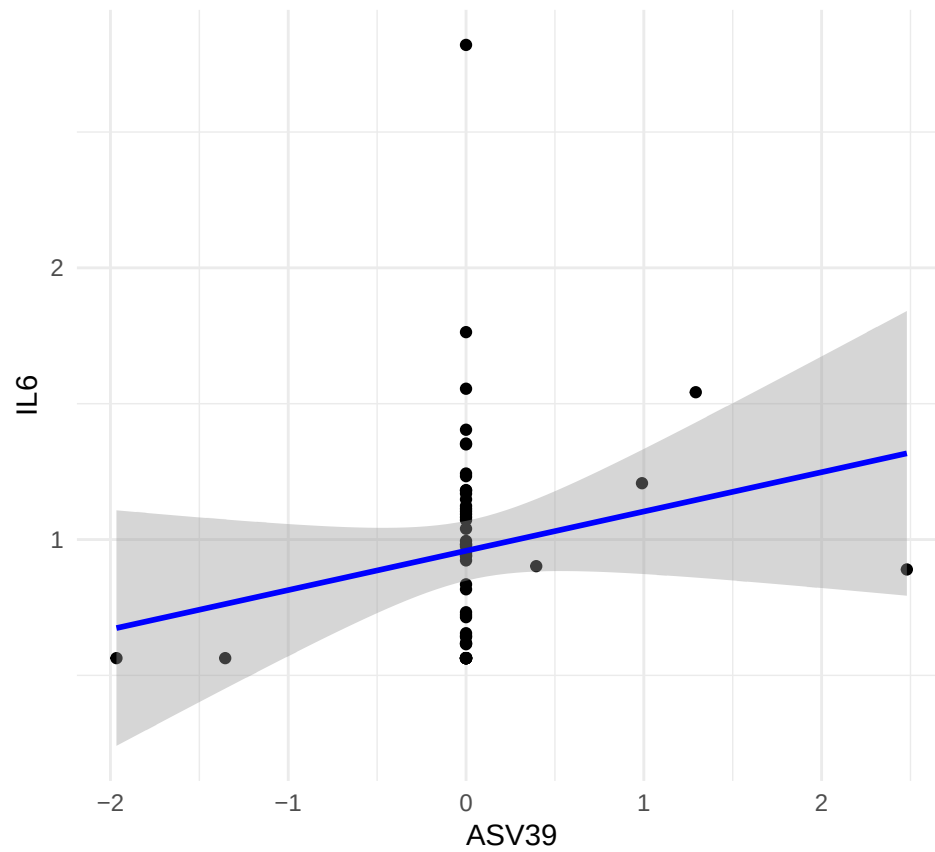
ASV86 vs CCL4.MIP1B.

$r = 0.303$ | $p = 0.028$



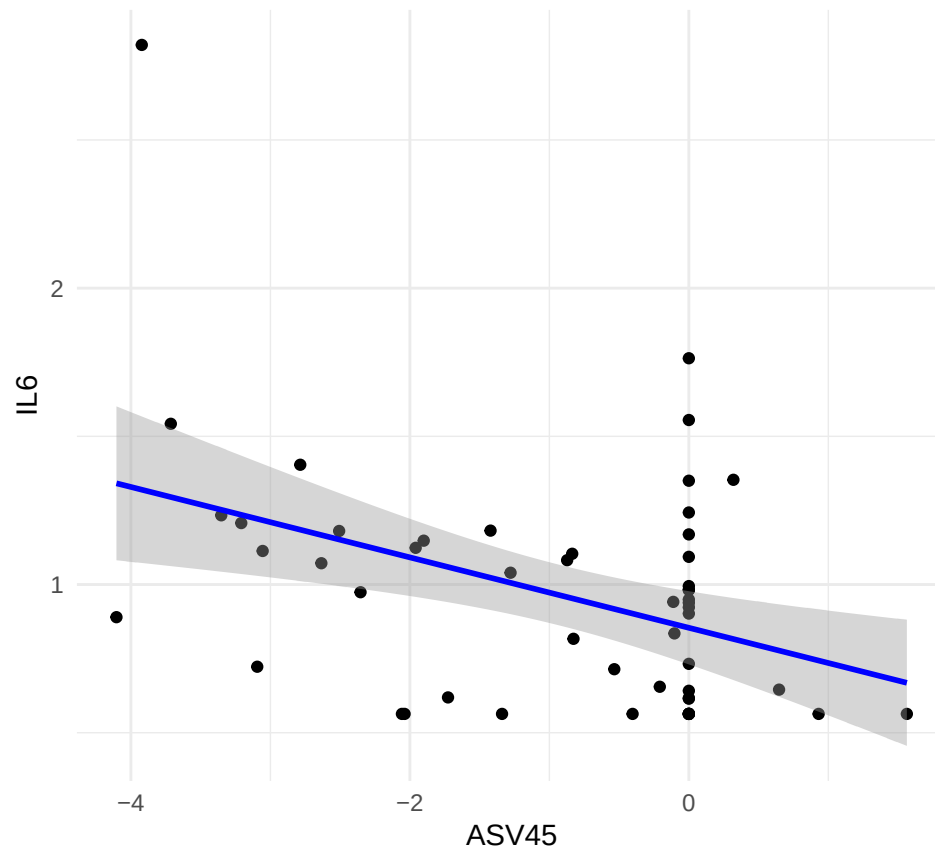
```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV39 vs IL6
 $r = 0.276$ | $p = 0.046$



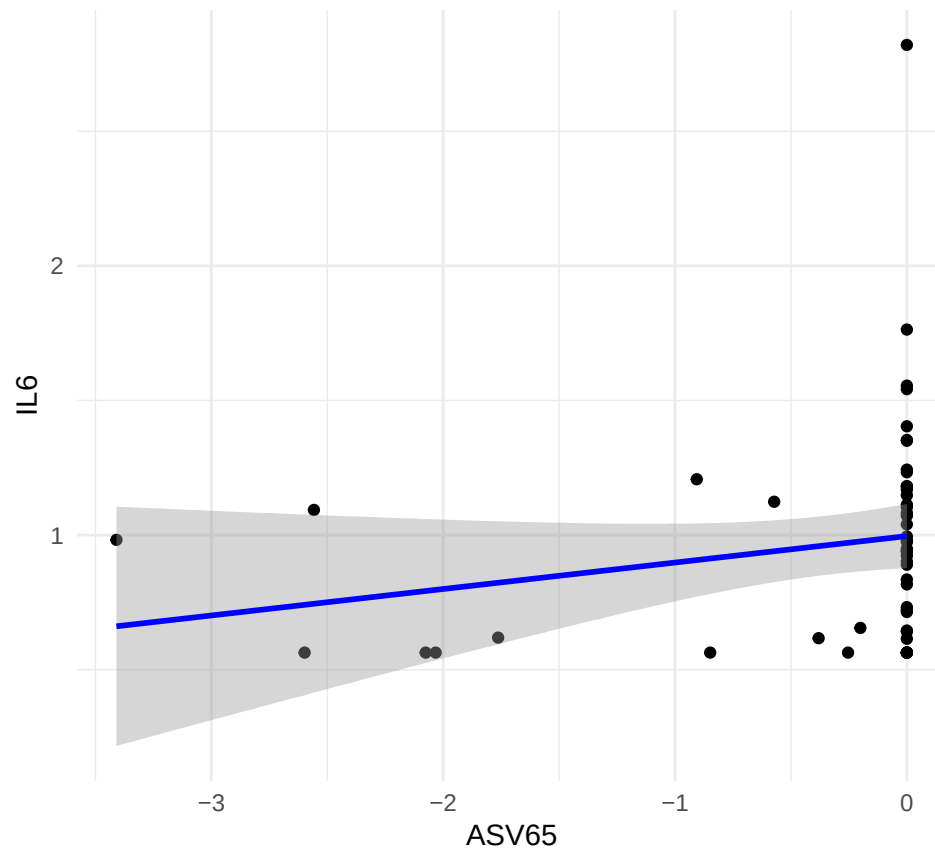
```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV45 vs IL6
 $r = -0.3 \mid p = 0.029$



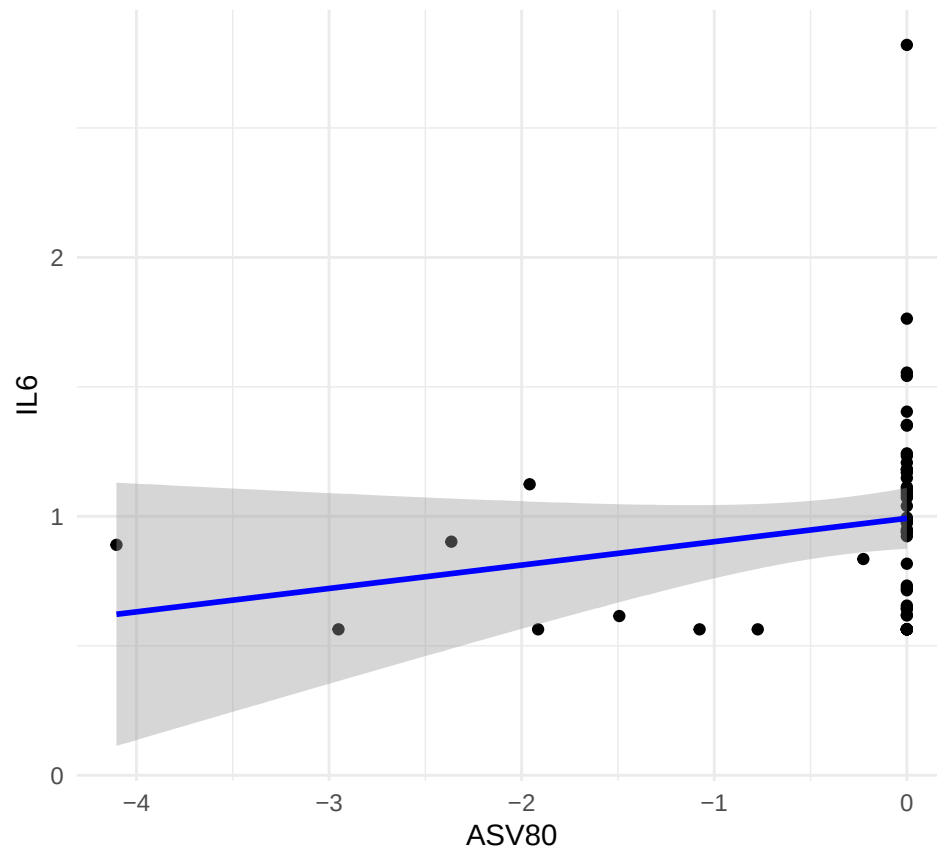
```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV65 vs IL6
 $r = 0.286$ | $p = 0.038$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

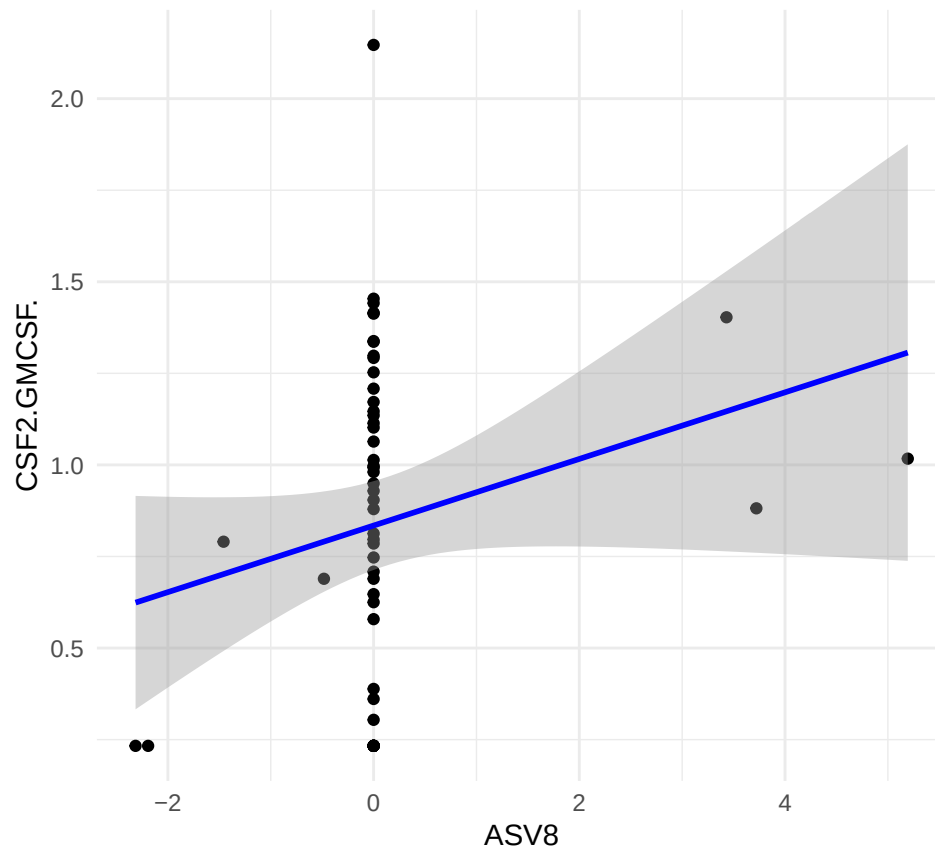
ASV80 vs IL6
 $r = 0.317$ | $p = 0.021$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV8 vs CSF2.GMCSF.

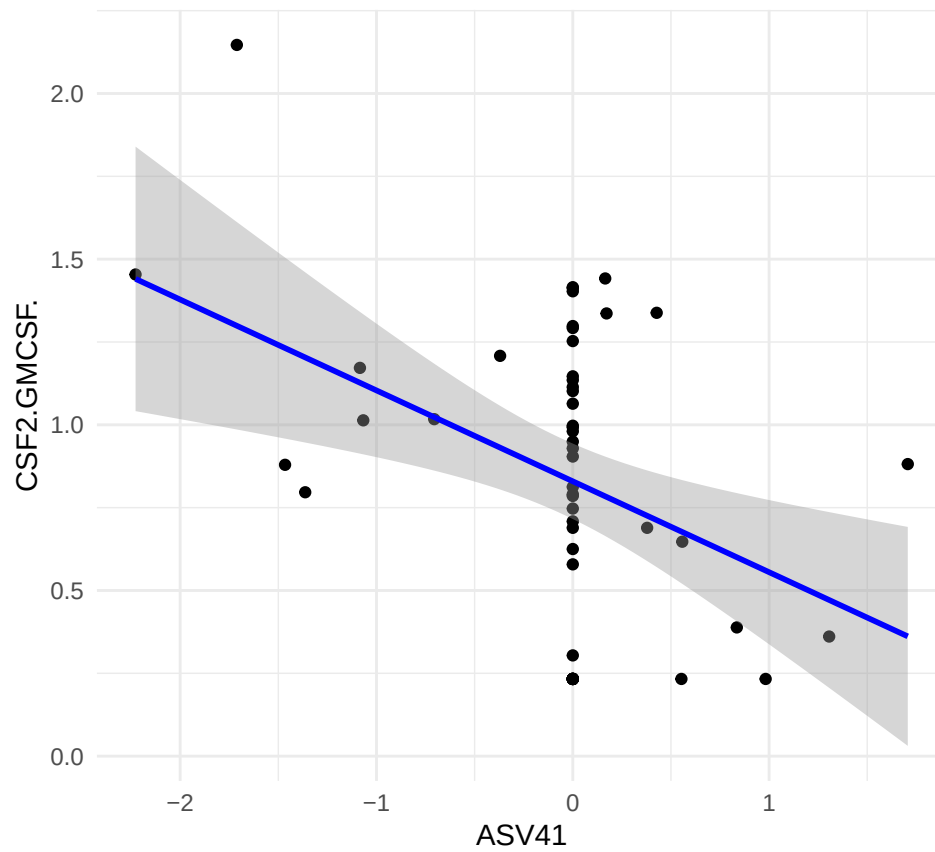
$r = 0.287$ | $p = 0.037$



```
## 'geom_smooth()' using formula = 'y ~ x'
```


ASV41 vs CSF2.GMCSF.

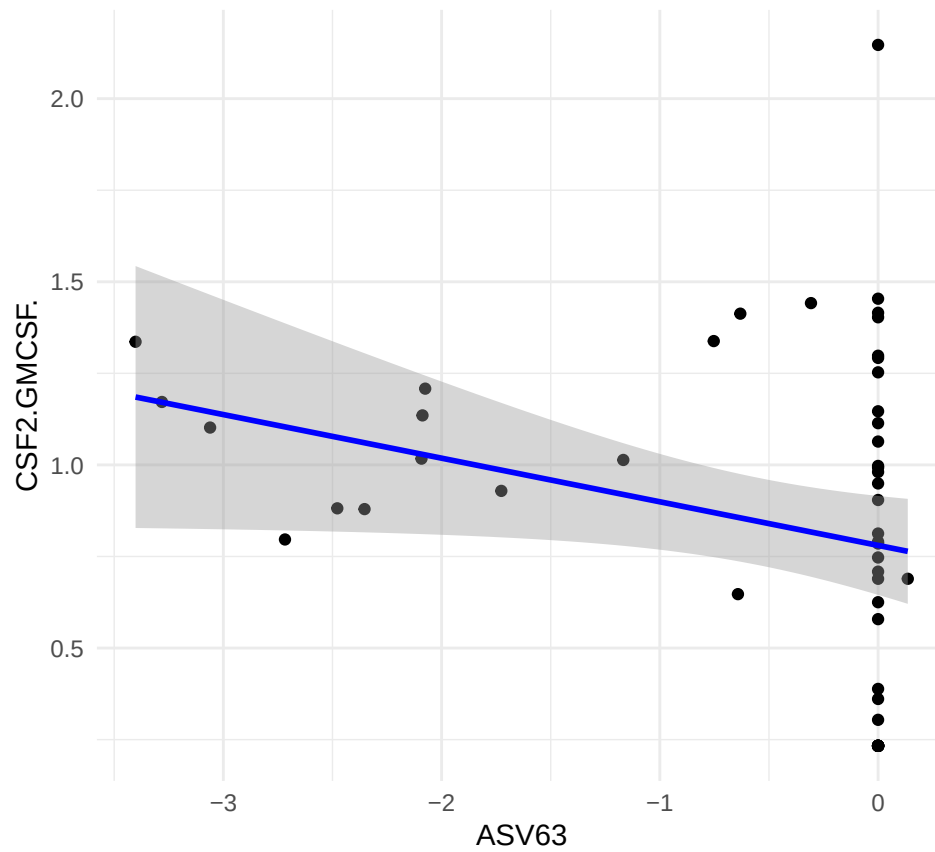
$r = -0.275$ | $p = 0.046$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV63 vs CSF2.GMCSF.

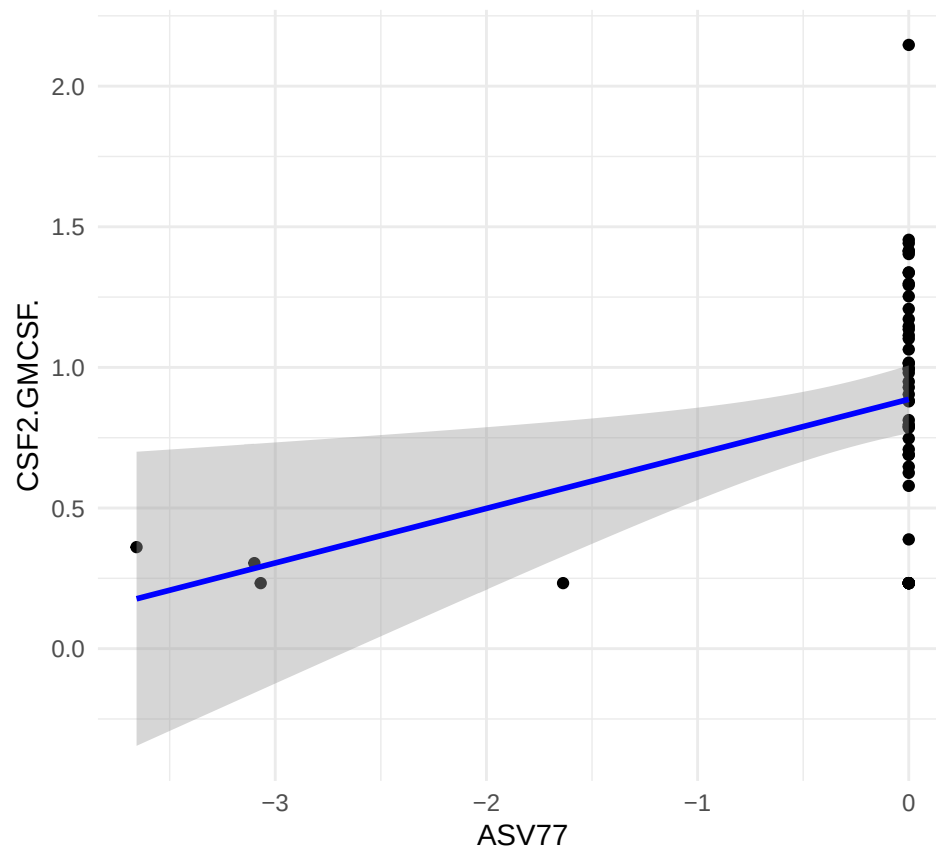
$r = -0.366$ | $p = 0.007$



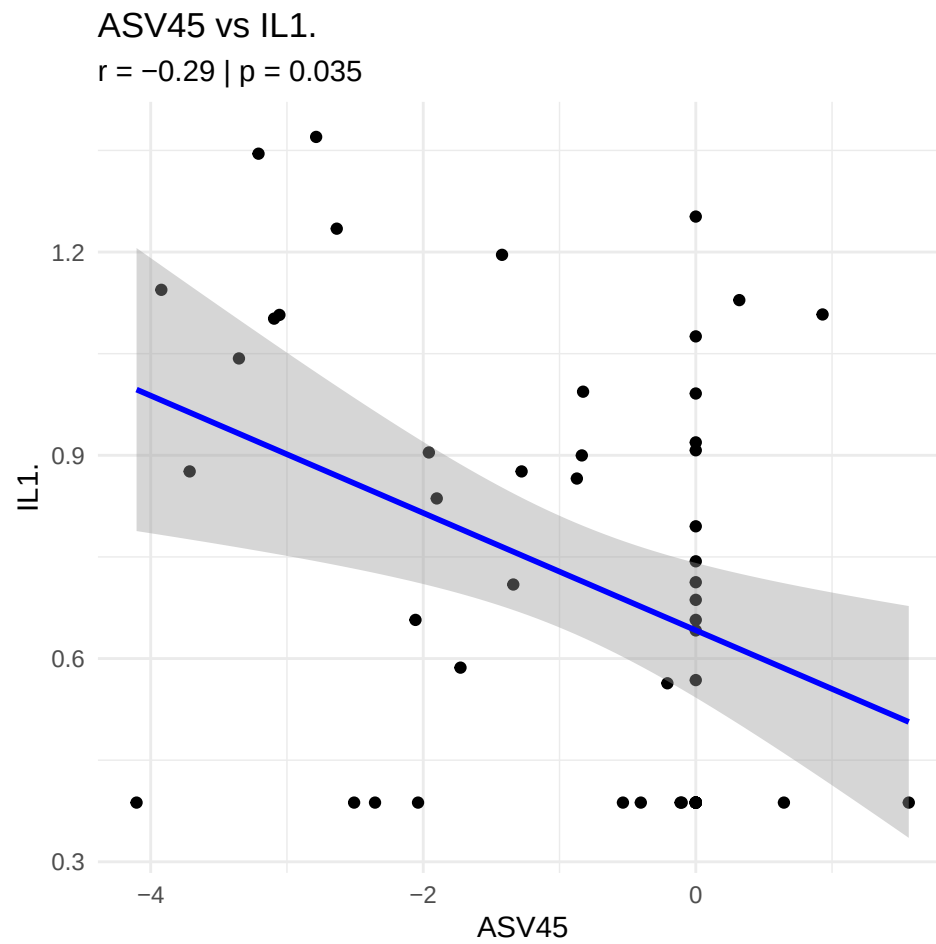
```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV77 vs CSF2.GMCSF.

$r = 0.33$ | $p = 0.016$



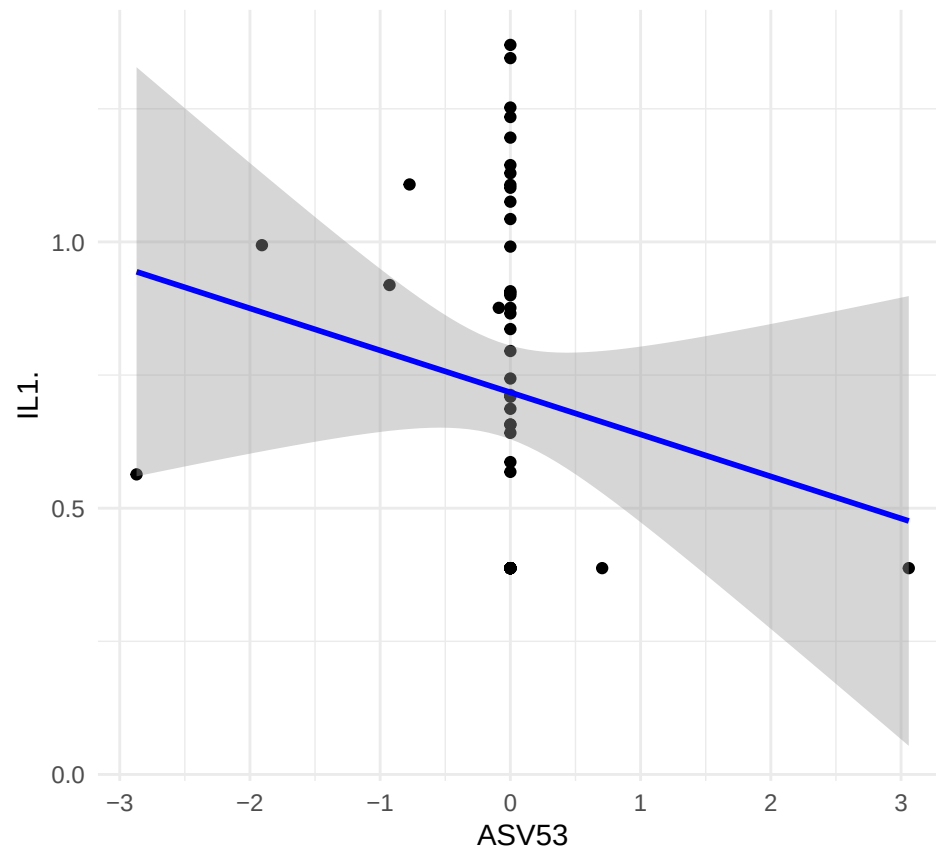
```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV53 vs IL1.

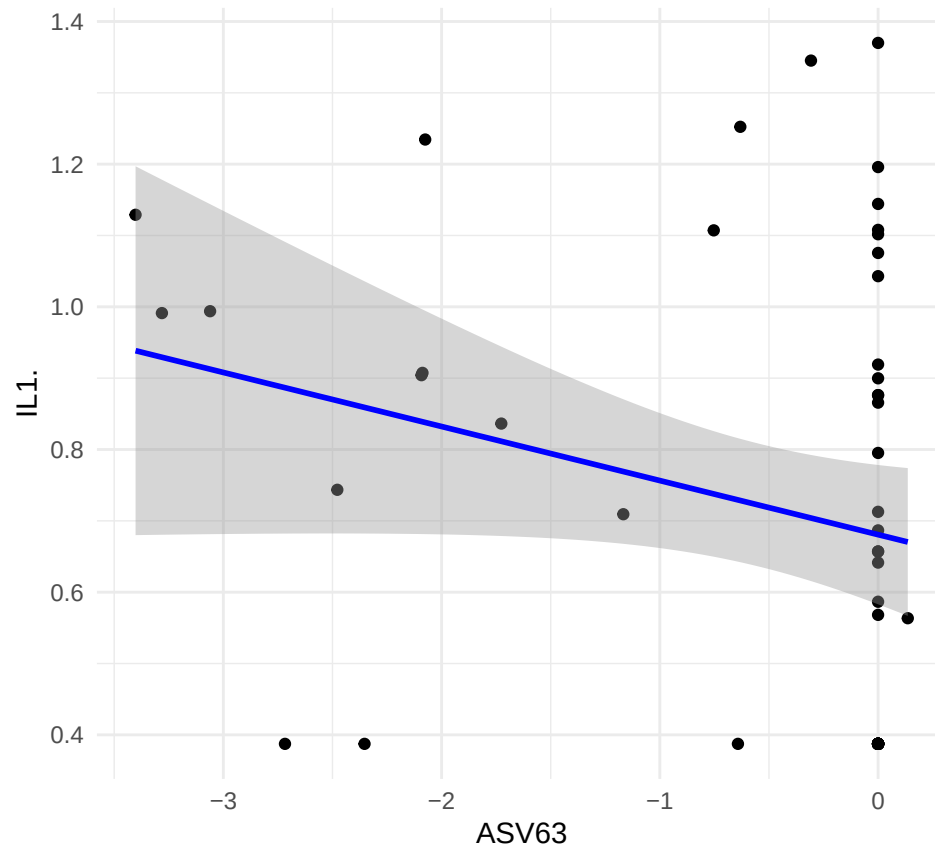
$r = -0.274$ | $p = 0.047$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV63 vs IL1.

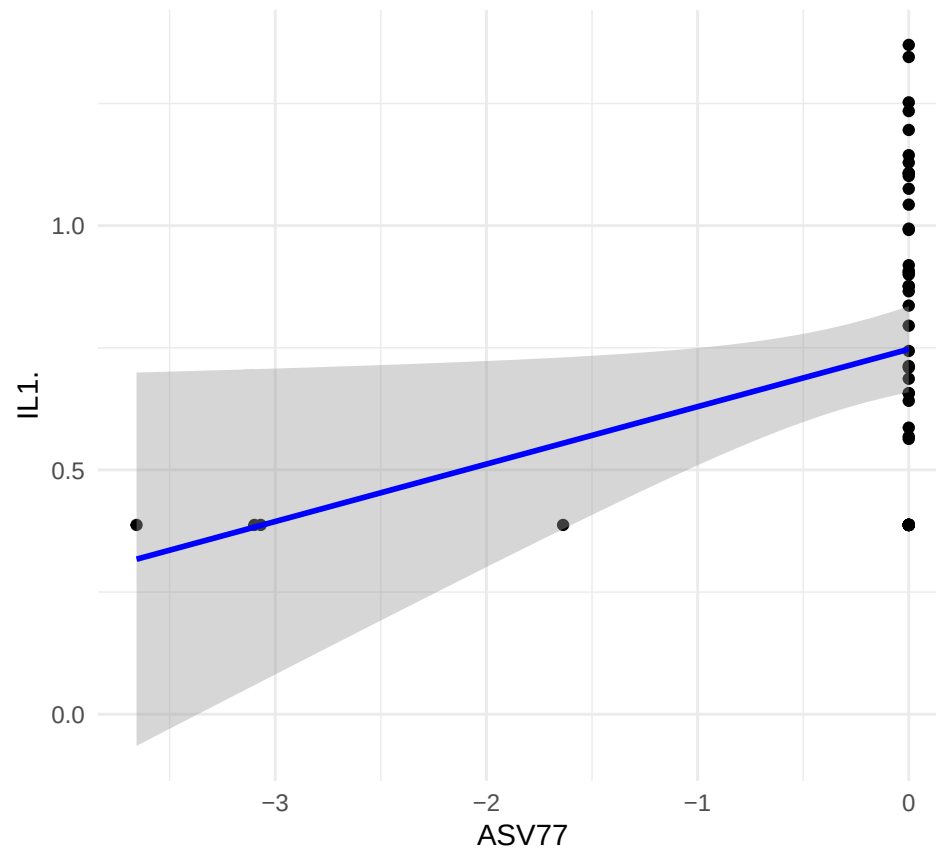
$r = -0.309$ | $p = 0.024$



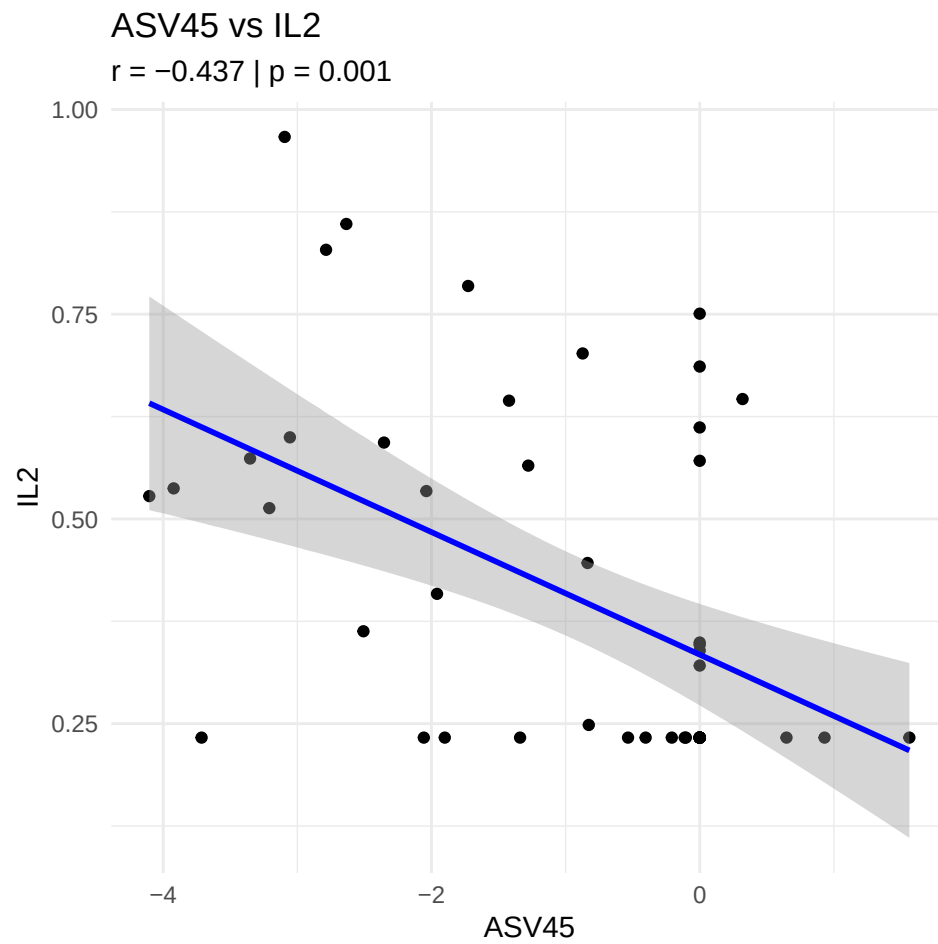
```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV77 vs IL1.

$r = 0.317$ | $p = 0.021$

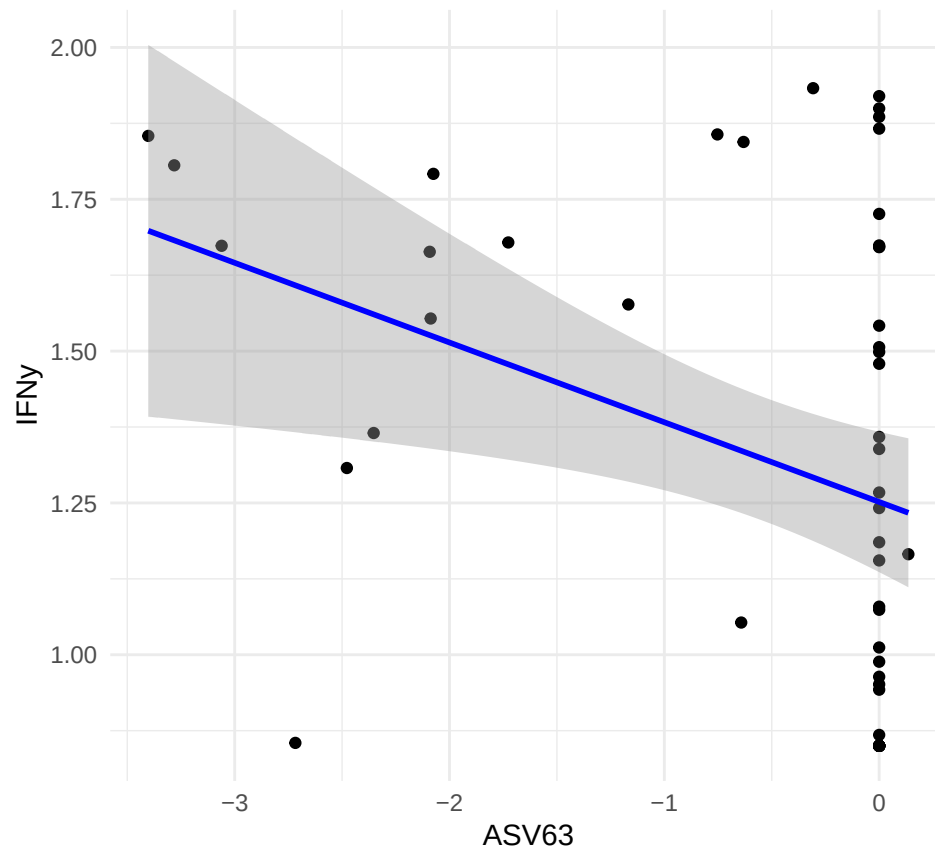


```
## 'geom_smooth()' using formula = 'y ~ x'
```

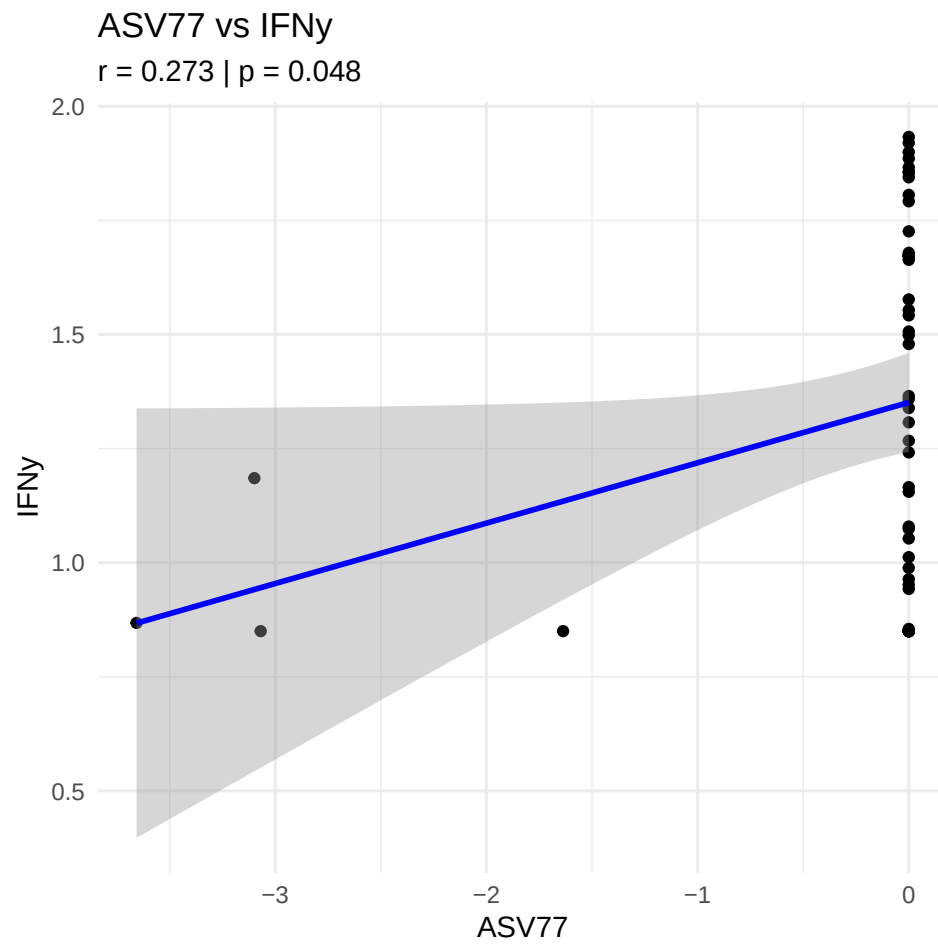


```
## 'geom_smooth()' using formula = 'y ~ x'
```


ASV63 vs IFNy
 $r = -0.398$ | $p = 0.003$



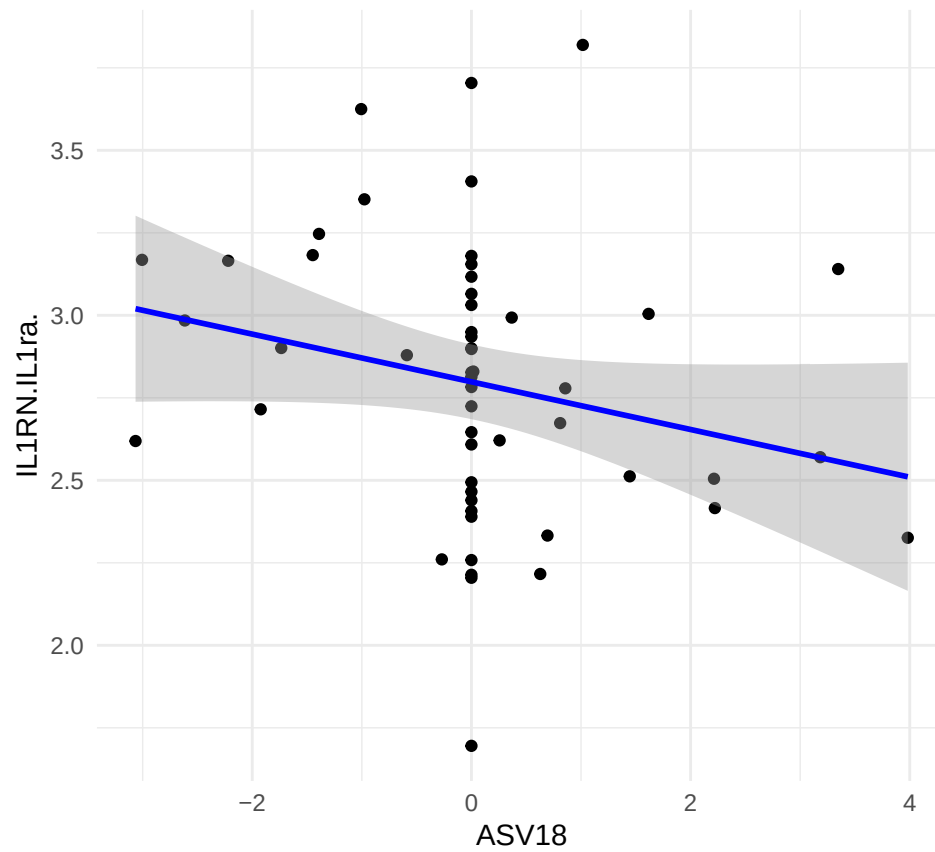
```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV18 vs IL1RN.IL1ra.

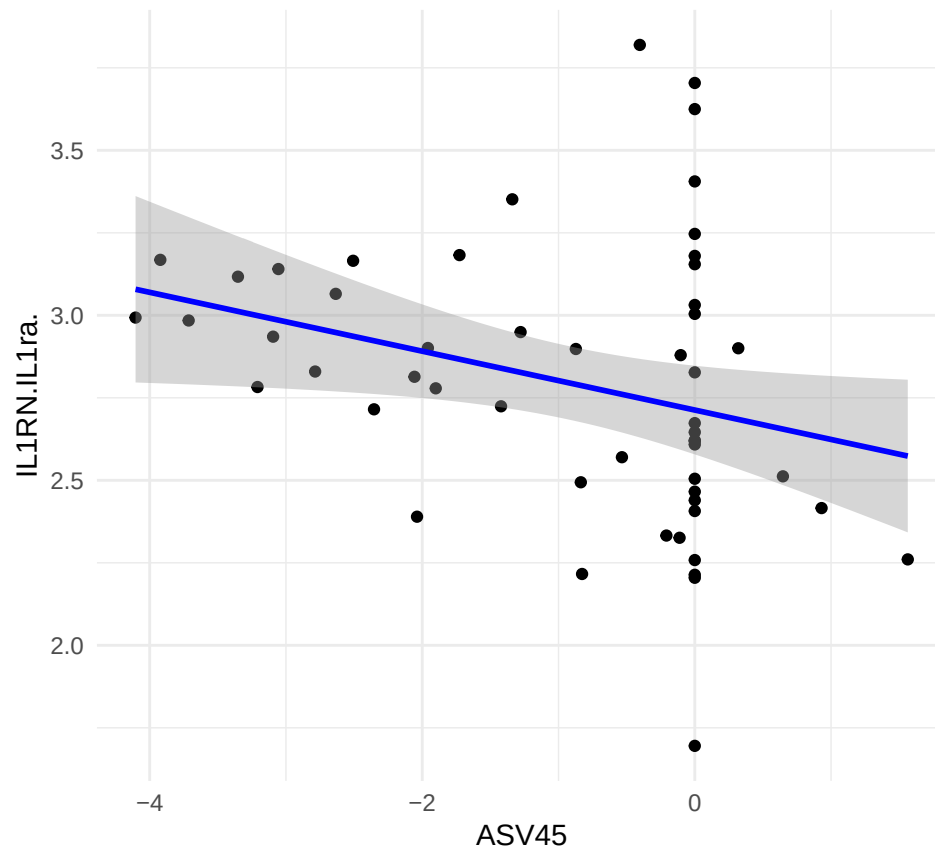
$r = -0.289$ | $p = 0.036$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV45 vs IL1RN.IL1ra.

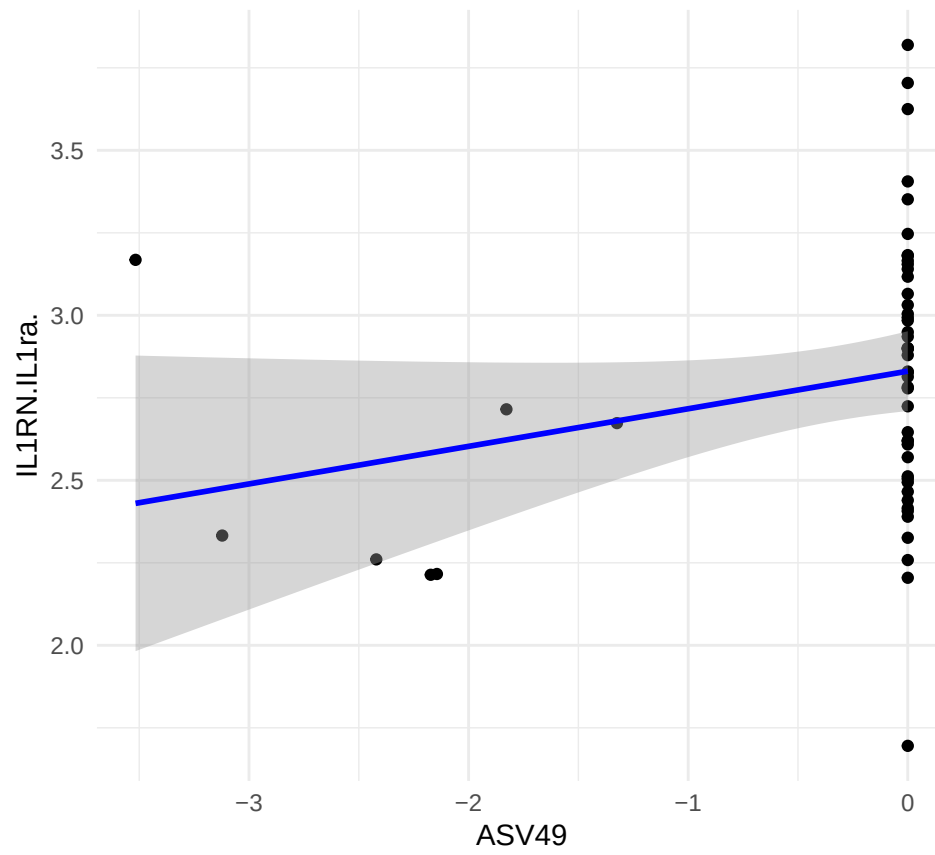
$r = -0.316$ | $p = 0.021$



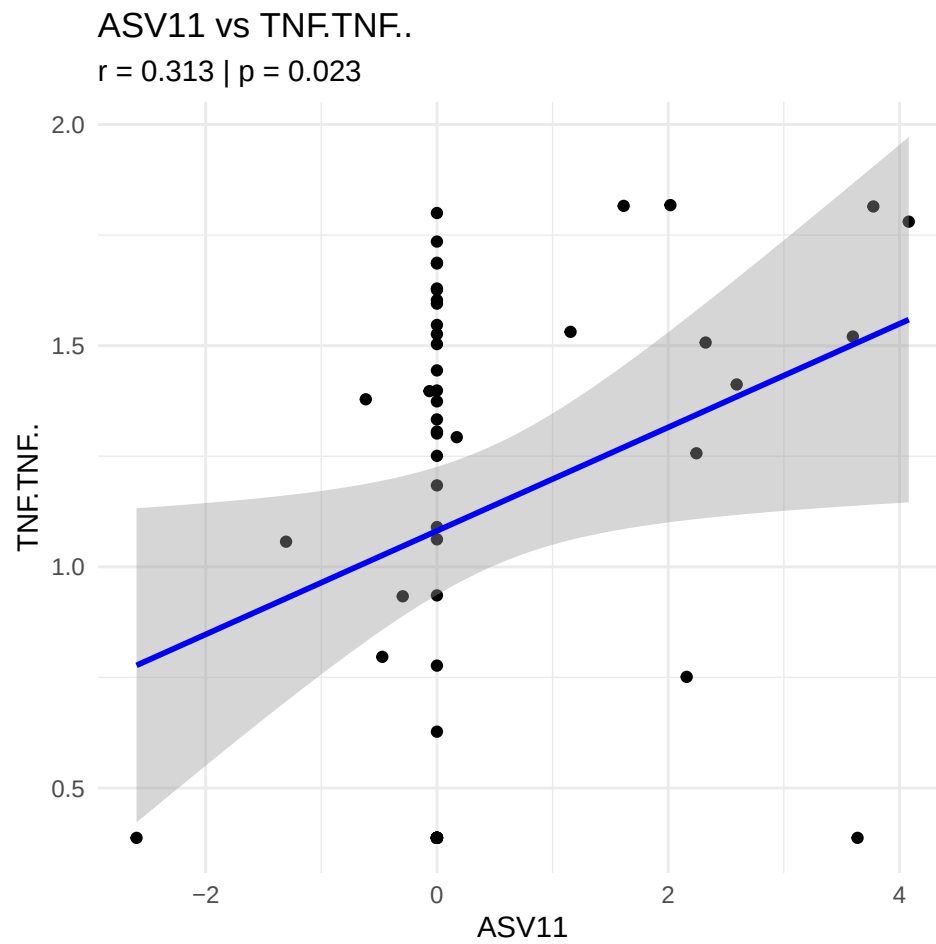
```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV49 vs IL1RN.IL1ra.

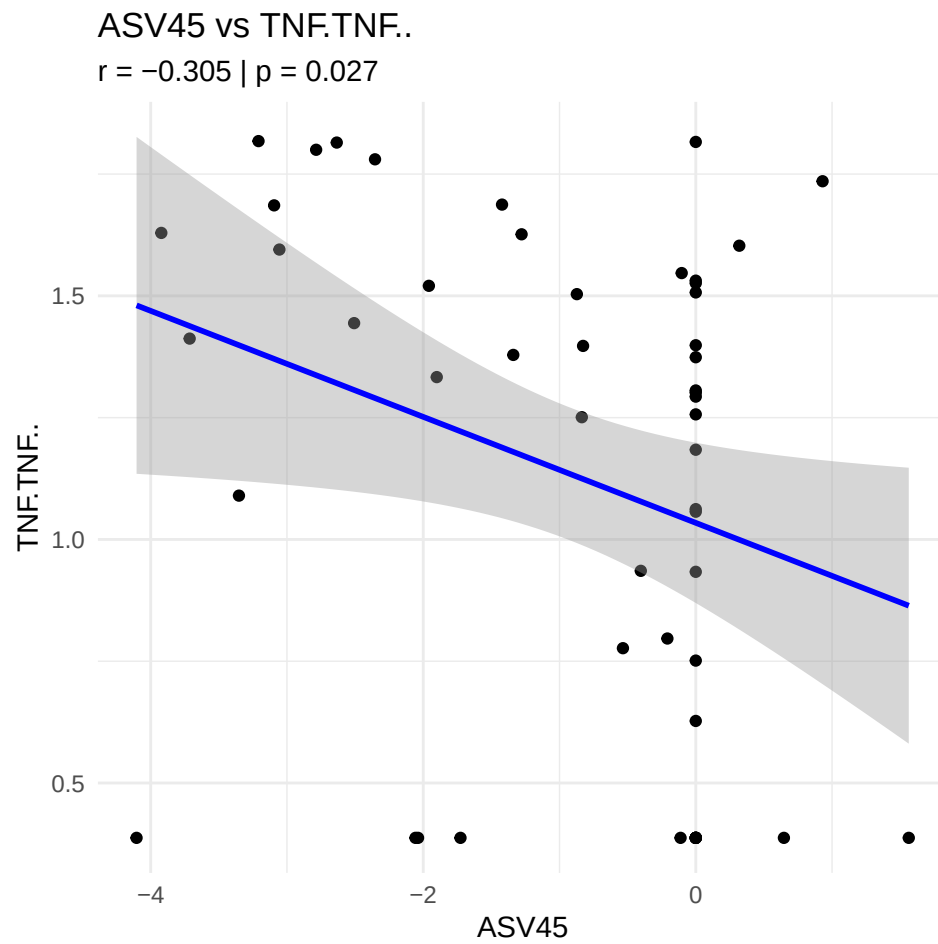
$r = 0.278$ | $p = 0.044$



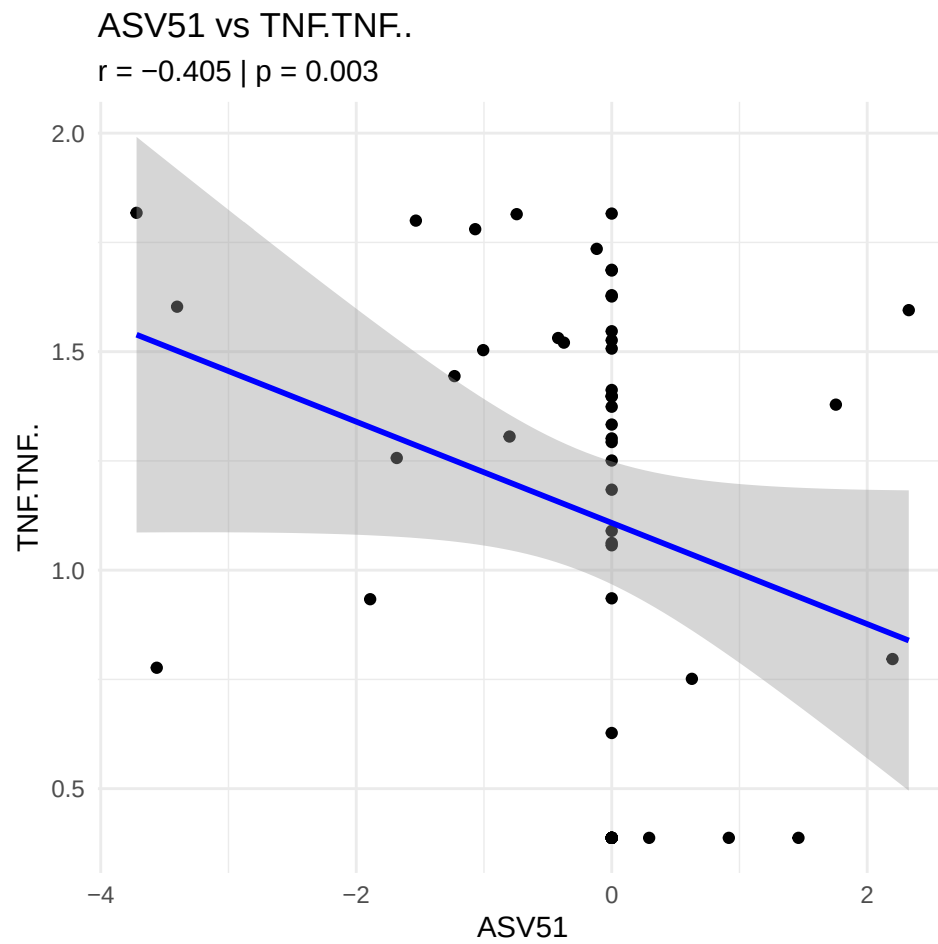
```
## 'geom_smooth()' using formula = 'y ~ x'
```



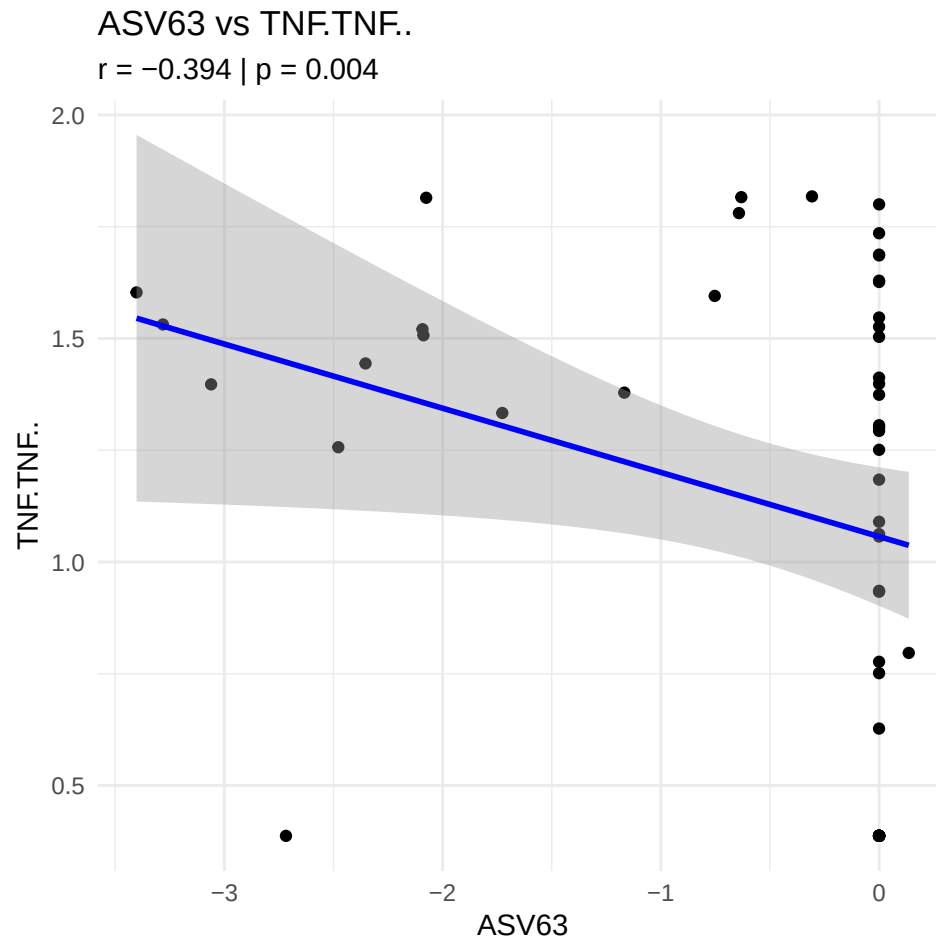
```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```



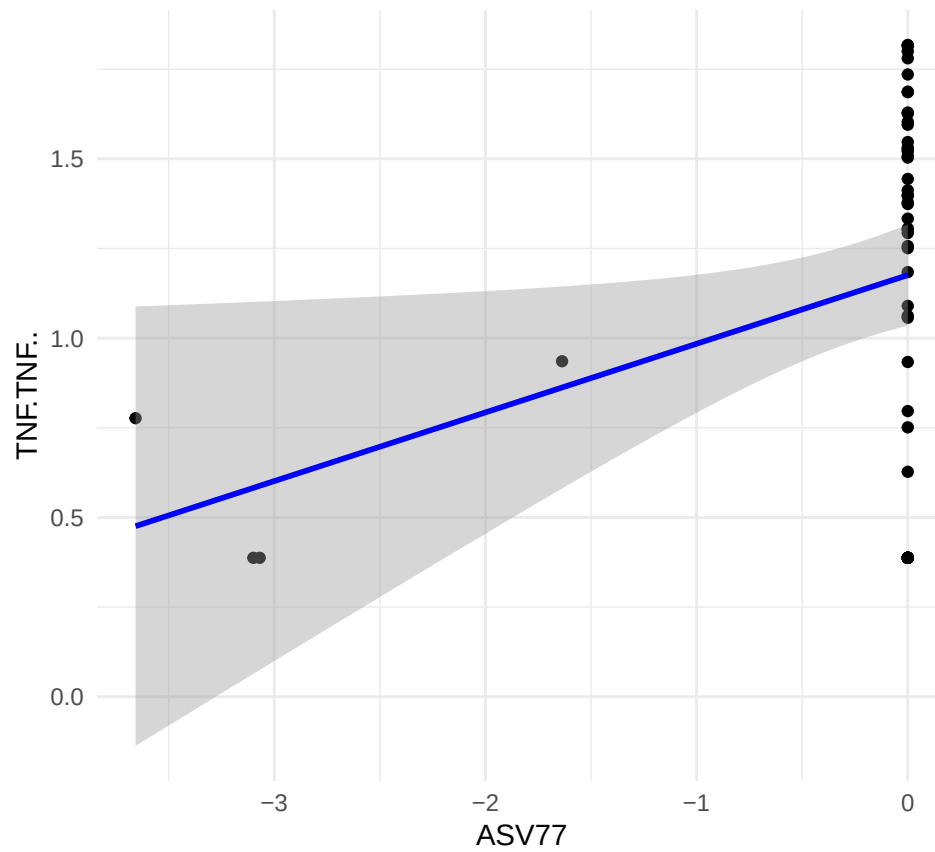
```
## 'geom_smooth()' using formula = 'y ~ x'
```

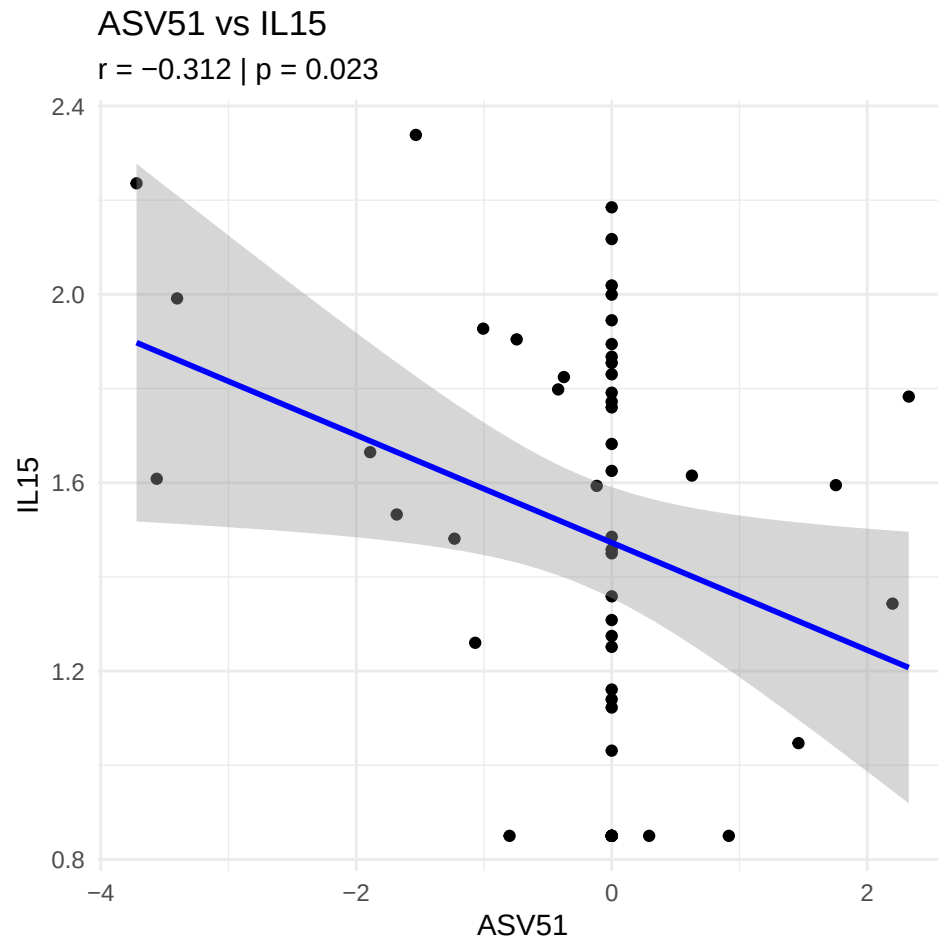
```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV77 vs TNF.TNF..

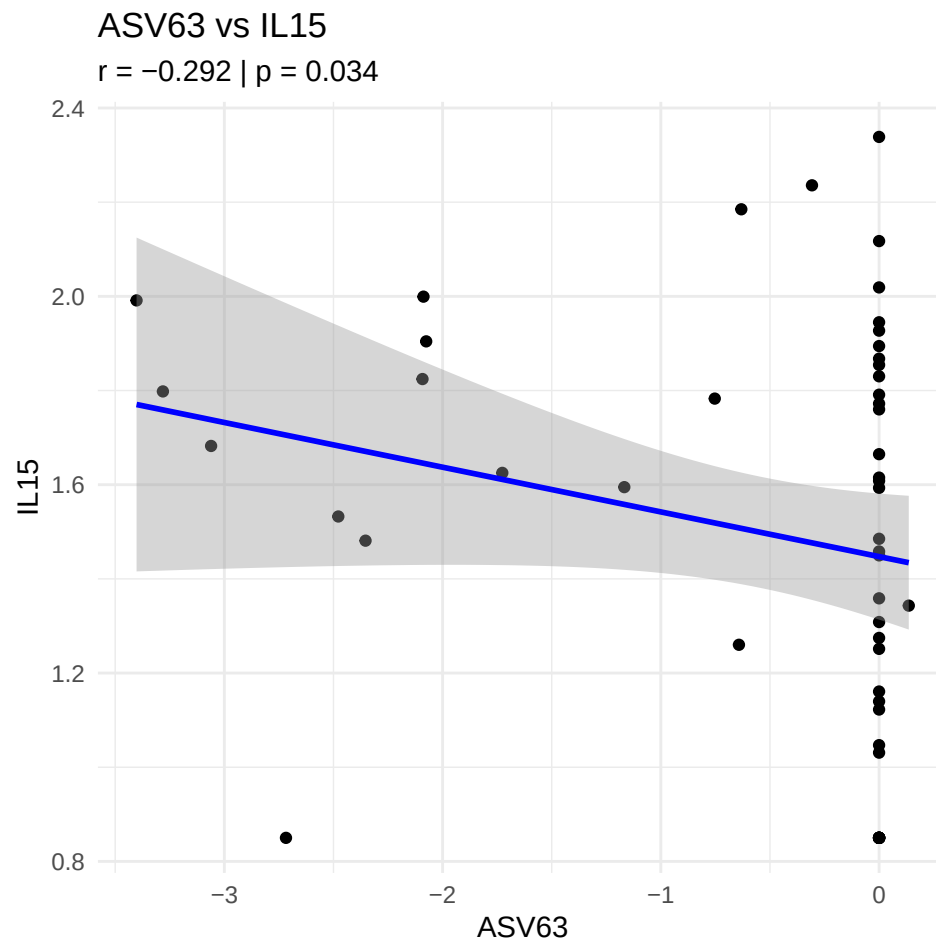
$r = 0.278$ | $p = 0.044$



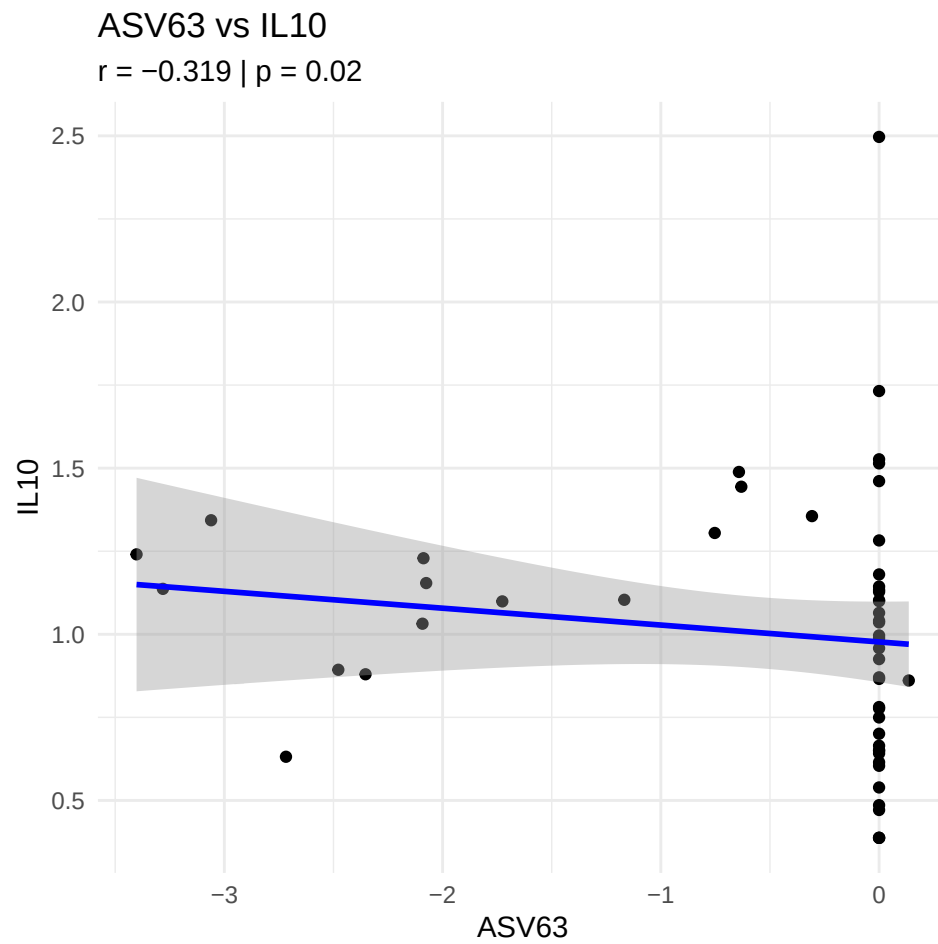
```
## 'geom_smooth()' using formula = 'y ~ x'
```



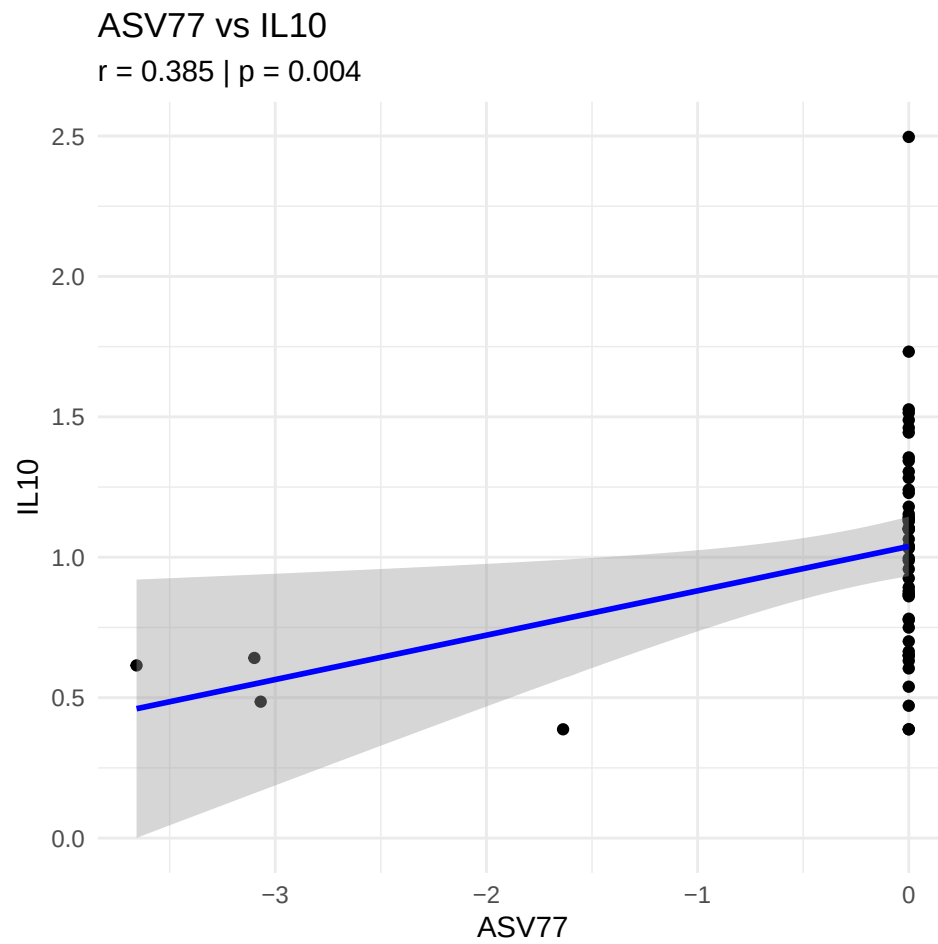
```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```



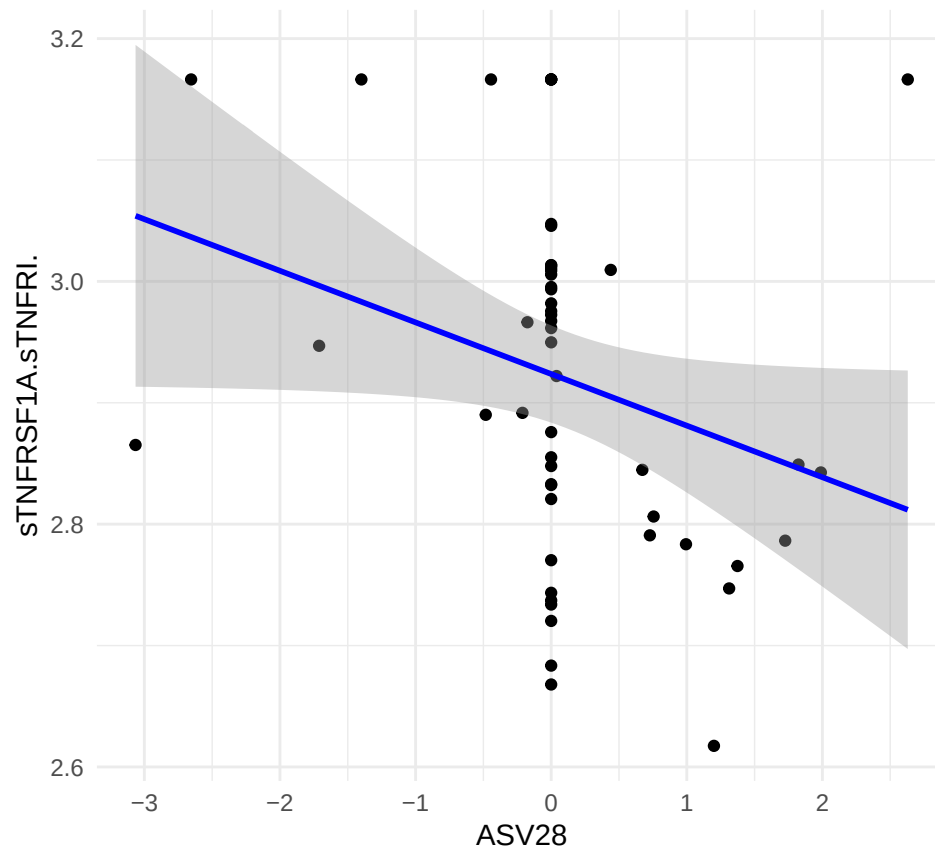
```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV28 vs sTNFRSF1A.sTNFRI.

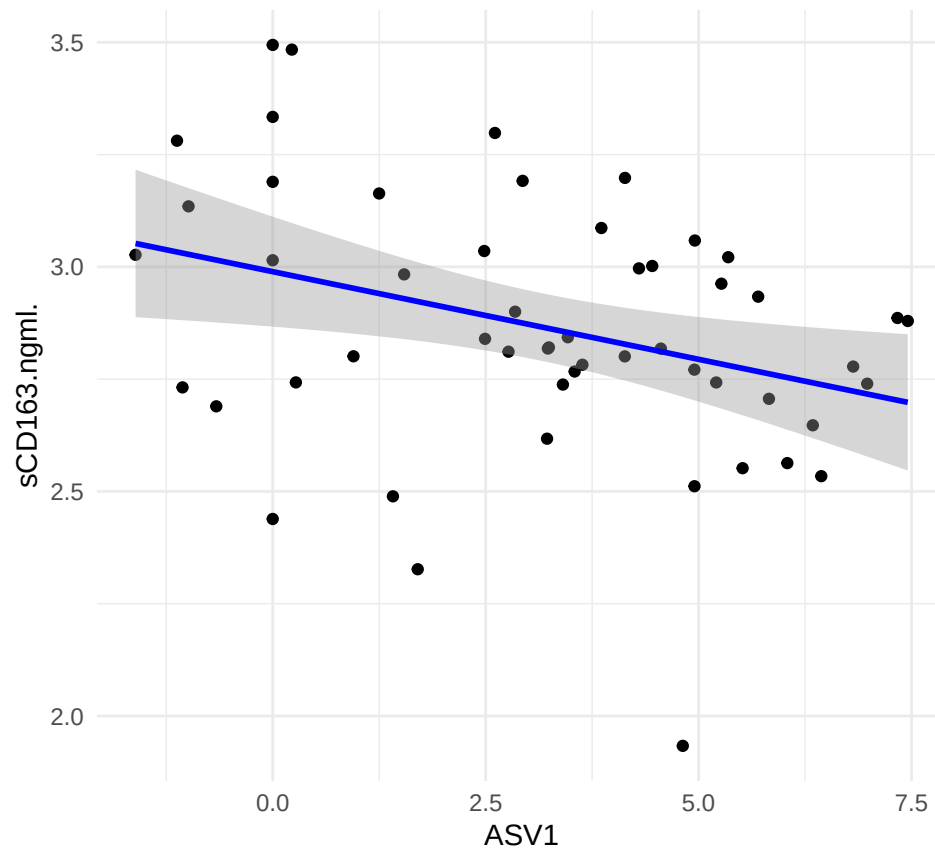
$r = -0.342$ | $p = 0.012$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV1 vs sCD163.ngml.

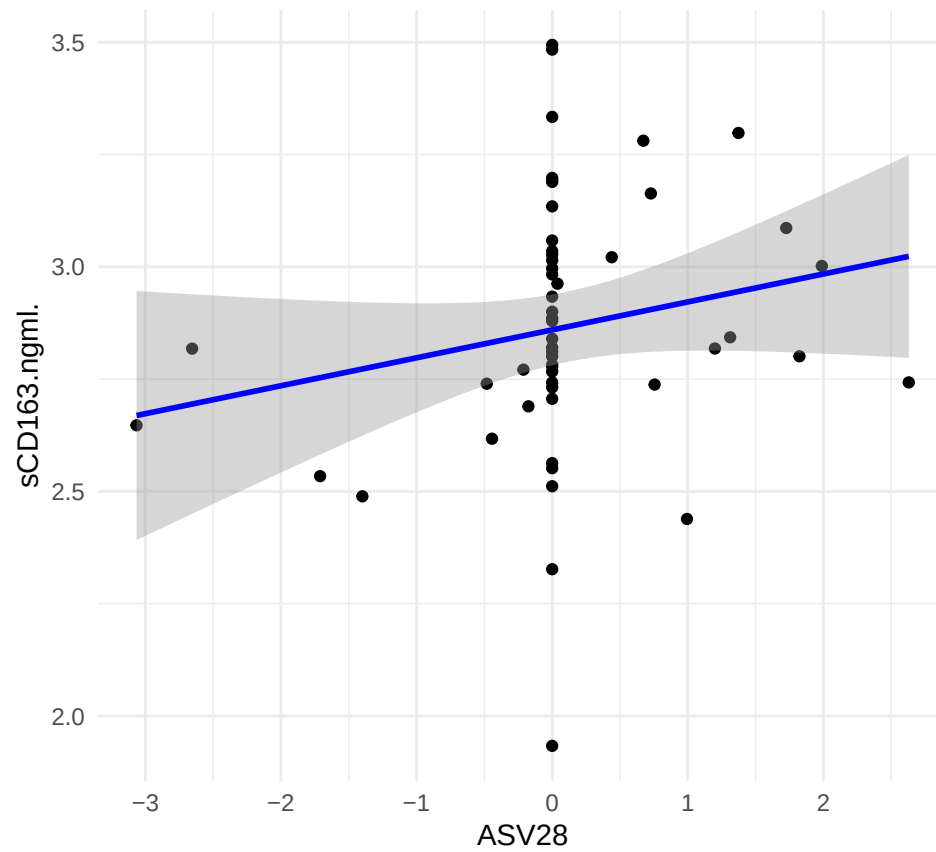
$r = -0.321$ | $p = 0.019$



```
## 'geom_smooth()' using formula = 'y ~ x'
```


ASV28 vs sCD163.ngml.

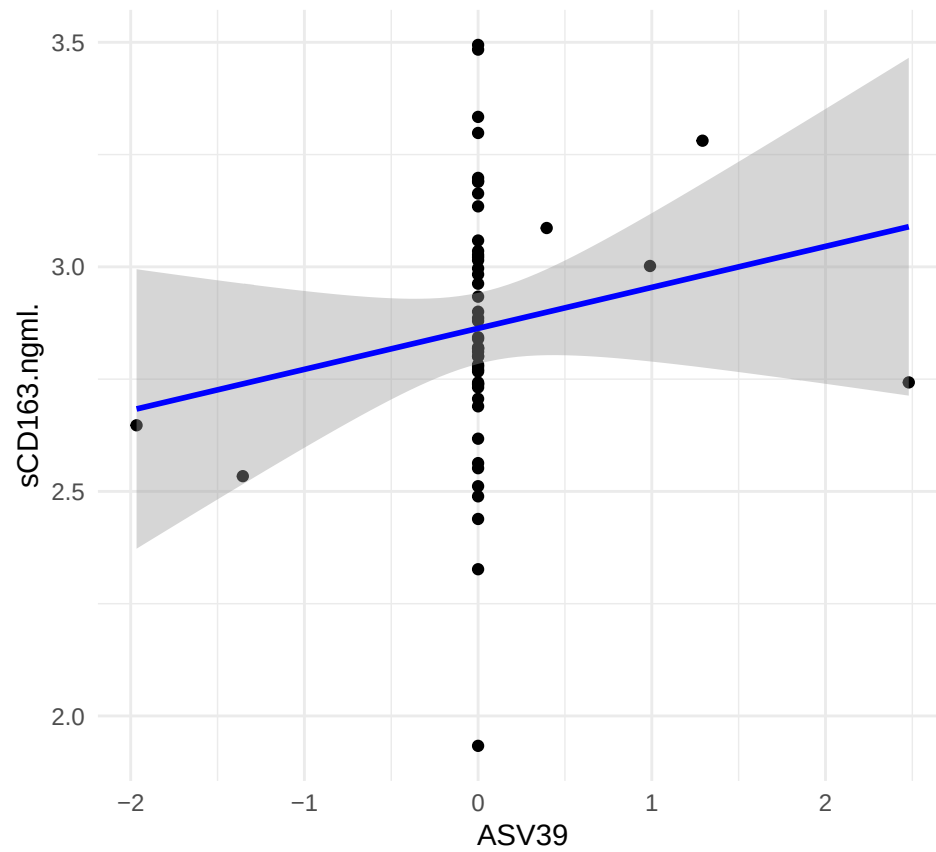
$r = 0.321$ | $p = 0.019$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV39 vs sCD163.ngml.

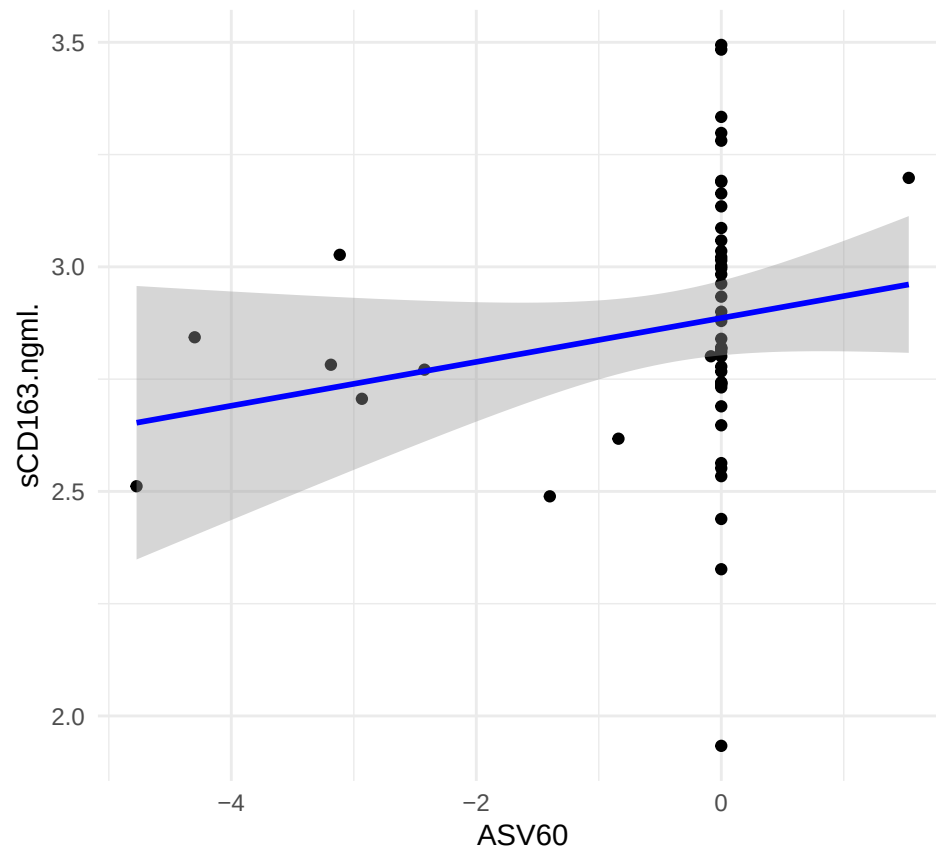
$r = 0.273$ | $p = 0.048$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV60 vs sCD163.ngml.

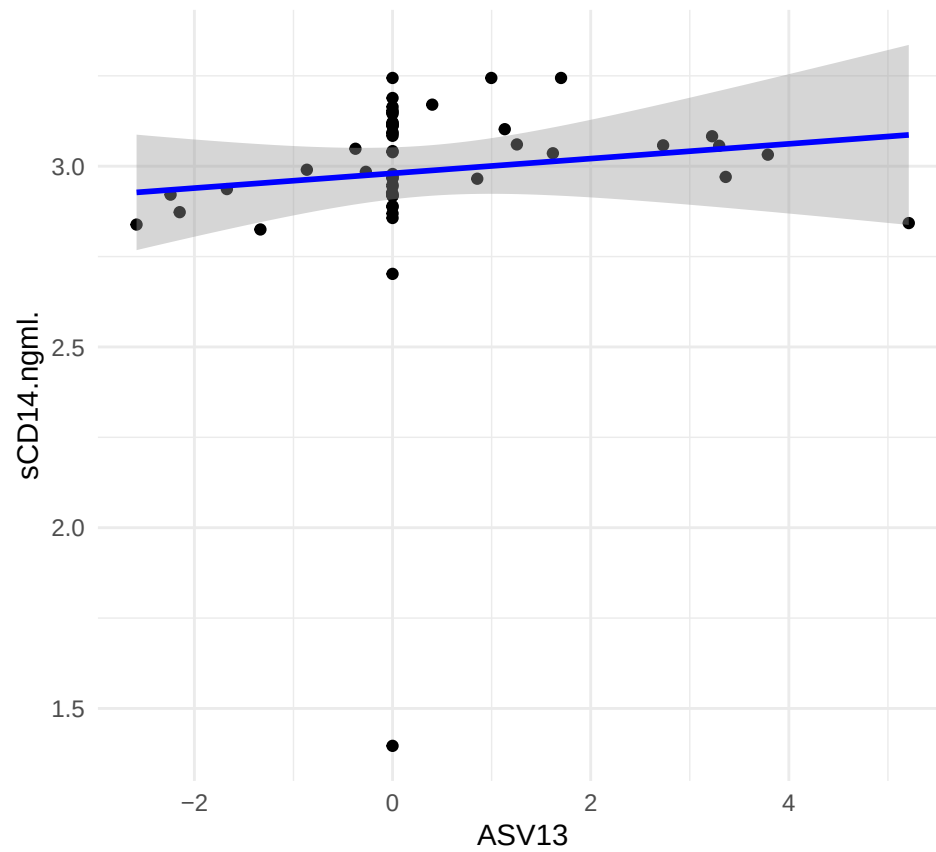
$r = 0.298$ | $p = 0.03$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV13 vs sCD14.ngml.

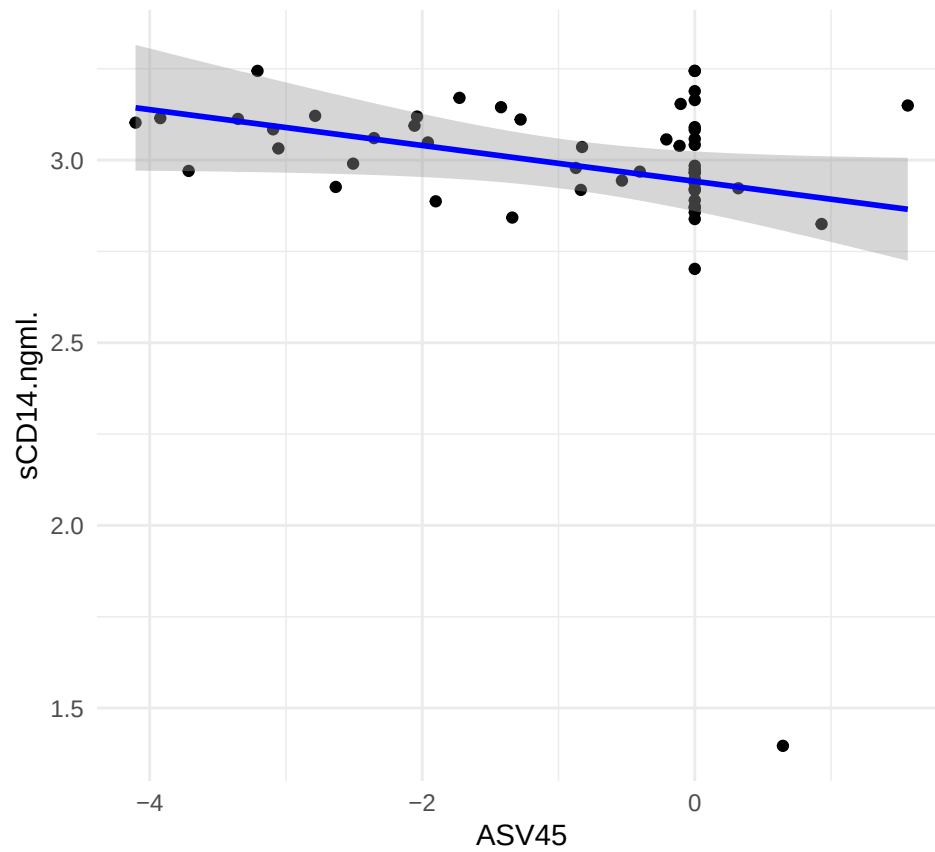
$r = 0.315$ | $p = 0.022$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV45 vs sCD14.ngml.

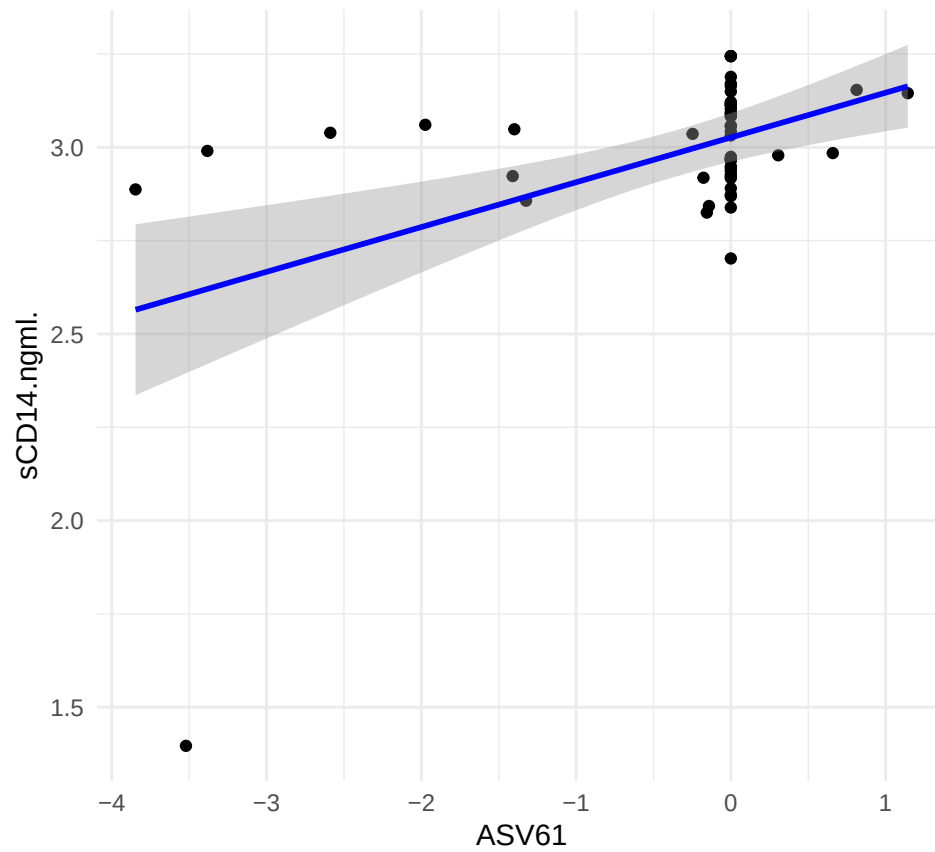
$r = -0.345$ | $p = 0.011$



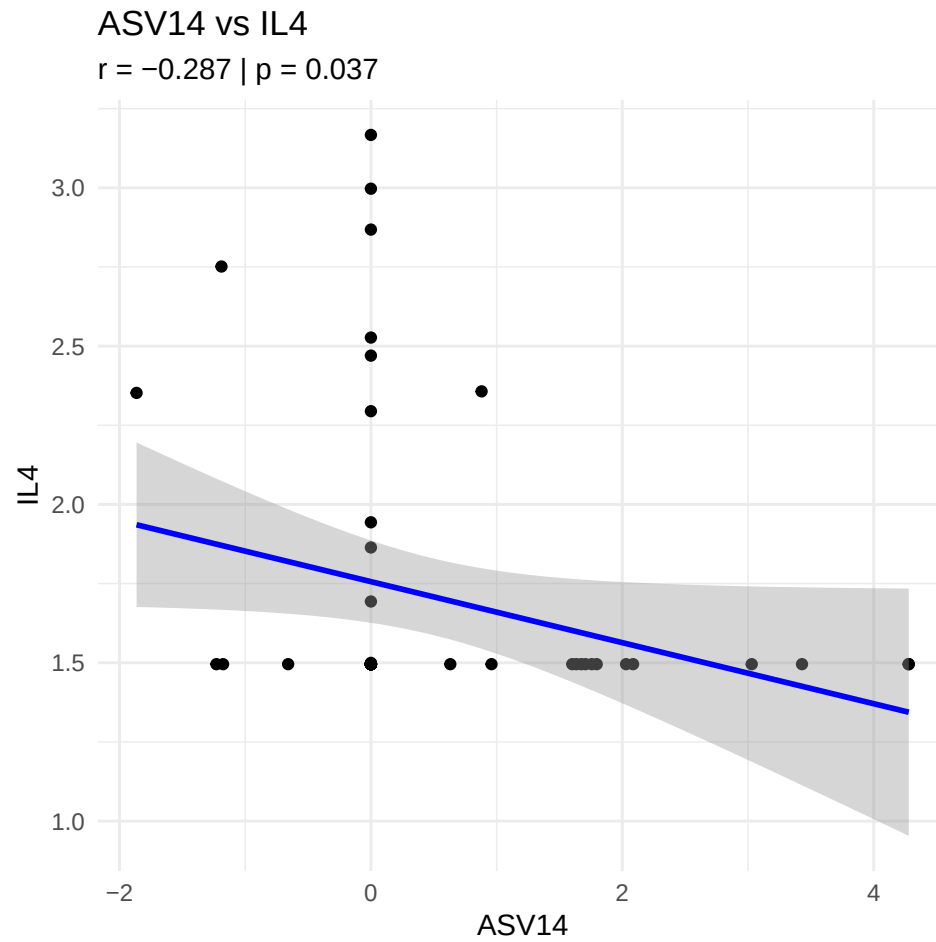
```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV61 vs sCD14.ngml.

$r = 0.368$ | $p = 0.007$



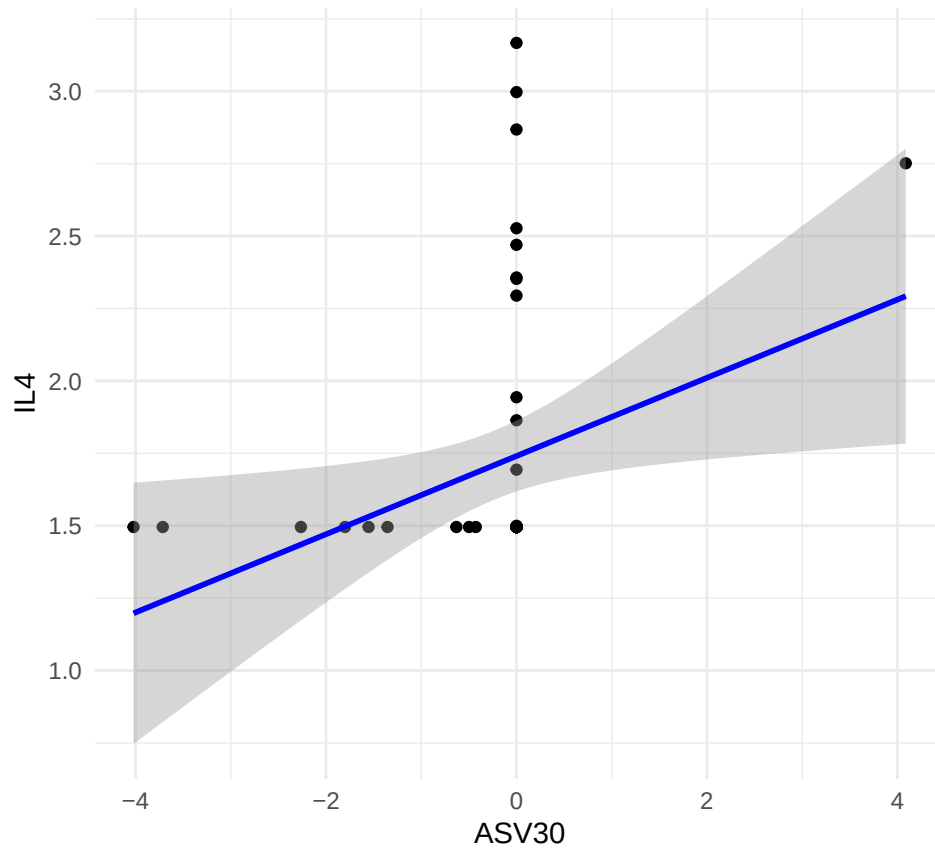
```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

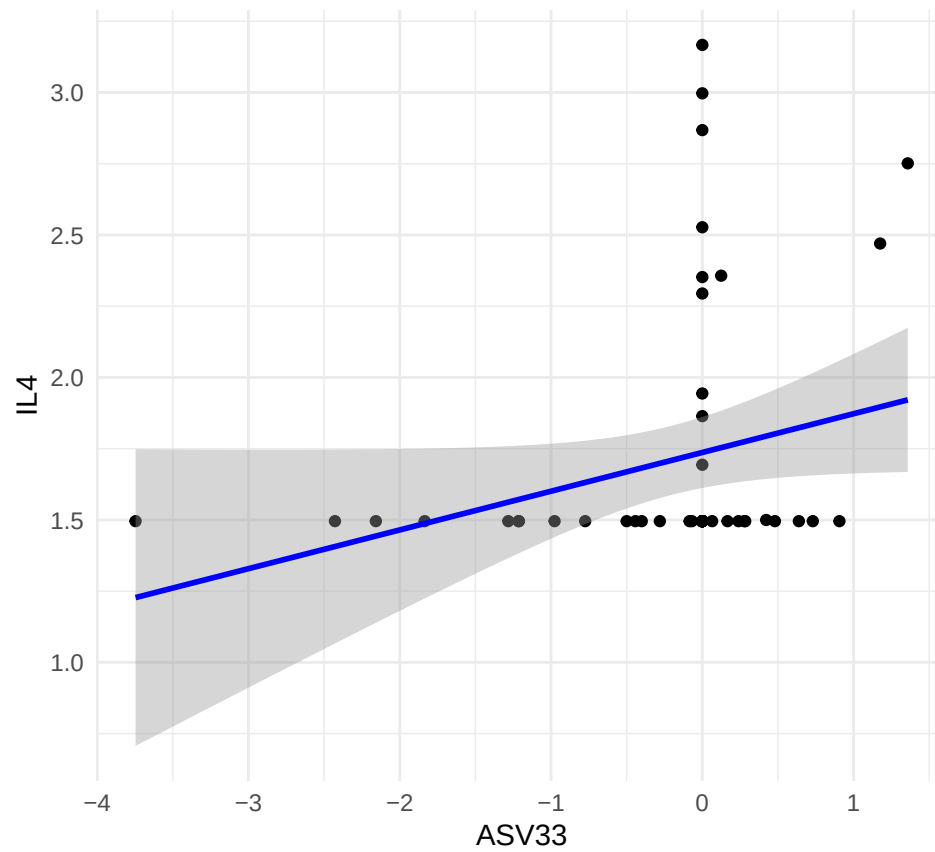
ASV30 vs IL4

$r = 0.317 \mid p = 0.021$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

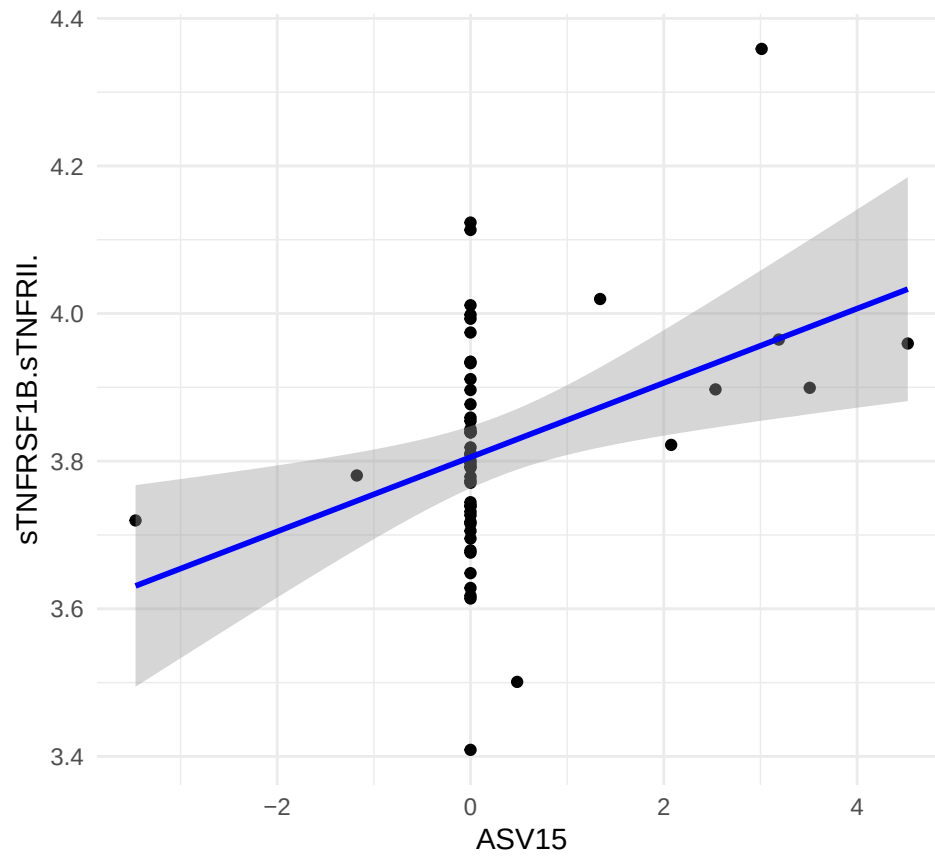

ASV33 vs IL4
 $r = 0.282 \mid p = 0.04$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV15 vs sTNFRSF1B.sTNFRII.

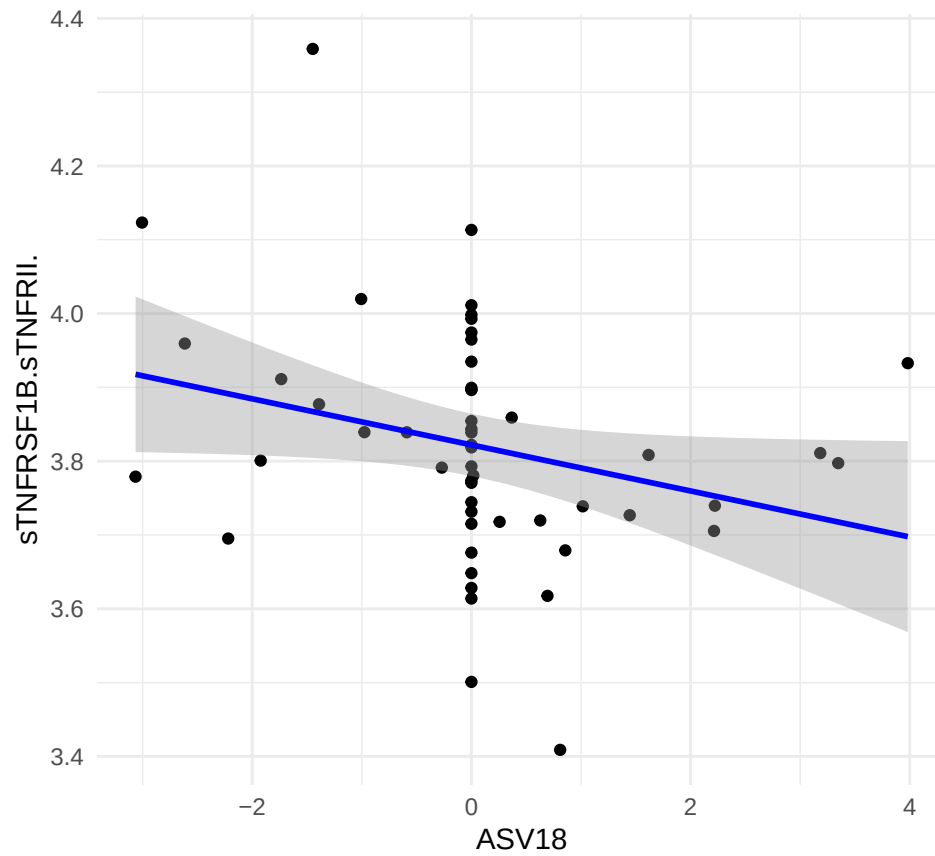
$r = 0.331$ | $p = 0.015$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV18 vs sTNFRSF1B.sTNFRII.

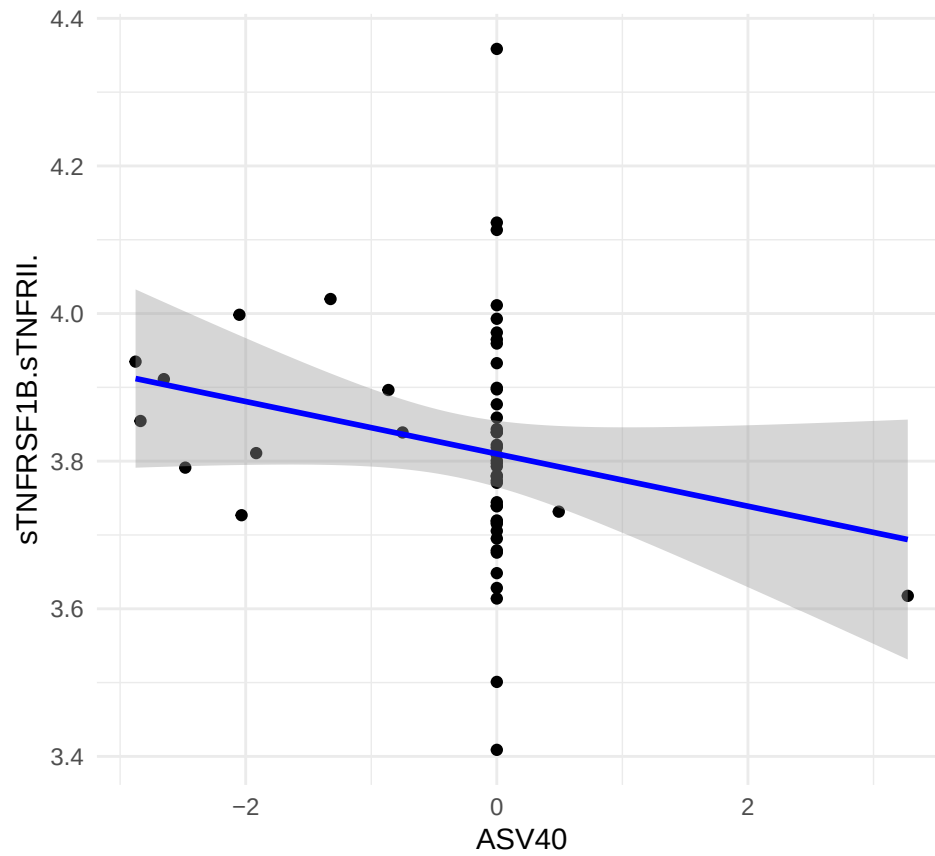
$r = -0.363$ | $p = 0.008$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV40 vs sTNFRSF1B.sTNFRII.

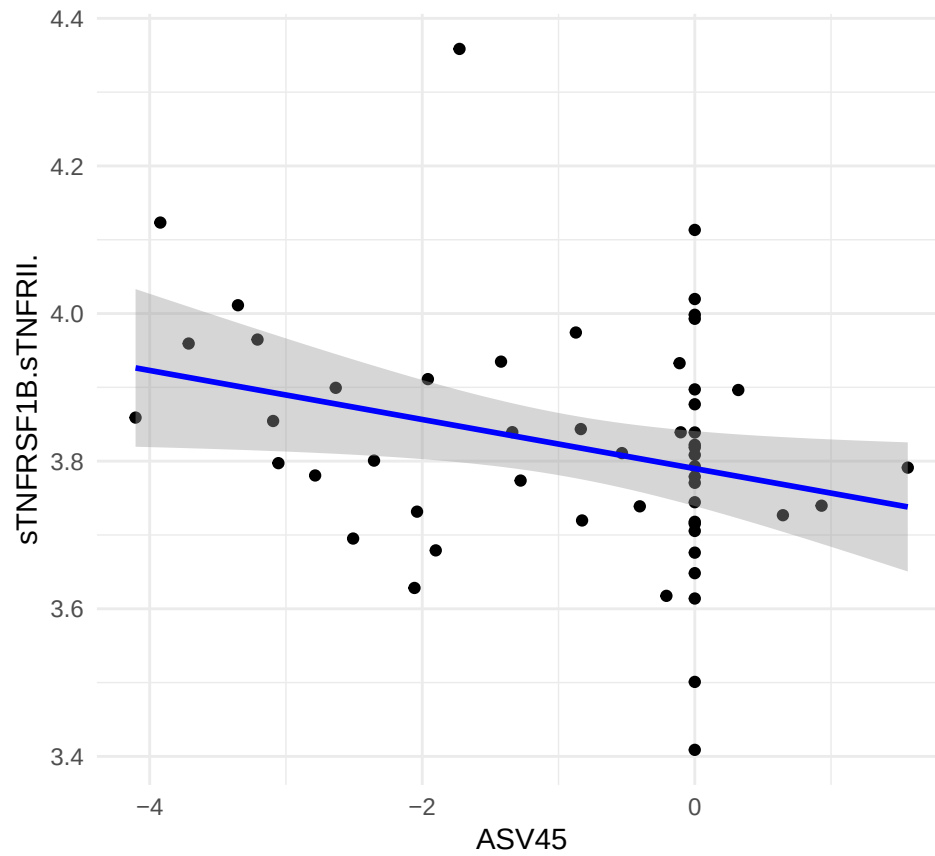
$r = -0.314$ | $p = 0.022$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV45 vs sTNFRSF1B.sTNFRII.

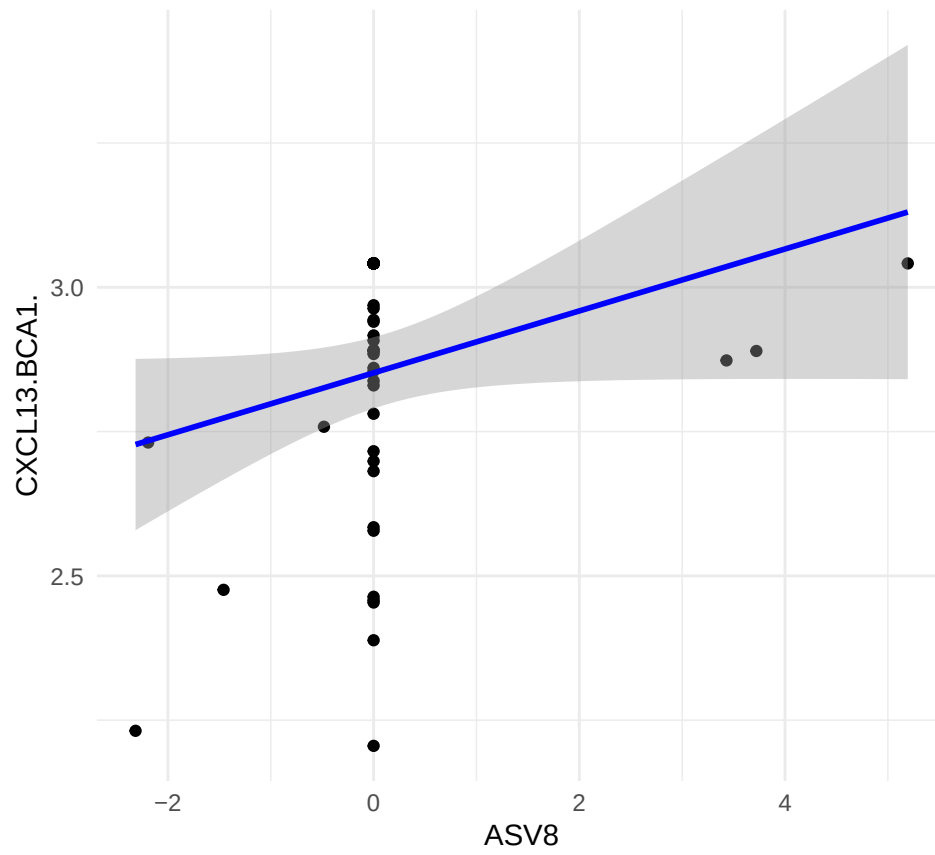
$r = -0.275$ | $p = 0.046$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV8 vs CXCL13.BCA1.

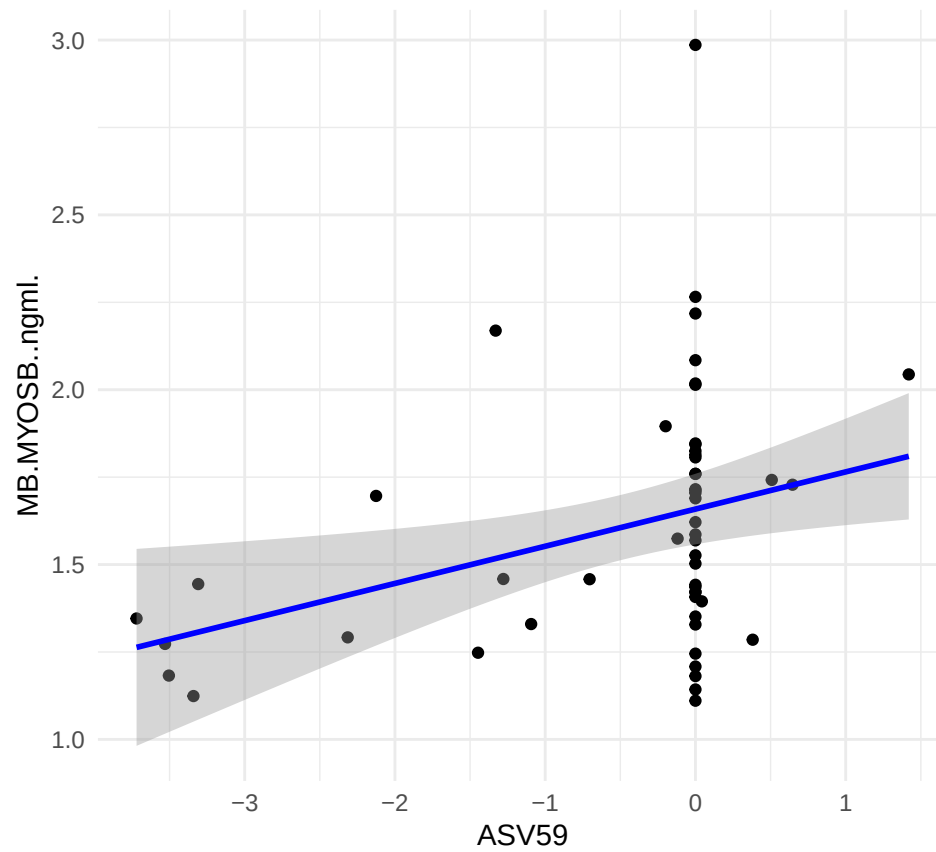
$r = 0.277$ | $p = 0.045$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV59 vs MB.MYOSB..ngml.

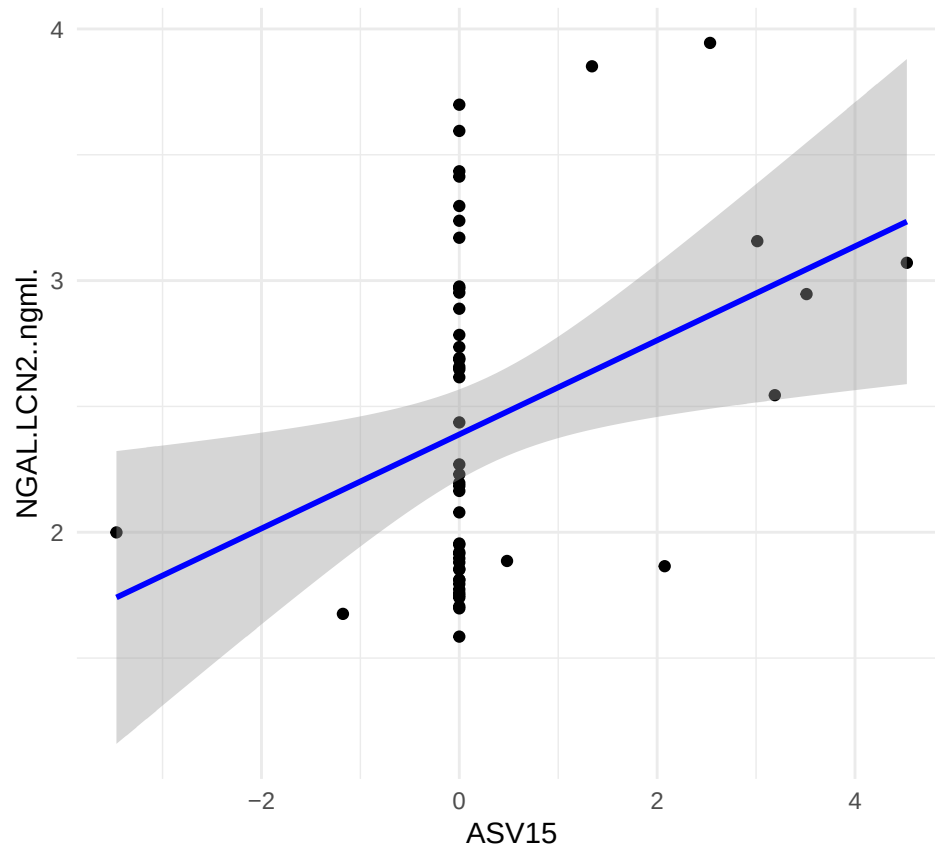
$r = 0.281$ | $p = 0.041$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV15 vs NGAL.LCN2..ngml.

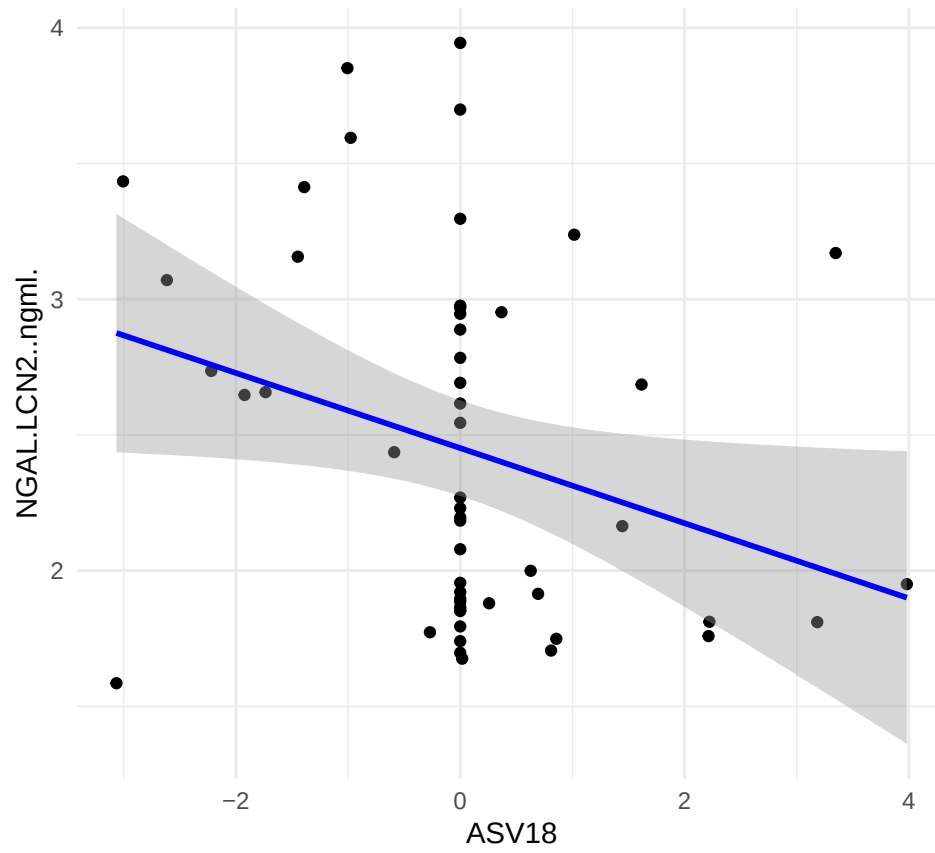
$r = 0.314$ | $p = 0.022$



```
## 'geom_smooth()' using formula = 'y ~ x'
```


ASV18 vs NGAL.LCN2..ngml.

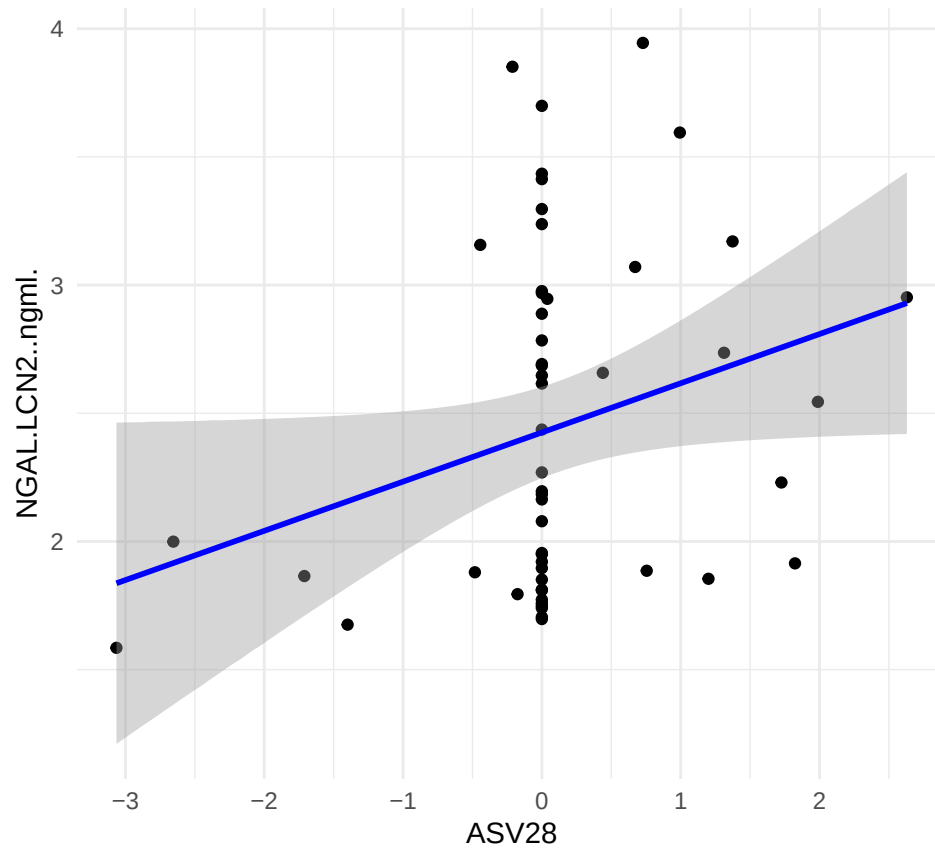
$r = -0.325$ | $p = 0.017$



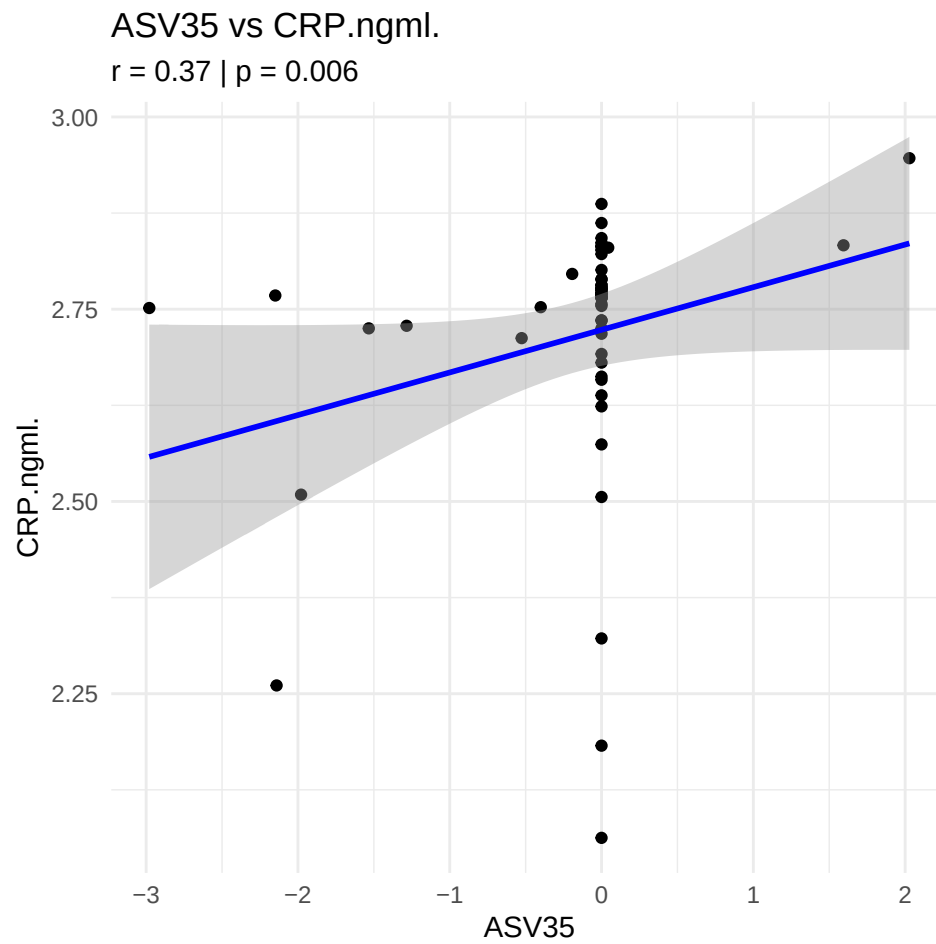
```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV28 vs NGAL.LCN2..ngml.

$r = 0.294$ | $p = 0.033$



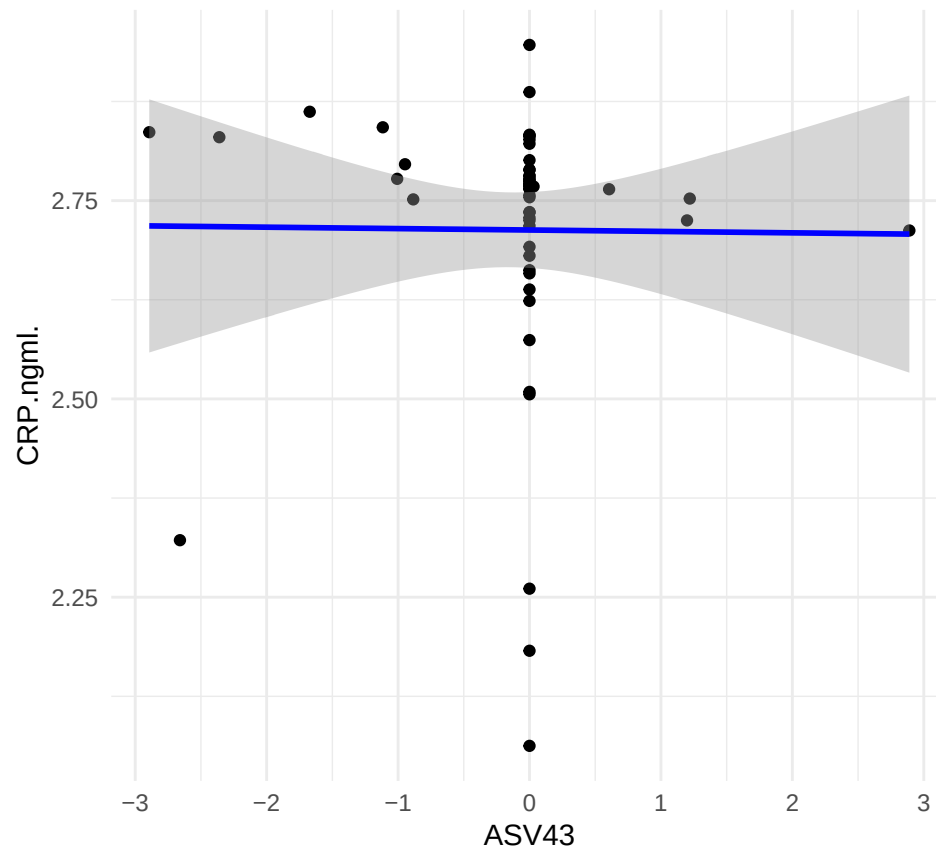
```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV43 vs CRP.ngml.

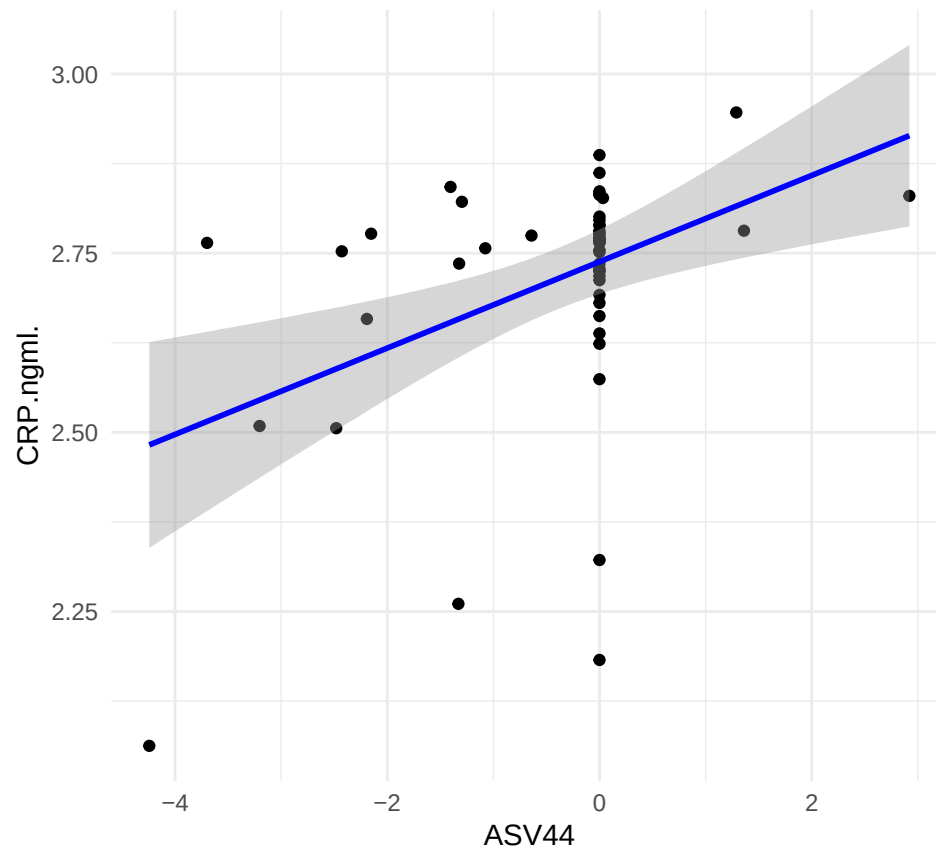
$r = -0.283$ | $p = 0.04$



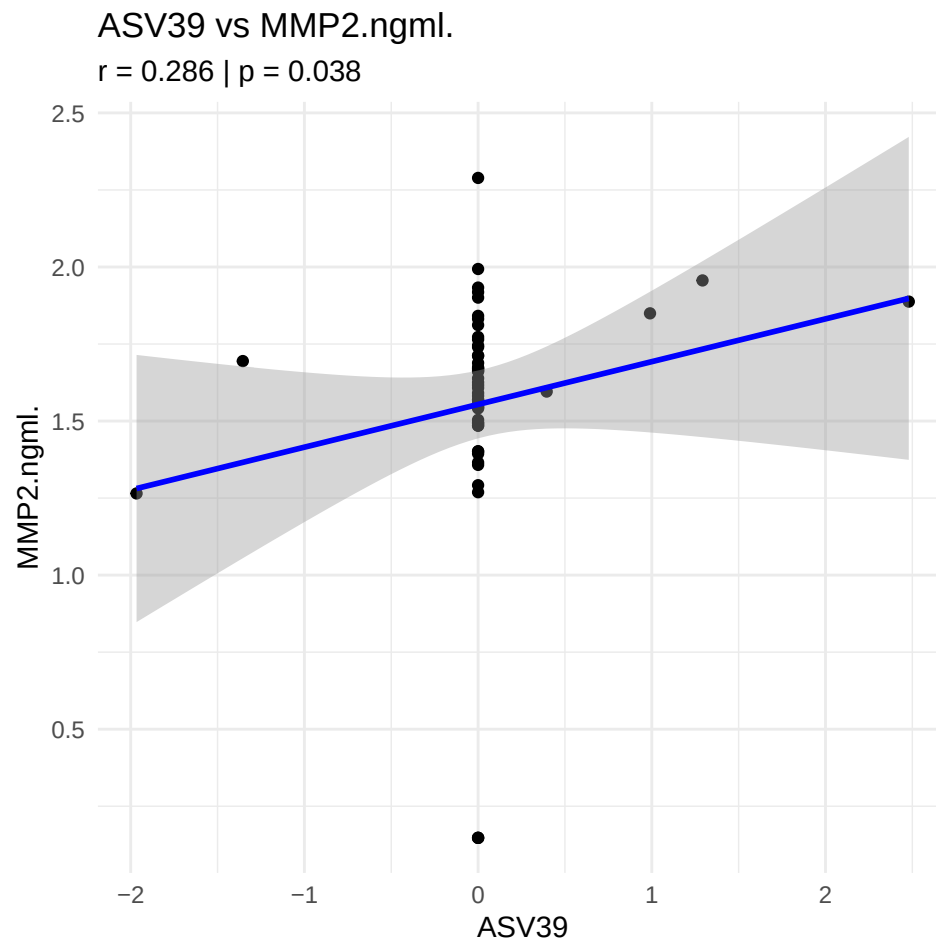
```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV44 vs CRP.ngml.

$r = 0.353$ | $p = 0.01$



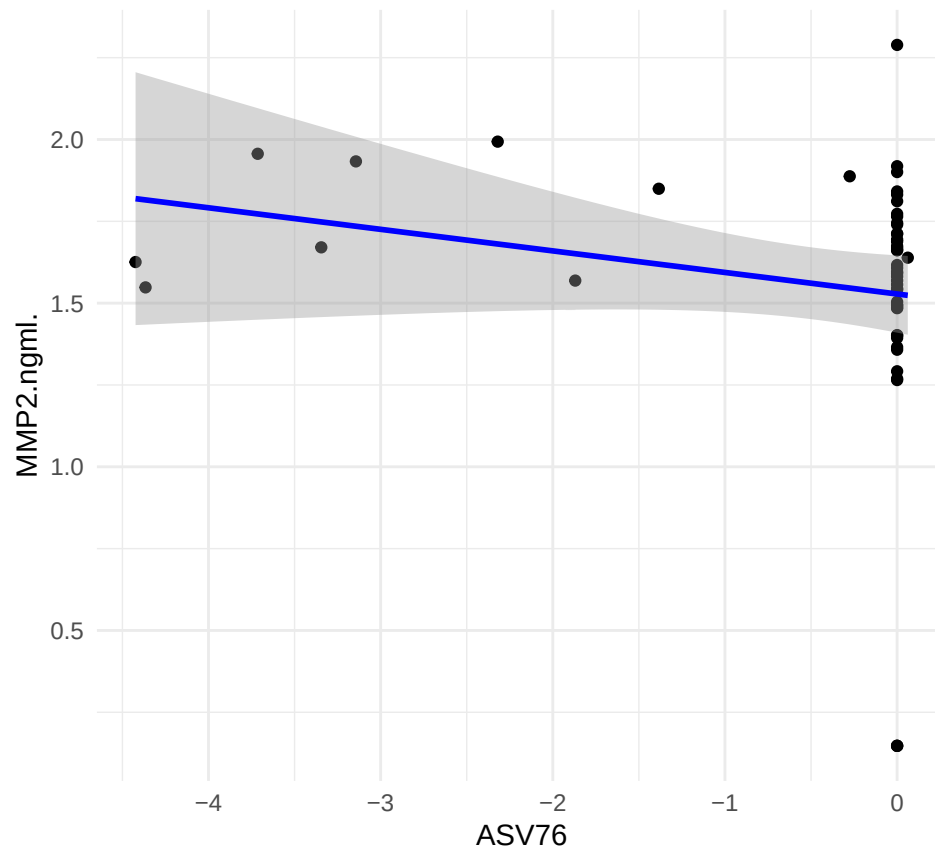
```
## 'geom_smooth()' using formula = 'y ~ x'
```



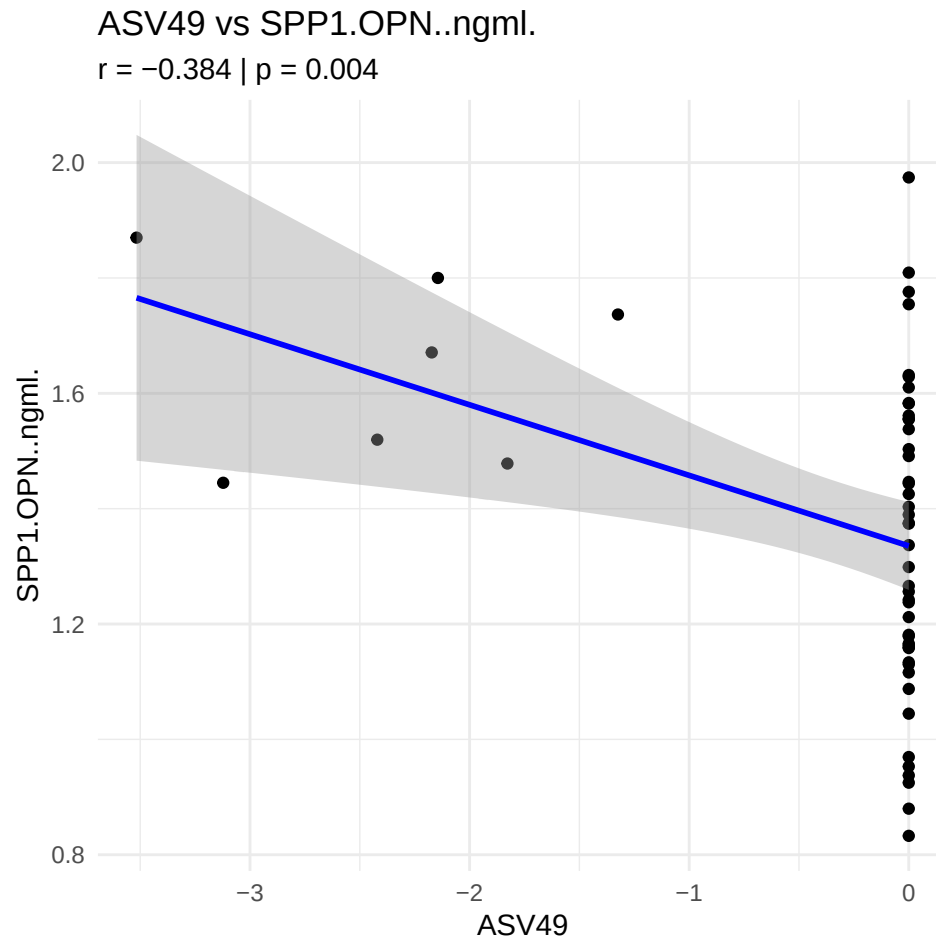
```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV76 vs MMP2.ngml.

$r = -0.3$ | $p = 0.029$



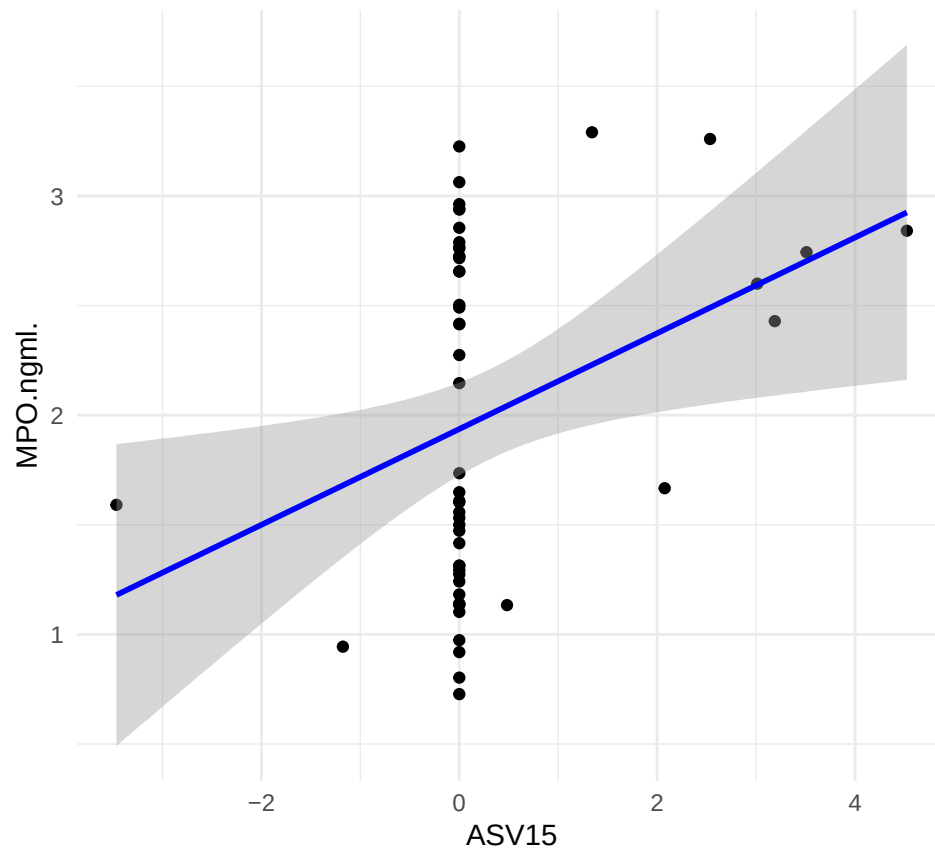
```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```


ASV15 vs MPO.ngml.

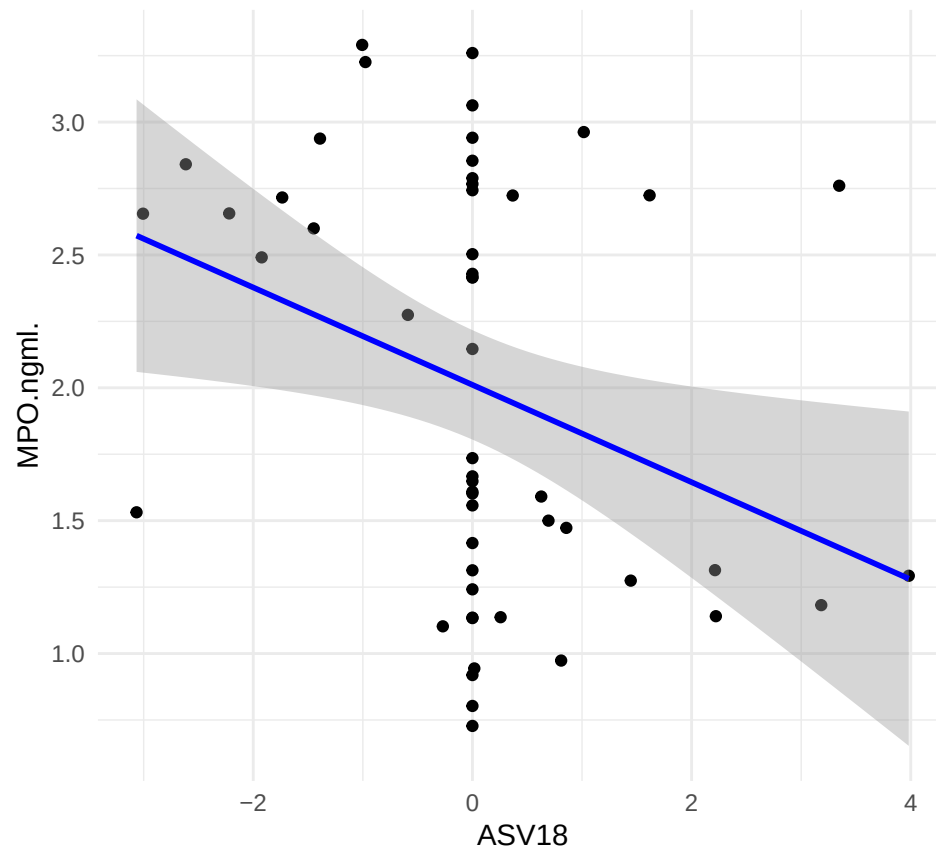
$r = 0.311$ | $p = 0.023$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV18 vs MPO.ngml.

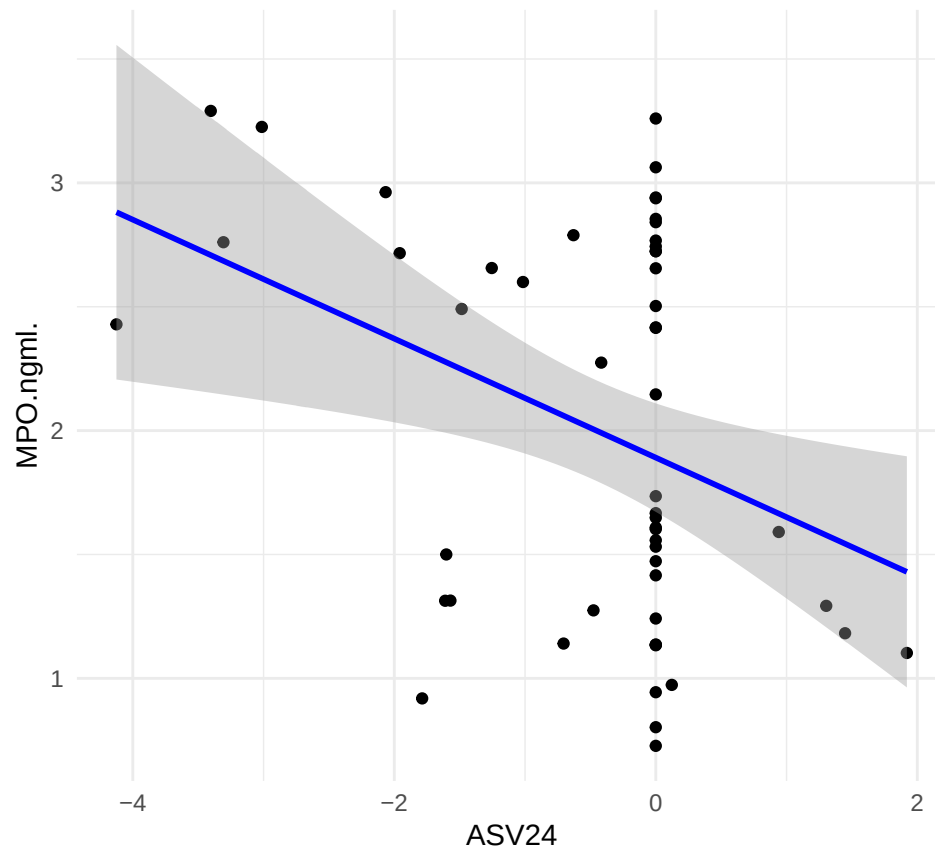
$r = -0.329$ | $p = 0.016$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV24 vs MPO.ngml.

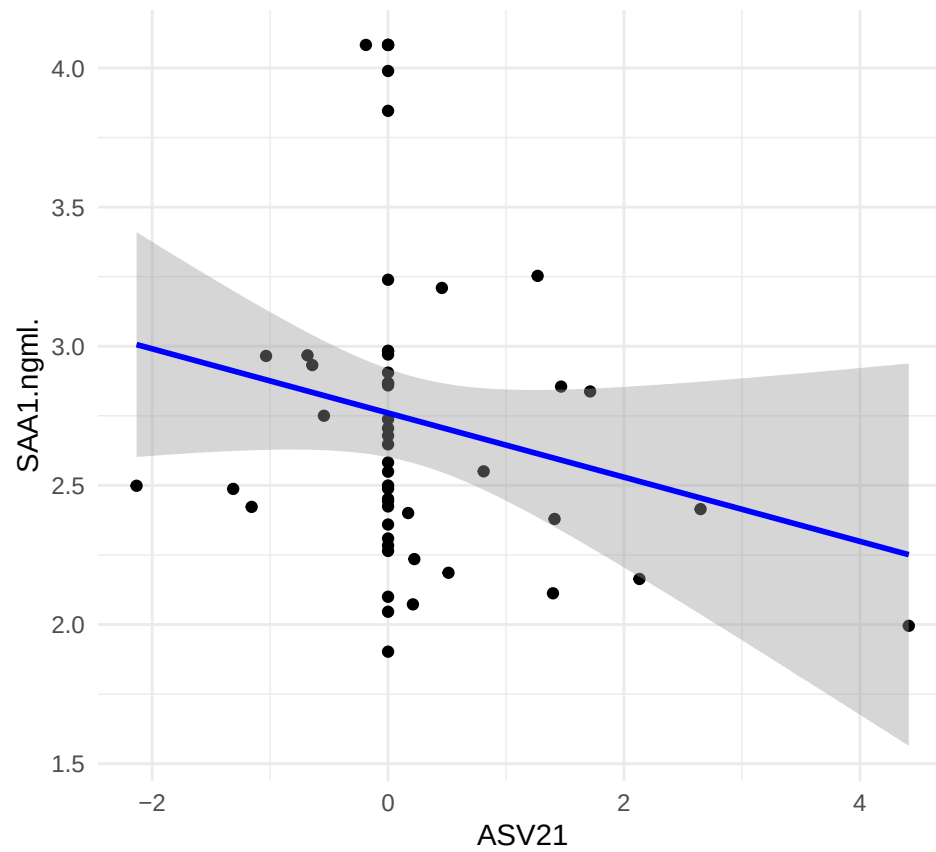
$r = -0.3 \mid p = 0.029$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV21 vs SAA1.ngml.

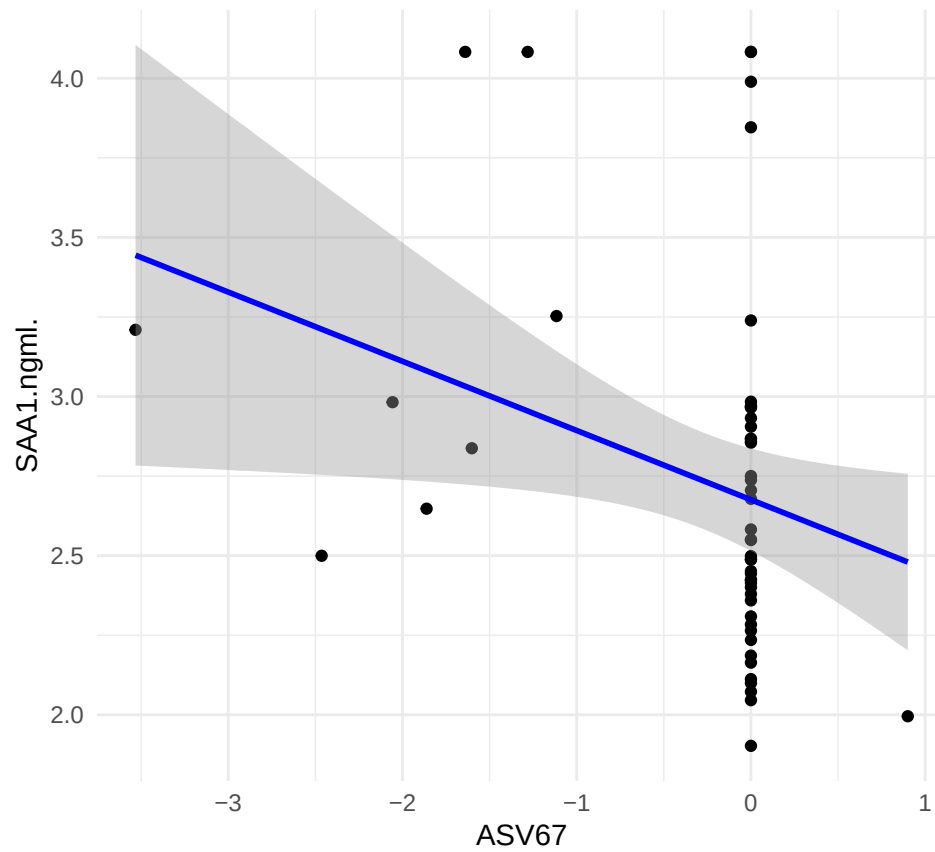
$r = -0.302$ | $p = 0.028$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV67 vs SAA1.ngml.

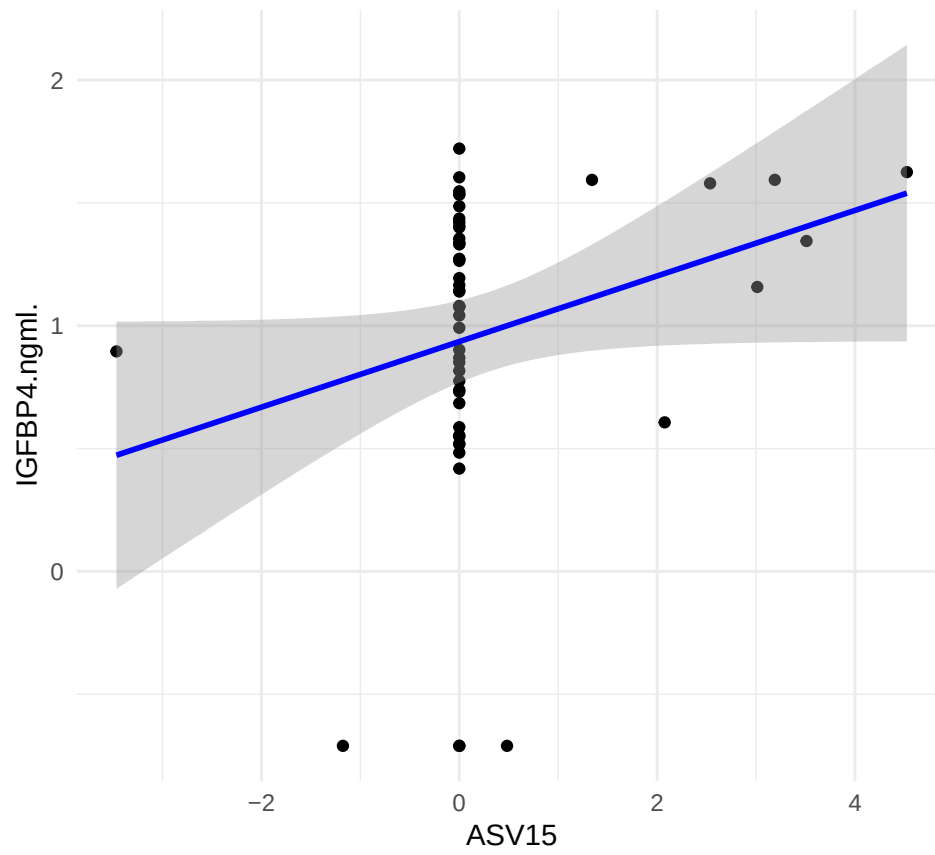
$r = -0.403$ | $p = 0.003$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV15 vs IGFBP4.ngml.

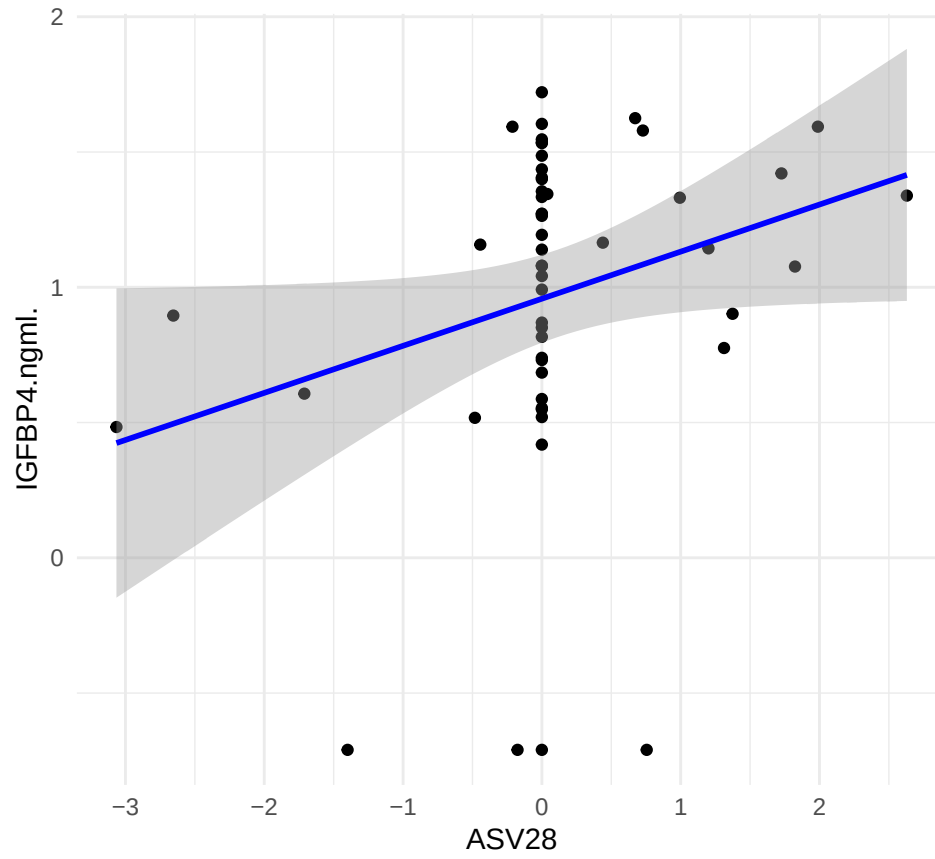
$r = 0.3 \mid p = 0.029$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV28 vs IGFBP4.ngml.

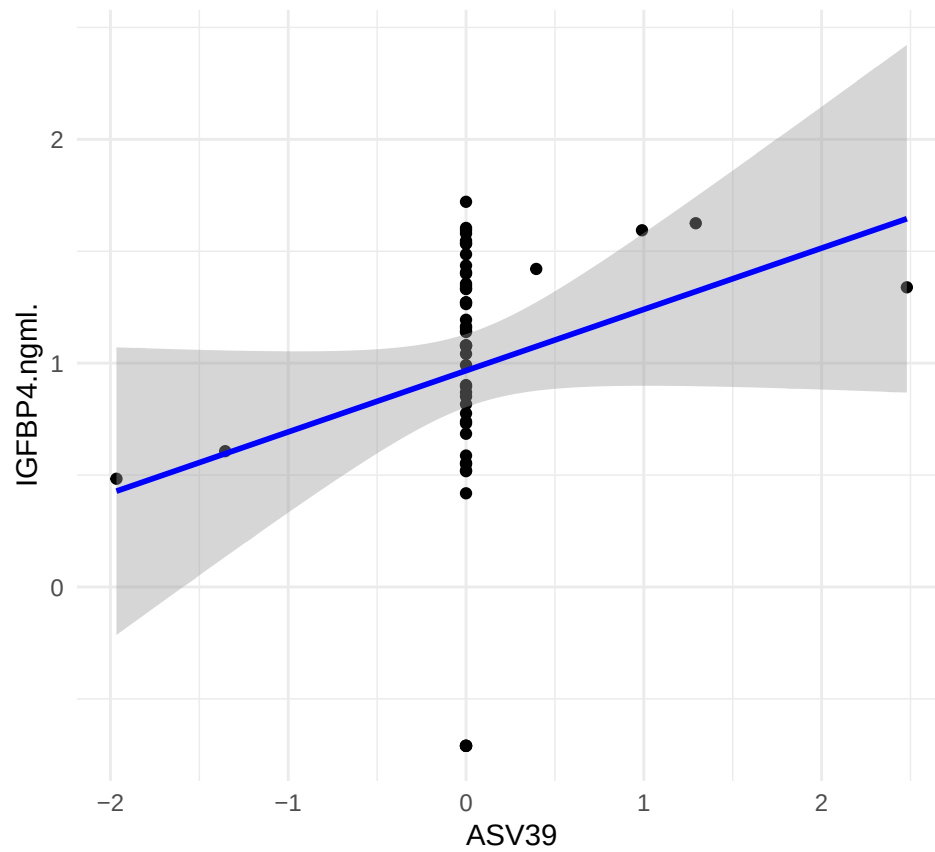
$r = 0.316$ | $p = 0.021$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV39 vs IGFBP4.ngml.

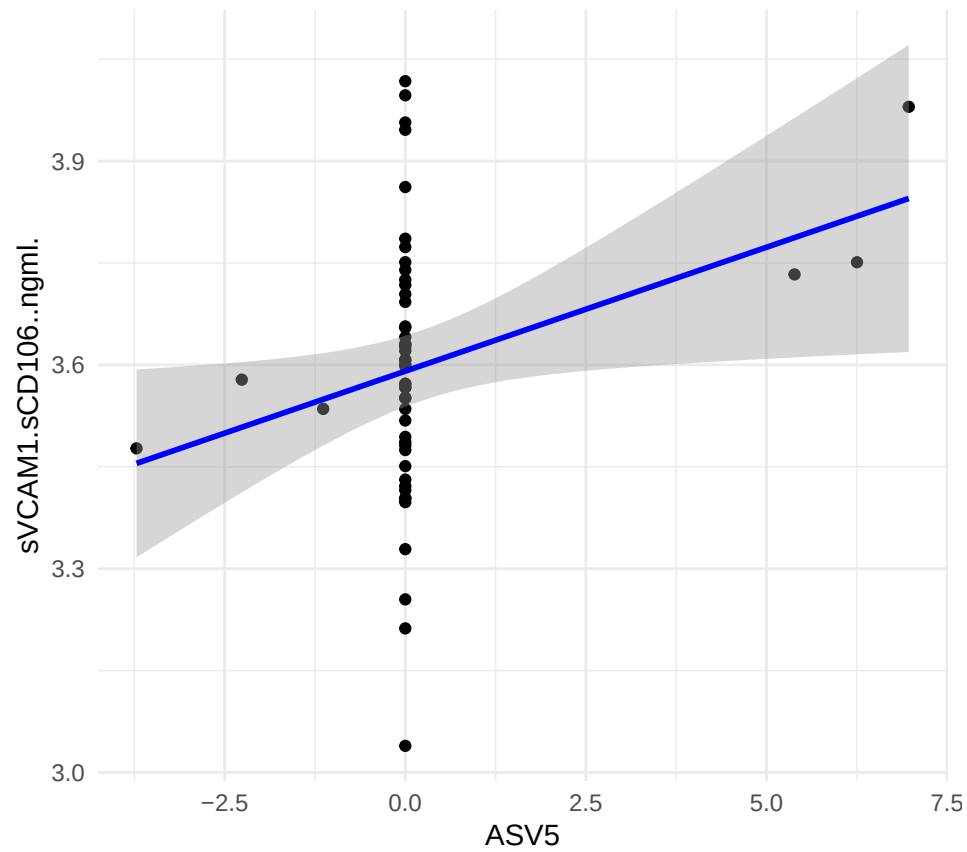
$r = 0.402$ | $p = 0.003$



```
## 'geom_smooth()' using formula = 'y ~ x'
```


ASV5 vs sVCAM1.sCD106..ngml.

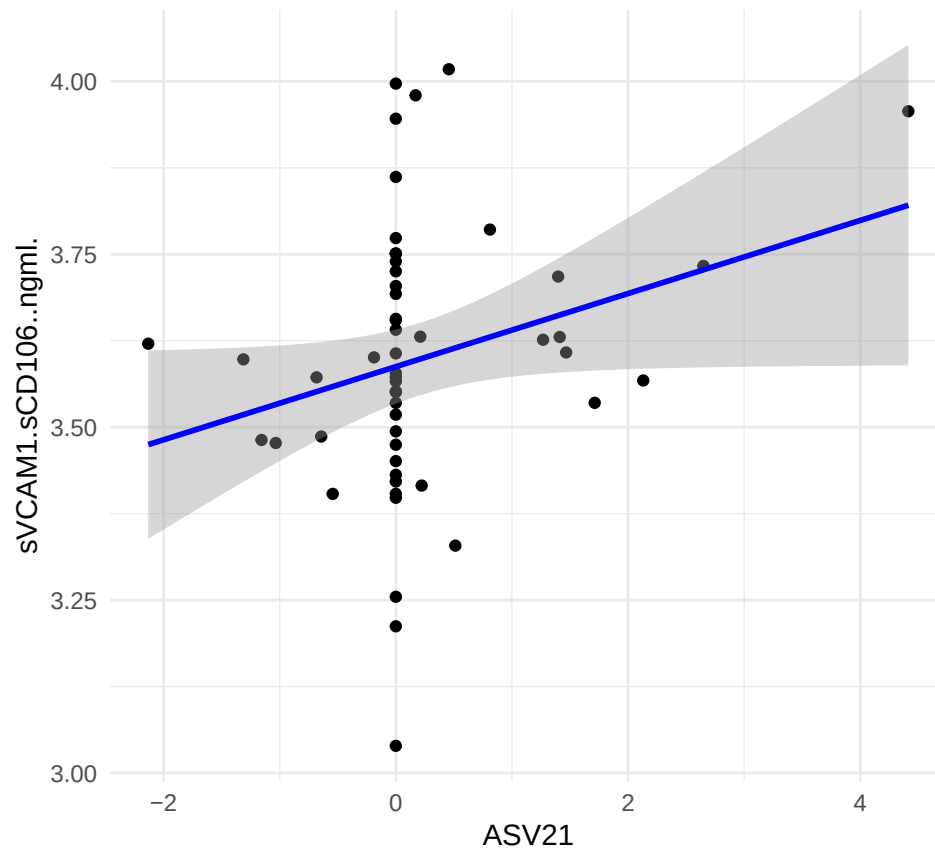
$r = 0.292$ | $p = 0.034$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV21 vs sVCAM1.sCD106..ngml.

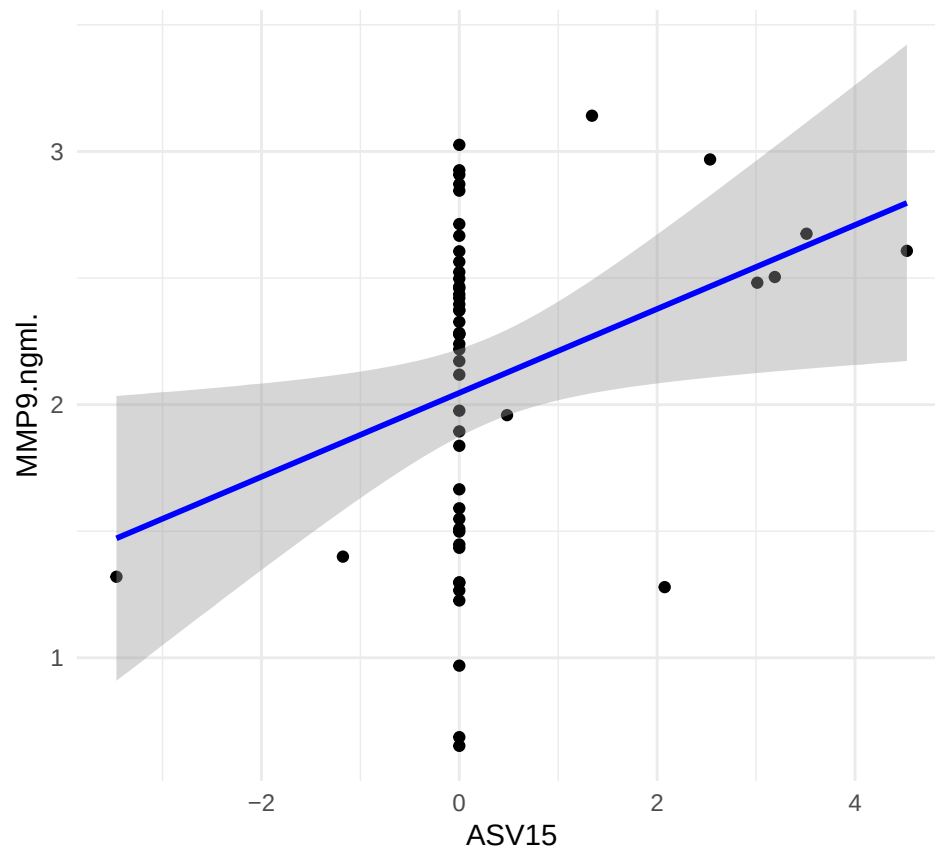
$r = 0.278$ | $p = 0.043$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV15 vs MMP9.ngml.

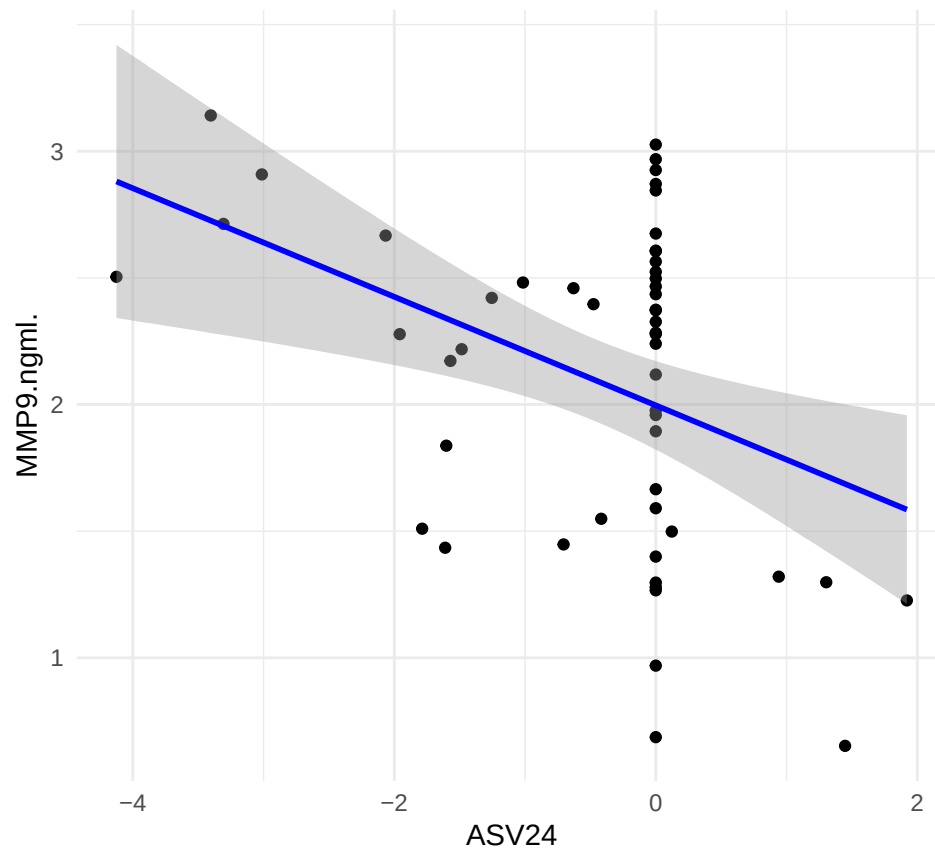
$r = 0.34$ | $p = 0.013$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV24 vs MMP9.ngml.

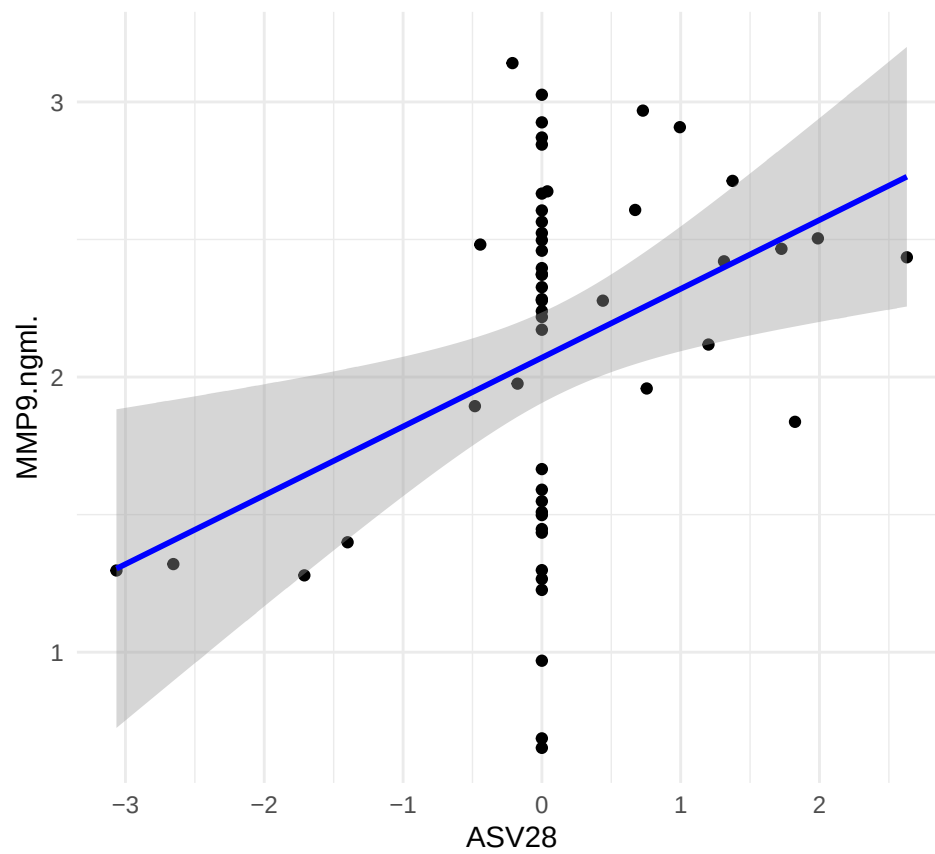
$r = -0.328$ | $p = 0.016$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV28 vs MMP9.ngml.

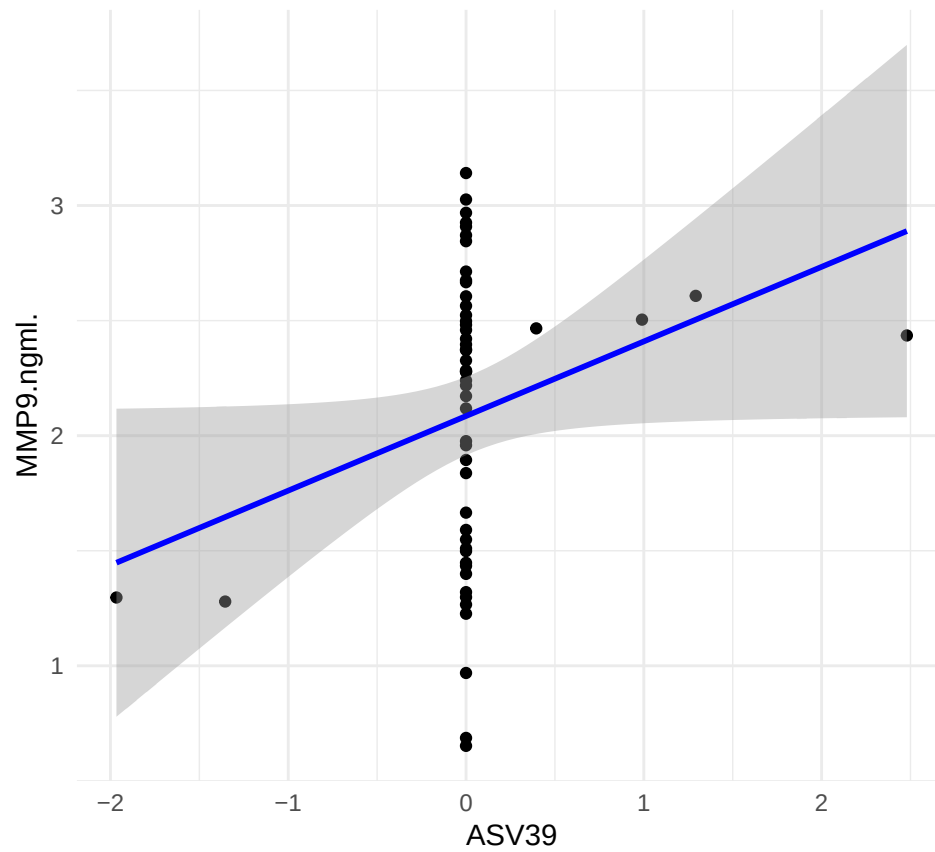
$r = 0.336$ | $p = 0.014$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV39 vs MMP9.ngml.

$r = 0.311$ | $p = 0.023$



```
## 'geom_smooth()' using formula = 'y ~ x'
```

ASV76 vs CST3.CystatinC..ngml.

$r = -0.322$ | $p = 0.019$

